

Relative entropy of mixtures - Numerics

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1 From conformal fields to explicit lattice states

1.1 What is being constructed and measured?

A periodic chain contains N two-level sites. A basis configuration is $|n_0 \cdots n_{N-1}\rangle$, with $n_j = 0, 1$. A wavefunction is a table of complex amplitudes, $|a\rangle = \sum_{\mathbf{n}} \psi_a(\mathbf{n})|\mathbf{n}\rangle$, normalized so that $\sum_{\mathbf{n}} |\psi_a(\mathbf{n})|^2 = 1$. An interval A consists of the first ℓ consecutive sites; its complement is $\bar{A}$. We trace out $\bar{A}$ to form

$$(\rho_A^a)_{uv} = \sum_w \psi_a(u, w) \psi_a(v, w)^*, \quad x = \ell/N. \quad (1)$$

Here u, v label interval configurations, w labels the complement, and the star denotes complex conjugation. Throughout this report,

$$m_{ab} = \frac{1}{2}(\rho_A^a + \rho_A^b), \quad D_{a|b} = \text{Tr } \rho_A^a (\log \rho_A^a - \log m_{ab}). \quad (2)$$

Logarithms are natural, so relative entropy is measured in nats. The mixture probability is $p = 1/2$ and $0 \leq D_{a|b} \leq \log 2$. This is an incoherent mixture of density matrices, not a coherent sum of state vectors. The ordering $a | b$ identifies the numerator; reversing it gives a different observable unless a specific symmetry proves equality.

Conformal field theory (CFT) describes the long-distance limit of a critical chain. A *primary field* creates a basic excitation; its left and right scaling weights are $h, \bar{h}$, its scaling dimension is $h + \bar{h}$, and its conformal spin is $h - \bar{h}$. Descendants are obtained by acting with modes of the symmetry generators. A *conformal block* is a correlation amplitude with specified intermediate fusion channels, that is, specified outcomes when fields are brought together. We turn these amplitudes into wavefunction coefficients by associating the field components or fusion channels with the physical basis configurations. This construction chooses a lattice representative; it does not guarantee that every arbitrary field insertion is an eigenstate of a chosen lattice Hamiltonian.

The production constructions are given in Secs. 1.3 and 1.4. Independent state-quality checks and their error definitions are collected in Appendix E.

1.2 What the three supplied papers provide

Reference [6] constructs the $SU(2)_1$ spin chain using vertex fields and current-mode insertions, including fields at zero and infinity. Reference [7] explains how fusion channels themselves encode Ising spins, and tests the critical Ising ground-state block. Reference [8] relates $SU(n)_k$ blocks to projected fermion states: an auxiliary fermion state is projected onto the allowed on-site spin representation, facilitating conversion to a matrix product state (MPS). For $SU(2)_1$ this recovers the projected Fermi-sea/Haldane–Shastry construction. It does not supply an Ising σ eigenstate construction; $SU(2)_2$ has central charge $3/2$, whereas Ising has $1/2$.

Reference [9] relates the homogeneous Ising block vacuum to the exact fermion pairing state. The covariance reduction is described in Ref. [15]. Together these provide the basis for the Ising production construction in Sec. 1.4. Endpoint-block and projected-state constructions serve as independent validation and are documented in Appendix E.3.

1.3 Compact boson: from nonchiral primaries to vertex blocks

The target theory has independent left and right bosons $\phi_L(z)$ and $\phi_R(\bar{z})$, normalized by logarithmic two-point functions in z and $\bar{z}$, respectively; mixed left–right contractions vanish. We use the compactification convention $R = \sqrt{2}$, with momentum integer m and winding integer w :

$$Q_L = \frac{m}{R} + \frac{wR}{2}, \quad Q_R = \frac{m}{R} - \frac{wR}{2}. \quad (3)$$

Here R fixes the allowed charge lattice. The nonchiral vertex primary is

$$\mathcal{V}_{a,b}(z, \bar{z}) = : \exp \left[\frac{ia}{\sqrt{2}} \phi_L(z) + \frac{ib}{\sqrt{2}} \phi_R(\bar{z}) \right] :, \quad a = m + w, \quad b = m - w. \quad (4)$$

Colons denote normal ordering, which removes contractions of a field with itself at the same point. The state created at the origin by this field is denoted $|as, bs\rangle$, where $s = 1/\sqrt{2}$: $|as, bs\rangle = \mathcal{V}_{a,b}(0, 0)|0\rangle$. This is the state–operator correspondence: a field inserted at the origin specifies a state on a circle surrounding it. Each chiral charge Q contributes weight $Q^2/2$, hence $h = a^2/4$ and $\bar{h} = b^2/4$. This follows from the stress-tensor operator product $T(z)V_Q(0) \sim (Q^2/2)z^{-2}V_Q(0) + z^{-1}\partial V_Q(0)$, where T generates conformal transformations.

To obtain a finite spin-chain vector we use the auxiliary chiral endpoint construction of Ref. [6], with the chosen charge labels at zero and infinity. The resulting chain has a nonchiral continuum limit; the two endpoint labels specify the target charge sectors. The auxiliary chiral correlator is the wavefunction amplitude: we do not multiply it by its complex conjugate to construct a different state. Its absolute square is taken only when computing probabilities.

The chiral boson is normalized by $\langle \phi(z)\phi(w) \rangle = -\log(z - w)$. Its vertex field $V_q(z) = : \exp(iq\phi(z)) :$ has weight $q^2/2$. Gaussian contraction gives

$$\left\langle \prod_r V_{q_r}(z_r) \right\rangle = \delta_{\sum_r q_r, 0} \prod_{r < t} (z_r - z_t)^{q_r q_t}. \quad (5)$$

The neutrality symbol sets the amplitude to zero unless the charges sum to zero. It fixes the allowed occupation sector, reducing the computation. More explicitly, a pair of exponentials contributes $\exp[-q_r q_t \langle \phi(z_r)\phi(z_t) \rangle] = (z_r - z_t)^{q_r q_t}$, while integration over the spatially constant boson mode imposes the neutrality condition. Thus the product in Eq. (5) is determined by the two-point function.

1. Put the sites at $z_j = \exp(2\pi i j/N)$, $j = 0, \dots, N-1$. Associate occupation n_j with a vertex charge $s_j/\sqrt{2}$, where $s_j = 2n_j - 1$. The symbol s_j is a site sign; the state-label symbol $s = 1/\sqrt{2}$ used below is a continuum charge unit.
2. Insert charges $a/\sqrt{2}$ at the origin and $b/\sqrt{2}$ at infinity. The latter is defined by $V_q(\infty) = \lim_{W \rightarrow \infty} W^{q^2} V_q(W)$. Neutrality gives $M = \sum_j n_j = (N - a - b)/2$, leaving $\binom{N}{M}$ allowed configurations. These specify the state; the production contraction in Sec. 3 avoids enumerating them on large chains.
3. Multiply Eq. (5) by the Marshall sign $\chi(\mathbf{n}) = \exp(i\pi \sum_j j n_j)$, which fixes the spin-basis convention. After dropping factors independent of $\mathbf{n}$, the amplitude is

$$\psi_{a,b}(\mathbf{n}) \propto \chi(\mathbf{n}) \prod_{i < j} (z_i - z_j)^{s_i s_j / 2} \prod_j (-z_j)^{a s_j / 2} \prod_j (1 - z_j/W)^{b s_j / 2}. \quad (6)$$

Set the last product to one at strict infinity. To reproduce the original numerics, use $W = 10^8$.

4. Evaluate logarithms with one consistent principal branch for each unordered pair. Subtract the largest real log amplitude before exponentiating, then normalize. For the occupied set O , the implemented configuration-dependent logarithm is

$$L(O) = 2 \sum_{i < j \in O} \log(z_i - z_j) - \sum_{i \in O} \sum_{j \neq i} \log(z_{\min(i,j)} - z_{\max(i,j)}) \\ + i\pi \sum_{i \in O} i + a \sum_{i \in O} \log(-z_i) + b \sum_{i \in O} \log(1 - z_i/W). \quad (7)$$

5. Use $(a, b) = (0, 0), (1, 1), (3, 1)$ for the three states. Their continuum charges and weights are

State	(Q_L, Q_R)	$(h, \bar{h})$	M
$ 00\rangle$	$(0, 0)$	$(0, 0)$	$N/2$
$ ss\rangle$	$(1, 1)/\sqrt{2}$	$(1/4, 1/4)$	$N/2 - 1$
$ 3s, s\rangle$	$(3, 1)/\sqrt{2}$	$(9/4, 1/4)$	$N/2 - 2$

The different M values ensure exact orthogonality. No orthogonalization is applied to these states.

The charge-three vertex is a primary of the free-boson Virasoro algebra. In the extended $SU(2)_1$ current algebra it is a level-two descendant of the spin-half field, consistent with Ref. [6]. These are compatible uses of “primary” for different symmetry algebras. Algebraically expanding $s_j = 2n_j - 1$ in Eq. (6) gives the same Laughlin/Jastrow amplitude as the original generator. We use bit j for site j throughout the production calculation. The explicit numerical comparison, including conversion of the original basis ordering, is reported in Appendices E.2 and E.4.

1.3.1 Deriving the Laughlin/Jastrow form

The following reduction makes the boson construction a directly reproducible example of the conformal-block method. Let $O = \{i_1 < \dots < i_M\}$ be the occupied sites and write $d_{ij} = z_i - z_j$ for $i < j$.

1. Expand the site-charge exponent:

$$\frac{s_i s_j}{2} = 2n_i n_j - n_i - n_j + \frac{1}{2}. \quad (8)$$

The last term contributes the same factor to every configuration and can be absorbed into normalization. The first term produces $\prod_{\mu < \nu} (z_{i_\mu} - z_{i_\nu})^2$.

2. Collect the two terms linear in occupation into one factor per occupied site. Define

$$D_i = \prod_{j < i} (z_j - z_i) \prod_{j > i} (z_i - z_j). \quad (9)$$

Then the configuration-dependent pair product is $\prod_{\mu < \nu} (z_{i_\mu} - z_{i_\nu})^2 \prod_{i \in O} D_i^{-1}$. The endpoint factors similarly reduce to $\prod_{i \in O} (-z_i)^a (1 - z_i/W)^b$.

3. Use the fact that the z_j are the roots of $z^N - 1$:

$$\prod_{j \neq i} (z_i - z_j) = N z_i^{N-1}, \quad D_i = (-1)^i N z_i^{N-1}. \quad (10)$$

The factors $(-1)^i$ cancel the Marshall signs. Since $z_i^N = 1$, $z_i^{-(N-1)} = z_i$. For integer a , the remaining $(-1)^{aM}$ is a configuration-independent phase.

4. The explicit polynomial amplitude is therefore

$$\psi_{a,b}(O) = \frac{\mathcal{P}_{a,b}(O; W)}{\sqrt{\mathcal{Z}_{a,b}(W)}},$$

$$\mathcal{P}_{a,b}(O; W) = \prod_{\mu < \nu} (z_{i_\mu} - z_{i_\nu})^2 \prod_{\mu=1}^M z_{i_\mu}^{a+1} (1 - z_{i_\mu}/W)^b, \quad (11)$$

where $\mathcal{Z}_{a,b}(W) = \sum_{|O|=M} |\mathcal{P}_{a,b}(O; W)|^2$. An overall phase is immaterial. Equation (11) is the exponent-two Laughlin/Jastrow form used by the original method. At strict infinity, set every $(1 - z_i/W)^b$ to one.

Although b disappears from the explicit position factors at infinity, it still fixes $M = (N - a - b)/2$. Thus changing b changes the state. The block and polynomial formulas give the same normalized vector, up to one overall phase; taking an absolute square would destroy this equivalence.

1.3.2 Three canonical nonchiral examples

For each example, the displayed polynomial and its normalization $\mathcal{Z}$ specify the amplitudes in the common 2^N -dimensional spin basis. The production Pfaffian contraction uses this definition without storing the full vector. The expressions below are at strict infinity.

1. **Vacuum** $|00\rangle$. Choose $(a, b) = (0, 0)$, or momentum/winding $(m, w) = (0, 0)$. The continuum vertex is the identity, with $(h, \bar{h}) = (0, 0)$. Neutrality requires $M = N/2$, and

$$\mathcal{P}_{0,0}(O; \infty) = \prod_{\mu < \nu} (z_{i_\mu} - z_{i_\nu})^2 \prod_{\mu=1}^{N/2} z_{i_\mu}. \quad (12)$$

This is the half-filled Haldane–Shastry ground-state amplitude.

2. **Vertex primary** $|ss\rangle$. Choose $(a, b) = (1, 1)$, or $(m, w) = (1, 0)$. The continuum field is $:\exp[i(\phi_L + \phi_R)/\sqrt{2}]$, with $(h, \bar{h}) = (1/4, 1/4)$, dimension $1/2$, and conformal spin zero. Neutrality requires $M = N/2 - 1$, and

$$\mathcal{P}_{1,1}(O; \infty) = \prod_{\mu < \nu} (z_{i_\mu} - z_{i_\nu})^2 \prod_{\mu=1}^{N/2-1} z_{i_\mu}^2. \quad (13)$$

3. **Vertex primary** $|3s, s\rangle$. Choose $(a, b) = (3, 1)$, or $(m, w) = (2, 1)$. The continuum field is $:\exp[i(3\phi_L + \phi_R)/\sqrt{2}]$, with $(h, \bar{h}) = (9/4, 1/4)$, dimension $5/2$, and conformal spin two. Neutrality requires $M = N/2 - 2$, and

$$\mathcal{P}_{3,1}(O; \infty) = \prod_{\mu < \nu} (z_{i_\mu} - z_{i_\nu})^2 \prod_{\mu=1}^{N/2-2} z_{i_\mu}^4. \quad (14)$$

Its status as a Virasoro primary follows from the vertex-field stress-tensor product in Sec. 1.3. Calling it a descendant of the extended $SU(2)_1$ current algebra refers to a different symmetry algebra.

For the original data convention, multiply the last two polynomials by $\prod_{\mu} (1 - z_{i_\mu}/10^8)$ before normalization; the vacuum is unchanged. The three vectors have different occupation numbers and are exactly orthogonal without a Gram–Schmidt procedure. The total spin component in the convention $S_j^z = s_j/2$ is $S_{\text{tot}}^z = M - N/2 = -(a + b)/2$.

The production data use these same normalized vertex/Jastrow states, with $W = 10^8$ fixed to match the original implementation. For larger chains, their interval matrices are obtained by the exact Pfaffian contraction in Sec. 3; this preserves the amplitudes while avoiding full-chain enumeration. The independent polynomial/block comparison and the Haldane–Shastry sanity check are given in Appendix E.2.

1.4 Ising: exact fermion states

The nonchiral states required by Eqs. (4.13)–(4.18) of the supplied manuscript [5] are $I : (0, 0)$, $\sigma : (1/16, 1/16)$, and $\varepsilon : (1/2, 1/2)$, where each pair gives $(h, \bar{h})$. The energy primary ε contains both left and right fermions and has scaling dimension one. The production calculation uses the exact Jordan–Wigner solution for all three states.

The Hamiltonian that identifies these lattice states is

$$H = - \sum_{j=0}^{N-1} X_j X_{j+1} - \sum_{j=0}^{N-1} Z_j, \quad X_N = X_0, \quad (15)$$

with coupling and transverse field both one. These are periodic *spin* boundary conditions. Define $P = \prod_j Z_j$ and let $\mathbb{T}$ translate the spins by one site. The three desired states have $\mathbb{T} = 1$, corresponding to zero momentum, and are identified by

State	P	Energy ordering	Production sector
I	+1	Ground state	Antiperiodic vacuum
ε	+1	First excited even state	Two antiperiodic quasiparticles
σ	−1	Lowest odd state	Periodic vacuum; odd zero mode

Here $\mathbb{T}$ is distinct from the stress tensor $T(z)$.

The exact finite-chain energies of these sectors are

$$E_I = -2 \csc \frac{\pi}{2N}, \quad E_\sigma = -2 \cot \frac{\pi}{2N}, \quad E_\varepsilon = E_I + 8 \sin \frac{\pi}{2N}. \quad (16)$$

For the velocity $v = 2$ of Eq. (15), $E_a - E_I \sim (2\pi v/N)(h_a + \bar{h}_a)$ identifies the spin and energy primaries with scaling dimensions $1/8$ and 1 . We construct these sectors directly by their fermion occupations.

1.4.1 Exact fermion construction used for the production data

For all three Ising states, the production interval matrices use the exact Jordan–Wigner solution. It is efficient because a fermionic Gaussian state is specified by a $2N$ by $2N$ covariance matrix, rather than all 2^N amplitudes. For the spin-to-fermion solution see Ref. [9], Sec. VIII and Appendix C; for reduction by correlations see Ref. [15], Sec. 3.3(II), Eqs. (14)–(18) and the Majorana discussion immediately following them. The zero-mode parity selection below fixes the conventions of our implementation explicitly.

1. **Transform spins to Majorana fermions.** Define the Hermitian operators

$$a_{2j} = \left(\prod_{k < j} Z_k \right) X_j, \quad a_{2j+1} = \left(\prod_{k < j} Z_k \right) Y_j. \quad (17)$$

They satisfy $a_r a_s + a_s a_r = 2\delta_{rs}$. In a fixed parity sector, $H = (i/4) \sum_{r,s} \mathcal{A}_{rs} a_r a_s$, where the real antisymmetric matrix has $\mathcal{A}_{r,r+1} = 2$, $\mathcal{A}_{r+1,r} = -2$, and boundary entry $\mathcal{A}_{2N-1,0} = -2P$.

2. **Select the correct boundary sector.** Even spin parity gives antiperiodic fermions, with momenta $k = (2m + 1)\pi/N$. Odd spin parity gives periodic fermions, with $k = 2m\pi/N$. In both cases $m = 0, \dots, N - 1$, with momenta understood modulo 2π . These are two sectors of the same periodic *spin* Hamiltonian; we have not changed its physical boundary condition.
3. **Choose the occupation pattern for each primary.** Diagonalizing the quadratic Hamiltonian defines fermion annihilation operators γ_k with excitation energies $\omega_k = 4|\sin(k/2)|$. The identity

state is their vacuum in the antiperiodic sector: $\gamma_k|I_N\rangle = 0$. The energy primary occupies the two lowest modes in that sector:

$$|\varepsilon_N\rangle = \gamma_{\pi/N}^\dagger \gamma_{-\pi/N}^\dagger |I_N\rangle. \quad (18)$$

For σ , use the periodic sector, empty its nonzero-energy modes, and choose the zero-mode occupation that gives $P = -1$. The zero mode must be fixed: a covariance with an unresolved zero mode does not describe the required pure parity eigenstate.

4. **Construct the covariance in the implemented basis.** Let $i\mathcal{A} = U \text{diag}(\lambda_\mu) U^\dagger$. For the identity, assign mode signs $\nu_\mu = \text{sgn } \lambda_\mu$. For ε , reverse the signs for the two lowest positive-energy modes and their negative-energy partners. Then set

$$G = \text{Re}\left[iU \text{diag}(\nu_\mu) U^\dagger\right], \quad G_{rs} = \langle ia_r a_s \rangle \quad (r \neq s), \quad G_{rr} = 0. \quad (19)$$

The mode signs are bookkeeping for the covariance, not occupation probabilities. In the periodic sector start with $\nu_\mu = 0$ on the zero modes, take real orthonormal null vectors u, v of $\mathcal{A}$, and add either $uv^T - vu^T$ or its negative. Choose the sign such that $(-1)^N \text{Pf}(G) = -1$. The Pfaffian is the antisymmetric-matrix polynomial obeying $\text{Pf}(G)^2 = \det G$; its sign here fixes the physical fermion parity.

5. **Check purity and compute the interval matrix.** Verify $G^T = -G$ and $G^2 = -\mathbb{I}$. Compare the energy $E = \frac{1}{4} \sum_{r,s} \mathcal{A}_{rs} G_{rs}$ with Eq. (16). Restrict the covariance to the interval and reconstruct its density matrix using Eq. (20).

This calculation is performed by `majorana_covariance` and `gaussian_rdm`. The numerical purity and full-vector comparisons are documented in Appendices E.4 and E.6.

Restrict G to the first 2ℓ indices. An orthogonal transformation brings it into 2×2 blocks with off-diagonal entries g_r . If b_r are the correspondingly rotated Majoranas, the interval state is

$$\rho_A = \prod_{r=0}^{\ell-1} \frac{\mathbb{I} + g_r i b_{2r} b_{2r+1}}{2}. \quad (20)$$

We build this $2^\ell \times 2^\ell$ matrix explicitly. Each component state is Gaussian; their mixture generally is not. We diagonalize the actual matrix m_{ab} , not an averaged covariance matrix. For the compact boson, the production Pfaffian contraction evaluates Eq. (1) for the fixed-occupation Jastrow states without explicitly forming the full coefficient matrix. The connected large-chain procedures for both CFTs are developed in Sec. 3.

2 Coefficients available in the manuscript

We separate an explicit result in the supplied manuscript [5] from a coefficient estimated by fitting finite-chain data. Throughout this section, α denotes the leading power-law exponent. Every reported coefficient c_r multiplies x^r itself and includes all powers of π . For example, the boson vacuum–spin coefficient is $\pi^2/12$, not $1/12$. The same convention applies to y^β for large intervals. These expansions concern the continuum, small-interval limit; they are not exact formulas at fixed two-site or three-site intervals.

The previous report added derivations of boson x^4, x^6 coefficients and Ising spin-channel x^4 coefficients. Those were additional calculations, not coefficients stated in the original paper. They are excluded from the current theoretical comparisons and preserved separately in the task’s archive. No missing subleading coefficient is derived in this revision.

2.1 Ising: two explicit subleading coefficients

At $p = y = 1/2$, Eqs. (4.15)–(4.16) of the supplied manuscript [5] give

$$D_{I|\varepsilon} = \frac{2\pi^4}{15}x^4 - \frac{8\pi^6}{45}x^6 + \mathcal{O}(x^8), \quad (21)$$

$$D_{\varepsilon|I} = \frac{2\pi^4}{15}x^4 - \frac{8\pi^6}{21}x^6 + \mathcal{O}(x^8). \quad (22)$$

Here $D_{a|b} = S(\rho_A^a \| (\rho_A^a + \rho_A^b)/2)$, where ρ_A^a is the reduced density matrix of state a on the interval A . The substitution into the paper’s general-weight expressions is $(8/15)(1/4) = 2/15$ for the leading coefficient and $(32/315)(1/4)(8 - 15) = -8/45$ or $(32/315)(1/4)(8 - 23) = -8/21$ for the subleading coefficient. Thus the coefficients of x^6 itself are $-8\pi^6/45$ and $-8\pi^6/21$, respectively.

For each of $I \mid \sigma$, $\sigma \mid I$, $\sigma \mid \varepsilon$, and $\varepsilon \mid \sigma$, Eqs. (4.13), (4.14), (4.17), and (4.18) of the supplied manuscript [5] give only

$$D_{a|b} = \frac{\pi^2}{48}x^2 + \mathcal{O}(x^4). \quad (23)$$

The paper explicitly leaves the fourth-order four-energy-field contribution unevaluated. Its discussion identifies additional sectors at this order, but does not give a complete x^4 coefficient for these four directions. The shared leading coefficient does not assert equality of the complete directed relative entropies.

Direction	α	c_α	$c_{\alpha+2}$	Supplied equation
$I \mid \sigma$	2	$\pi^2/48$	—	(4.13)
$\sigma \mid I$	2	$\pi^2/48$	—	(4.14)
$I \mid \varepsilon$	4	$2\pi^4/15$	$-8\pi^6/45$	(4.15)
$\varepsilon \mid I$	4	$2\pi^4/15$	$-8\pi^6/21$	(4.16)
$\sigma \mid \varepsilon$	2	$\pi^2/48$	—	(4.17)
$\varepsilon \mid \sigma$	2	$\pi^2/48$	—	(4.18)

Table 1: Complete coefficients supplied for the six Ising directions at $p = 1/2$, in powers of x . A dash means that the complete coefficient is not supplied; it does not mean zero. Equation numbers in the last column refer to the supplied manuscript [5]. Higher coefficients beyond those shown are not supplied for these directed mixture formulas.

2.2 Compact boson: leading coefficients only

Write the left and right charge vector as $\mathbf{Q}_a$: $\mathbf{Q}_{00} = (0, 0)$, $\mathbf{Q}_{ss} = (1/\sqrt{2}, 1/\sqrt{2})$, and $\mathbf{Q}_{3s,s} = (3/\sqrt{2}, 1/\sqrt{2})$. These are the charges denoted by (Q_L, Q_R) in Sec. 1.3; α denotes the leading exponent. Define the squared charge separation $\kappa_{ab} = \|\mathbf{Q}_a - \mathbf{Q}_b\|^2$. Applying the paper's leading current-sector formula at $p = 1/2$ gives

$$D_{a|b} = \frac{\kappa_{ab}\pi^2}{12}x^2 + \mathcal{O}(x^4). \quad (24)$$

The original manuscript does not provide complete x^4 or x^6 coefficients for these six equal-mixture directions. General formulas for individual subleading sectors do not supply the sum of all contributions at the same order.

Directions, each separately	κ_{ab}	α	c_α
$00 \mid ss, \quad ss \mid 00$	1	2	$\pi^2/12$
$00 \mid 3s, s, \quad 3s, s \mid 00$	5	2	$5\pi^2/12$
$ss \mid 3s, s, \quad 3s, s \mid ss$	2	2	$\pi^2/6$

Table 2: Boson leading coefficients at $p = 1/2$ in powers of x . Each direction has an $\mathcal{O}(x^4)$ remainder whose complete coefficient is not supplied. Equal leading coefficients do not constrain finite-chain reverse directions to be equal.

2.3 Unknown coefficients in the fitting ansatz

To test corrections numerically without deriving their coefficients, we use only the leading and the next remainder power identified here:

$$F_1(x) = c_0 x^\alpha, \quad F_2(x) = c_0 x^\alpha + c_1 x^{\alpha+2}, \quad \alpha = \begin{cases} 4, & \{a, b\} = \{I, \varepsilon\}, \\ 2, & \text{all other requested pairs.} \end{cases} \quad (25)$$

The subscripts 0 and 1 label the two coefficients, not their exponents. The paper fixes the leading exponent and the next remainder order for these cases. A remainder does not establish a nonzero coefficient, and we do not assume an infinite sequence of even powers. Both coefficients in F_2 are fitted. Section 4 additionally fits a single power with its exponent free, to test whether the data recover α .

3 Computing larger chains and larger subsystems step by step

3.1 Step 1: choose both physical length scales

The small-interval expansion concerns an interval short compared with the whole system, while a continuum description also requires that interval to contain many lattice spacings. With lattice spacing set to one, these are the two conditions $\ell/N \ll 1$ and $\ell \gg 1$. Increasing N at fixed $\ell = 1$ only addresses the first condition. We retain the six-to-ten-site intervals for all current fits and figures. The previous reference grid was

$$\begin{aligned} \mathcal{N} &= \{32, 48, 64, 82, 96, 112, 128\} \\ &\cup \{144, 160, 176, 192, 208, 224, 240, 256\}, \\ N &\in \mathcal{N}, \quad \ell \in \{6, 7, 8, 9, 10\}, \quad x = \ell/N. \end{aligned} \quad (26)$$

The symbol $\mathcal{N}$ denotes this reference subset; $N = 82$ is used literally. Each CFT has $15 \times 5 = 75$ retained geometries on this subset; the 30 additional $\ell = 4, 5$ geometries remain archived. The new active grid is Eq. (52). The historical comparison on that subset used the windows

$$x \leq x_{\text{fit}} = 0.08 \quad \text{for fitting}, \quad x \leq x_{\text{show}} = 0.20 \quad \text{for figures.} \quad (27)$$

On this reference subset there are 51 fitted and 70 displayed observations per directed pair. The smallest retained ratio is $6/256 = 0.0234375$; the largest fitted and displayed ratios are $5/64 = 0.078125$ and $9/48 = 0.1875$, respectively. The sampled grid contains no point between $5/64$ and the historical fitting cutoff. The active mixture fits instead use the CFT-wide windows in Eq. (59) on the extended grid. The small-Ising cutoff excludes the reference subset, whereas the wider small-boson window includes part of it. Increasing N at fixed $\ell = 6, \dots, 10$ improves the small- x range while retaining the same lattice resolution; it does not by itself remove finite- ℓ cutoff effects.

3.2 Step 2: identify the matrix that must be computed

For a normalized primary state $|\psi_a^{(N)}\rangle$, split the spin basis into an interval configuration u on $A = \{0, \dots, \ell - 1\}$ and an environment configuration y on its complement B . If $\Psi_a(u, y) = \langle u, y | \psi_a^{(N)} \rangle$, the exact partial trace is

$$(\rho_{a,A})_{uv} = \sum_y \Psi_a(u, y) \Psi_a(v, y)^*, \quad \text{Tr } \rho_{a,A} = 1. \quad (28)$$

We need the full interval matrix, including its off-diagonal entries, because the equal mixture and directed relative entropy are

$$\mu_{ab,A} = \frac{1}{2}(\rho_{a,A} + \rho_{b,A}), \quad D_{a|b} = \text{Tr } \rho_{a,A} \log \rho_{a,A} - \text{Tr } \rho_{a,A} \log \mu_{ab,A}. \quad (29)$$

The logarithm is natural and $p = 1/2$. A full-chain vector has 2^N entries, whereas $\rho_{a,A}$ has dimension only 2^ℓ , at most 1024 here. The next two steps evaluate Eq. (28) without constructing that vector: fermion covariance matrices for Ising, and an exact sum of Jastrow amplitudes for the compact boson.

3.3 Step 3: obtain the Ising interval matrix from its primary sector

The ingredients are the Jordan–Wigner solution in Ref. [9], Sec. VIII and Appendix C, and Gaussian reduction in Ref. [15], Sec. 3.3(II). The states remain those specified in Sec. 1.4, with independent validation in Appendix E.4. To make the large-chain calculation explicit, start from $H = -\sum_j X_j X_{j+1} - \sum_j Z_j$ with

periodic spin boundaries. The Jordan–Wigner Majoranas are $a_{2j} = (\prod_{k<j} Z_k)X_j$ and $a_{2j+1} = (\prod_{k<j} Z_k)Y_j$. They obey $\{a_r, a_s\} = 2\delta_{rs}$ and give

$$H = \frac{i}{4} \sum_{r,s=0}^{2N-1} \mathcal{A}_{rs} a_r a_s, \quad \mathcal{A}_{r,r+1} = 2, \quad \mathcal{A}_{2N-1,0} = -2P, \quad \mathcal{A}^T = -\mathcal{A}, \quad (30)$$

where $P = \prod_j Z_j = \pm 1$ is the global parity. All other independent entries vanish. The following operations determine the state and its reduction.

1. **Select the primary and diagonalize the quadratic Hamiltonian.** For I, ε take $P = +1$, which gives antiperiodic fermions; for σ take $P = -1$, which gives periodic fermions. Diagonalize the Hermitian matrix $i\mathcal{A} = U \text{diag}(\lambda_\mu)U^\dagger$. For I , use signs $\nu_\mu = \text{sgn } \lambda_\mu$. For ε , reverse the signs of the two smallest positive modes and their negative-energy partners. This occupies the quasiparticles at momenta $\pm\pi/N$, with excitation energy $8 \sin[\pi/(2N)]$.
2. **Construct the real covariance and fix the spin state.** Set $G = \text{Re}[iU \text{diag}(\nu_\mu)U^\dagger]$; its off-diagonal entries are $G_{rs} = \langle ia_r a_s \rangle$. In the periodic sector, initially set $\nu_\mu = 0$ on the zero modes, choose real orthonormal null vectors u, v of $\mathcal{A}$, and add $\pm(uv^T - vu^T)$. Choose the sign giving $(-1)^N \text{Pf}(G) = -1$. This resolves the zero-mode occupation and produces the required pure odd-parity σ state. Check $G^T = -G$ and $G^2 = -\mathbb{I}$. The independent finite-chain energy checks are

$$E_I = -2 \csc \frac{\pi}{2N}, \quad E_\sigma = -2 \cot \frac{\pi}{2N}, \quad E_\varepsilon = E_I + 8 \sin \frac{\pi}{2N}, \quad E = \frac{1}{4} \sum_{rs} \mathcal{A}_{rs} G_{rs}. \quad (31)$$

3. **Restrict to the interval and reconstruct its density matrix.** Take $G_A = G_{0:2\ell,0:2\ell}$. A real Schur decomposition gives an orthogonal matrix O such that

$$O^T G_A O = \bigoplus_{j=0}^{\ell-1} \begin{pmatrix} 0 & g_j \\ -g_j & 0 \end{pmatrix}, \quad |g_j| \leq 1. \quad (32)$$

Define interval Majorana matrices using the same spin strings, and rotate them as $b_r = \sum_s O_{sr} a_s$. The commuting operators $ib_{2j} b_{2j+1}$ have eigenvalues ± 1 ; therefore

$$\rho_{a,A} = 2^{-\ell} \prod_{j=0}^{\ell-1} [\mathbb{I} + g_j ib_{2j} b_{2j+1}]. \quad (33)$$

Each factor supplies probabilities $(1 \pm g_j)/2$ for one canonical mode, so the product is normalized and has the required covariance. It is the unique Gaussian reduction of the chosen primary. Because the interval starts at site zero, all its Jordan–Wigner strings lie inside A , making this also the spin reduced density matrix.

The largest full covariance has dimension 512. The global spectral step has cubic cost and quadratic storage in N , while reconstructing Eq. (33) depends exponentially on ℓ , not N . We compute the ten-site reduction once for each state and obtain every smaller requested interval by the further partial trace in Sec. 3.5.

3.4 Step 4: sum the compact-boson amplitudes exactly

Direct enumeration of $\binom{N}{M}$ occupation configurations becomes impractical as N increases. We instead sum the same Jastrow amplitudes algebraically. This changes the contraction algorithm, not the primary state or its endpoint convention. Pfaffian expressions for diagonal observables in the Haldane–Shastry Jastrow

state are established in Ref. [14], Appendix A.1, Eqs. (A5)–(A11), and Appendix A.2, Eqs. (A14)–(A15). The off-diagonal reduced-state formula in Eq. (42) is an implementation derivation in this report.

Here are the steps in a form that specifies every matrix used in the calculation. The three charge codes are $(a, b) = (0, 0), (1, 1), (3, 1)$; $M = (N - a - b)/2$ is the number of occupied sites. To specify one basis configuration $|n_0 \cdots n_{N-1}\rangle$, with $n_j = 0, 1$, it suffices to list the sites whose bit is one. This list is the *occupied set*

$$O = \{j \in \{0, \dots, N-1\} : n_j = 1\}, \quad |O| = \sum_{j=0}^{N-1} n_j = M. \quad (34)$$

Thus O contains site indices, and $|O|$ denotes their number. In the boson spin convention $s_j = 2n_j - 1$, “occupied” means $s_j = +1$. For example, on an illustrative six-site chain, $|010101\rangle$ has $O = \{1, 3, 5\}$ and $M = 3$, with sites numbered from zero. Every allowed configuration corresponds to one subset of M sites; this is why there are $\binom{N}{M}$ amplitudes.

The coordinate of site j is $z_j = e^{2\pi i j/N}$. Write the unnormalized coefficient of the basis configuration specified by O as

$$P(O) = \Delta(O)^2 \prod_{j \in O} z_j^{a+1} g(z_j), \quad \Delta(O) = \prod_{i < j \in O} (z_j - z_i), \quad g(z) = (1 - z/W)^b, \quad W = 10^8. \quad (35)$$

This is the polynomial form of the previously validated vertex block. Let the interval be $A = \{0, \dots, \ell-1\}$ and its complement be B . With $Z = \sum_{|O|=M} |P(O)|^2$, the normalized amplitude is $\Psi(O) = P(O)/\sqrt{Z}$. To label a matrix element, let $U, V \subset A$ be the occupied sets of its row and column configurations, respectively, and let $Y \subset B$ be the occupied set in the traced-out environment. The two full-chain configurations are therefore $U \cup Y$ and $V \cup Y$. The partial trace uses the same Y in both amplitudes and sums over all such environment configurations. For $|U| = |V| = k$, fixed total occupation requires $|Y| = M - k$. Thus Eq. (28) becomes

$$(\rho_A)_{UV} = \frac{1}{Z} \sum_{\substack{Y \subset B \\ |Y|=M-k}} P(U \cup Y) P(V \cup Y)^*, \quad |U| = |V| = k. \quad (36)$$

For unequal $|U|, |V|$ the matrix element is zero, because no common environment can satisfy the fixed total occupation M . The remaining steps evaluate this numerator and its normalization without enumeration.

1. **Convert the environment sum into a moment matrix.** Set $w(z) = z^{-2(M-1)} |g(z)|^2$. For indices $r, s = 0, \dots, 2M-1$, define

$$A_{rs} = (s - r) \sum_{j=0}^{N-1} w(z_j) z_j^{r+s-1}. \quad (37)$$

The matrix is antisymmetric. Its Pfaffian, denoted Pf , is the signed sum over all pairings of its indices; for a 2×2 matrix it is the upper-right entry, and its square is the determinant. The empty Pfaffian is one. A confluent Vandermonde determinant uses two columns $v(z), v'(z)$ for each summed site, where $v_r(z) = z^r$ and $v'_r(z) = rz^{r-1}$ (zero at $r = 0$). Expanding this determinant and summing over site subsets gives

$$Z = \sum_{|O|=M} |P(O)|^2 = \text{Pf}(A). \quad (38)$$

Indeed, on the unit circle $|P(O)|^2 = \Delta(O)^4 \prod_{j \in O} w(z_j)$. Each summed site supplies a pair of columns, so its contribution to the moment matrix is $w(z)[v(z)v'(z)^T - v'(z)v(z)^T]$. This is the discrete weighted form of the confluent-determinant sum in Ref. [14], Appendix A.1. That reference uses L for our site number N , and N for our particle number M .

2. **Remove the interval sites with a small matrix update.** Let T have columns $v(z_j), v'(z_j)$ for $j \in A$, in site order, and let C be block diagonal with blocks $-w(z_j) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then $A_B = A + TCT^T$ sums only the environment sites. All following small matrices have dimension 2ℓ :

$$\begin{aligned} K_0 &= T^T A^{-1} T, & Q &= C^{-1} + K_0, \\ p_\emptyset &= \text{Pf}(Q) \prod_{j \in A} w(z_j), & K_B &= K_0 - K_0 Q^{-1} K_0. \end{aligned} \quad (39)$$

Here $p_\emptyset = \text{Pf}(A_B)/Z$ is the probability that the interval is empty; it is a reference for the algebra, not a conditioning applied to the final physical state. All transposes in these antisymmetric identities are ordinary transposes. Equation (39) follows from two explicit matrix identities:

$$\begin{aligned} A_B^{-1} &= A^{-1} - A^{-1} T Q^{-1} T^T A^{-1}, \\ \frac{\text{Pf}(A + TCT^T)}{\text{Pf}(A)} &= (-1)^\ell \text{Pf}(C) \text{Pf}(Q) = \left[\prod_{j \in A} w(z_j) \right] \text{Pf}(Q). \end{aligned} \quad (40)$$

The first identity can be verified by multiplying by A_B ; left and right multiplication by T^T, T gives the stated K_B . In the second, $\text{Pf}(C) = (-1)^\ell \prod_j w(z_j)$. The empty-interval probability is nonzero on the grid, so the reference inverse exists even when some individual occupation blocks are small.

3. **Retain the bra and ket configurations explicitly.** An interval matrix element is labelled by occupied sets $U, V \subset A$. It vanishes unless $|U| = |V| = k$, because the global particle number is fixed. Set $m_j = \mathbf{1}_{j \in U} + \mathbf{1}_{j \in V}$. A site with $m_j = 1$ selects its evaluation column in T ; one with $m_j = 2$ selects both its evaluation and derivative columns. Denote the resulting ordered column indices by I_{UV} and define

$$\Delta_{UV} = \prod_{i < j \in A} (z_j - z_i)^{m_i m_j}. \quad (41)$$

The exact normalized reduced-state element is

$$(\rho_A)_{UV} = P(U)P(V)^* \prod_{v \in V} z_v^{-2(M-k)} \frac{(-1)^k p_\emptyset \text{Pf}[(K_B)_{I_{UV}, I_{UV}}]}{\Delta_{UV}}. \quad (42)$$

To see the phase and powers, write the common occupied environment as $Y \subset B$, $|Y| = M - k$. The product of amplitudes factors as

$$\begin{aligned} P(U \cup Y)P(V \cup Y)^* &= P(U)P(V)^* \prod_{v \in V} z_v^{-2(M-k)} \Delta(Y)^4 \prod_{y \in Y} w(z_y) \\ &\times \prod_{u \in U, y \in Y} (z_u - z_y)^2 \prod_{v \in V, y \in Y} (z_v - z_y)^2. \end{aligned} \quad (43)$$

The fixed columns for U, V and the paired columns for Y give a confluent Vandermonde determinant with an extra factor Δ_{UV} . For clarity, let T_{UV} consist of those fixed columns and let $\mathcal{S}_{UV}$ denote the sum of the environment-dependent factors on the right-hand side of Eq. (43). Summing each remaining column pair gives the bordered Pfaffian identity

$$\begin{aligned} \Delta_{UV} \mathcal{S}_{UV} &= (-1)^k \text{Pf} \begin{pmatrix} A_B & T_{UV} \\ -T_{UV}^T & 0 \end{pmatrix} \\ &= (-1)^k \text{Pf}(A_B) \text{Pf}(T_{UV}^T A_B^{-1} T_{UV}). \end{aligned} \quad (44)$$

The first equality is the same pairing expansion as Eq. (38), with $2k$ unsummed columns; moving these fixed columns to the end produces $(-1)^k$. The second follows by block elimination with

determinant-one triangular matrices. Substituting $K_B = T^T A_B^{-1} T$ and $\text{Pf}(A_B)/Z = p_\emptyset$ yields Eq. (42). Thus both off-diagonal phases and normalization are carried through the environment sum explicitly. The Pfaffian elimination uses $\text{Pf}(BAB^T) = \det(B) \text{Pf}(A)$, given in Ref. [16], Eq. (5); this identity also fixes the sign rather than extracting a square root of a determinant.

4. **Exploit the roots of unity and check precision.** The identity $\sum_j z_j^n = N$ when n is a multiple of N , and zero otherwise, makes AR tridiagonal; $R_{rs} = \delta_{r, 2M-1-s}$ reverses the basis order. Put $L = 2M - 1$. Its nonzero entries are

$$\begin{aligned} (AR)_{rr} &= N(L - 2r)(1 + b/W^2), \\ (AR)_{r, r+1} &= (AR)_{r+1, r} = -N(L - 2r - 1)b/W. \end{aligned} \quad (45)$$

Solving this tridiagonal system gives $A^{-1}T$ without a dense inverse. Only the 2ℓ matrices and Pfaffians in Eq. (39) remain. For fixed $\ell \leq 10$, the dependence on N is linear; the dependence on interval length is still exponential. At $\ell = 10$ the same principal Pfaffian appears in many bra-ket pairs. We compute it once and reuse it. For an ordered even index set $I = \{i_1, \dots, i_{2k}\}$, the recursion is

$$f(I) = \sum_{j=2}^{2k} (-1)^j (K_B)_{i_1 i_j} f(I \setminus \{i_1, i_j\}), \quad f(\emptyset) = 1. \quad (46)$$

It is the first-row Pfaffian expansion in Ref. [16], Eq. (4), with memoization added here. The denominator Δ_{UV} and the Pfaffian depend on the site multiplicities, so their ratio is also reused for pairs with identical $U \cup V, U \cap V$. There are at most $(3^\ell + 1)/2$ even multiplicity patterns; intermediate index sets in the recursion are cached separately. This reduces repeated work without changing Eq. (42). The reference calculation through $N = 256$ uses 100 decimal digits internally to handle cancellation in the empty-interval update and Pfaffian recursion. Before converting to complex double precision, the trace defect and imaginary part of every diagonal element must be below 10^{-25} . Neither a normalization repair nor eigenvalue clipping is used to pass this check. Independent 130-digit calculations test the final output accuracy. No sampling or Gaussian approximation of the boson state is used.

3.5 Step 5: reduce shorter intervals and evaluate the mixture

The ten-site matrix from either model contains every requested smaller interval. With site zero the least significant bit, write a ten-site configuration as $u + 2^\ell h$, where u labels the retained sites and h the sites to trace out. Then

$$(\rho_{a,\ell})_{uv} = \sum_{h=0}^{2^{10-\ell}-1} (\rho_{a,10})_{u+2^\ell h, v+2^\ell h}. \quad (47)$$

This makes the matrices at different interval lengths mutually consistent and avoids repeating the most expensive construction.

To evaluate Eq. (29) efficiently, use its exact local symmetry. A global Ising parity eigenstate gives a reduced state commuting with $P_A = \prod_{j \in A} Z_j$; tracing out B removes terms of odd local parity. A fixed-number boson state gives a reduced state commuting with the interval number $Q_A = \sum_{j \in A} n_j$, since the same environment must occur in its bra and ket. Both primaries and their mixture therefore have a common direct-sum decomposition

$$\rho_{a,A} = \bigoplus_q \rho_{a,q}, \quad \mu_{ab,A} = \bigoplus_q \mu_q, \quad \mu_q = \frac{1}{2}(\rho_{a,q} + \rho_{b,q}). \quad (48)$$

For Ising q labels even/odd parity; for the boson it labels occupation number $0, \dots, \ell$. At $\ell = 10$, the Ising blocks have dimension 512 and the largest boson block has dimension $\binom{10}{5} = 252$. The blocks are *not* separately normalized: their traces are the physical probabilities of their symmetry sectors.

Let $r_{a,q,i}$ be the eigenvalues of $\rho_{a,q}$ and diagonalize $\mu_q = V_q \text{diag}(w_{q,j}) V_q^\dagger$. Define $d_{a,q,j} = (V_q^\dagger \rho_{a,q} V_q)_{jj}$. The complete matrix relative entropy is

$$D_{a|b} = \sum_{q,i} r_{a,q,i} \log r_{a,q,i} - \sum_{q,j} d_{a,q,j} \log w_{q,j}. \quad (49)$$

One diagonalization of each unordered mixture serves both directions: $d_{b,q,j} = 2w_{q,j} - d_{a,q,j}$. This is an exact block evaluation of the matrix logarithm; the mixture is not assumed Gaussian.

Numerically, use $0 \log 0 = 0$ and omit eigenvalues at or below 10^{-15} . Reject a primary or mixture eigenvalue below -2×10^{-12} and reject a discarded mixture-support weight $\sum_{q,j: w_{q,j} \leq 10^{-15}} d_{a,q,j}$ above 10^{-12} . Check Hermiticity, unit total trace, and $0 \leq D_{a|b} \leq \log 2$, allowing roundoff. The upper bound follows from $\mu_{ab,A} \geq \rho_{a,A}/2$. Repeating the entropy calculation with threshold 10^{-16} measures sensitivity to the numerical cutoff.

3.6 Step 6: validate the contraction before interpreting a fit

The validation protocol follows the same chain of operations as the production calculation: identify the primary, construct its interval matrix, take shorter partial traces, and evaluate the mixture entropy. We compare against full-vector reductions where enumeration is feasible, vary arithmetic precision and eigenvalue thresholds, and compare direct and symmetry-block evaluations of the same matrix logarithm. The numerical results, error definitions and tested sizes are collected in Appendix E.6; the independent primary-state comparisons are in Appendices E.2 and E.4. These checks assess finite-lattice numerical accuracy. The new $N = 256$ endpoint is checked before generating the intervening sizes. Its covariance has 512×512 entries, and its interval matrix still has only 1024×1024 entries at $\ell = 10$. For the boson, the moment solve grows with N while the cached minors depend on ℓ . The 100-digit production contraction is compared with 130 digits at the new endpoint. These checks and the additional production diagnostics are reported in Appendix E.6.1. The following step addresses the separate comparison with CFT.

3.7 Step 7: compare with the small-interval prediction

To compare each datum with the leading CFT prediction without adding a fitting law, define the dimensionless ratio

$$R_{a|b}(N, \ell) = \frac{D_{a|b}(N, \ell)}{c_\alpha^{\text{CFT}}(\ell/N)^\alpha}. \quad (50)$$

The CFT leading term corresponds to $R = 1$. Increasing ℓ at fixed N changes both the lattice cutoff sensitivity and x , so the ratio need not move monotonically toward one. The retained curves in Fig. 1 compare $\ell = 6, \dots, 10$ at each size. Doubling N reduces x at fixed interval length; the remaining variation with ℓ shows why smaller x alone does not establish the continuum coefficient. The excluded $\ell = 4, 5$ values remain in the raw data and do not appear in the figures or tables.

A complementary comparison holds x fixed and scales both lengths. The following directly computed ratios all have $x = 1/16$:

N	ℓ	Ising $R_{\sigma \varepsilon}$	Boson $R_{ss 3s,s}$
32	2	1.09914	0.78366
96	6	1.22690	0.97729
128	8	1.23071	0.98529
160	10	1.23216	0.98886

The first row is a historical reference and is excluded from the present fit. The successive changes shrink for both pairs. For Ising the values stabilize away from one: convergence with lattice resolution at fixed x need not equal the leading term alone. A coarse interval can appear closer to that term because lattice

and higher-order corrections partly cancel. These observations motivate the larger-subsystem selection; they do not specify a functional form for a lattice correction.

The active analysis uses the extended grid of Eq. (52) with free powers, F_1 , and F_2 . Section 4 specifies estimation, residuals and size cross-validation. No missing analytical subleading coefficient is derived or declared zero.

3.8 Step 8: extend the chain while retaining numerical precision

$$\mathcal{D} = \{2 \text{round}[30 \cdot 100^{j/48}] : j = 0, \dots, 48\}, \quad \mathcal{R} = \{200 + 4j : j = 0, \dots, 200\}. \quad (51)$$

$$\begin{aligned} \mathcal{N}_{\text{ising}} = & \mathcal{N} \cup \mathcal{D} \cup \mathcal{R} \cup \{4096 + 256j : j = 0, \dots, 16\} \\ & \cup \{152, 1024, 1118, 1146, 1232, 1356, 1492, 1528, 1604, 1644\} \\ & \cup \{1808, 1852, 1972, 2048, 2190, 2354, 2412, 2654, 2890, 3000\} \\ & \cup \{3140, 3216, 3294, 3436, 3500, 3552, 3606, 3660, 3784, 3854\} \\ & \cup \{3926, 4000, 4158, 4222, 4286, 4426, 4670, 4734, 4798, 5000\} \\ & \cup \{5246, 5310, 5542, 5758, 6270, 6526, 6782, 7038, 7294, 7550\} \\ & \cup \{7806, 8000, 8016, 8032, 8048, 8064, 8080, 8096, 8112, 8128\} \\ & \cup \{8144\}, \\ \mathcal{N}_{\text{boson}} = & \mathcal{N} \cup \mathcal{D} \cup \{100, 104, 168, 292, 322, 354, 428, 472, 512, 762\} \\ & \cup \{924, 1024, 1118, 1356, 1492, 1644, 1808, 2048, 2412, 2922\} \\ & \cup \{3540, 3750, 3916, 4096, 4234, 4376, 4608, 4720, 4834, 5000\} \\ & \cup \{5222, 5626, 5810, 6250, 6466, 6688, 6918, 7156, 7402, 7656\} \\ & \cup \{7920, 8192, 8436, 8688, 8948, 9216, 9462, 9714, 9974, 10240\} \\ & \cup \{10488, 10740, 10998, 11264, 11764, 12288, 12800, 13312, 13814, 14336\} \\ & \cup \{14840, 15360, 15864, 16384\}. \end{aligned} \quad (52)$$

All indices j in these set definitions are integers within the stated bounds. The earlier grid in Eq. (26) remains a reference subset. The active mixture fits use Eq. (52); no intervals longer than ten sites are added. The primary-to-primary sanity check in Appendix F retains its original $N \leq 256$ grid. The methods below change how the same finite-chain states and entropies are evaluated, not their definitions.

3.8.1 Fourier construction and local Ising transition solves

At criticality the Majorana matrix is uniform apart from its boundary twist, so its eigenmodes are known Fourier waves. Let $L = 2N$ count Majoranas, with indices $j = 0, \dots, L - 1$, and set $k_m = 2\pi(m + t)/L$. The twist is $t = 1/2$ for I, ε and $t = 0$ for σ . Define $w_m = -\text{sgn}(\sin k_m)$. For the energy state reverse w_m at $m = 0, N - 1, N, L - 1$. For the spin state set the two zero-mode weights w_0, w_N to zero and supply their parity explicitly. Then

$$(\Gamma_a)_{ij} = \frac{i}{L} \sum_{m=0}^{L-1} w_m e^{ik_m(i-j)} + \begin{cases} [(-1)^i - (-1)^j]/L, & a = \sigma, \\ 0, & a = I, \varepsilon. \end{cases} \quad (53)$$

The imaginary notation in the Fourier sum yields a real antisymmetric matrix. The last term is $vu^T - uv^T$, with $u_j = L^{-1/2}$ and $v_j = (-1)^j L^{-1/2}$; it selects the same odd Ramond state as Sec. 3.3. This is an exact finite-chain construction, using the free-fermion mode description of Refs. [9, 15]. Its energies are

$$E_I = -2 \csc \frac{\pi}{2N}, \quad E_\sigma = -2 \cot \frac{\pi}{2N}, \quad E_\varepsilon = E_I + 8 \sin \frac{\pi}{2N}. \quad (54)$$

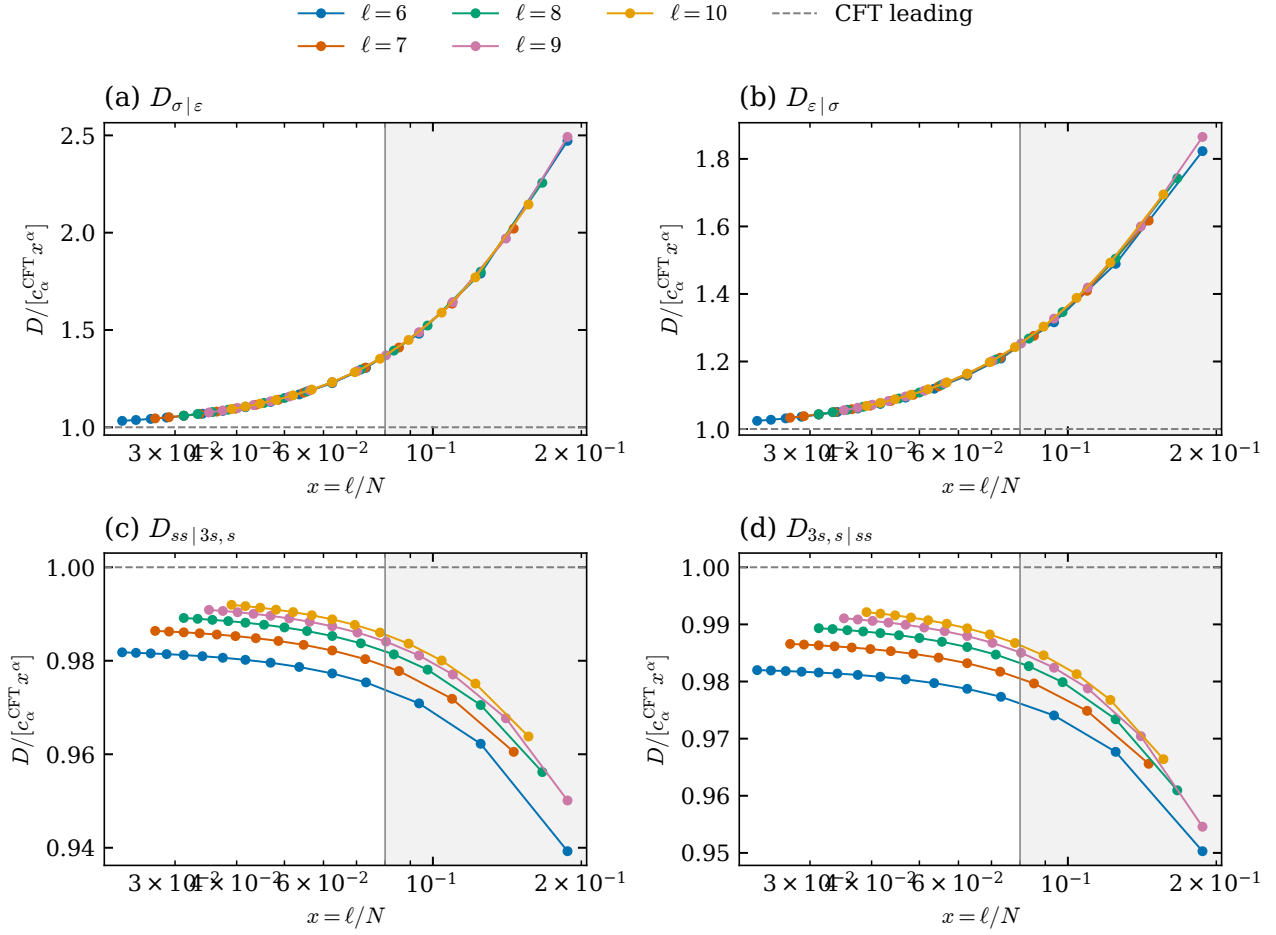


Figure 1: Historical smaller- x comparison, without fitting a lattice correction. The ordinate is R in Eq. (50), with $\alpha = 2$, $c_2^{\text{CFT}} = \pi^2/48$ in (a,b), and $c_2^{\text{CFT}} = \pi^2/6$ in (c,d). All data use $p = 1/2$, natural logarithms, the 15 sizes through $N = 256$ in Eq. (26), $\ell = 6, \dots, 10$, and $\ell/N \leq 0.20$. Colors identify fixed ℓ ; line segments only connect the computed sizes, not fitted curves. The horizontal dashed line is the CFT leading prediction. Ising uses the critical periodic Hamiltonian $H = -\sum XX - \sum Z$ and exact covariance matrices; the boson uses charge codes $(1, 1), (3, 1)$, $q = 2$, $z_j = e^{2\pi i j/N}$, $w_1 = 0$, $W = 10^8$, and the same normalized vertex/Jastrow states as the original method. The small- x ends of the curves compare intervals containing different numbers of lattice sites; every panel contains 70 observations. The vertical line marks $x_{\text{fit}} = 0.08$ and the shaded region contains data excluded from fitting. The ratios themselves use no fitted parameters.

A fast Fourier transform applies Γ_a to a vector in $O(N \log N)$ work. Thus the Hamiltonian eigensolve is unnecessary.

For the anchored transition in Sec. 5.3, let Γ'_b be the covariance after applying the local anchor. Form $K = \mathbb{I} - \Gamma_a \Gamma'_b$ by applying the Fourier transform to batches of columns. Factor the real matrix K once with pivoted LU (lower and upper triangular factors). Let E_A be the first 2ℓ columns of the identity. Solve only

$$KZ = (\mathbb{I} - i\Gamma_a)E_A, \quad \Gamma_{ab,A} = i\{E_A^T(\mathbb{I} - i\Gamma'_b)Z - \mathbb{I}_{2\ell}\}. \quad (55)$$

The real and imaginary right-hand sides reuse the same LU factors. The anchor overlap is $|\det K|^{1/4}/2^{N/2}$; its logarithm is obtained from the LU diagonal. Reconstruct the local Gaussian operator and restore the anchor, exactly as in the earlier Gram calculation. We never construct a full complex transition covariance or its inverse. The remaining dense factorization costs $O(N^3)$ arithmetic and the stored real matrices cost $O(N^2)$ memory. The interval density matrices still have dimension at most $2^{10} = 1024$.

3.8.2 Direct evaluation of small entropy remainders

At very small fractions, subtracting $\log 2 - D^B$ or two almost equal entropy traces loses relative precision. Instead, evaluate their remainder directly with matrix-function divided differences [18]. The following identity is an algebraic numerical device, not a CFT expansion or a new analytical entropy coefficient.

Set $f(t) = t \log t$, with $f(0) = 0$. In a basis where a positive reference matrix is $D = \text{diag}(d_i)$, write $H = D + E = V \text{diag}(\lambda_k) V^\dagger$. The first-order matrix derivative has diagonal $f'(d_i)E_{ii}$. For the remainder diagonal, use

$$\mathcal{R}_i = [f(H) - f(D)]_{ii} - f'(d_i)E_{ii} = \sum_k |(V^\dagger E)_{ki}|^2 \frac{\lambda_k \log(\lambda_k/d_i) - \lambda_k + d_i}{(\lambda_k - d_i)^2}. \quad (56)$$

To verify it, expand $[f(H)]_{ii} = \sum_k |V_{ik}|^2 f(\lambda_k)$, use $\sum_k |V_{ik}|^2 = 1$ and $E_{ii} = \sum_k |V_{ik}|^2 (\lambda_k - d_i)$, and factor the scalar Taylor remainder. Finally, $(\lambda_k - d_i)V_{ik} = (EV)_{ik}$ converts its squared factor to the numerator weight in Eq. (56). At equal arguments the quotient is $1/(2d_i)$. With $u = (\lambda - d)/d$, compute it as

$$\frac{1}{d} \frac{(1+u) \log(1+u) - u}{u^2} = \frac{1}{d} \sum_{n=0}^{\infty} \frac{(-u)^n}{(n+1)(n+2)}. \quad (57)$$

The code uses ten terms only for $|u| < 0.02$, where the omitted tail is below double-precision accuracy, and the closed expression otherwise. This series stabilizes a scalar matrix-function evaluation; it is unrelated to a small-interval power-law fit.

For D the eigenvalue matrix of the mixture on A and H the source state in that basis, equal unit traces give $S(\rho||\mu) = \sum_i \mathcal{R}_i$. For the complement, first define the local transition $T_{ab}^A = \text{Tr}_B |a\rangle\langle b|$. The unweighted Gram matrix has blocks $H = \begin{pmatrix} \rho_a^A & T_{ab}^A \\ (T_{ab}^A)^\dagger & \rho_b^A \end{pmatrix}$. Its relation $H = 2\mathcal{G}^*$ to the weighted Gram matrix is derived in Sec. 5.2. Transform the two diagonal blocks to their separate Schmidt eigenbases and let D contain their eigenvalues on its diagonal. Then E contains only the off-diagonal transition blocks, so its first-order diagonal vanishes. If P_a, P_b select the two state blocks, the Gram entropy identity gives the deficit $\delta_{a|b} \equiv \log 2 - D_{a|b}^B$ directly:

$$\delta_{a|b} = \text{Tr } P_a [f(H) - f(D)] = \sum_{i \in a} \mathcal{R}_i, \quad \delta_{b|a} = \sum_{i \in b} \mathcal{R}_i. \quad (58)$$

Complex conjugating the entire Gram matrix preserves these real block traces. Thus even a deficit much smaller than machine precision relative to $\log 2$ can be stored accurately on its own.

The identities are exact on a positive support. In double precision we retain reference eigenvalues above $\tau = 10^{-15}$ in each symmetry sector. The discarded tails are a numerical approximation; they are

checked by varying τ to 10^{-14} and 10^{-16} at every new size for $\ell = 4, 10$. We additionally compare with the original trace-log and Gram formulas and with a high-precision, noncommuting two-dimensional example. These checks, together with the larger-size state-contraction tests, are reported in Appendix B.5.

3.8.3 Boson scaling and the practical boundary

The boson continues to use the exact sparse moment and small Pfaffian reductions in Sec. 3.4 and Appendix B.2. The dense high-precision minors depend on ℓ , not on N . At the added sizes the moment algebra uses 120 decimal digits before its final conversion to double precision. Independent sampled entries use 160 digits and direct pivoted Pfaffians instead of cached recursive minors. Although the physical moment matrices are sparse, the current Python pivot searches scan remaining indices. Their bookkeeping can cost $O(N^2)$ time, in addition to the arithmetic and the $O(N\ell)$ stored right-hand sides. This becomes the main practical obstacle before the 2^ℓ interval dimension changes. The measured limits and timings below refer to this implementation on this machine; they are not mathematical maximum chain lengths.

Table 3: Measured largest completed chains, with three diagonal states, three transitions, both directions, and every $\ell = 4, \dots, 10$. Wall time includes both interval calculations and endpoint cutoff checks. Memory is the process high-water resident set (GiB), not the size of one matrix; it may include allocations retained from preceding stages. The host has 32 GiB RAM. These are validated endpoints, not absolute mathematical limits.

CFT	Largest N	Seconds	Peak GiB	Minimum retained fraction
ising	8192	93.46	9.0882	7.324×10^{-4}
boson	16384	506.34	0.73204	3.662×10^{-4}

For Ising, doubling the largest tested N would multiply the dominant stored matrix sizes by four, giving an estimated 36.4 GiB with the present allocation pattern, above this host's RAM. The dense LU arithmetic would grow by about eight. For the boson, sparse index scans still permit further growth but become costly: doubling N can quadruple that bookkeeping. Such extrapolations are estimates, not executed benchmarks. No larger size is claimed as validated.

4 Small-interval fits and the leading-power test

4.1 Data selection and coefficient convention

The active mixture data use the grid in Eq. (52), $\ell = 6, \dots, 10$, and $p = 1/2$. Let $x = \ell/N$ for the small interval A , and $y = \ell/N$ for the small complement of the large interval B . The common fitting windows are

$$\begin{aligned} W_x^{\text{Ising}} &= (0, 0.02], & W_x^{\text{boson}} &= (0, 0.1], \\ W_y^{\text{Ising}} &= (0, 0.002], & W_y^{\text{boson}} &= (0, 0.005]. \end{aligned} \quad (59)$$

Each window is shared by all six ordered pairs and by Free, F_1 , and F_2 for its CFT and interval type. Zero is an open lower bound; there is no observation at zero. The two small-interval windows are frozen. The large-boson upper cutoff is prescribed as 0.005; the updated coefficient and neighboring-cutoff checks are documented in Appendix H.4. In this revision, the large-boson cutoff changes from 0.0047 to 0.005; the other three windows are unchanged. Every full fit and every omitted-size training set has at least forty distinct (N, ℓ) observations. Exact counts and sampled endpoints are in Appendix D. The overview figures show data through $x, y = 0.20$ so the entire fitted range is visible, with excluded data shaded. Per-pair figures in Sec. 6 also provide fitting-window zooms. Historical primary-to-primary comparisons on $N \leq 256$ retain their separately stated fitting window.

Table 4: Active mixture geometries, with all $\ell = 6, \dots, 10$ at each listed size. The full grids are Eq. (52). Counts are per direction. The four common fitting windows are Eq. (59); all figures show fractions through 0.2. Historical $\ell = 11, 12$ values remain archived and are excluded from this revision.

Interval	CFT	$N_{\max}$	Retained	Fit	Shown	Minimum fraction
small	ising	8192	1650	1255	1645	7.324×10^{-4}
small	boson	16384	630	598	625	3.662×10^{-4}
large	ising	8192	1650	260	1645	7.324×10^{-4}
large	boson	16384	630	317	625	3.662×10^{-4}

All coefficient tables report amplitudes of x^r or y^r itself. Thus, if an archived file writes $\tilde{C}(\pi x)^r$, its amplitude in this report is $C = \pi^r \tilde{C}$. For a free exponent the conversion uses the *fitted* exponent, whereas the theoretical amplitude uses the theoretical exponent. Omitting this distinction changes the coefficient error whenever those exponents differ.

4.2 Free powers, one-term fits, and two-term fits

The overview figures, the per-pair figures, and the principal tables in Appendices A and B all use Eq. (59). Consequently every comparison of Free, F_1 , and F_2 within a CFT and interval type uses the same observations, including both directions of each primary pair.

To state the same procedure for both intervals, let z denote x for the small interval or y for its large complement. Let q_i be the observed $D_{a|b}^A$ or $\delta_{a|b}$, respectively, at z_i . The theoretical leading power is $r_0 = \alpha_{\text{CFT}}$ in the first case and $r_0 = \beta_{\text{CFT}}$ in the second; r_1 denotes the next power listed in the corresponding theory table. We compare

$$\begin{aligned} F_{\text{free}}(z) &= C z^r, & C > 0, \quad r \text{ free}, \\ F_1(z) &= c_0 z^{r_0}, & F_2(z) &= c_0 z^{r_0} + c_1 z^{r_1}. \end{aligned} \quad (60)$$

The subscripts on F_1, F_2 count fitted coefficients. They have one and two fitted parameters, respectively; the free-power fit also has two. The exponent of F_1 is imposed, so its agreement cannot validate that

exponent independently. F_2 holds both powers fixed and fits both coefficients without sign constraints. There is no constant term. The CFT curve has no fitted parameters: it contains the supplied x^4, x^6 terms for $I \leftrightarrow \varepsilon$, the leading term for the other small-interval pairs, and only the leading term for large intervals.

For every fitted model, use unweighted least squares in the observable:

$$r_i = q_i - M(z_i), \quad \hat{\theta} = \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^m r_i^2. \quad (61)$$

Here m is the retained sample count and θ the fitted parameters. The symbol r_i with a sample index denotes a residual, while r without an index in Eq. (60) is an exponent. For the free fit we write $F = \exp(u)(z/z_\star)^r$, where $z_\star$ is the fitting cutoff; $C = \exp(u)z_\star^{-r}$. A logarithmic regression of the data supplies only an initial guess. No theoretical exponent or coefficient enters this optimization. An independent scalar search profiles out the amplitude at each exponent and checks the predictions. For F_1, F_2 , a scaled design matrix $X_{ij} = (z_i/z_\star)^{r_j}$ gives a linear least-squares solve; its coefficients are divided by $z_\star^{r_j}$ before reporting.

4.3 Residuals and validation measures

For the theory amplitude c_0^{CFT} , define

$$\begin{aligned} \text{SSE} &= \sum_i r_i^2, & \text{RMSE} &= \sqrt{\text{SSE}/m}, & R_i[\%] &= 100r_i/q_i, \\ \text{RMSrel}[\%] &= \sqrt{m^{-1} \sum_i R_i^2}, & \text{MAPE}[\%] &= m^{-1} \sum_i |R_i|, \\ \eta_c[\%] &= 100(\hat{c}_0/c_0^{\text{CFT}} - 1), & \eta_r[\%] &= 100(\hat{r}/r_0 - 1), \\ G[\%] &= 100(1 - \text{RMSE}_M/\text{RMSE}_{F_1}). \end{aligned} \quad (62)$$

For the free fit, $\hat{c}_0 = C$. An imposed exponent has no measured exponent deviation. Every residual ordinate is linear and shows ordinary percentage tick values; only the horizontal interval fraction is logarithmic. All reported indicators use the fitting window, even though figures also show excluded data. A negative G means a larger residual than F_1 . Because F_1 is nested in F_2 , a smaller training residual for F_2 alone does not establish predictive improvement.

We therefore omit every point at one chain size, refit the remaining sizes, and predict the omitted points. With $M^{(-N_i)}$ denoting that refit, define

$$\text{CVRMSE} = \sqrt{m^{-1} \sum_i [q_i - M^{(-N_i)}(z_i)]^2}, \quad G_{\text{CV}}[\%] = 100(1 - \text{CVRMSE}_M/\text{CVRMSE}_{F_1}). \quad (63)$$

The CSV retains every omitted-size fit, the maximum absolute residual, parameter ranges across folds, and descriptive standard errors from $[\text{SSE}/(m-k)](J^T J)^{-1}$, where k is the parameter count and J is the prediction Jacobian. These deterministic data do not support interpreting those errors as statistical confidence intervals.

4.4 What the free-power test establishes

An unweighted fit over a fixed window is most sensitive to its larger observations. Adding tiny- x points need not move its fitted exponent to the asymptotic value. Appendix D records the retained fraction range and enforces the minimum sample count; smaller trial windows with fewer than forty points are excluded. The exponent is estimated without changing the objective or imposing the predicted power. A stable exponent close to CFT supports the leading-power prediction; a remaining amplitude offset at fixed ℓ can still reflect finite lattice resolution. Increasing N alone does not take $\ell \rightarrow \infty$. Using full-window

free-exponent deviations range from -0.348% to $+1.188\%$. F_2 improves size-holdout RMSE over F_1 in 6 of six directions. These statements use the prescribed fitting window and the sample-count requirements in Appendix D.

Compact-boson full-window free-exponent deviations range from -1.756% to -0.087% . F_2 improves size-holdout RMSE over F_1 in 4 of six directions. These statements use the prescribed fitting window and the sample-count requirements in Appendix D.

The supplied next coefficients provide a separate test: the adopted common-window $x \leq 0.02$ F_2 fits give $c_1 = -170.71, -355.38$ for $I \mid \varepsilon$ and $\varepsilon \mid I$, respectively, whereas CFT gives $-8\pi^6/45$ and $-8\pi^6/21$. Their signed next-coefficient discrepancies are approximately -0.122% and -2.966% . Table 10 gives all parameters from these same fits. Agreement of the leading exponent does not establish agreement of the next-order amplitude.

For the difficult Ising spin–energy pair, the window $x \leq 0.001$ gives free exponents 1.999324, 1.999296, compared with two. The corresponding full amplitudes are 0.204756, 0.204713, compared with $\pi^2/48$. The order is $\sigma \mid \varepsilon$, then $\varepsilon \mid \sigma$. No analytical exponent is imposed in these fits.

The complete small-interval parameters and residual comparisons are in Appendix A. The plots below show every retained model; curves outside the fitting window are extrapolations of a truncated description. Nonpositive extrapolated values cannot be drawn on a logarithmic entropy ordinate and are omitted from that curve only; their signed residuals remain displayed.

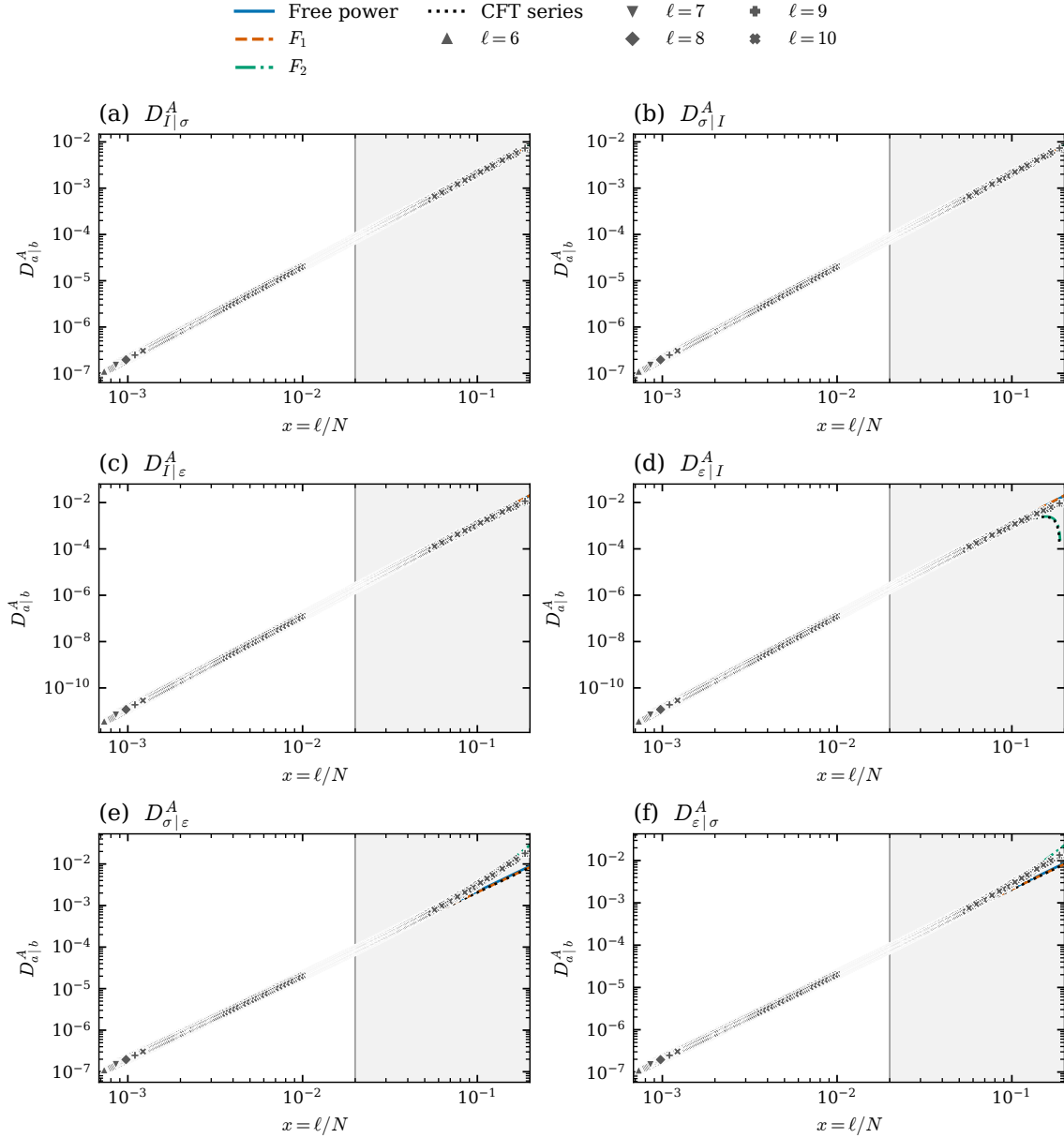


Figure 2: Both axes are logarithmic. Nonpositive values of an extrapolated truncated curve are omitted only from this logarithmic panel, while residual panels retain their predictions. Ising mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 8192$. There are 1645 displayed geometries per direction at $x \leq 0.2$, of which 1255 satisfy $x \leq 0.02$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The periodic Ising Hamiltonian is $H = -\sum_j X_j X_{j+1} - \sum_j Z_j$; primaries use exact fermionic covariances, including the odd Ramond ground state for σ . Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the supplied CFT series (zero). Powers and amplitudes use Eqs. (60) and (25); all coefficients include powers of π . Fits minimize unweighted squared residuals in D^A . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 10 and 11 also have per-pair full-range and fitting-window views in Sec. 6.4.

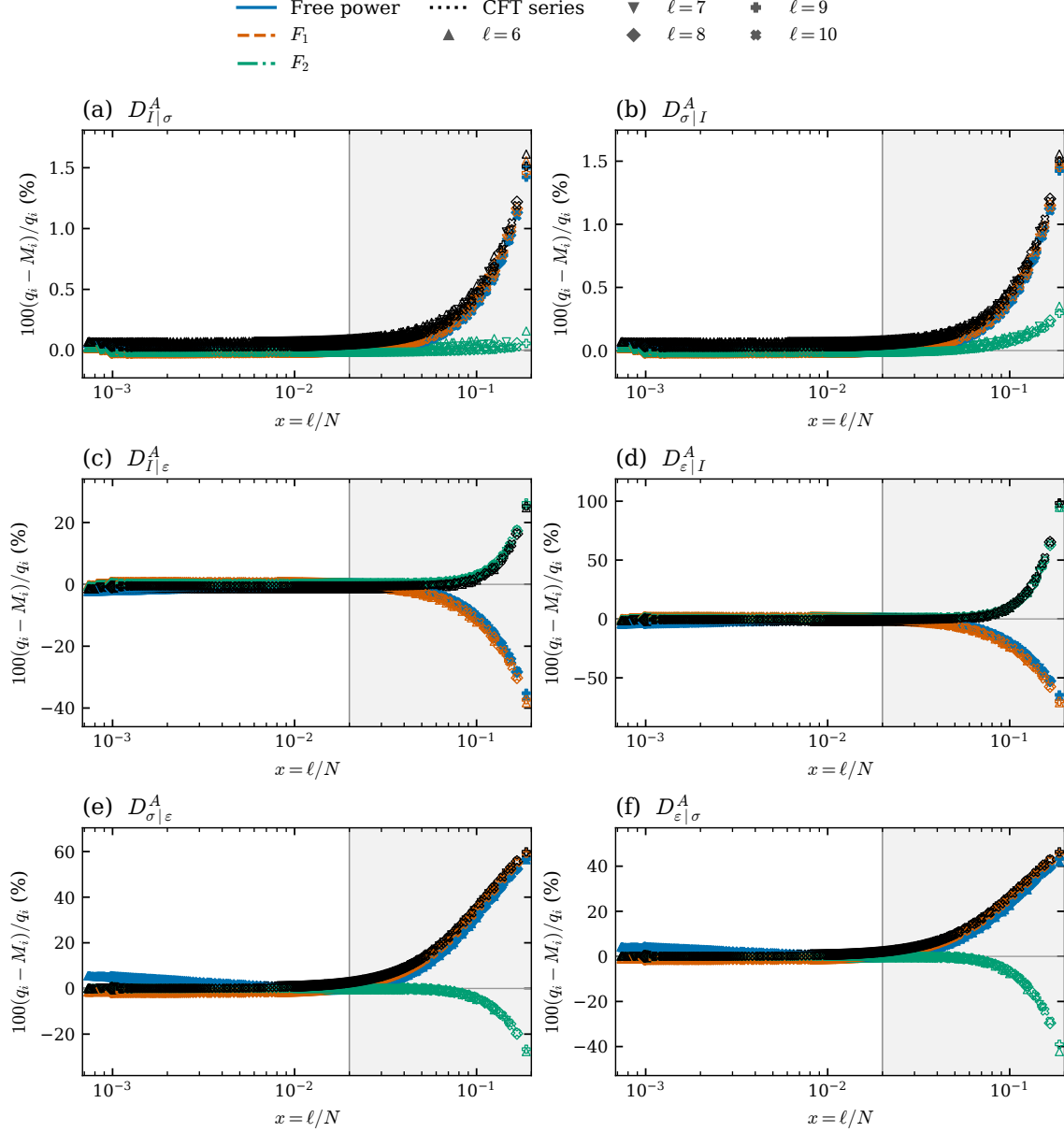


Figure 3: The ordinate is $100(q_i - M_i)/q_i$ percent, with $q_i = D_i^A$ for a small interval and $q_i = \delta_i$ for a large interval; the horizontal zero line aids comparison. All four models have residual markers, colored as their curves. The residual ordinate is linear; the fraction axis is logarithmic. Using mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 8192$. There are 1645 displayed geometries per direction at $x \leq 0.2$, of which 1255 satisfy $x \leq 0.02$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The periodic Ising Hamiltonian is $H = -\sum_j X_j X_{j+1} - \sum_j Z_j$; primaries use exact fermionic covariances, including the odd Ramond ground state for σ . Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the supplied CFT series (zero). Powers and amplitudes use Eqs. (60) and (25); all coefficients include powers of π . Fits minimize unweighted squared residuals in D^A . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 10 and 11 also have per-pair full-range and fitting-window views in Sec. 6.4.

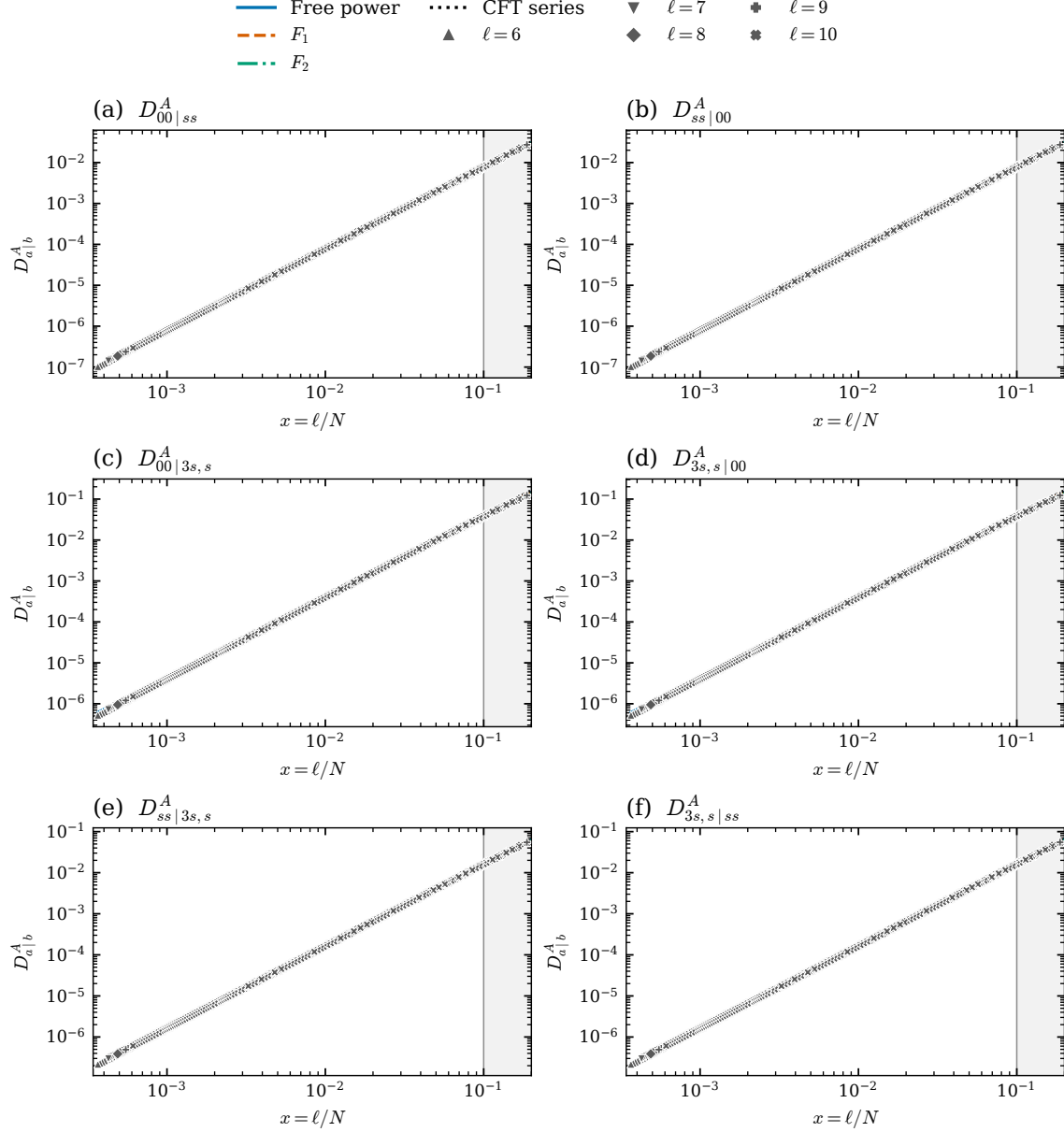


Figure 4: Both axes are logarithmic. Nonpositive values of an extrapolated truncated curve are omitted only from this logarithmic panel, while residual panels retain their predictions. Compact-boson mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 16384$. There are 625 displayed geometries per direction at $x \leq 0.2$, of which 598 satisfy $x \leq 0.1$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The boson uses $s = 1/\sqrt{2}$, $q = 2$, charge codes $(0, 0), (1, 1), (3, 1)$, $z_j = e^{2\pi i j/N}$, $w_1 = 0$, and $W = 10^8$; moment algebra uses 100 digits through $N = 256$ and 120 digits at the added sizes. Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the supplied CFT series (zero). Powers and amplitudes use Eqs. (60) and (25); all coefficients include powers of π . Fits minimize unweighted squared residuals in D^A . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 12 and 13 also have per-pair full-range and fitting-window views in Sec. 6.5.

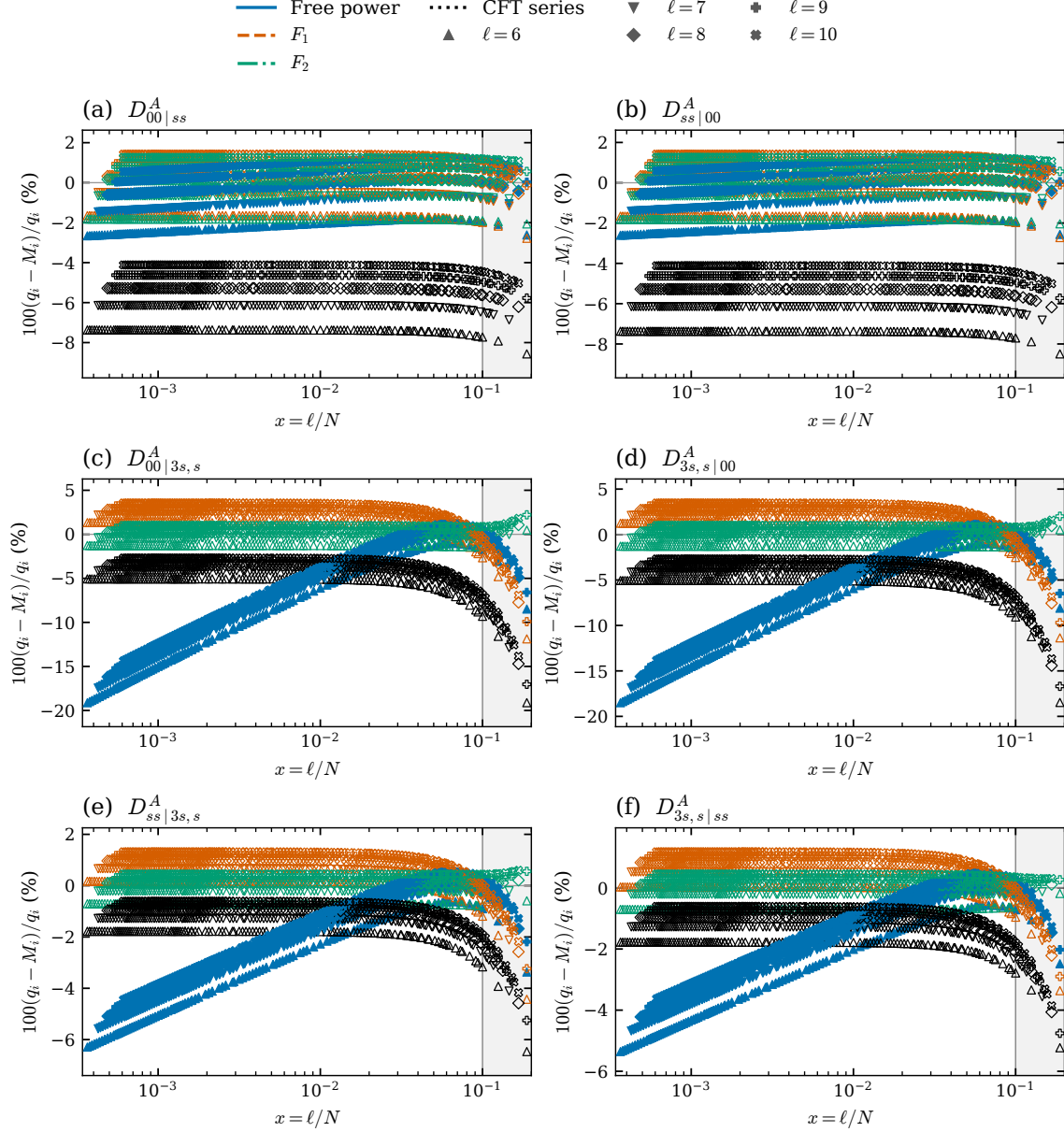


Figure 5: The ordinate is $100(q_i - M_i)/q_i$ percent, with $q_i = D_i^A$ for a small interval and $q_i = \delta_i$ for a large interval; the horizontal zero line aids comparison. All four models have residual markers, colored as their curves. The residual ordinate is linear; the fraction axis is logarithmic. Compact-boson mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 16384$. There are 625 displayed geometries per direction at $x \leq 0.2$, of which 598 satisfy $x \leq 0.1$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The boson uses $s = 1/\sqrt{2}$, $q = 2$, charge codes $(0, 0), (1, 1), (3, 1)$, $z_j = e^{2\pi i j/N}$, $w_1 = 0$, and $W = 10^8$; moment algebra uses 100 digits through $N = 256$ and 120 digits at the added sizes. Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the supplied CFT series (zero). Powers and amplitudes use Eqs. (60) and (25); all coefficients include powers of π . Fits minimize unweighted squared residuals in D^A . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 12 and 13 also have per-pair full-range and fitting-window views in Sec. 6.5.

5 Large complementary intervals

Throughout this report, α denotes the leading small-interval power and β the leading large-interval deficit power. For the latter, β_{CFT} is the analytical prediction and $\hat{\beta}$ is the freely fitted exponent.

5.1 Step 1: specify the interval and the observable

The primaries and their lattice representatives are the same as in Secs. 1 and 3.1. Let $A = \{0, \dots, \ell - 1\}$ contain ℓ consecutive sites, and let $B = \{\ell, \dots, N - 1\}$ be its complement on the periodic chain. Define

$$y = \frac{\ell}{N}, \quad x_B = 1 - y, \quad \rho_a^B = \text{Tr}_A |a\rangle\langle a|, \quad \bar{\rho}_{ab}^B = \frac{\rho_a^B + \rho_b^B}{2}. \quad (64)$$

Here a, b label distinct primaries, $p = 1/2$ is the mixing probability, and y always denotes the small complement fraction in this section. The theory manuscript uses y for a mixing probability in some formulas; that manuscript variable is set to $1/2$ here. Its small parameter $\epsilon = 1 - x$ is our y . For each ordered pair we calculate

$$D_{a|b}^B = \text{Tr} \rho_a^B (\log \rho_a^B - \log \bar{\rho}_{ab}^B), \quad \delta_{a|b}(y) = \log 2 - D_{a|b}^B. \quad (65)$$

All logarithms are natural. The two directions use the same incoherent mixture; the source state changes. Positivity and $\bar{\rho}_{ab}^B \geq \rho_a^B/2$ imply $0 \leq D_{a|b}^B \leq \log 2$. The manuscript predicts $D_{a|b}^B \rightarrow \log 2$ as $y \rightarrow 0$. The deficit resolves the approach to this limit more clearly than $D_{a|b}^B$ alone.

The active mixture grid is Eq. (52), with $\ell = 6, \dots, 10$. All models and all six directions use $0 < y \leq 0.002$ for Ising or $0 < y \leq 0.005$ for the boson, as in Eq. (59). The boson window is prescribed; its coefficient comparisons are in Appendix H.4. On the adopted large-interval windows, F_2 improves the leading coefficient over F_1 in 6 of six Ising directions and 4 of six compact-boson directions. Omitted-size prediction improves in 6 of six Ising directions and 2 of six compact-boson directions. The boson uses $0 < y \leq 0.005$; Appendix H.4 gives the full comparison. The displayed range is $y \leq 0.20$, or $x_B \geq 0.80$. Fits use $x_B \geq 0.998$ for Ising and $x_B \geq 0.995$ for the boson. Counts are in Table 4. Previously computed $\ell = 4, 5, 11, 12$ observations remain archived and are excluded. The overview figures and summary tables use the same fits as the per-pair views in Sec. 6.

5.2 Step 2: replace the large matrix by a Gram matrix

Although ρ_a^B acts on $2^{N-\ell}$ basis states, a pure global state has Schmidt rank at most 2^ℓ across this cut. A mixture of two such states has rank at most $2^{\ell+1}$. To use this fact we need both the individual matrices on A and the transition matrix

$$\rho_a^A = \text{Tr}_B |a\rangle\langle a|, \quad T_{ab}^A = \text{Tr}_B |a\rangle\langle b|. \quad (66)$$

The transition is generally neither Hermitian nor a density matrix. It records how the two Schmidt supports overlap on B ; the individual ρ_a^A, ρ_b^A do not determine that overlap.

For the derivation only, write $|a\rangle = \sum_{u,v} X_a(v, u) |u\rangle_A |v\rangle_B$, where u and v enumerate spin configurations. Let $Z = (X_a, X_b)/\sqrt{2}$. Then $ZZ^\dagger = \bar{\rho}_{ab}^B$, while its small Gram matrix is

$$\mathcal{G} = Z^\dagger Z = \frac{1}{2} \begin{pmatrix} (\rho_a^A)^T & (T_{ab}^A)^* \\ (T_{ab}^A)^T & (\rho_b^A)^T \end{pmatrix}. \quad (67)$$

The transpose and complex conjugation follow from assigning the A configuration to the column of X_a ; they must not be dropped. The nonzero eigenvalues λ_k of $\mathcal{G}$ are exactly those of $\bar{\rho}_{ab}^B$. If v_k is its normalized eigenvector, define $P_a = \text{diag}(\mathbb{I}_{2^\ell}, 0)$ and $P_b = \text{diag}(0, \mathbb{I}_{2^\ell})$. The corresponding normalized

large-space vector is $w_k = Zv_k/\sqrt{\lambda_k}$. Using $X_a = \sqrt{2}ZE_a$, where E_a embeds the first block, gives $\langle w_k | \rho_a^B | w_k \rangle = 2\lambda_k v_k^\dagger P_a v_k$. Purity equates the nonzero spectra of ρ_a^A and ρ_a^B . Therefore the complete entropy is

$$D_{a|b}^B = \text{Tr}(\rho_a^A \log \rho_a^A) - 2 \sum_{\lambda_k > 0} \lambda_k (v_k^\dagger P_a v_k) \log \lambda_k. \quad (68)$$

Replace P_a by P_b and ρ_a^A by ρ_b^A for the other direction. This form avoids division by small eigenvalues. Multiplying T_{ab}^A by one constant phase only conjugates $\mathcal{G}$ by a block-diagonal unitary and leaves both entropies unchanged.

We diagonalize common quantum-number blocks on B . An Ising block has parity $\mathbf{p}_a(-1)^{|u|}$, with $\mathbf{p}_I = \mathbf{p}_\varepsilon = 1$, $\mathbf{p}_\sigma = -1$, and $|u|$ the number of occupied sites in u . The scalar parity $\mathbf{p}_a$ is distinct from the block projector P_a . A boson block has $M_a - |u|$ occupied sites on B , where M_a is the total occupation in state a . At $\ell = 12$ the full Gram dimension is 8192; the largest symmetry blocks are 4096 for Ising and 1716 for the boson pairs. We construct each common- B block directly, avoiding the full Gram allocation. The production calculation constructs no state or matrix of dimension $2^{N-\ell}$.

For the new $\ell = 11, 12$ data, an additional numerical reduction retains the occupied Schmidt vectors. In each local symmetry sector, diagonalize ρ_a^A and collect eigenvectors with eigenvalues $\mu_{a,j} > \tau_s = 10^{-16}$ in U_a ; let $\Lambda_a = \text{diag}(\mu_{a,j})$ contain the retained values. Use U_b, Λ_b for the other state. Here these matrices collect the retained vectors and values from all local sectors; the calculation still diagonalizes each common- B sector separately. With $V = \text{diag}(U_a^*, U_b^*)$, the projected Gram is

$$\mathcal{G}_s = V^\dagger \mathcal{G} V = \frac{1}{2} \begin{pmatrix} \Lambda_a & (U_a^\dagger T_{ab}^A U_b)^* \\ (U_a^\dagger T_{ab}^A U_b)^T & \Lambda_b \end{pmatrix}. \quad (69)$$

Use the same source-weight formula in Eq. (68), with projectors onto the two retained blocks. The diagonal $\text{Tr} \rho_a^A \log \rho_a^A$ is evaluated from the corresponding eigenvalues with the original entropy threshold 10^{-15} . This discards small Schmidt weights and is a numerical approximation. We record $|1 - \text{Tr} \Lambda_a|$ and check the entropies against the full Gram calculation and against $\tau_s = 10^{-17}, 10^{-15}$ in Sec. B.4. The $N = 256$ benchmark data themselves use the full Gram matrices.

5.3 Step 3: obtain Ising transitions from anchored Gaussian states

Use the exact parity-resolved covariance G_a of Sec. 3.3 for each of I, σ, ε . In our convention $\langle \gamma_r \gamma_s \rangle = \delta_{rs} - i(G_a)_{rs}$ and $G_a^2 = -\mathbb{I}$. The γ_r are Jordan–Wigner Majorana operators, with site zero first. Distinct primaries are orthogonal, so a transition formula that divides by $\langle b|a \rangle$ is singular. We remove that singularity by acting on the bra state with a local Hermitian unitary:

$$U_A = \begin{cases} \gamma_0 = X_0, & \text{pairs containing } \sigma, \\ i\gamma_0\gamma_1 = -Z_0, & I, \varepsilon \text{ pair,} \end{cases} \quad |b'\rangle = U_A |b\rangle. \quad (70)$$

Here X_0, Z_0 are Pauli matrices on site zero. The first choice changes parity; the second couples the two even states. Both give a nonzero overlap with $|a\rangle$ on the tested grid. They are computational devices and do not change the production primary states.

Conjugation gives $G_{b'} = R G_b R^T$, where R is diagonal with minus signs at index 0 for the first case or at 0, 1 for the second case, and plus signs elsewhere. In the first case the true Majorana reflection differs by an overall minus sign, which cancels in the covariance. The Gaussian overlap and product rules [3, 4] give, in our convention,

$$\begin{aligned} K &= \mathbb{I} - G_a G_{b'}, & o &= |\langle b'|a \rangle| = 2^{-N/2} \det(K)^{1/4}, \\ Q &= (\mathbb{I} - iG_{b'})K^{-1}(\mathbb{I} - iG_a), & G_* &= i(Q - \mathbb{I}). \end{aligned} \quad (71)$$

The inverse denotes a linear solve. We evaluate $\log o = \frac{1}{4} \sum_j \log s_j(K) - \frac{N}{2} \log 2$ from the positive singular values $s_j(K)$, avoiding a large determinant. The matrix G_* is complex antisymmetric and describes the trace-one operator $|a\rangle\langle b'|/\langle b'|a\rangle$. It need not describe a physical density matrix.

Restrict G_* to its first 2ℓ Majoranas. For any complex antisymmetric matrix H on those modes define the operator

$$\mathcal{R}(H) = 2^{-\ell} \sum_{\substack{S \subset \{0, \dots, 2\ell-1\} \\ |S| \text{ even}}} i^{|S|/2} \text{Pf}(H_S) \prod_{r \in S}^{\rightarrow} \gamma_r, \quad \text{Pf}(H_\emptyset) = 1. \quad (72)$$

H_S is the principal submatrix on S , the product is in increasing index order, and Pf denotes the Pfaffian introduced in Sec. 3.4. This is Wick's theorem expressed in the Majorana operator basis. Undoing the local anchor gives

$$T_{ab}^A = o \mathcal{R}((G_*)_A) U_A \quad \text{up to one overall phase.} \quad (73)$$

The code evaluates all principal Pfaffians recursively and converts the Pauli-string coefficients to matrix entries with a Walsh transform. It requires $O(\ell^2 4^\ell)$ local work and a $2N$ -dimensional covariance solve, without a full-chain wavefunction. For the added diagonal matrices at $\ell = 12$, the same reconstruction gives $\rho_a^A = \mathcal{R}((G_a)_A)$. It avoids forming and rotating all 24 dense Majorana matrices; its reduction to $\ell = 10$ is checked against the existing Schur-based diagonal matrices. The anchor and this reconstruction are project adaptations of the published Gaussian identities. Their explicit-wavefunction checks are reported in Sec. B.

5.4 Step 4: contract different boson occupation sectors exactly

The same conformal-block/Jastrow amplitudes of Eq. (35) are used, with endpoint $W = 10^8$. To distinguish charge labels from state names, denote ket charges by (a_1, b_1) and bra charges by (a_2, b_2) , with $(0, 0), (1, 1), (3, 1)$ representing $00, ss, 3s, s$. Their occupied-site counts are $M_i = (N - a_i - b_i)/2$. Let $U, V \subset A$ be the occupied sites in the ket and bra configurations; $Y \subset B$ is their common occupied environment. A transition element can be nonzero only when

$$m = M_1 - |U| = M_2 - |V|, \quad 0 \leq m \leq N - \ell. \quad (74)$$

Thus different occupation numbers of the global primaries are fully compatible with a nonzero transition on A .

Set $z_j = e^{2\pi i j/N}$, $g_i(z) = (1 - z/W)^{b_i}$, and define the unnormalized local polynomial

$$L_i(U) = \prod_{j \in U} z_j^{a_i+1} g_i(z_j) \prod_{\substack{j < k \\ j, k \in U}} (z_j - z_k)^2. \quad (75)$$

Splitting the full occupied set into $U \cup Y$ in the ket and $V \cup Y$ in the bra, and using $\overline{z_i - z_j} = -(z_i - z_j)/(z_i z_j)$, yields

$$\begin{aligned} (T_{12}^A)_{UV} &= \frac{L_1(U) \overline{L_2(V)}}{\sqrt{\mathcal{Z}_1 \mathcal{Z}_2}} \prod_{j \in V} z_j^{-2m} \mathcal{E}_{UV}, \\ \mathcal{E}_{UV} &= \sum_{\substack{Y \subset B \\ |Y|=m}} \prod_{\substack{i < j \\ i, j \in Y}} (z_i - z_j)^4 \prod_{i \in Y} \left[w(z_i) \prod_{j \in A} (z_j - z_i)^{2\nu_j} \right], \\ \nu_j &= \mathbf{1}_{j \in U} + \mathbf{1}_{j \in V}, \\ w(z) &= z^\kappa g_1(z) \overline{g_2(z)}, \quad \kappa = a_1 - a_2 - 2M_2 + 2. \end{aligned} \quad (76)$$

Here $\mathcal{Z}_i$ is the full norm sum of state i . The key identity is $2m + \sum_j \nu_j = M_1 + M_2 =: d$: the moment-matrix dimension d is independent of U, V .

To evaluate the environment sum, introduce the column $f(z) = (1, z, \dots, z^{d-1})^T$ and its derivative $f'(z)$. The skew moment matrix on the environment is

$$D_B = \sum_{j \in B} w(z_j) [f(z_j) f'(z_j)^T - f'(z_j) f(z_j)^T]. \quad (77)$$

At each site $j \in A$, include no column if $\nu_j = 0$, $f(z_j)$ if $\nu_j = 1$, and the two columns $f(z_j), f'(z_j)$ if $\nu_j = 2$. Put these columns, in site order, in E_{UV} . With $q = \sum_j \nu_j$ and $\Delta_{\text{ext}} = \prod_{i < j \in A} (z_j - z_i)^{\nu_i \nu_j}$, the discrete confluent-Vandermonde identity gives

$$\mathcal{E}_{UV} = \frac{(-1)^{q(q-1)/2}}{\Delta_{\text{ext}}} \text{Pf} \begin{pmatrix} D_B & E_{UV} \\ -E_{UV}^T & 0 \end{pmatrix}. \quad (78)$$

This follows by expanding the moment Pfaffian over distinct environment sites. The confluent determinant contains their fourth-power Vandermonde, the external factor Δ_{ext} , and the cross factors in Eq. (76). It extends the norm-sum construction of Stéphan and Pollmann [14] to a bra and ket with different charges. It is an exact contraction of the interacting spin wavefunction.

Computing a new d -dimensional Pfaffian for every U, V would waste work. We instead use full-circle root-of-unity moments, remove the ℓ sites of A by a rank- 2ℓ update, and cache the resulting small Pfaffian minors. Odd d and singular even cross moments require the exact border/regularizer construction in Sec. B.2. All large- N moment work then uses sparse matrices; the final minors have at most $2\ell + 1$ indices. Through $N = 256$ we evaluate the cancellation-prone moment algebra at 100 decimal digits before converting T_{12}^A to double precision.

5.5 Step 5: test leading and next powers from CFT

The analytical source is `manuscript_theory/main3.tex` [2]. Its compiled `main.pdf` is an older version without the worked examples used here. In the mixing operator product expansion, let Δ be the lowest exchanged scaling dimension and c_{ab} its normalized coefficient. At $p = 1/2$ the leading deficit is

$$\delta_{a|b}(y) = B_{ab} y^{\beta_{\text{CFT}}} + \dots, \quad \beta_{\text{CFT}} = 2\Delta, \quad B_{ab} = \pi^{2\Delta} |c_{ab}|^2 \frac{\sqrt{\pi} \Gamma(1 + \Delta)}{4\Gamma(3/2 + \Delta)}. \quad (79)$$

Γ is Euler's gamma function. In particular, B_{ab} includes $\pi^{\beta_{\text{CFT}}}$. Each leading coefficient is shared by the two directions; their higher-order coefficients need not coincide.

CFT	Pair (both directions)	β_{CFT}	B_{ab}	Next power
Ising	I, σ	1/4	$\pi^{3/4} \Gamma(9/8) / [4\Gamma(13/8)]$	1/2
Ising	I, ε	2	$\pi^2/3$	4
Ising	σ, ε	1/4	$\pi^{3/4} \Gamma(9/8) / [16\Gamma(13/8)]$	1/2
Boson	$00, ss$	1	$\pi^2/8$	2
Boson	$00, 3s, s$	5	$5\pi^6/64$	6
Boson	$ss, 3s, s$	2	$\pi^2/3$	3 (allowed)

Table 5: Leading large-interval amplitudes multiplying y^β and next powers used in F_2 , at $p = 1/2$ and $s = 1/\sqrt{2}$. The Ising spin-energy channel includes $|c_{\sigma\varepsilon}|^2 = 1/4$. For boson charge codes (a_i, b_i) , $\Delta = [(a_1 - a_2)^2 + (b_1 - b_2)^2]/4$ and $|c_{ab}| = 1$. The last next power follows from the manuscript's general three-point sector rather than a separately evaluated example; its coefficient may vanish. No next-order coefficient is supplied or derived here.

The manuscript's Ising examples explicitly identify four-spin terms of order $y^{1/2}$ and later two-spin-energy terms of order $y^{5/4}$; the first correction retained in F_2 is $y^{1/2}$. The identity-energy example

specifies a remainder of order y^4 . The boson vacuum examples specify orders y^2 and y^6 . For the remaining boson pair, the general subsection “Three point function” contains powers $2\Delta_{L'} + \Delta_L$: the exchanged vertex has $\Delta_{L'} = 1$ and a current has $\Delta_L = 1$, giving the candidate power three. This identifies an allowed sector, not a proved nonzero total coefficient after replica continuation. We consequently interpret that pair’s F_2 as a test of the next allowed power. This qualification also prevents an $O(y^r)$ remainder from being mistaken for a known coefficient. No universal increment of two is imposed for large intervals.

$$F_{\text{free}}(y) = Cy^{\hat{\beta}}, \quad F_1(y) = c_0 y^{\beta_{\text{CFT}}}, \quad F_2(y) = c_0 y^{\beta_{\text{CFT}}} + c_1 y^{\beta_1}. \quad (80)$$

Here β_1 is the next power in Table 5. Only the exponents of F_1, F_2 are imposed; all their coefficients are fitted separately in each direction. “CFT leading” always means $B_{ab}y^{\beta_{\text{CFT}}}$, with neither parameter fitted. For completeness, the one-term solution is

$$c_0 = \frac{\sum_i y_i^{\beta_{\text{CFT}}} \delta_i}{\sum_i y_i^{2\beta_{\text{CFT}}}}. \quad (81)$$

The fits use the CFT-wide windows in Eq. (59): $y \leq 0.002$ for Ising and $y \leq 0.005$ for the boson. Every direction uses the same window as its reverse. Figures show data through $y \leq 0.20$, with fitting-window zooms in the per-pair views. The objective, coefficient convention, residuals and size cross-validation are exactly Eqs. (61), (62) and (63).

$$R_i = 100[\delta_i - M(y_i)]/\delta_i, \quad \eta_\beta = 100(\hat{\beta}/\beta_{\text{CFT}} - 1). \quad (82)$$

$$\text{CVRMSE} = \sqrt{m^{-1} \sum_i [\delta_i - M^{(-N_i)}(y_i)]^2}. \quad (83)$$

The residual sign in the entropy $D^B = \log 2 - \delta$ is opposite to the deficit residual. Tables in Appendix B separate Free, F_1 , F_2 and CFT rows for all twelve directions.

5.6 Results

Ising full-window free-exponent deviations range from -0.074% to $+5.861\%$. F_2 improves size-holdout RMSE over F_1 in 6 of six directions. These statements use the prescribed fitting window and the sample-count requirements in Appendix D.

Compact-boson full-window free-exponent deviations range from -0.438% to -0.074% . F_2 improves size-holdout RMSE over F_1 in 2 of six directions. These statements use the prescribed fitting window and the sample-count requirements in Appendix D.

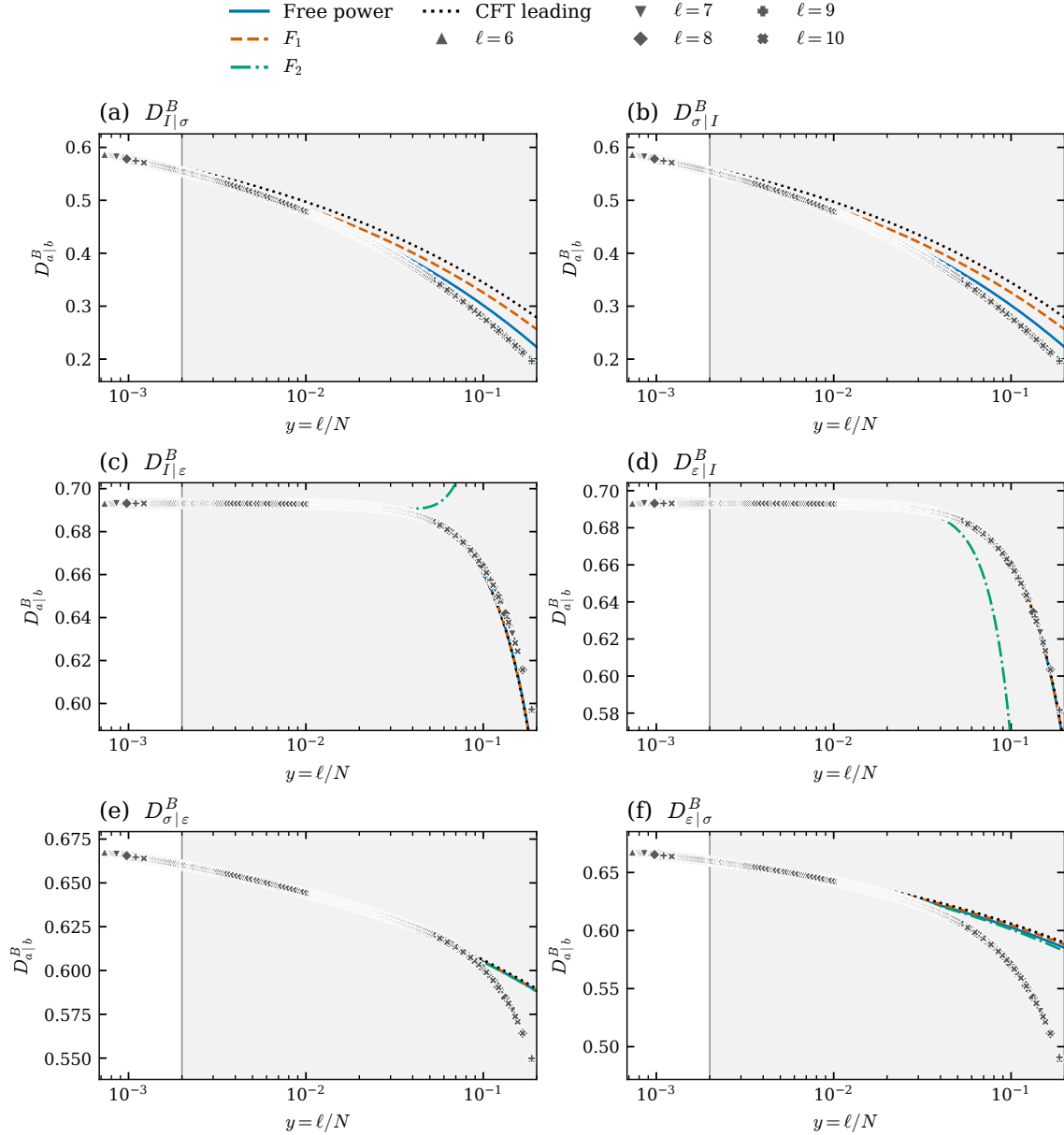


Figure 6: The ordinate is $D^B = \log 2 - \delta$, on a linear scale; tick labels give its value directly, without an additive offset. The fraction axis is logarithmic. Ising mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 8192$. There are 1645 displayed geometries per direction at $y \leq 0.2$, of which 260 satisfy $y \leq 0.002$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The periodic Ising Hamiltonian is $H = -\sum_j X_j X_{j+1} - \sum_j Z_j$; primaries use exact fermionic covariances, including the odd Ramond ground state for σ . Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the analytical CFT leading term (zero). Powers and amplitudes use Eqs. (60) and (80); all coefficients include powers of π . Fits minimize unweighted squared residuals in δ . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 14 and 15 also have per-pair full-range and fitting-window views in Sec. 6.6.

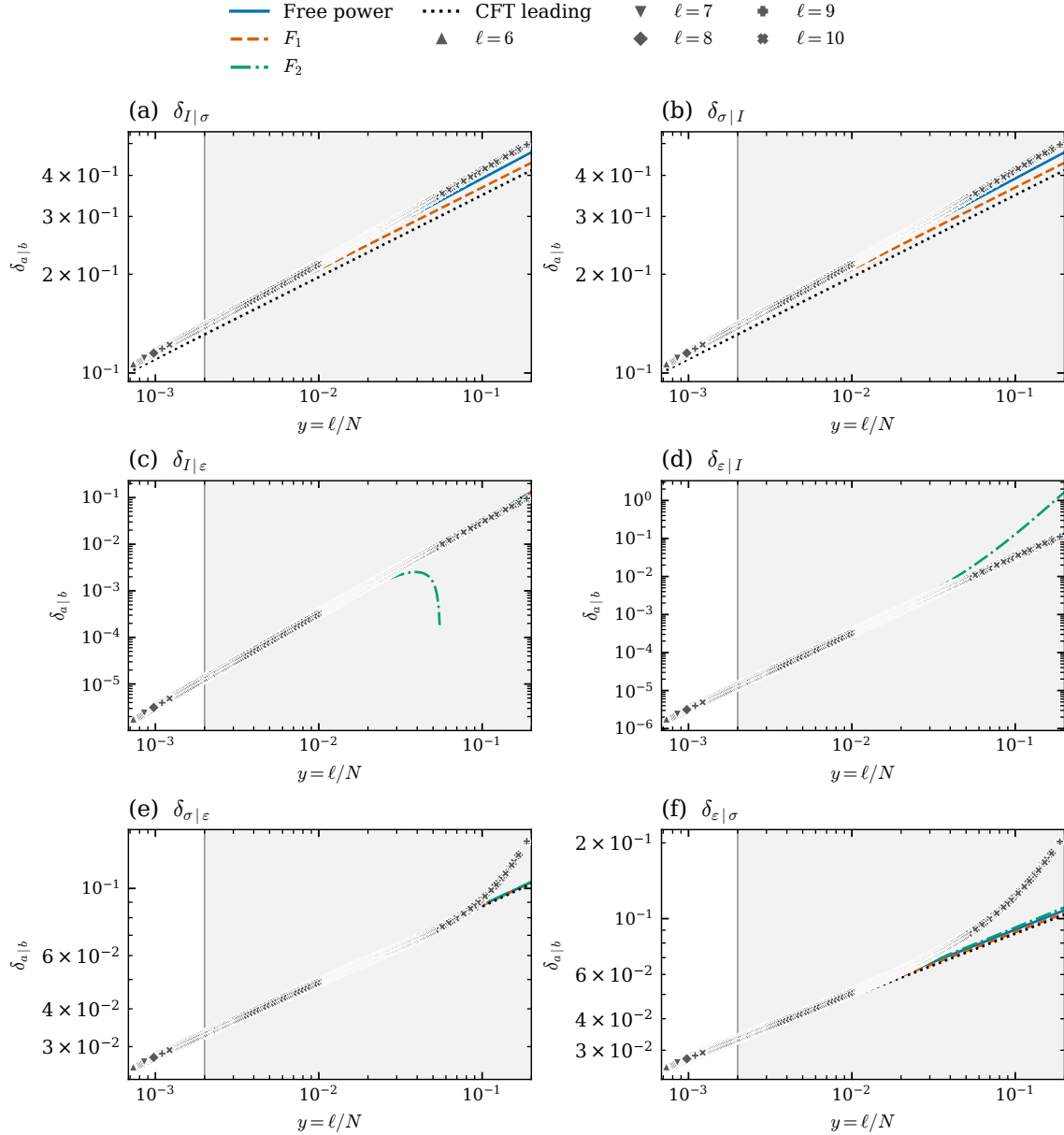


Figure 7: Both axes are logarithmic. Nonpositive values of an extrapolated truncated curve are omitted only from this logarithmic panel, while residual panels retain their predictions. Using mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 8192$. There are 1645 displayed geometries per direction at $y \leq 0.2$, of which 260 satisfy $y \leq 0.002$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The periodic Ising Hamiltonian is $H = -\sum_j X_j X_{j+1} - \sum_j Z_j$; primaries use exact fermionic covariances, including the odd Ramond ground state for σ . Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the analytical CFT leading term (zero). Powers and amplitudes use Eqs. (60) and (80); all coefficients include powers of π . Fits minimize unweighted squared residuals in δ . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 14 and 15 also have per-pair full-range and fitting-window views in Sec. 6.6.

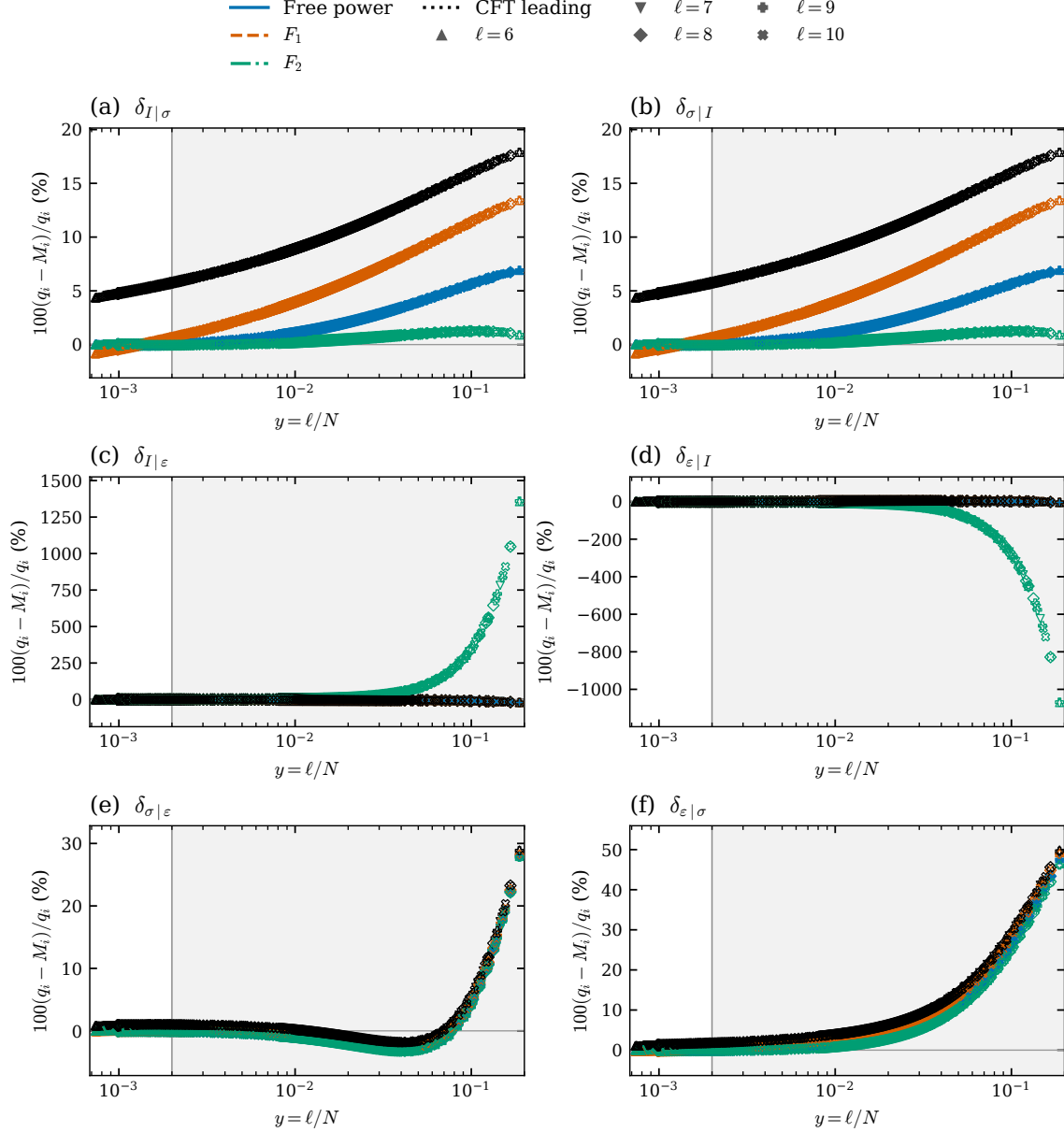


Figure 8: The ordinate is $100(q_i - M_i)/q_i$ percent, with $q_i = D_i^A$ for a small interval and $q_i = \delta_i$ for a large interval; the horizontal zero line aids comparison. All four models have residual markers, colored as their curves. The residual ordinate is linear; the fraction axis is logarithmic. Using mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 8192$. There are 1645 displayed geometries per direction at $y \leq 0.2$, of which 260 satisfy $y \leq 0.002$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The periodic Ising Hamiltonian is $H = -\sum_j X_j X_{j+1} - \sum_j Z_j$; primaries use exact fermionic covariances, including the odd Ramond ground state for σ . Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the analytical CFT leading term (zero). Powers and amplitudes use Eqs. (60) and (80); all coefficients include powers of π . Fits minimize unweighted squared residuals in δ . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 14 and 15 also have per-pair full-range and fitting-window views in Sec. 6.6.

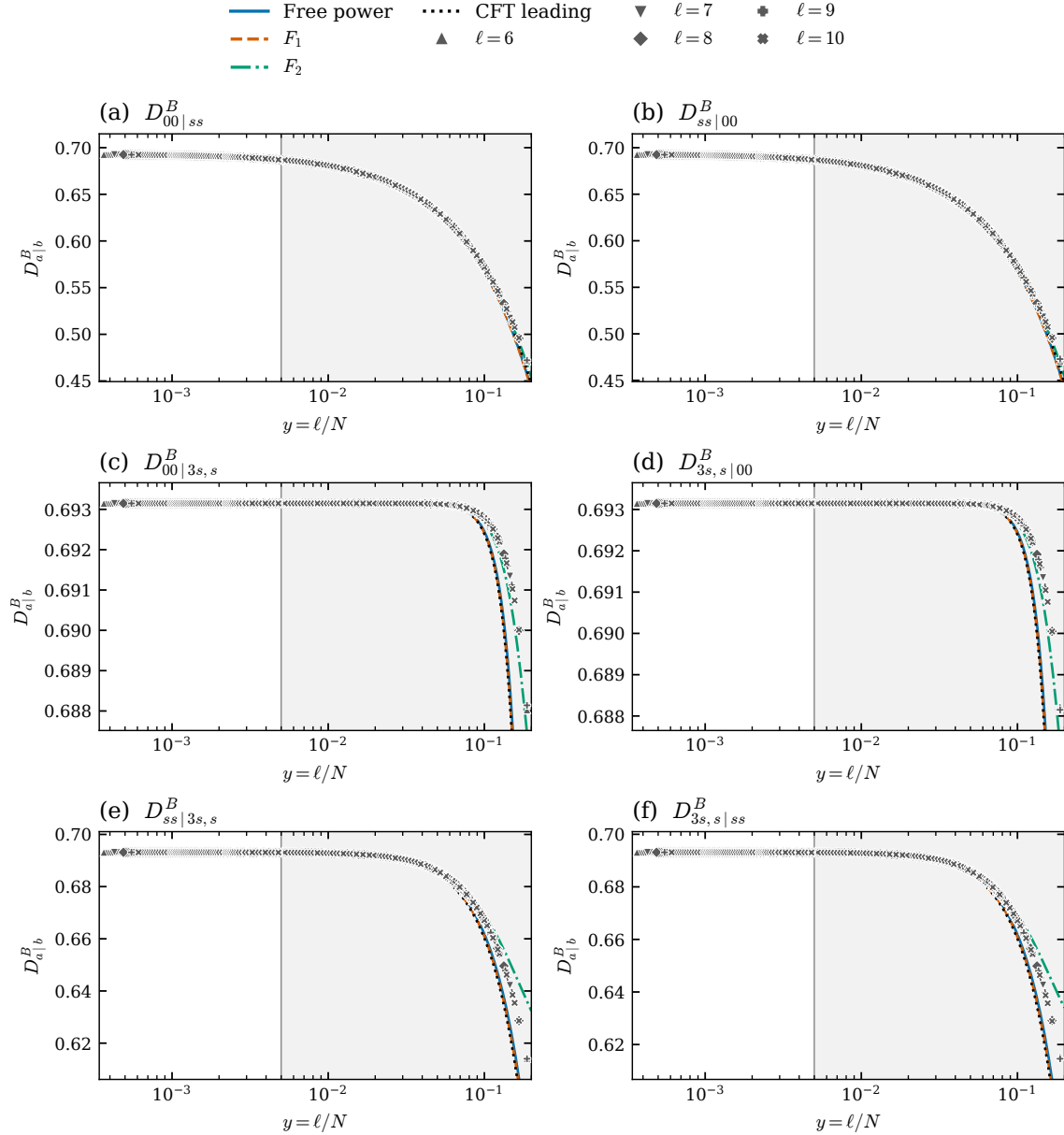


Figure 9: The ordinate is $D^B = \log 2 - \delta$, on a linear scale; tick labels give its value directly, without an additive offset. The fraction axis is logarithmic. Compact-boson mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 16384$. There are 625 displayed geometries per direction at $y \leq 0.2$, of which 317 satisfy $y \leq 0.005$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The boson uses $s = 1/\sqrt{2}$, $q = 2$, charge codes $(0, 0), (1, 1), (3, 1)$, $z_j = e^{2\pi i j/N}$, $w_1 = 0$, and $W = 10^8$; moment algebra uses 100 digits through $N = 256$ and 120 digits at the added sizes. Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the analytical CFT leading term (zero). Powers and amplitudes use Eqs. (60) and (80); all coefficients include powers of π . Fits minimize unweighted squared residuals in δ . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 16 and 17 also have per-pair full-range and fitting-window views in Sec. 6.7.

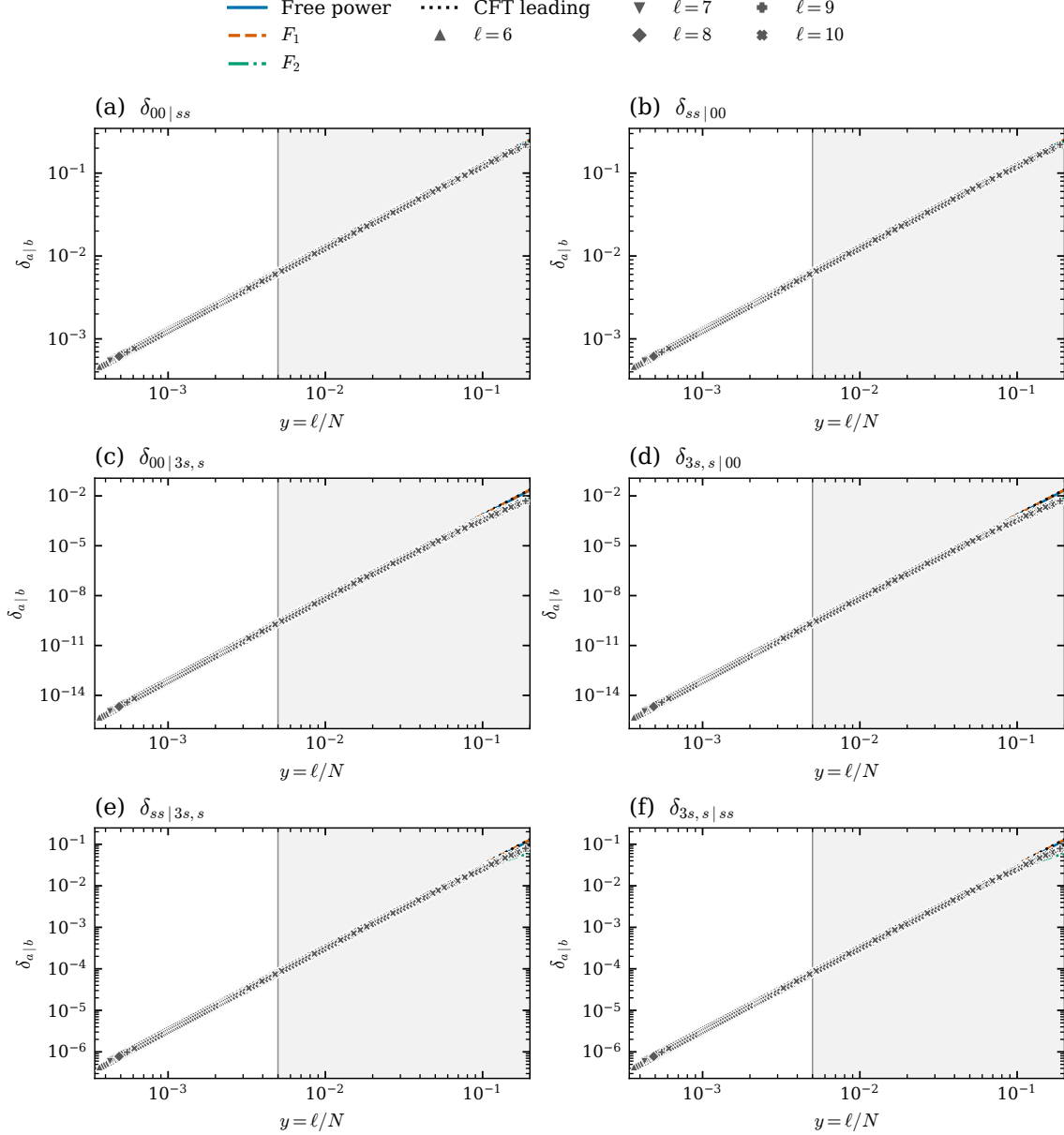


Figure 10: Both axes are logarithmic. Nonpositive values of an extrapolated truncated curve are omitted only from this logarithmic panel, while residual panels retain their predictions. Compact-boson mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 16384$. There are 625 displayed geometries per direction at $y \leq 0.2$, of which 317 satisfy $y \leq 0.005$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The boson uses $s = 1/\sqrt{2}$, $q = 2$, charge codes $(0, 0), (1, 1), (3, 1)$, $z_j = e^{2\pi i j/N}$, $w_1 = 0$, and $W = 10^8$; moment algebra uses 100 digits through $N = 256$ and 120 digits at the added sizes. Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the analytical CFT leading term (zero). Powers and amplitudes use Eqs. (60) and (80); all coefficients include powers of π . Fits minimize unweighted squared residuals in δ . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 16 and 17 also have per-pair full-range and fitting-window views in Sec. 6.7.

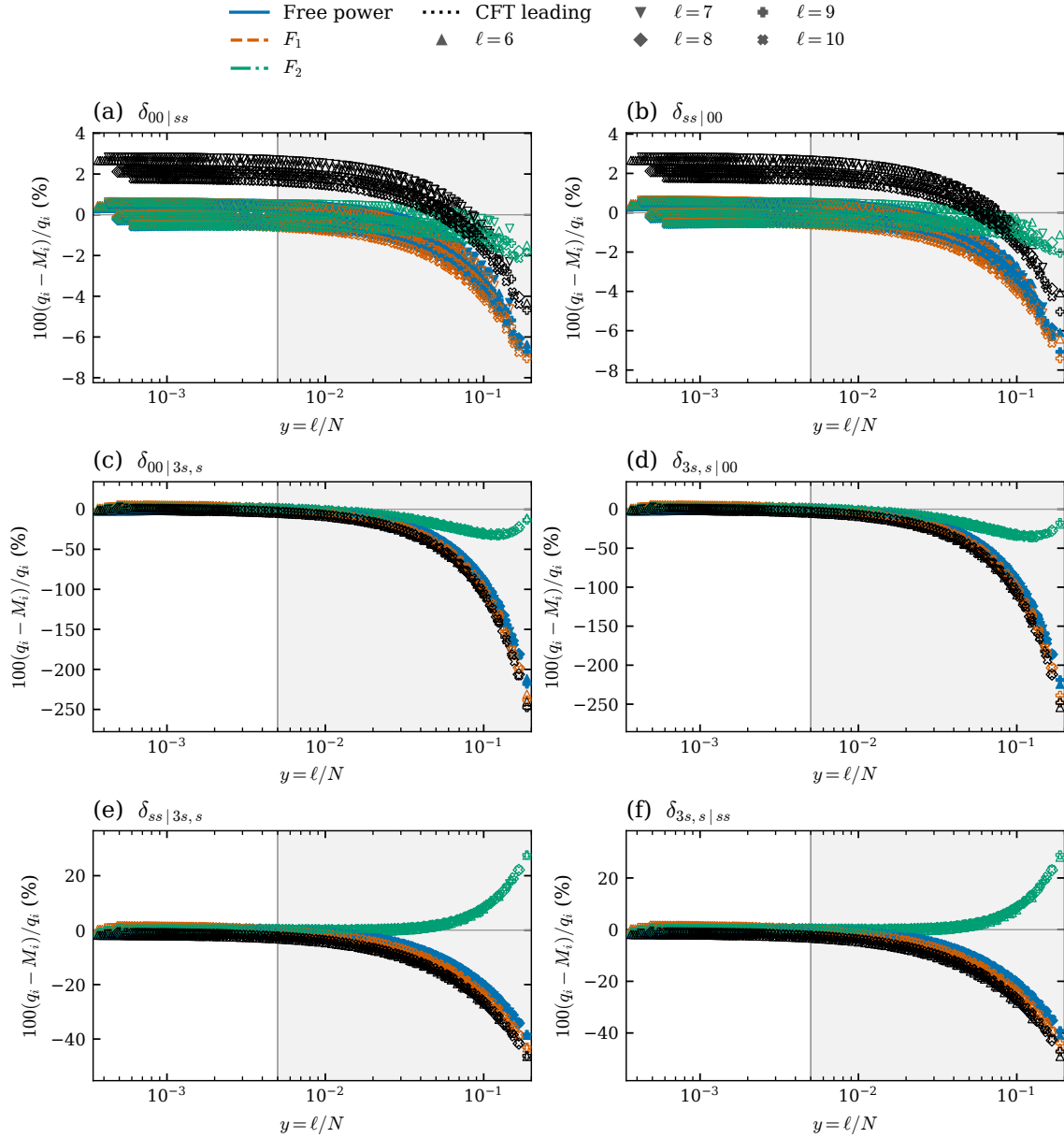


Figure 11: The ordinate is $100(q_i - M_i)/q_i$ percent, with $q_i = D_i^A$ for a small interval and $q_i = \delta_i$ for a large interval; the horizontal zero line aids comparison. All four models have residual markers, colored as their curves. The residual ordinate is linear; the fraction axis is logarithmic. Compact-boson mixtures at $p = 1/2$, $\ell = 6, 7, 8, 9, 10$, and the chain-size set in Eq. (52), with $32 \leq N \leq 16384$. There are 625 displayed geometries per direction at $y \leq 0.2$, of which 317 satisfy $y \leq 0.005$ and enter every fit. Marker shape identifies ℓ ; the vertical boundary and gray shading mark the fitting cutoff and excluded data. The boson uses $s = 1/\sqrt{2}$, $q = 2$, charge codes $(0, 0)$, $(1, 1)$, $(3, 1)$, $z_j = e^{2\pi i j/N}$, $w_1 = 0$, and $W = 10^8$; moment algebra uses 100 digits through $N = 256$ and 120 digits at the added sizes. Natural logarithms apply. Added sizes use the direct entropy remainder in Sec. 3.8.2, with support cutoff 10^{-15} . Blue solid is the free-power fit (two fitted parameters), orange dashed is F_1 (one), green dash-dot is F_2 (two), and black dotted is the analytical CFT leading term (zero). Powers and amplitudes use Eqs. (60) and (80); all coefficients include powers of π . Fits minimize unweighted squared residuals in δ . These curves use `scaling_fit_summary.csv`. The identical fits in Tables 16 and 17 also have per-pair full-range and fitting-window views in Sec. 6.7.

6 Visual comparison of fits for each ordered pair

6.1 Question, data, and models

For each ordered pair we display the fitting models, their actual intervals, their residuals, and the supplied analytical curve together. Each figure compares the principal summary-table fits in its left column with the varying-window fits in its right column. We compare Free 1, F_1 , and F_2 . Each column uses one common interval for all three models, so adding the second term can be assessed on identical observations.

We use every previously computed mixture geometry in Eq. (52), with $p = 1/2$ and $\ell = 6, \dots, 10$. The raw $\ell = 4, 5$ observations remain archived but enter neither estimation nor display. This restores the five-length analysis on its original chain grid; the subsequent $\ell = 11, 12$ extension and complete-seven-length-block study remain separate archived comparisons. In the current analysis, each observation with $\ell \in \{6, 7, 8, 9, 10\}$ is selected individually using its fraction ℓ/N and the group's fitting window; a size need not contribute every length inside that window. The dense scan adds 235 Ising and 47 boson sizes, all below the previously validated maxima; see Appendix H. Write $z = x = \ell/N$ and $q = D^A$ for small intervals; write $z = y = \ell/N$ and $q = \delta = \log 2 - D^B$ for large complementary intervals. Thus α always denotes a small-interval leading power and β a large-interval leading power. All active models use the four common windows in Eq. (59); the small-interval choices are frozen and the large-boson upper cutoff is prescribed as 0.005. The same fits appear in the overview figures and the principal tables in Appendices A and B. The earlier $(0, 0.001]$ free-power fits remain a separate diagnostic.

For a window $W = [u, v]$, we compare

$$\begin{aligned} F_1(z) &= c_1 z^{r_0}, & F_{\text{free},1}(z) &= c_1 z^{r^{(1)}}, \\ F_2(z) &= c_1 z^{r_0} + c_2 z^{r_{\text{next}}}. \end{aligned} \tag{84}$$

Here r_0, r_{next} are the pair-specific manuscript powers; F_2 fixes both, since a general separation between freely estimated leading and next powers is not established. Free 1 estimates only the leading power. In this section c_1 is the leading amplitude and c_2 the next amplitude, corresponding to c_0, c_1 in Eq. (60). They multiply z^r itself and include all powers of π . There are one, two, and two fitted parameters in F_1 , Free 1, and F_2 , respectively. The objective is always

$$\text{SSE}_W = \sum_{i: z_i \in W} [q_i - M(z_i)]^2. \tag{85}$$

All geometries receive equal weight, including distinct (N, ℓ) that have the same fraction. No logarithmic residual objective or extra lattice-correction term is fitted.

The theory comparison uses the source `manuscript_theory/main3.tex` [2], especially its non-chiral compact-boson and Ising examples. The small-interval powers and complete supplied coefficients are in Tables 1 and 2: $r_{\text{next}} = \alpha + 2$. Only $I \mid \varepsilon$ and $\varepsilon \mid I$ have complete supplied next coefficients. Large intervals use Table 5; no common increment is assumed. For large boson $ss \leftrightarrow 3s, s$, power three is an allowed sector, not a proved nonzero coefficient. An empty analytical cell in the new tables means that the complete coefficient is unavailable.

6.2 Fitting formulas and plotted quantities

For fixed powers, the coefficients follow from linear least squares. Use $t_i = \log(z_i/v)$ and $A_j = c_j v^{r_j}$. Define

$$X_{ij} = \exp(r_j t_i), \quad \hat{A} = X^+ \mathbf{q}, \quad c_j = \hat{A}_j / v^{r_j}. \tag{86}$$

X^+ is the Moore–Penrose pseudoinverse, computed by singular-value decomposition. For F_1 the design matrix has one column with power r_0 ; for F_2 it has two columns with powers r_0, r_{next} . Free 1 uses one column and minimizes the residual over its single trial power. Eliminating the linear coefficient before

this search is variable projection [1]. Dividing all q_i by their root-mean-square value improves numerical scaling and does not change the least-squares optimum.

In the figures, the upper panels show q and the fitted curves. Solid colored segments lie inside the corresponding interval W ; thin dotted segments show extrapolation. The lower residual panels show only the observations used by each fit, with $R_i = 100[q_i - M(z_i)]/q_i$ on a linear ordinate. The interval bars and parameter block specify W separately for every model. A zero lower endpoint is open; it includes the smallest positive observation. The left column shows all available data through $z = 0.20$; the right column zooms into the common fitting window using the same curves. Both residual columns show the same fitted observations on linear ordinates. The shared parameter block lists each fit once.

For the numerical-error tables, define

$$E_{\text{fit}} = 100 \sqrt{\frac{\sum_{i \in W} [q_i - M(z_i)]^2}{\sum_{i \in W} q_i^2}}, \quad E_{\text{CV}} = 100 \sqrt{\frac{\sum_{i \in W} [q_i - M^{(-N_i)}(z_i)]^2}{\sum_{i \in W} q_i^2}}. \quad (87)$$

Here $M^{(-N_i)}$ is refitted after omitting all observations at chain size N_i , as in Eq. (63). These aggregate percentages differ from the pointwise percentages in the residual panels. Every displayed fit and omitted-size training set contains at least forty observations. Different fitting intervals contain different observations, so their absolute RMSEs should not be ranked directly.

6.3 Comparison with theory and results

Compare the fitted parameters with the manuscript using

$$\eta_{c,j} = 100 \left(\frac{c_j}{c_j^{\text{CFT}}} - 1 \right), \quad \eta_{r,j} = 100 \left(\frac{r^{(j)}}{r_j^{\text{CFT}}} - 1 \right), \quad j = 1, 2. \quad (88)$$

An unknown analytical coefficient leaves both its comparison cell and its percentage discrepancy empty. Fixed exponents are identified as imposed rather than estimated. All models and all six directions within one CFT and interval type use the same window. Both columns show the same fitted models. The current omitted-size controls are in `data/f2_summary_cv.csv`; the principal-table gains use Eq. (89). The known small-Ising next coefficients provide a direct comparison: F_2 gives $c_2 = -170.71, -355.38$ for $I \mid \varepsilon$ and $\varepsilon \mid I$, whereas the manuscript gives $-8\pi^6/45$ and $-8\pi^6/21$. The signed discrepancies are -0.12% , -2.97% . These fits use the common small-Ising window in both columns; Appendix H examines the stability of both coefficients. The tables retain every fitted value and compare all supplied theoretical powers and coefficients.

Table 6: Ising small intervals: fitting intervals for the models shown in the right-hand panels of Sec. 6.4. All three models share the displayed interval, providing an identical-data comparison. Zero is an open lower bound, with no observation at zero.

Direction Free 1 and F_1 F_2		
$I \mid \sigma$	(0, 0.02]	(0, 0.02]
$\sigma \mid I$	(0, 0.02]	(0, 0.02]
$I \mid \varepsilon$	(0, 0.02]	(0, 0.02]
$\varepsilon \mid I$	(0, 0.02]	(0, 0.02]
$\sigma \mid \varepsilon$	(0, 0.02]	(0, 0.02]
$\varepsilon \mid \sigma$	(0, 0.02]	(0, 0.02]

Table 7: Compact boson small intervals: fitting intervals for the models shown in the right-hand panels of Sec. 6.5. All three models share the displayed interval, providing an identical-data comparison. Zero is an open lower bound, with no observation at zero.

Direction Free 1 and F_1 F_2		
00 ss	(0, 0.1]	(0, 0.1]
ss 00	(0, 0.1]	(0, 0.1]
00 $3s, s$	(0, 0.1]	(0, 0.1]
$3s, s$ 00	(0, 0.1]	(0, 0.1]
ss $3s, s$	(0, 0.1]	(0, 0.1]
$3s, s$ ss	(0, 0.1]	(0, 0.1]

Table 8: Ising large intervals: fitting intervals for the models shown in the right-hand panels of Sec. 6.6. All three models share the displayed interval, providing an identical-data comparison. Zero is an open lower bound, with no observation at zero.

Direction Free 1 and F_1 F_2		
I σ	(0, 0.002]	(0, 0.002]
σ I	(0, 0.002]	(0, 0.002]
I ε	(0, 0.002]	(0, 0.002]
ε I	(0, 0.002]	(0, 0.002]
σ ε	(0, 0.002]	(0, 0.002]
ε σ	(0, 0.002]	(0, 0.002]

Table 9: Compact boson large intervals: fitting intervals for the models shown in the right-hand panels of Sec. 6.7. All three models share the displayed interval, providing an identical-data comparison. Zero is an open lower bound, with no observation at zero.

Direction Free 1 and F_1 F_2		
00 ss	(0, 0.005]	(0, 0.005]
ss 00	(0, 0.005]	(0, 0.005]
00 $3s, s$	(0, 0.005]	(0, 0.005]
$3s, s$ 00	(0, 0.005]	(0, 0.005]
ss $3s, s$	(0, 0.005]	(0, 0.005]
$3s, s$ ss	(0, 0.005]	(0, 0.005]

The full amplitudes, powers, and theoretical comparisons are in Appendix G. Each figure below treats one ordered pair. Its parameter block lists every displayed fit and the common window used in both columns. Window sensitivity is examined separately in Appendix H. The analytical curve has no fitted parameters. All retained-model results remain visible without window classifications or acceptance labels.

6.4 Ising, small intervals

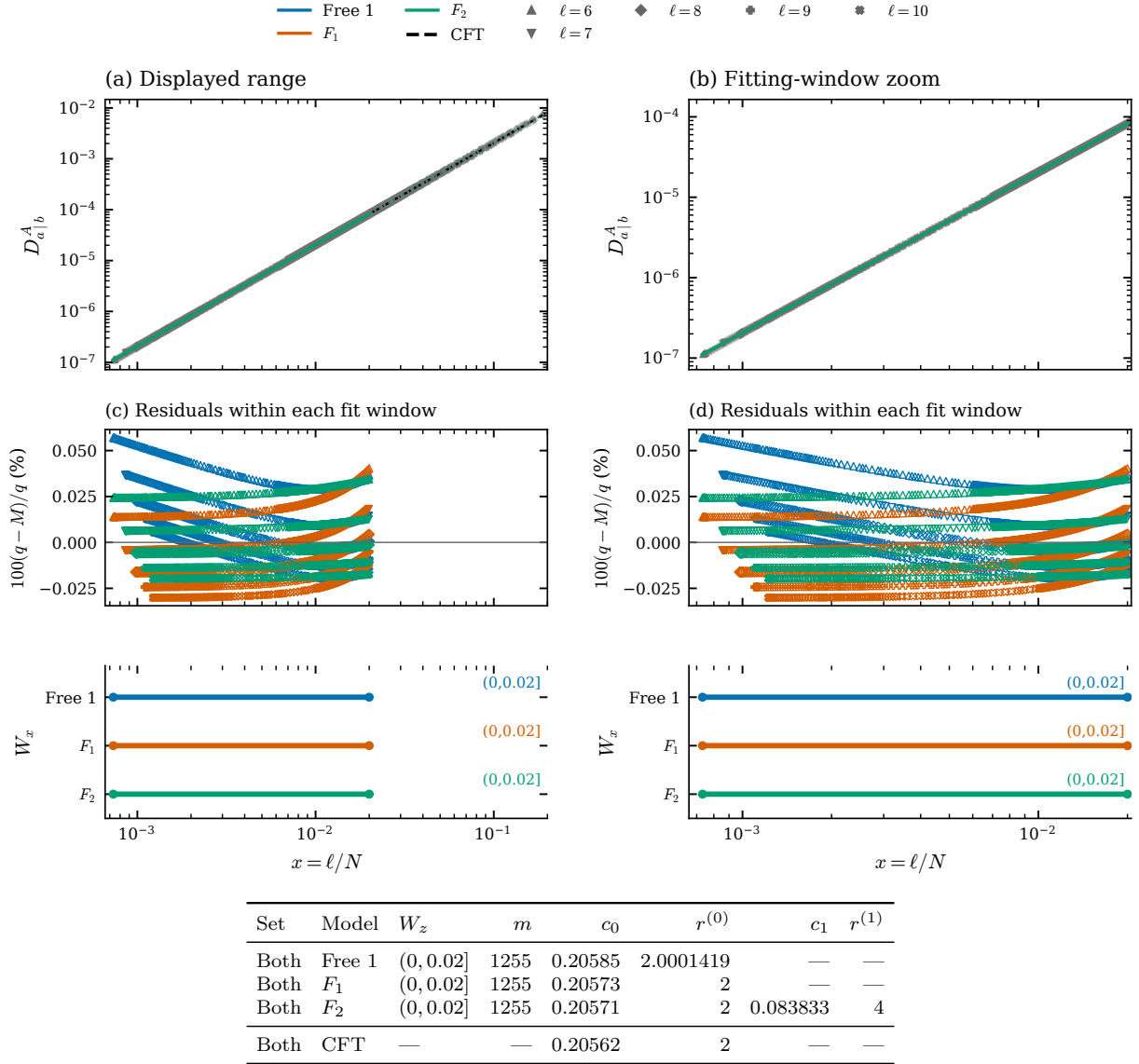


Figure 12: Ising, small intervals, $I \mid \sigma$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

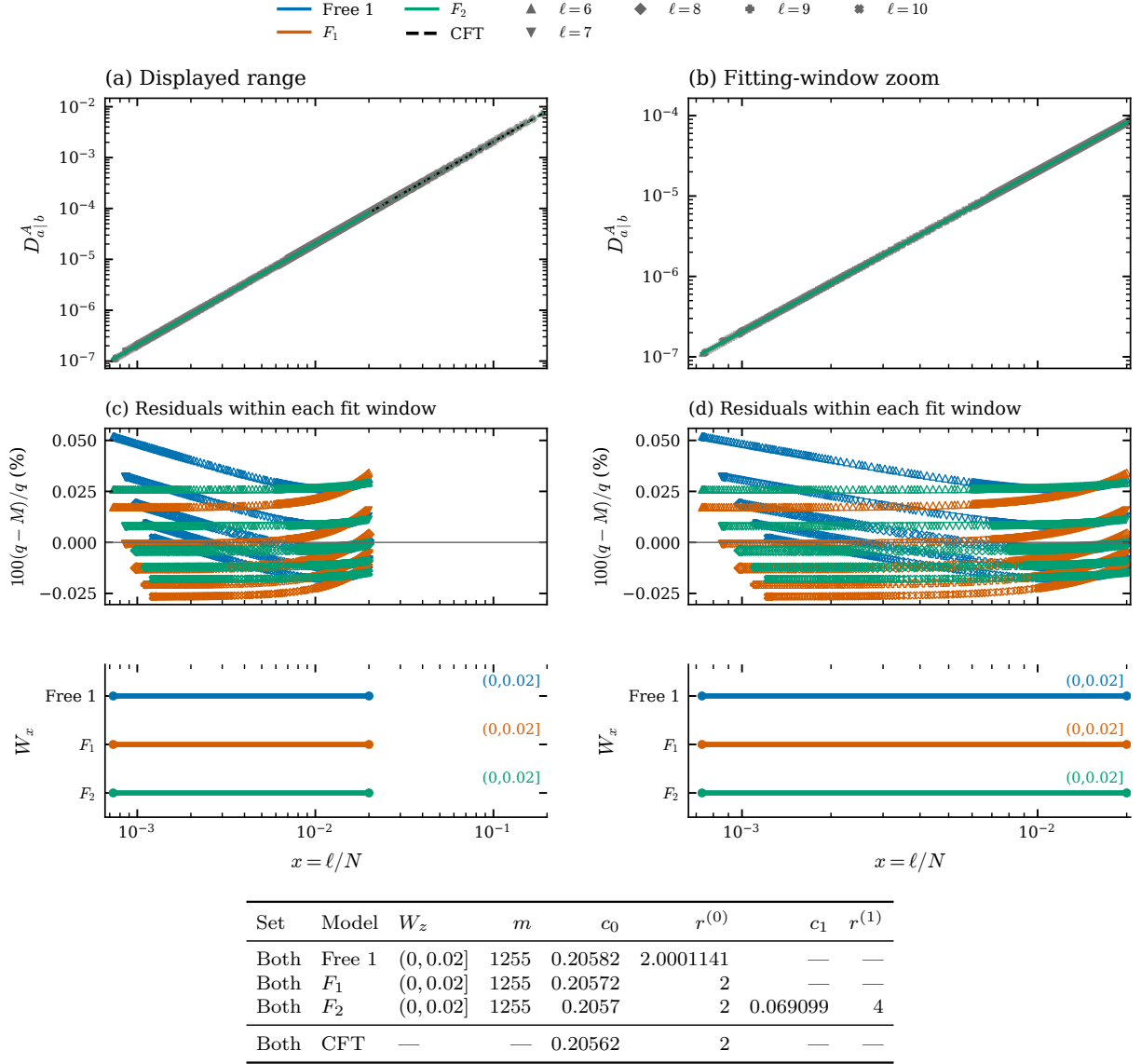


Figure 13: Ising, small intervals, $\sigma \mid I$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

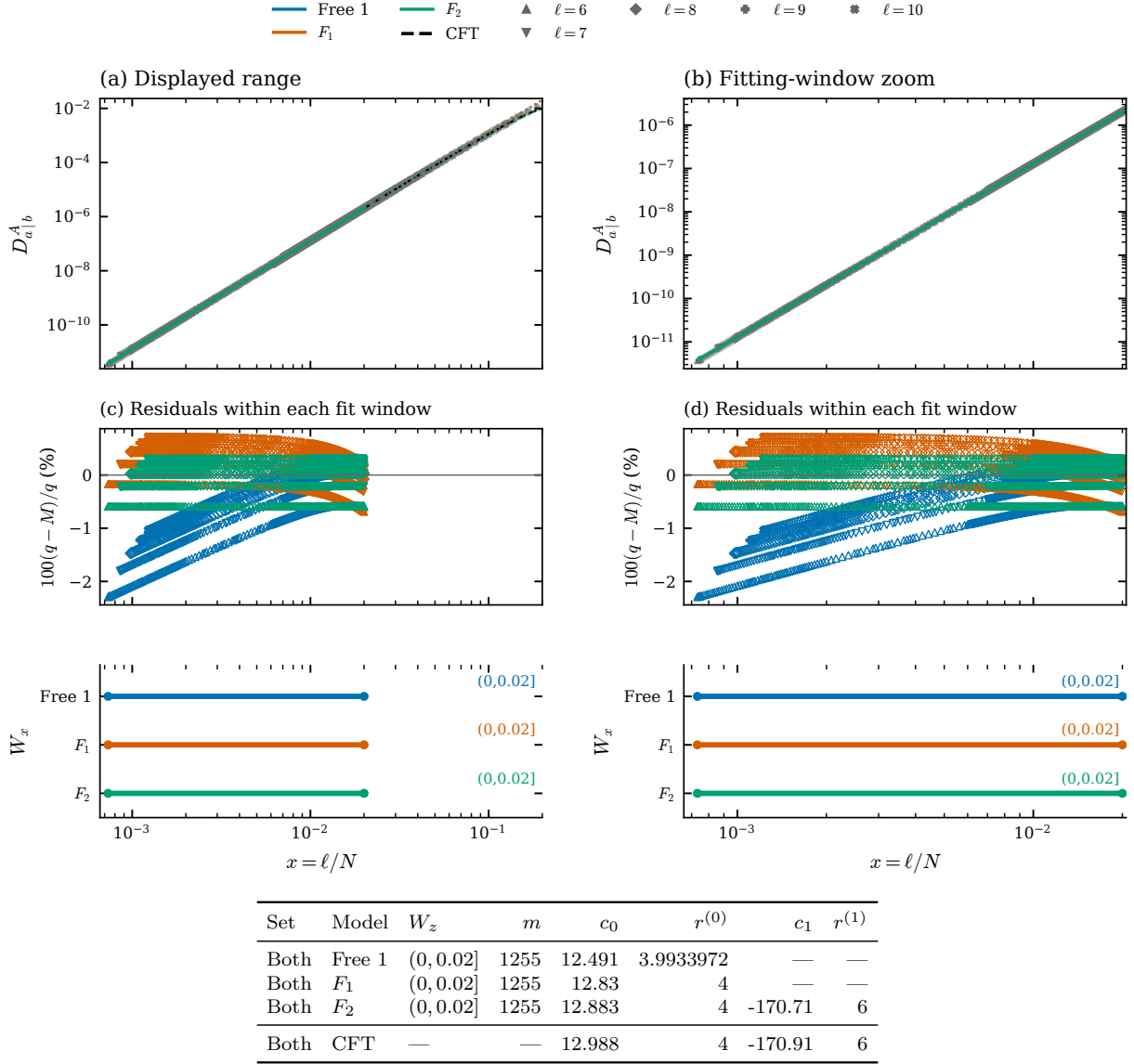


Figure 14: Ising, small intervals, $I \mid \varepsilon$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

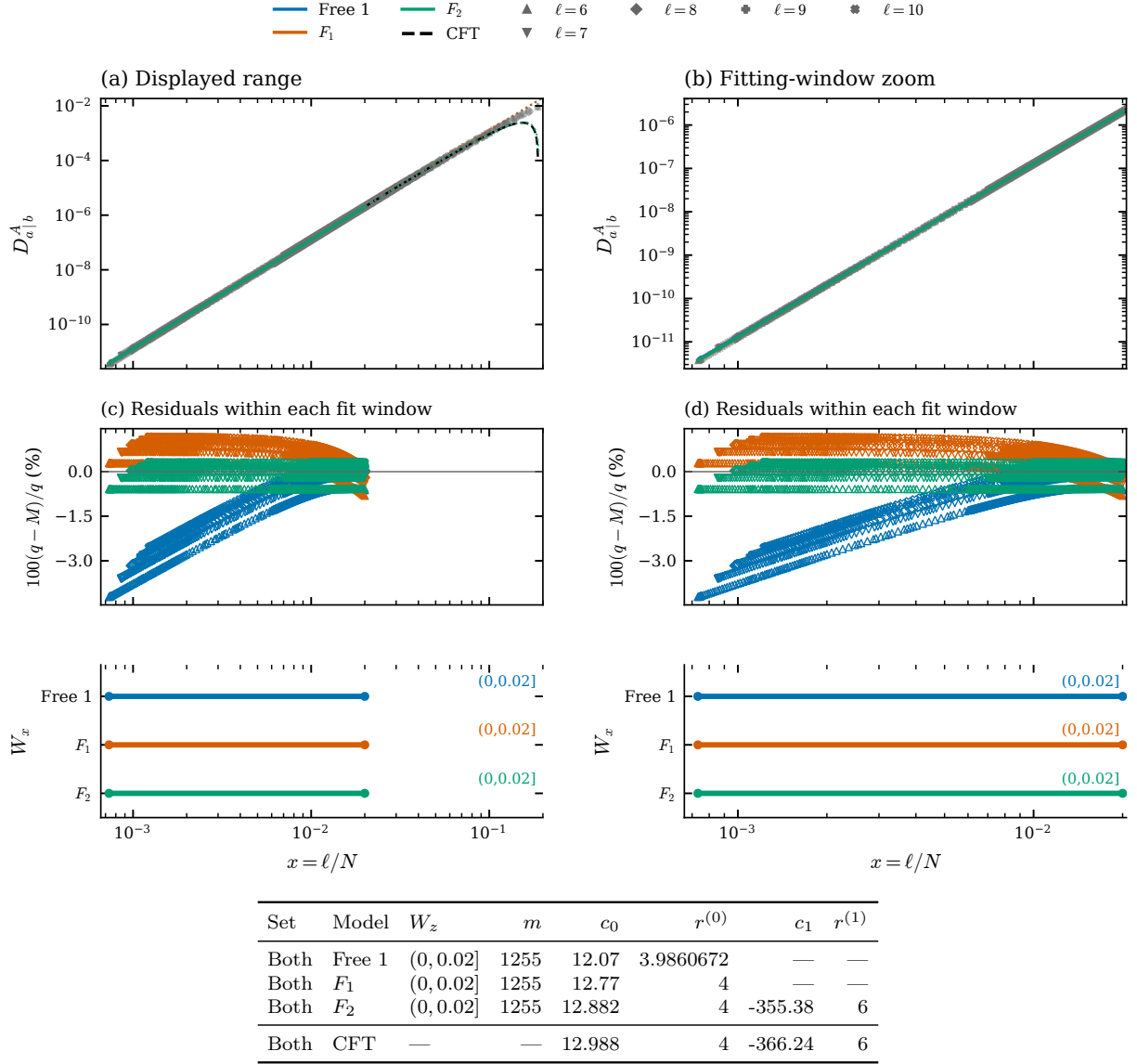


Figure 15: Ising, small intervals, $\varepsilon \mid I$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

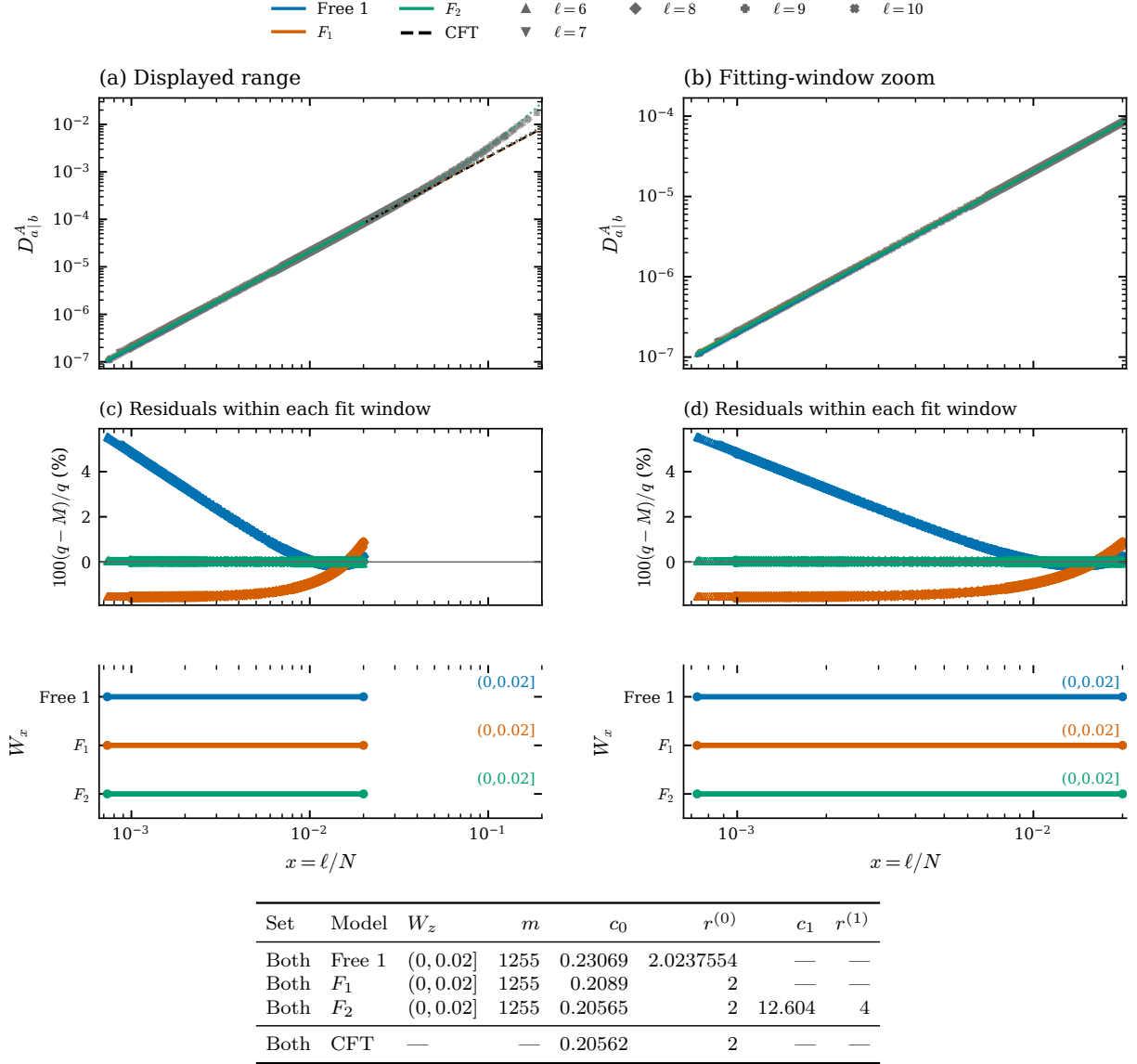


Figure 16: Ising, small intervals, $\sigma \mid \varepsilon$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

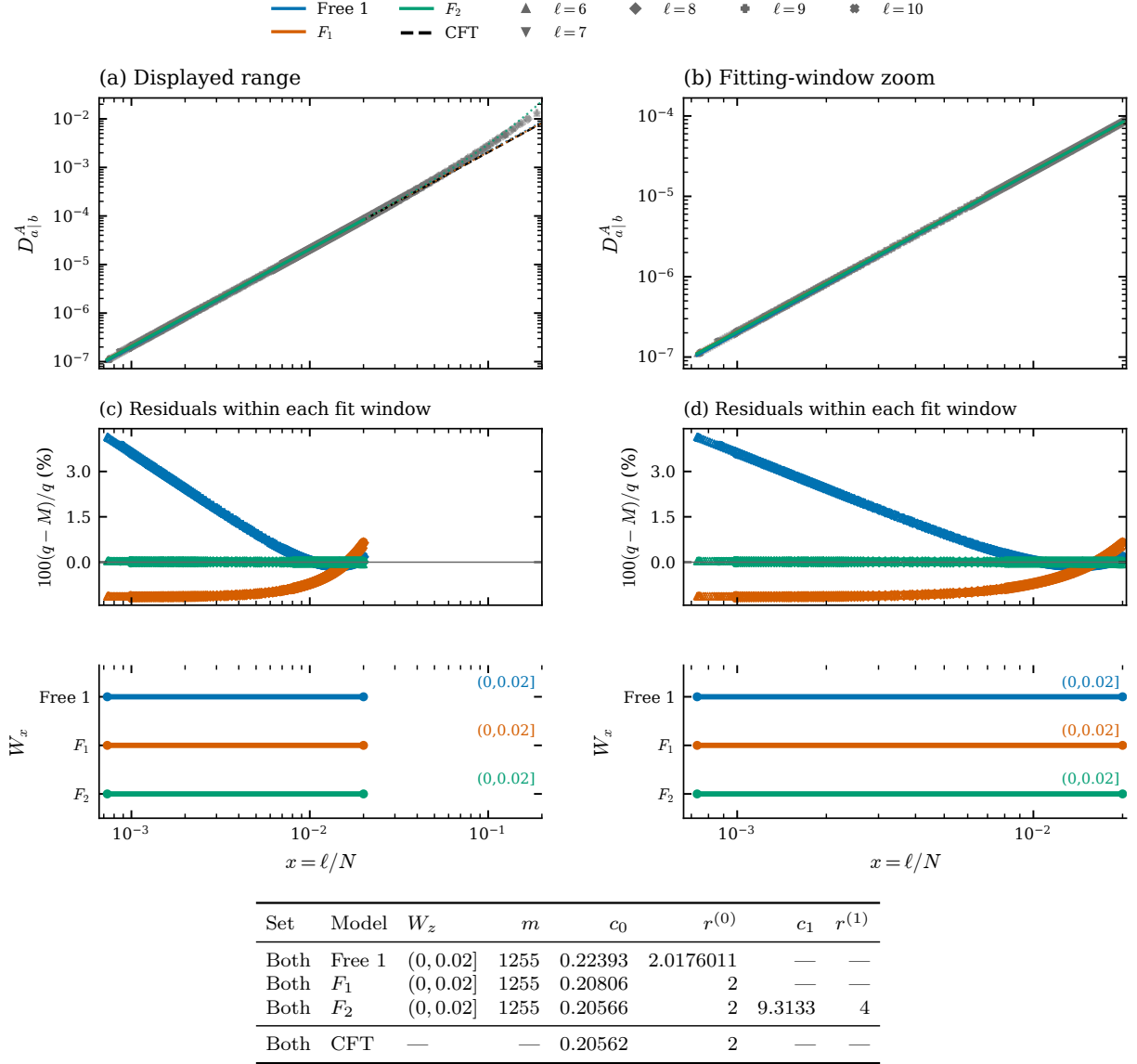
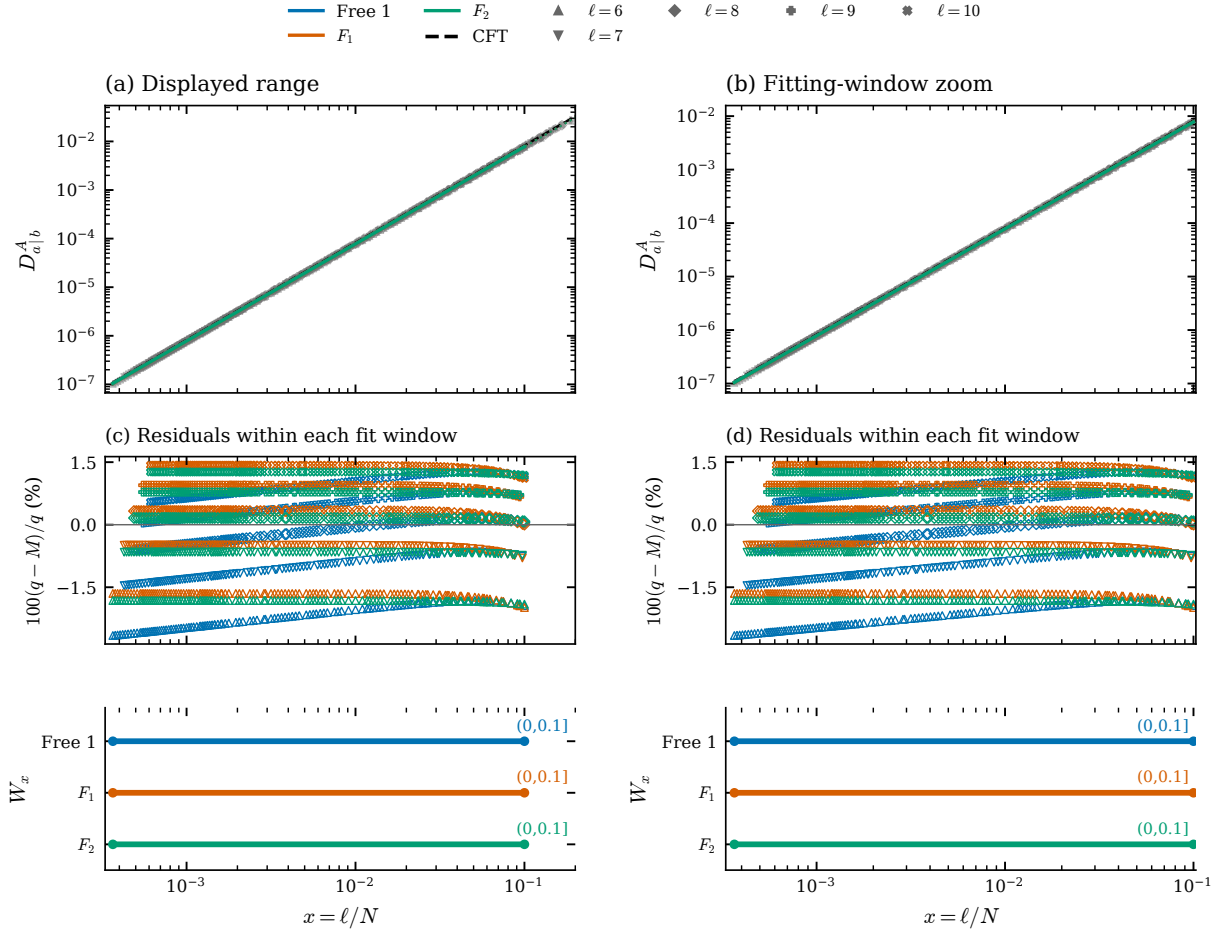


Figure 17: Ising, small intervals, $\varepsilon \mid \sigma$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

6.5 Compact boson, small intervals



Set	Model	W_z	m	c_0	$r^{(0)}$	c_1	$r^{(1)}$
Both	Free 1	(0, 0.1]	598	0.77501	1.998159	—	—
Both	F_1	(0, 0.1]	598	0.77866	2	—	—
Both	F_2	(0, 0.1]	598	0.77996	2	-0.1917	4
Both	CFT	—	—	0.82247	2	—	—

Figure 18: Compact boson, small intervals, $00 \mid ss$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

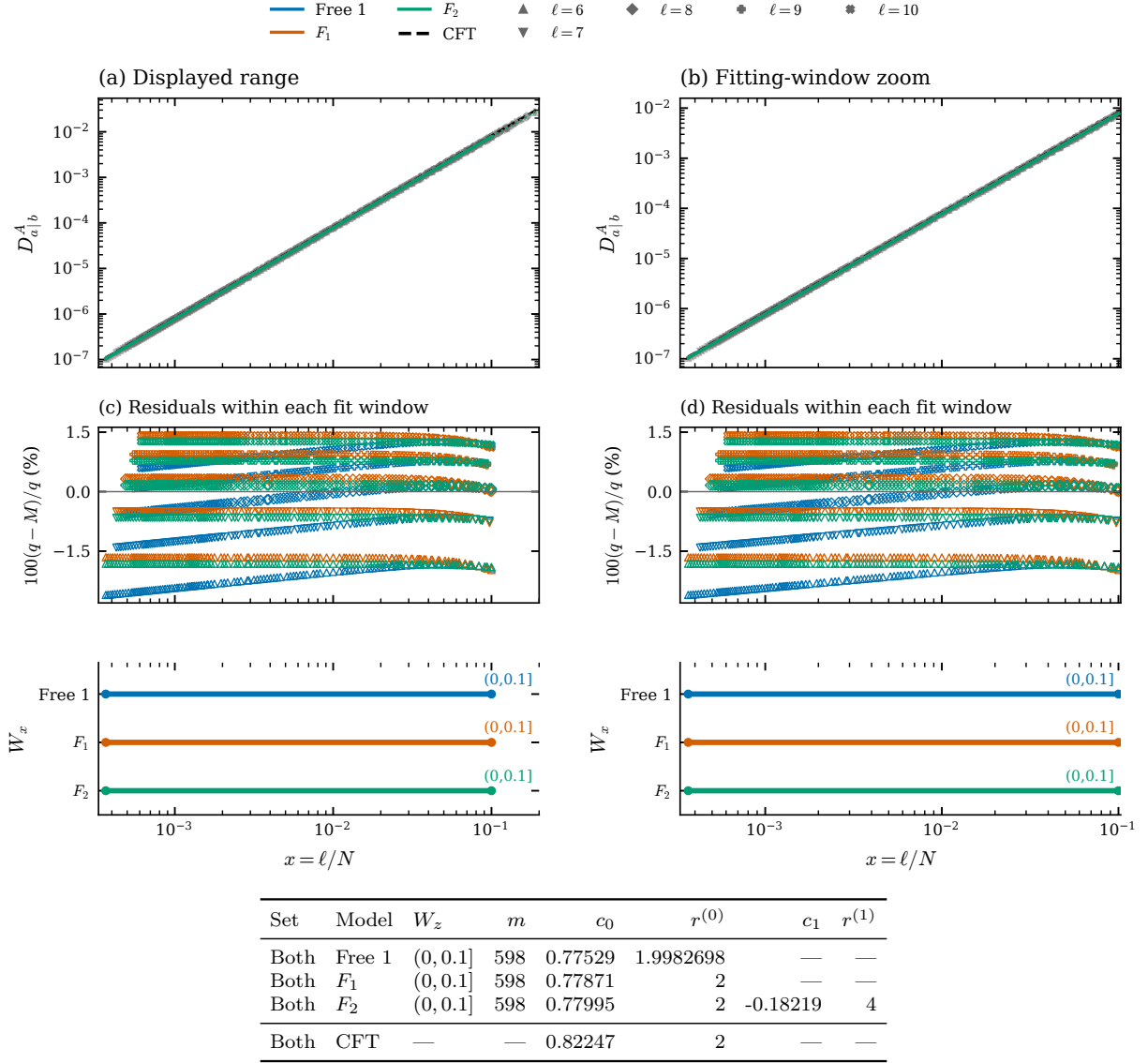


Figure 19: Compact boson, small intervals, $ss \mid 00$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

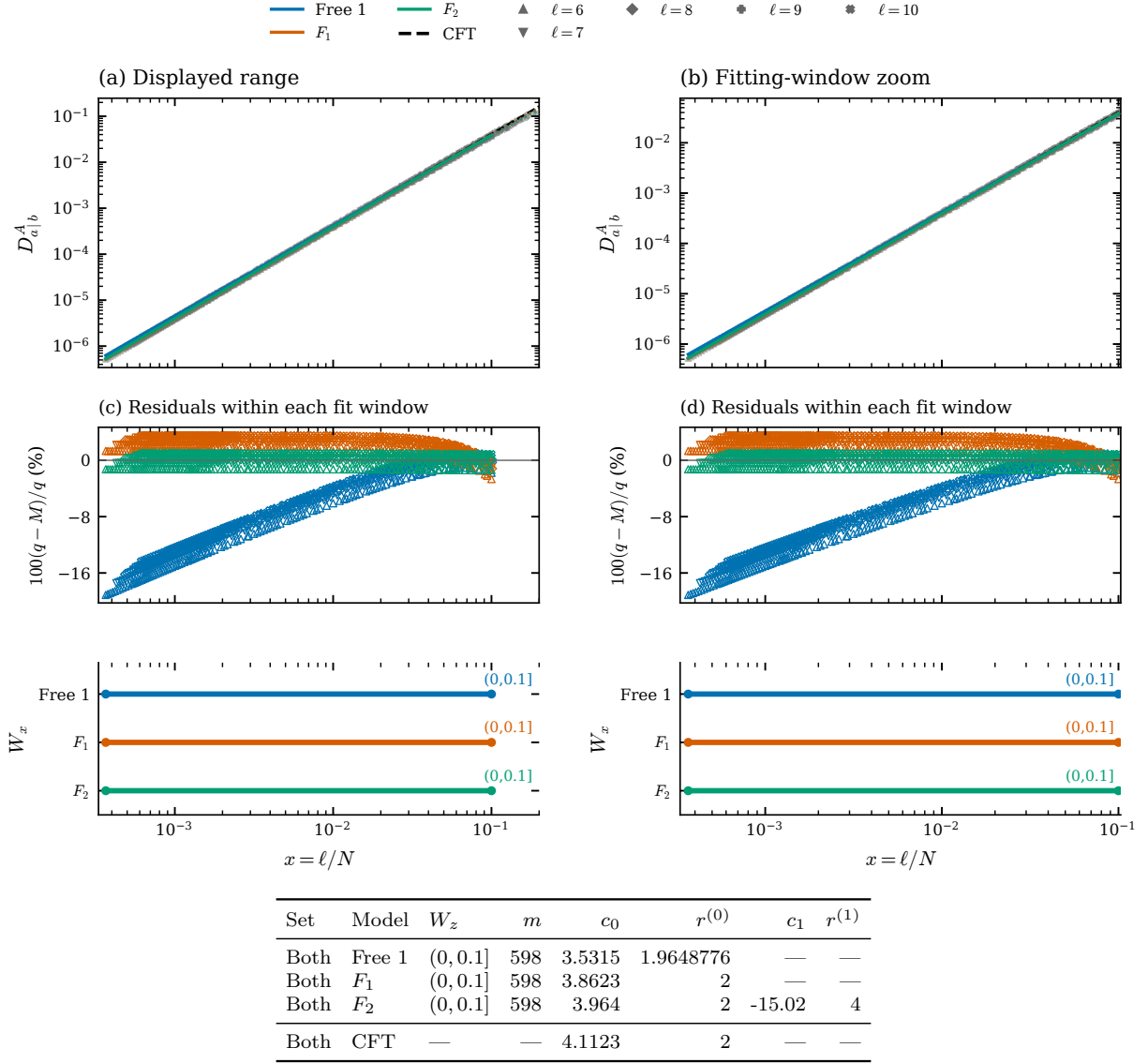


Figure 20: Compact boson, small intervals, $00 \mid 3s, s$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

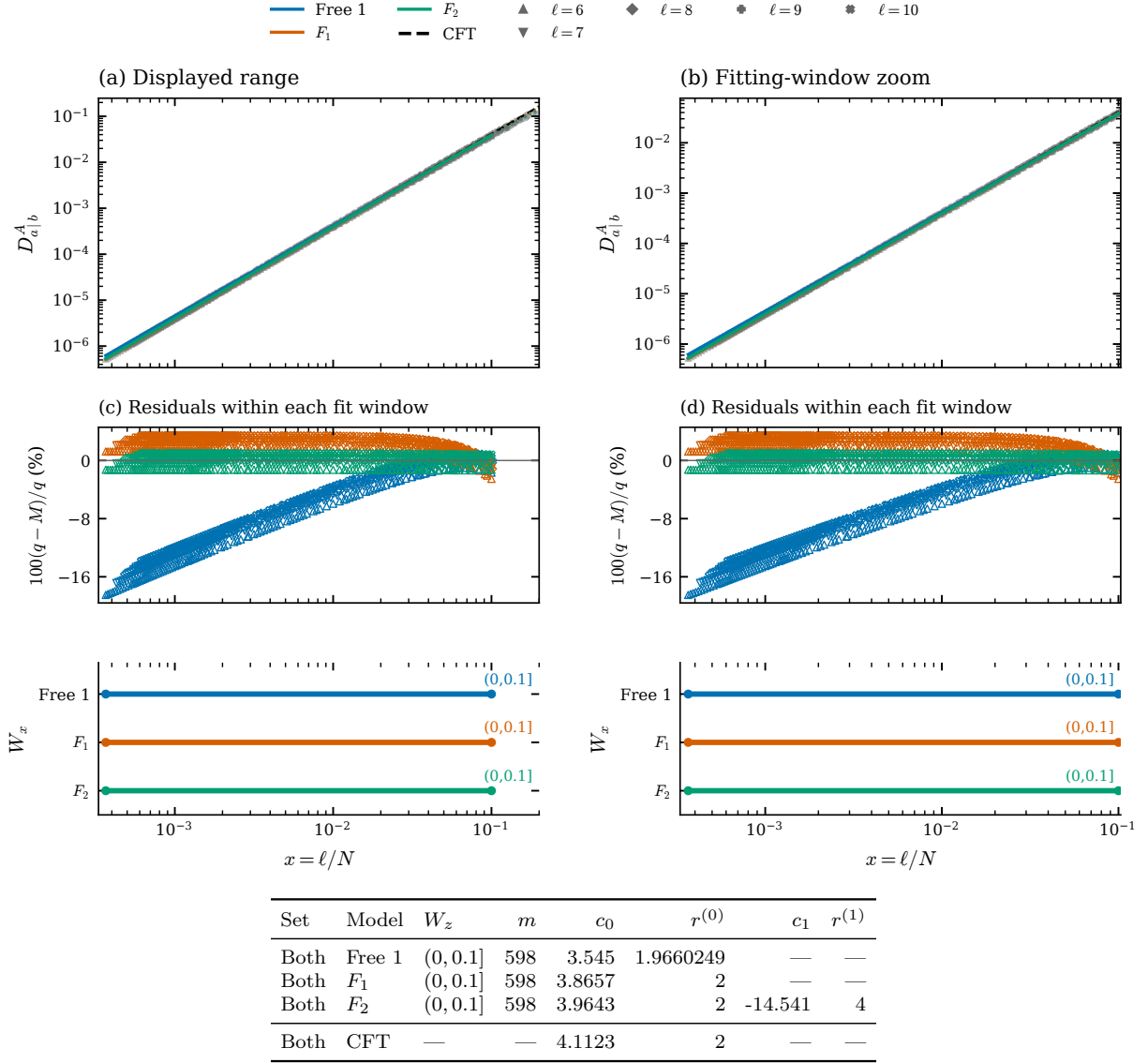


Figure 21: Compact boson, small intervals, $3s, s \mid 00$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

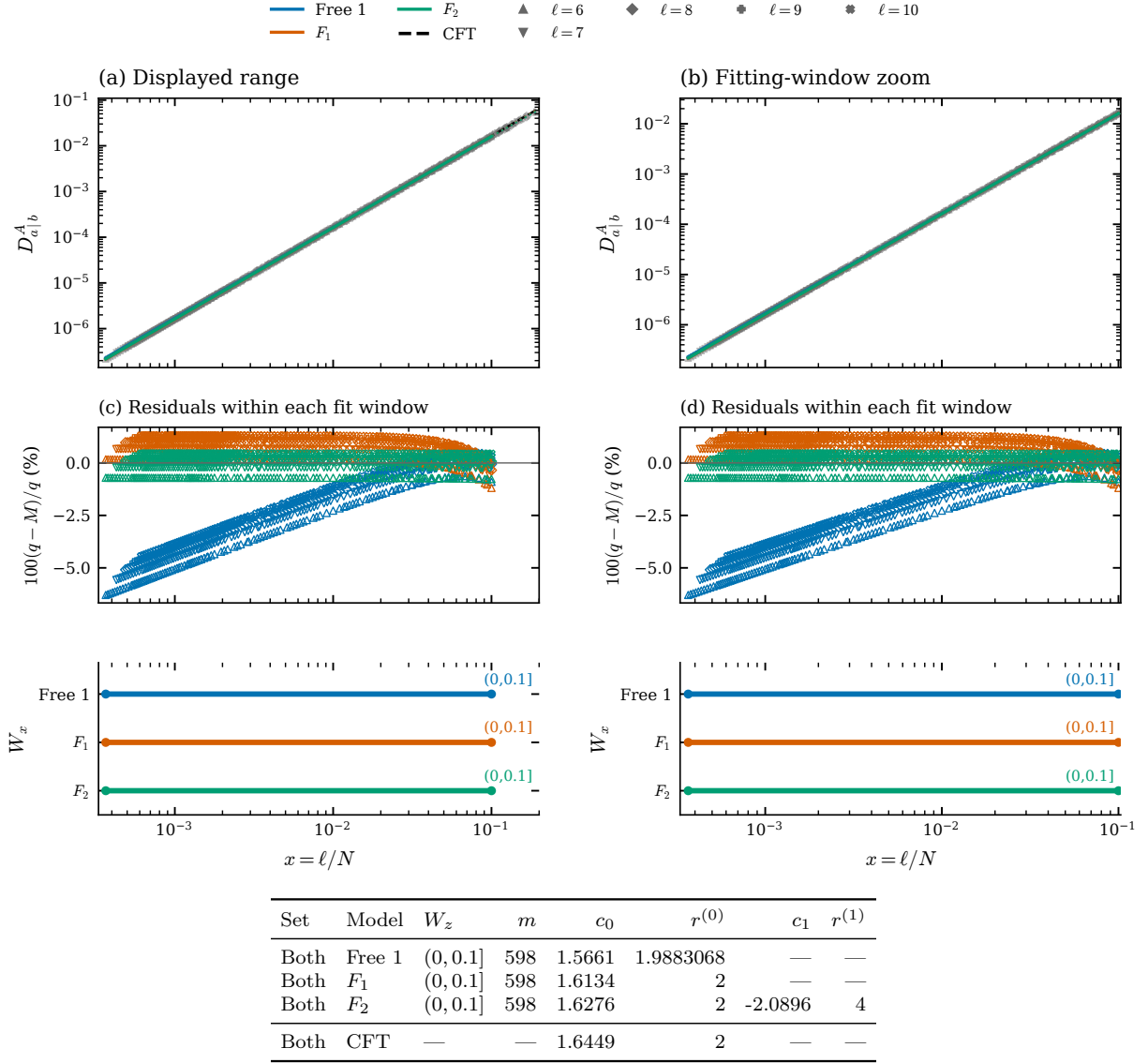


Figure 22: Compact boson, small intervals, $ss \mid 3s, s$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

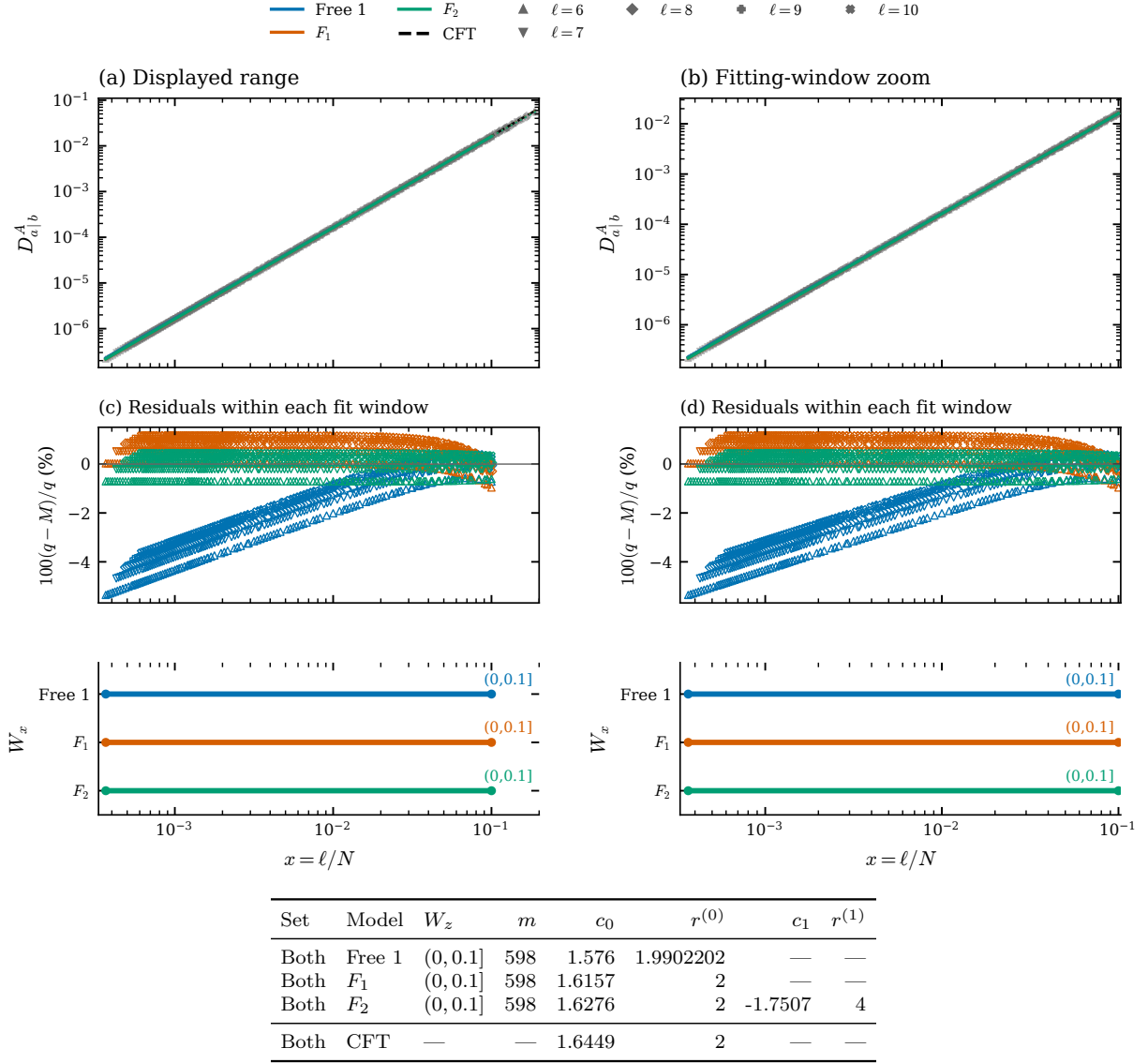


Figure 23: Compact boson, small intervals, $3s, s \mid ss$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = x$ and $q = D^A$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the supplied CFT series. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

6.6 Ising, large intervals

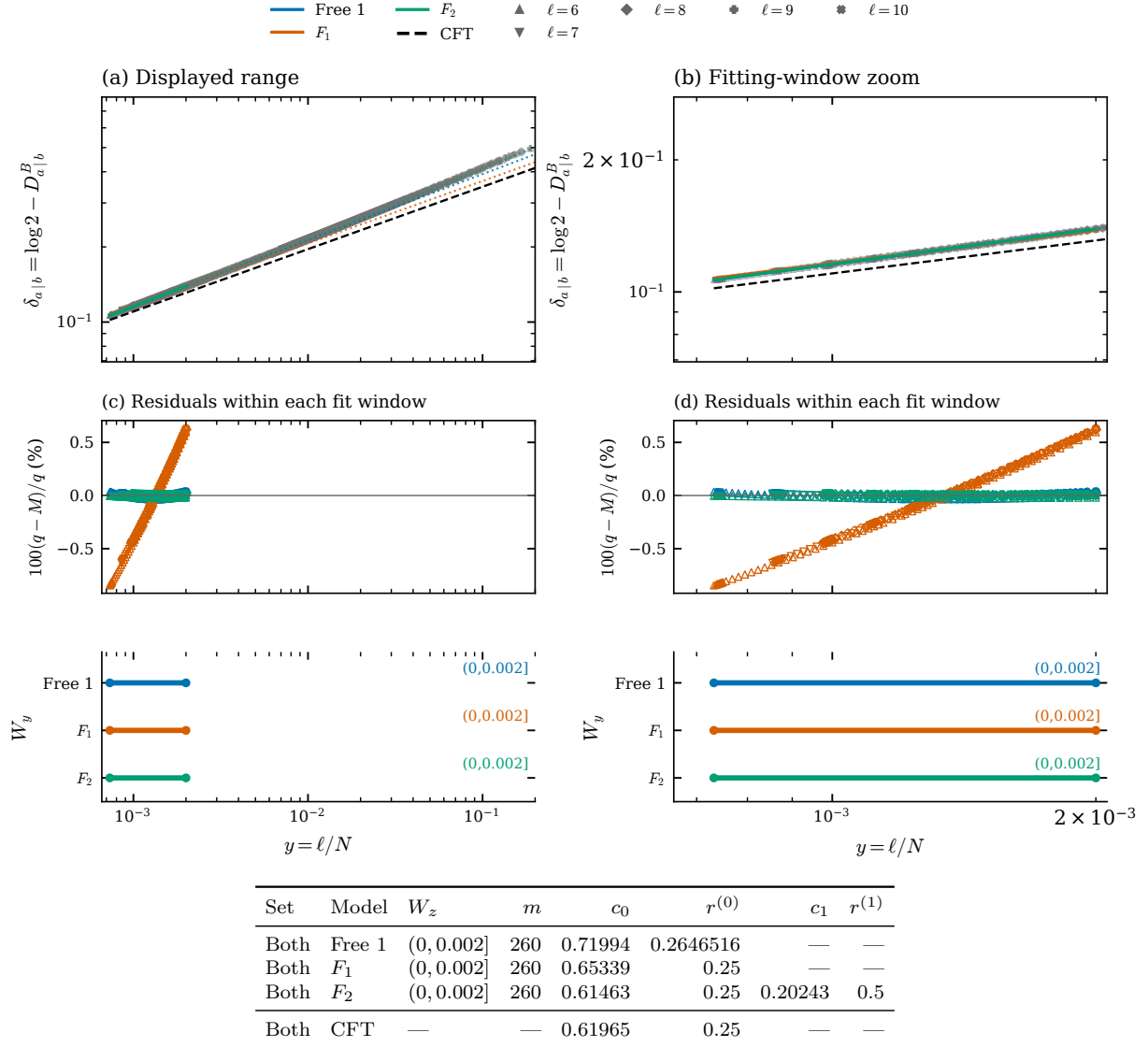


Figure 24: Ising, large intervals, $I \mid \sigma$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

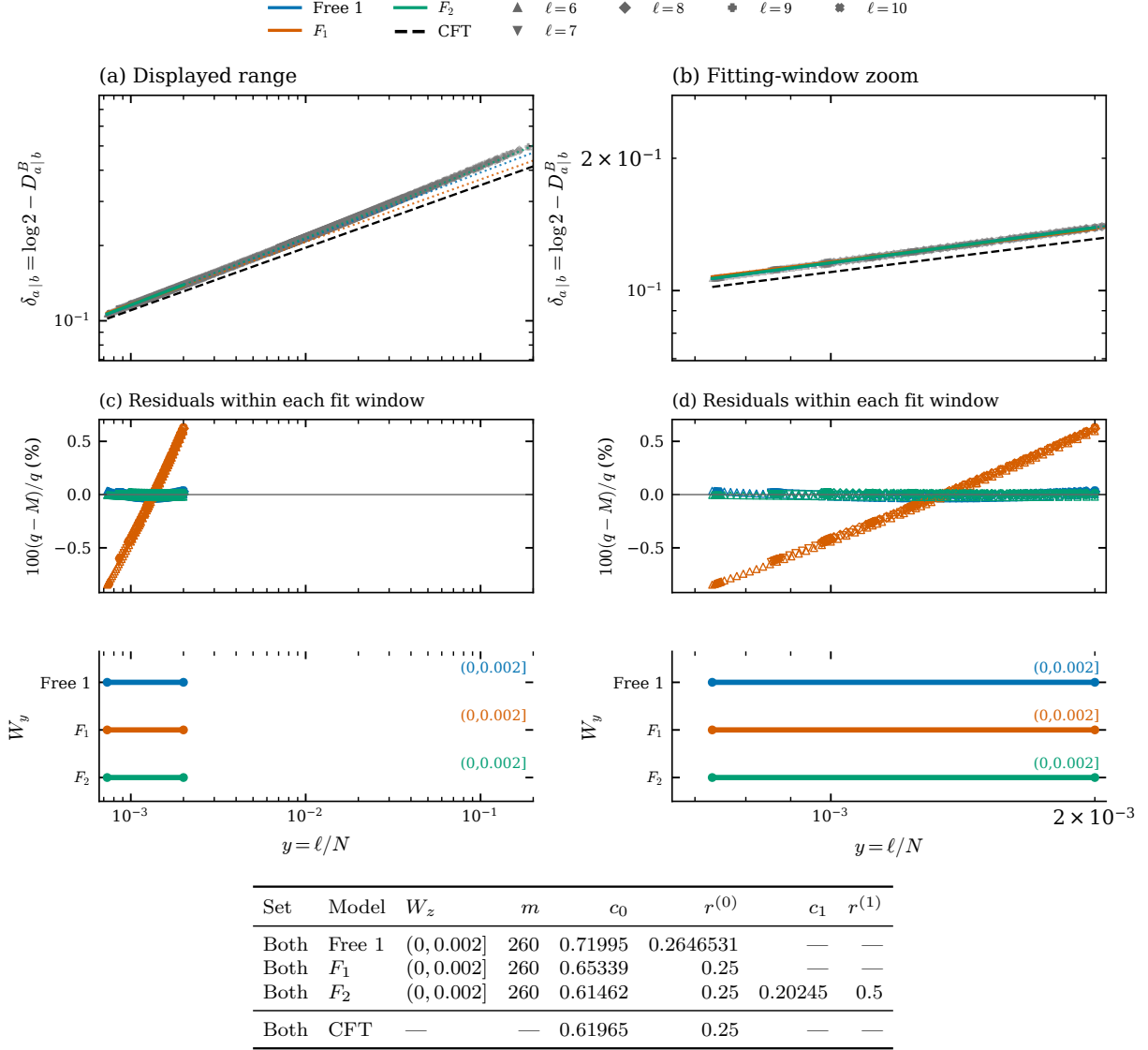


Figure 25: Ising, large intervals, $\sigma \mid I$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

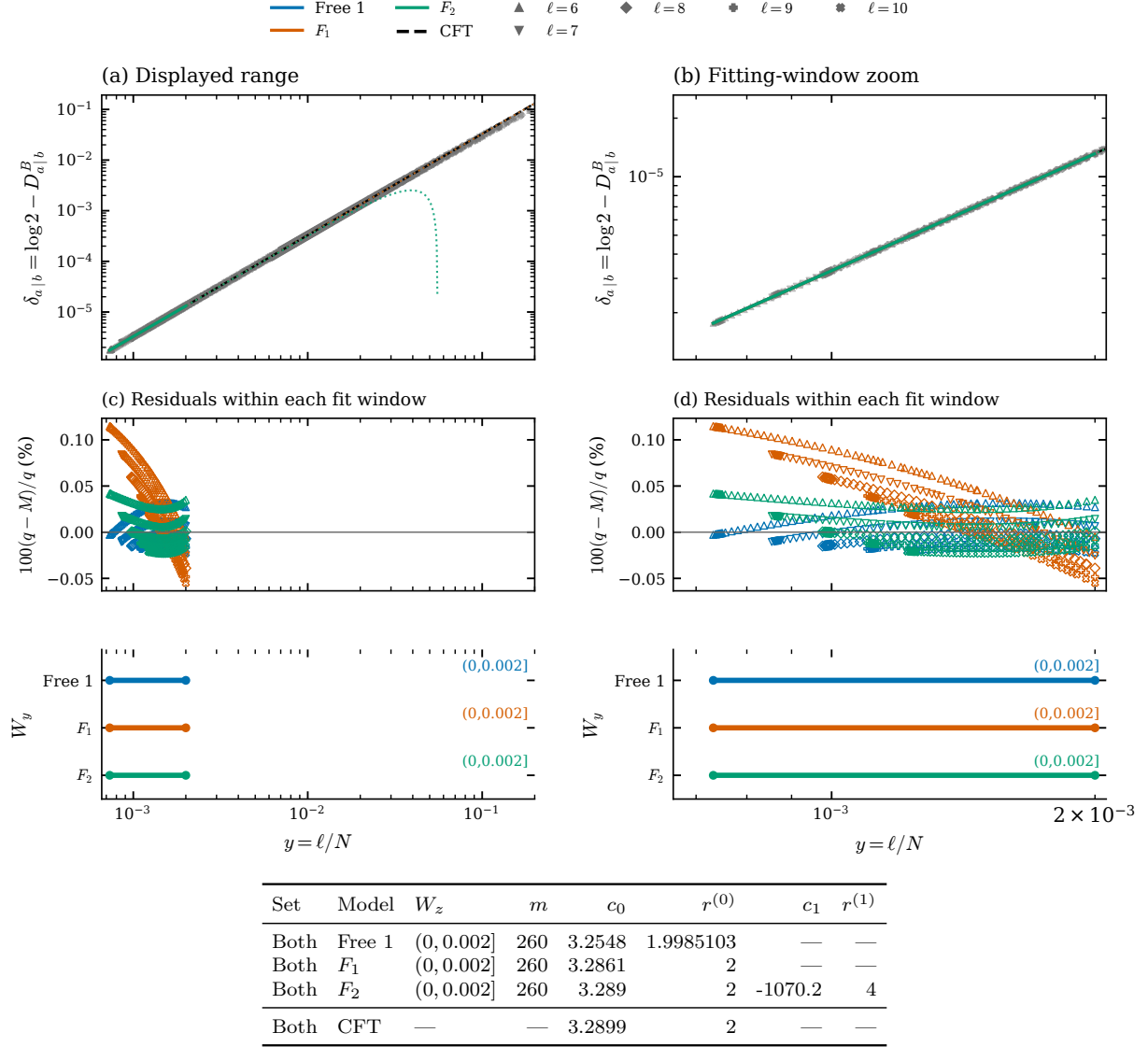


Figure 26: Ising, large intervals, $I \mid \varepsilon$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

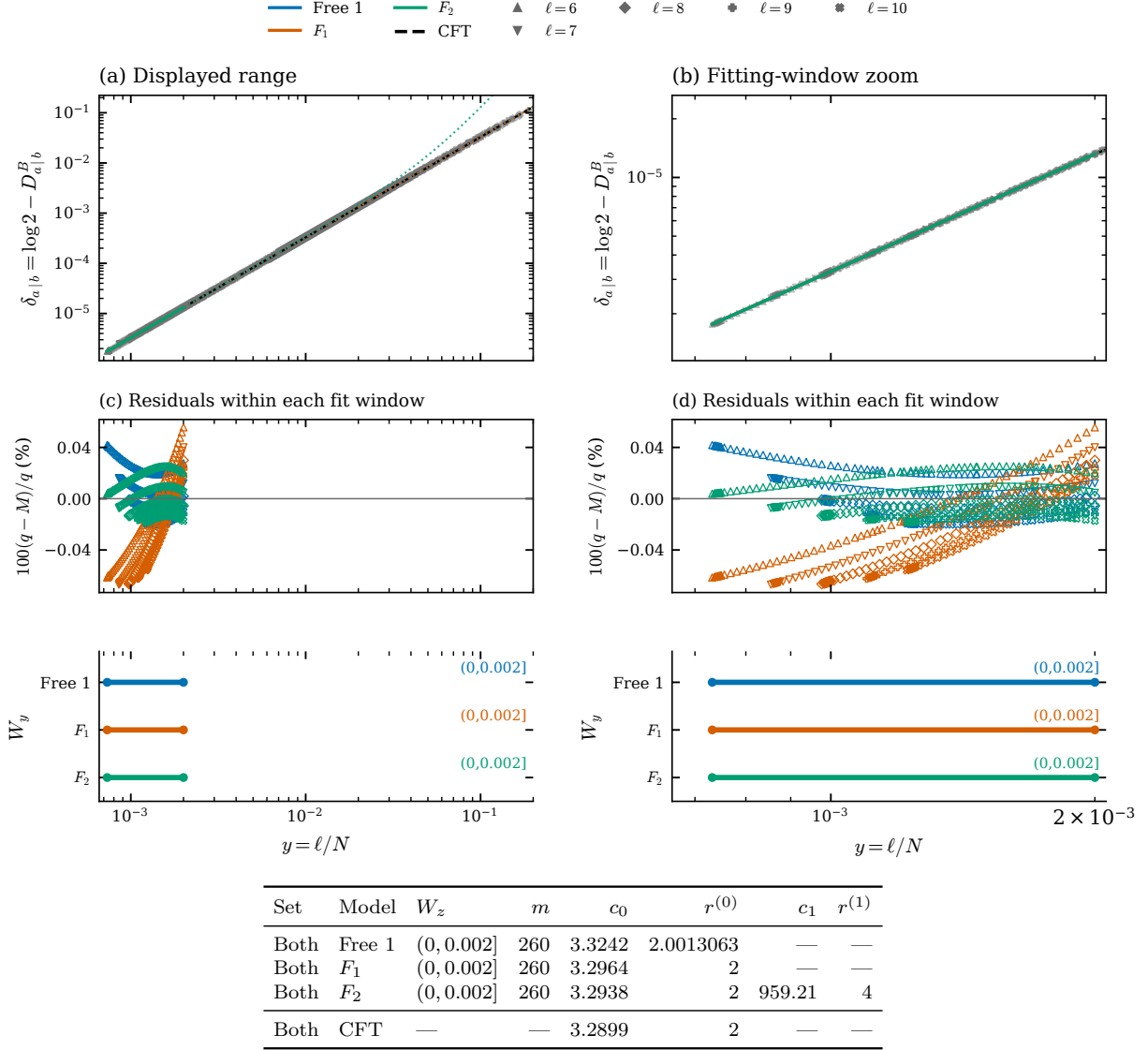
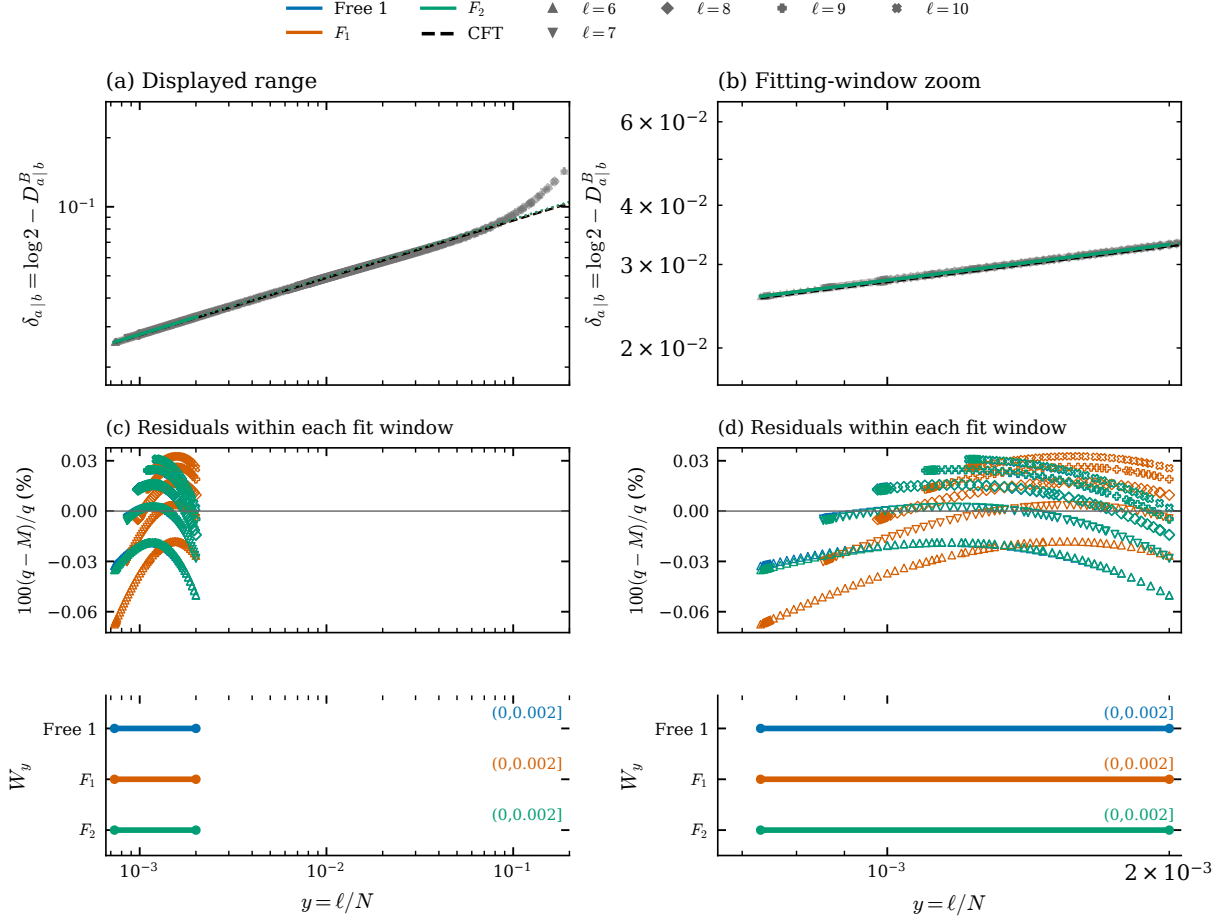


Figure 27: Ising, large intervals, $\varepsilon \mid I$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.



Set	Model	W_z	m	c_0	$r^{(0)}$	c_1	$r^{(1)}$
Both	Free 1	(0, 0.002]	260	0.15691	0.2505807	—	—
Both	F_1	(0, 0.002]	260	0.15631	0.25	—	—
Both	F_2	(0, 0.002]	260	0.15595	0.25	0.0018716	0.5
Both	CFT	—	—	0.15491	0.25	—	—

Figure 28: Ising, large intervals, $\sigma \mid \varepsilon$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

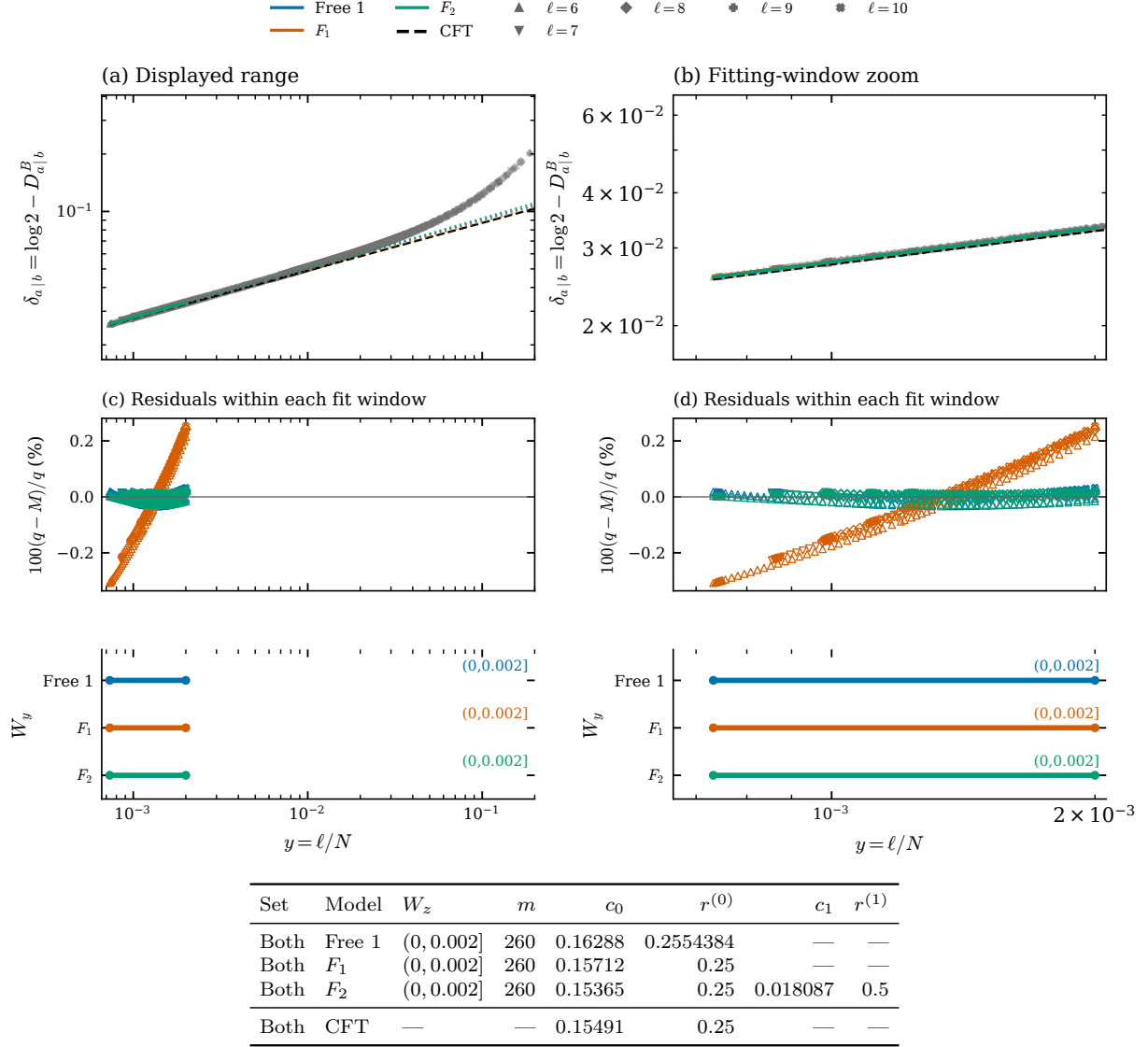


Figure 29: Ising, large intervals, $\varepsilon \mid \sigma$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}$, $z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

6.7 Compact boson, large intervals

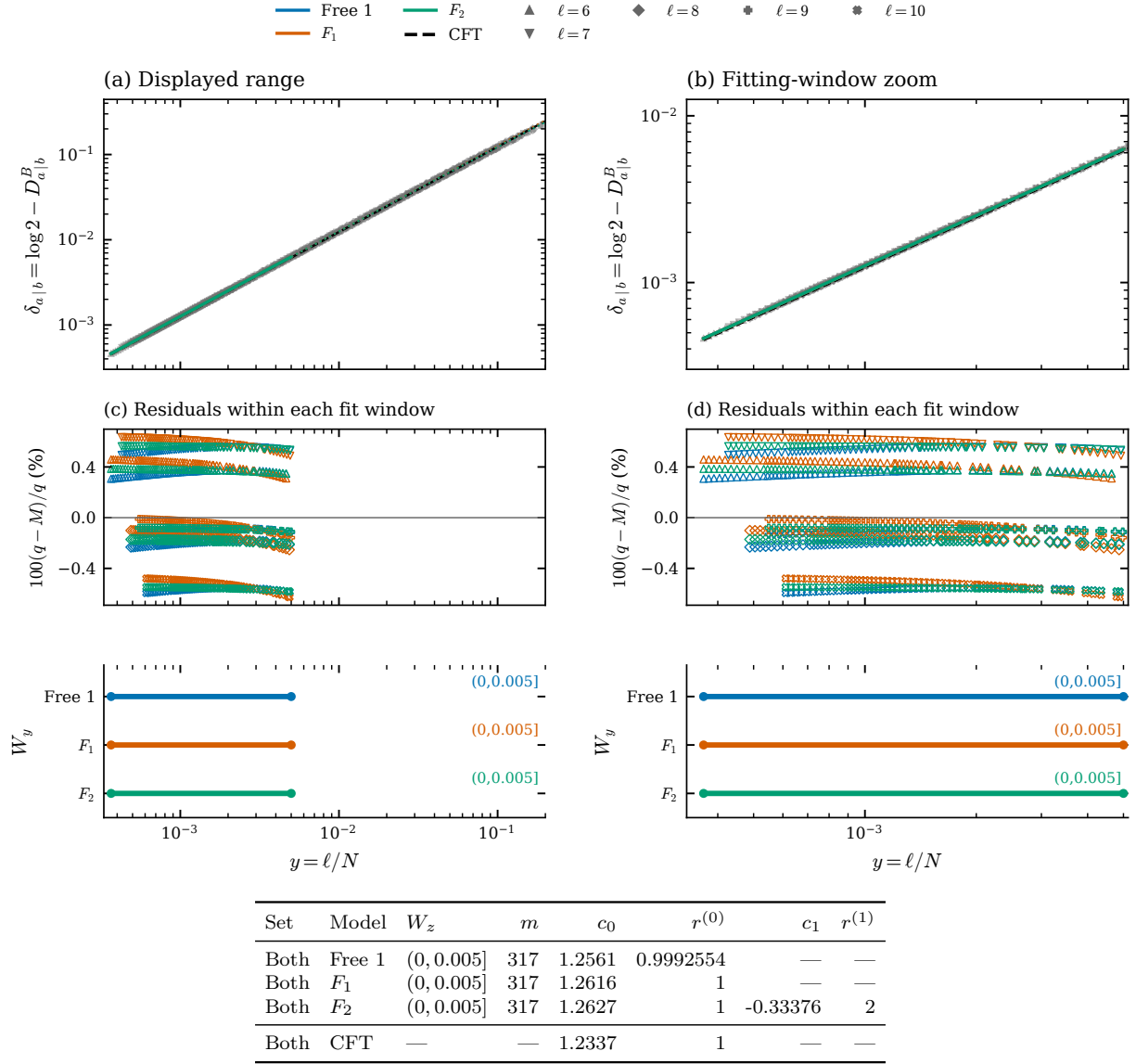


Figure 30: Compact boson, large intervals, $00 \mid ss$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

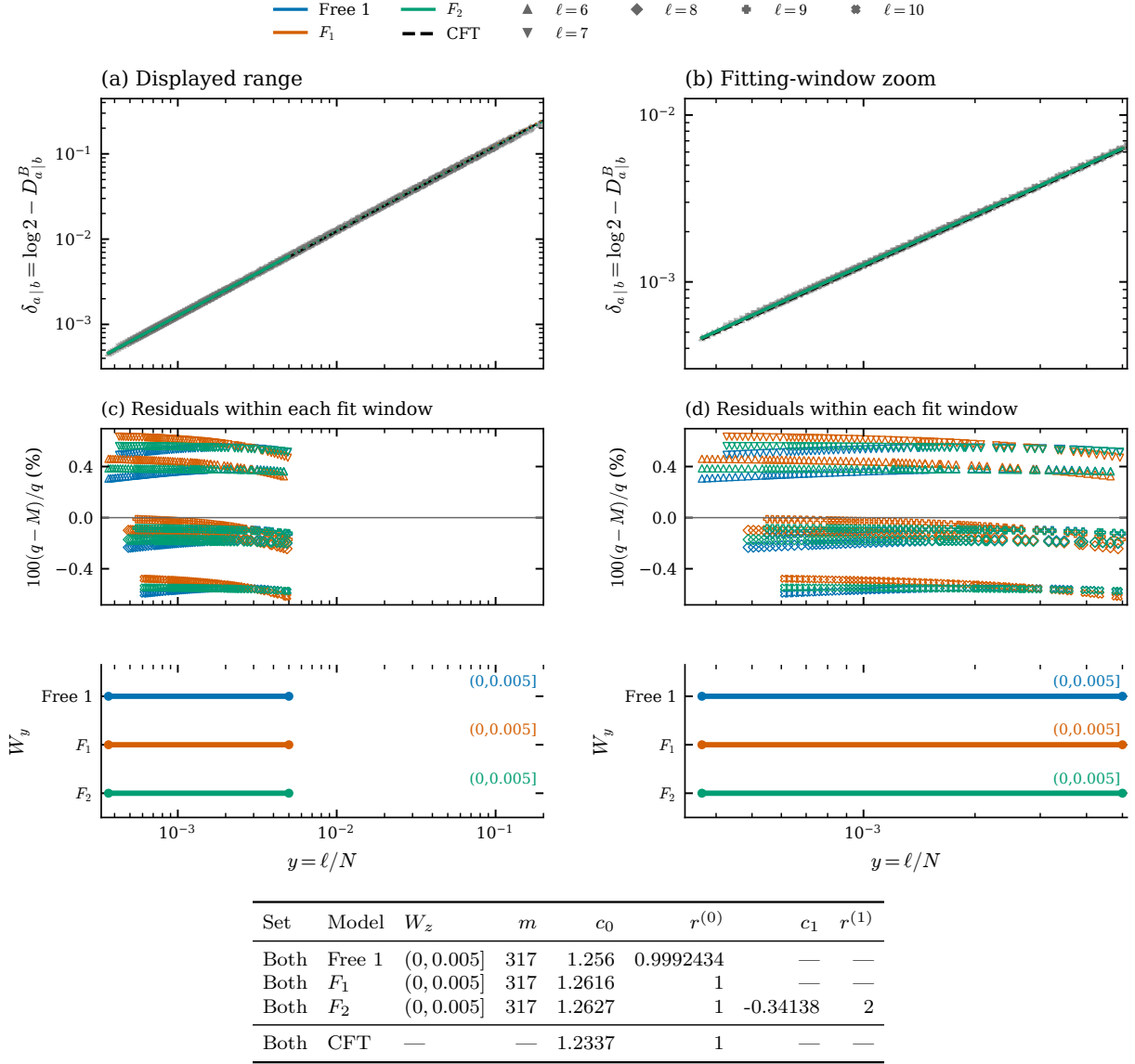


Figure 31: Compact boson, large intervals, $ss \mid 00$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

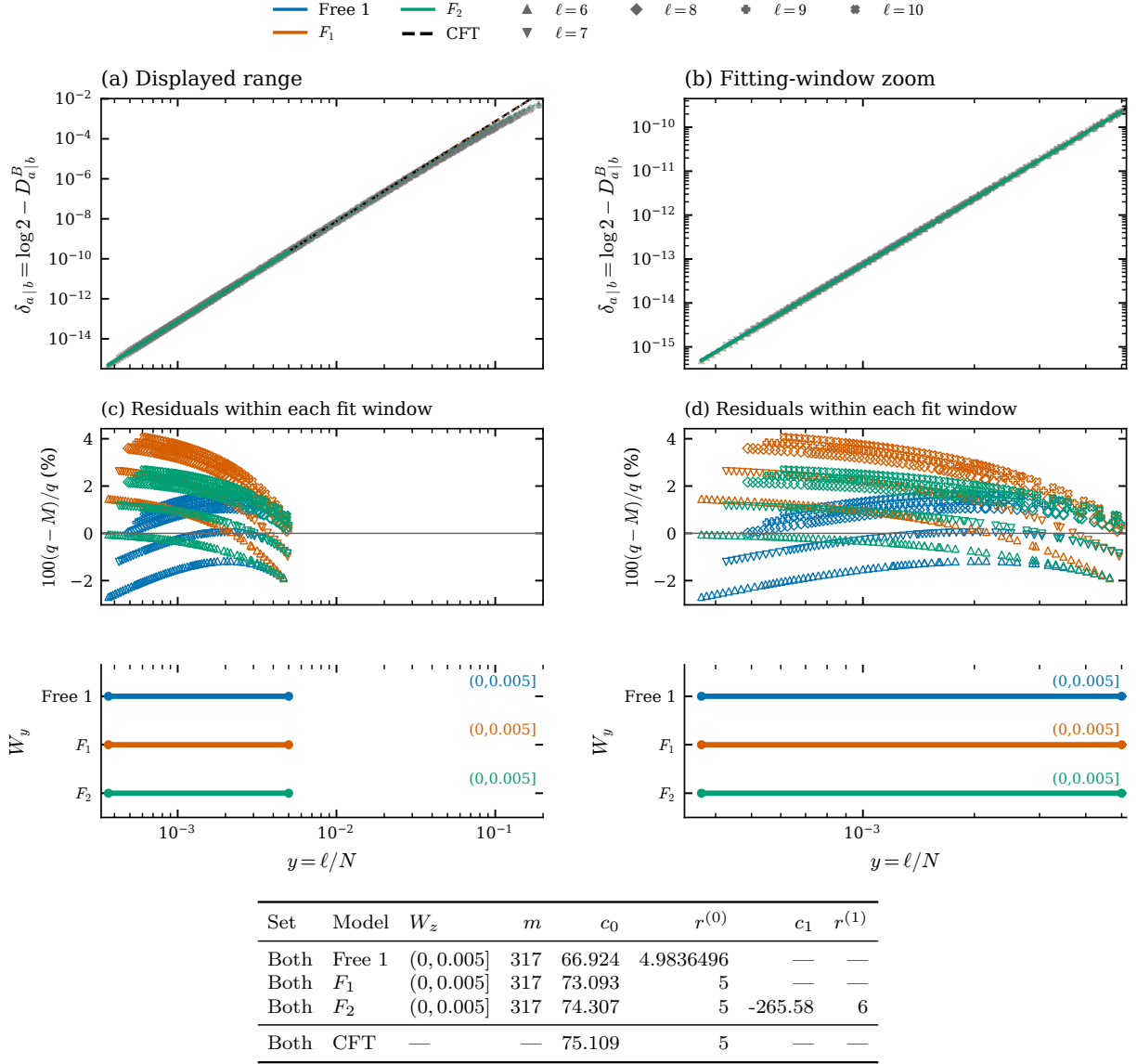


Figure 32: Compact boson, large intervals, $00 \mid 3s, s$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

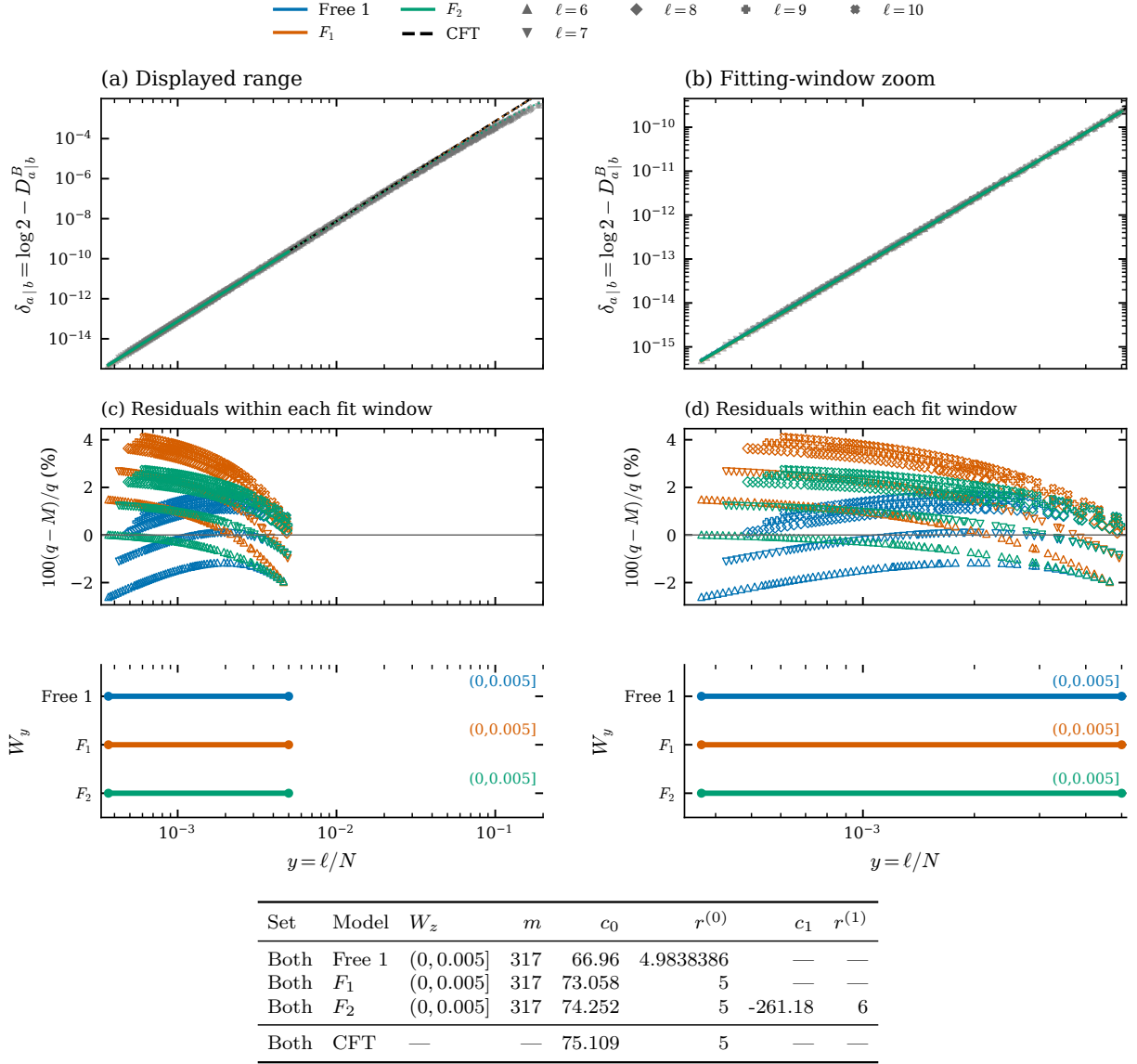


Figure 33: Compact boson, large intervals, $3s, s \mid 00$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

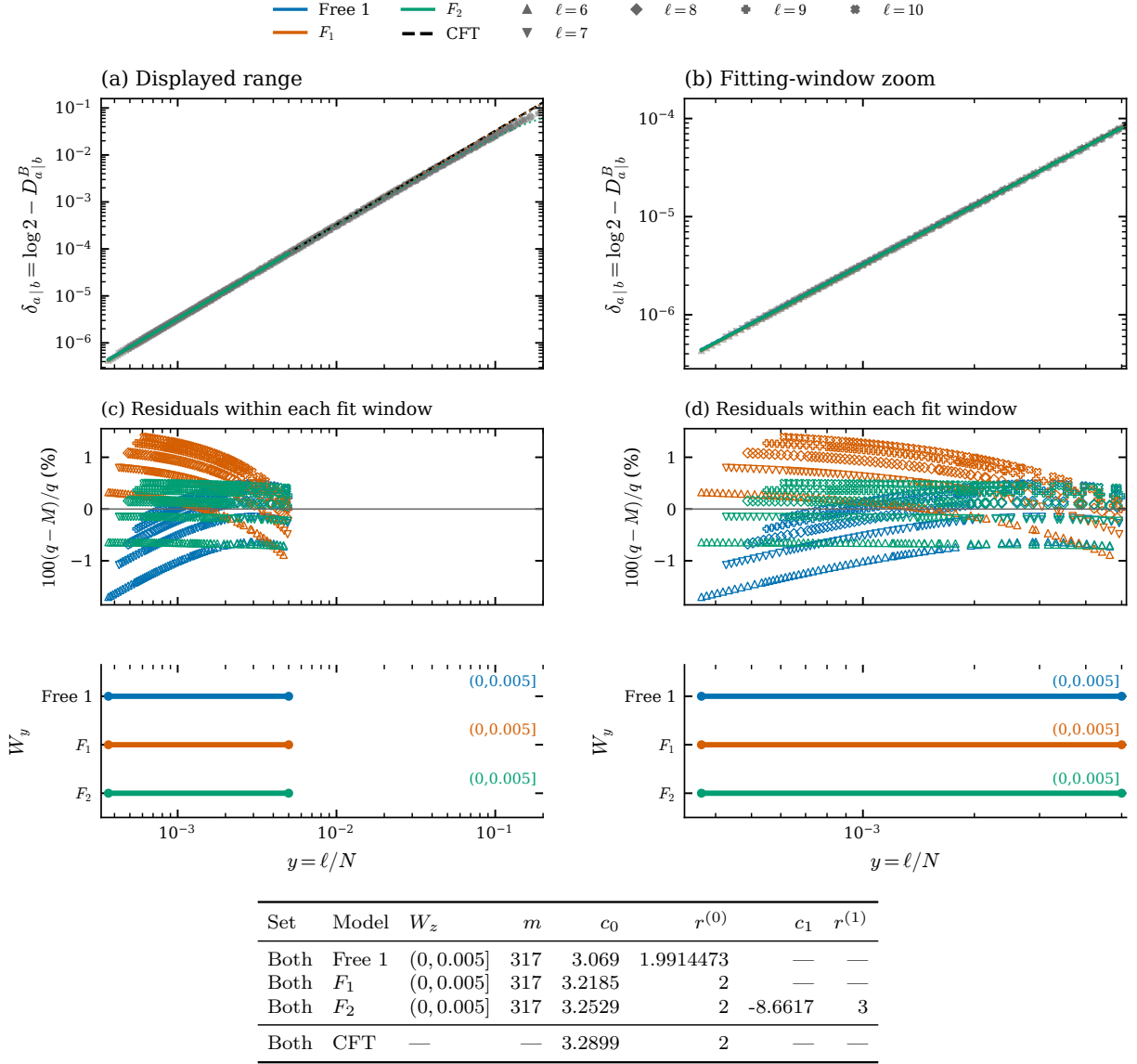


Figure 34: Compact boson, large intervals, $ss \mid 3s, s$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

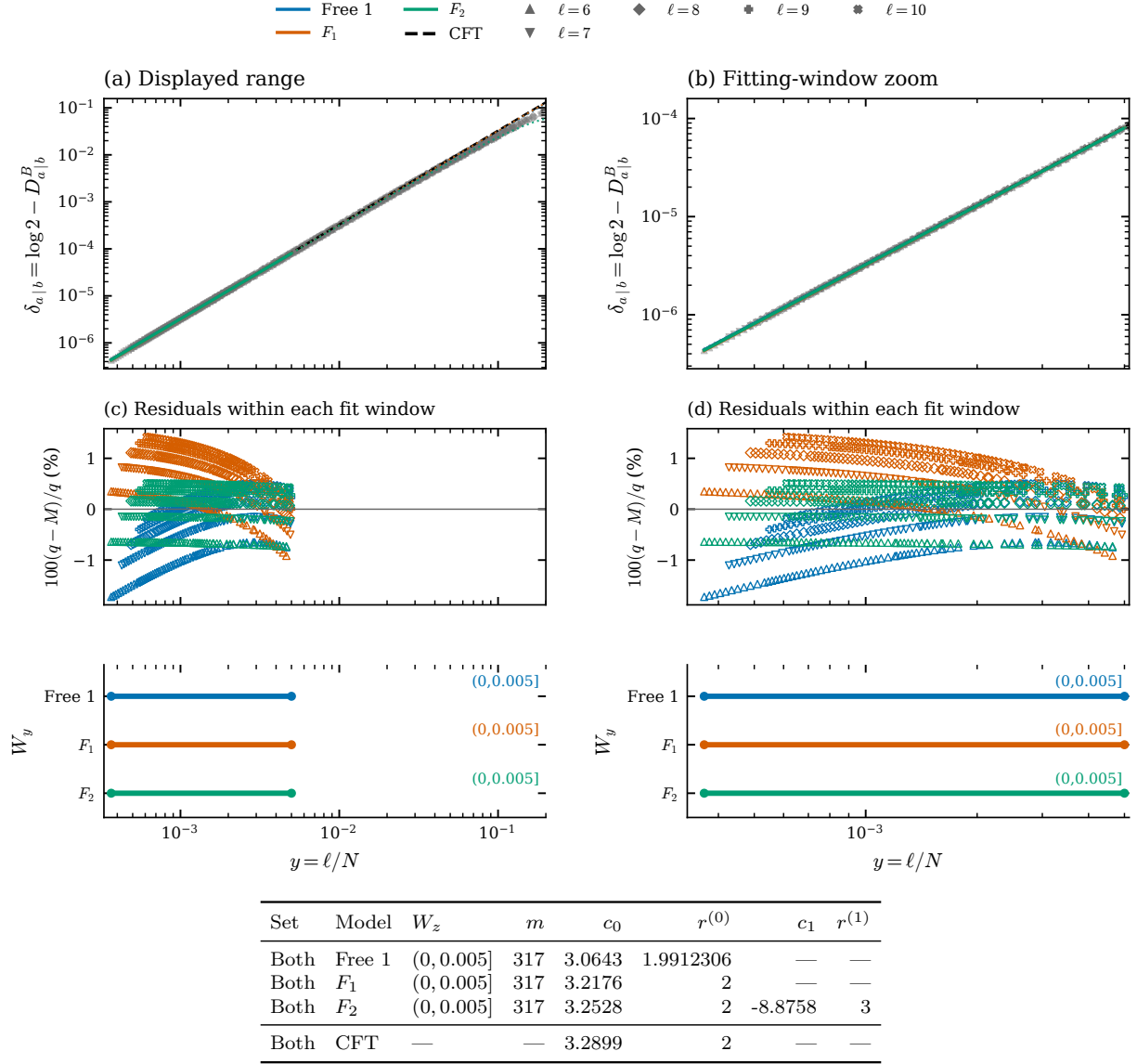


Figure 35: Compact boson, large intervals, $3s, s \mid ss$, at $p = 1/2$, $\ell = 6, \dots, 10$, and the complete size grid in Eq. (52). Here $z = y$ and $q = \delta = \log 2 - D^B$. Left: all computed data through $z = 0.20$; right: a zoom into the common fitting window. Both columns show the same Free 1, F_1 , and F_2 fits. All six directions of this CFT and interval type use the same window. Gray markers denote the data. Colored solid segments lie inside each model's fitting interval; dotted segments are extrapolations. Black dashed is the analytical CFT leading deficit. The observable axes are logarithmic and span the measured data; nonpositive predictions cannot appear there and out-of-range extrapolations leave the panel. Residual ordinates are linear and show $100(q - M)/q$ only on each model's fitted observations. Bars give the exact fitting intervals. The parameter block follows Eq. (60), with c_0, c_1 multiplying $z^{r^{(0)}}, z^{r^{(1)}}$; all coefficients include powers of π . A dash means an absent fitted term, or no fitting window for CFT. Values shown are rounded; curves use full precision.

A Small-interval fit summaries

The active mixture fits compare a free power law with F_1, F_2 and the supplied analytical CFT series. The models, parameter conversion, unweighted objective and residual measures are defined in Sec. 4. Every coefficient includes its power of π . The parameter tables group Free, F_1 , F_2 , and CFT consecutively for each ordered pair and give each fitted model's selection interval. They explicitly give the theoretical leading exponent α_{CFT} and the analytical coefficients $c_0^{\text{CFT}}, c_1^{\text{CFT}}$ beside the model coefficients. Only $I \mid \varepsilon$ and $\varepsilon \mid I$ have a supplied next-order coefficient, respectively $-8\pi^6/45$ and $-8\pi^6/21$ at power six; see Table 1. A dash in the analytical next-order column means not supplied, never an assumed zero. Free, F_1 , and F_2 use the same CFT-wide window in Eq. (59) for every ordered pair. Sec. 6 gives a visual comparison for each pair. All parameters retain full precision in `data/f2_summary_fits.csv`; the identical overview-figure fits are in `data/scaling_fit_summary.csv`. The displayed rounding does not enter predictions or residuals. For a model M with window W_M , the quality tables use

$$G_M[\%] = 100 \left[1 - \frac{\text{RMSE}(M; W_M)}{\text{RMSE}(F_1^{[W_M]}; W_M)} \right],$$

$$G_{M,\text{CV}}[\%] = 100 \left[1 - \frac{\text{CVRMSE}(M; W_M)}{\text{CVRMSE}(F_1^{[W_M]}; W_M)} \right]. \quad (89)$$

Here $F_1^{[W_M]}$ means that the one-term model is refitted on the same observations as M , also within each omitted-size fold. For fitted models, its denominator equals the displayed F_1 row because the current windows coincide. The analytical CFT residual uses these same observations. Raw RMSEs from different windows are not directly comparable.

A.1 Ising

Table 10: Ising small-interval parameters at $p = 1/2$, $\ell = 6, \dots, 10$, and the grid in Eq. (52). Models are grouped by ordered pair. The column W_x gives each fitted model's interval; CFT adjusts no parameter and has no fitting window. Free, F_1 , and F_2 use the same displayed interval for all six ordered pairs of this CFT and interval type, so the additional term is tested on identical observations. Every model is visualized for each ordered pair in Sec. 6.4. All amplitudes multiply x^r itself and include powers of π . Columns c_0, c_1 multiply the displayed powers $r^{(0)}, r^{(1)}$; a dash indicates an absent term. Analytical columns give the complete supplied coefficients. A dash in c_1^{CFT} means that the manuscript does not supply that coefficient, not that it vanishes. Percentage discrepancies follow Eq. (62); an imposed power has no estimated exponent discrepancy.

Direction	Model	W_x	α_{CFT}	$r^{(0)}$	c_0^{CFT}	c_0	$r^{(1)}$	c_1^{CFT}	c_1	η_c [%]	η_r [%]
$I \mid \sigma$	Free	(0, 0.02]	2	2.0001	0.20562	0.20585	—	—	—	0.1146	0.0070964
$I \mid \sigma$	F_1	(0, 0.02]	2	2	0.20562	0.20573	—	—	—	0.055265	—
$I \mid \sigma$	F_2	(0, 0.02]	2	2	0.20562	0.20571	4	—	0.083833	0.044771	—
$I \mid \sigma$	CFT	—	2	2	0.20562	0.20562	—	—	—	0	—
$\sigma \mid I$	Free	(0, 0.02]	2	2.0001	0.20562	0.20582	—	—	—	0.099239	0.0057066
$\sigma \mid I$	F_1	(0, 0.02]	2	2	0.20562	0.20572	—	—	—	0.051526	—
$\sigma \mid I$	F_2	(0, 0.02]	2	2	0.20562	0.2057	4	—	0.069099	0.042877	—
$\sigma \mid I$	CFT	—	2	2	0.20562	0.20562	—	—	—	0	—
$I \mid \varepsilon$	Free	(0, 0.02]	4	3.9934	12.988	12.491	—	-170.91	—	-3.8239	-0.16507
$I \mid \varepsilon$	F_1	(0, 0.02]	4	4	12.988	12.83	—	-170.91	—	-1.219	—
$I \mid \varepsilon$	F_2	(0, 0.02]	4	4	12.988	12.883	6	-170.91	-170.71	-0.80492	—
$I \mid \varepsilon$	CFT	—	4	4	12.988	12.988	6	-170.91	-170.91	0	—

Direction	Model	W_x	α_{CFT}	$r^{(0)}$	c_0^{CFT}	c_0	$r^{(1)}$	c_1^{CFT}	c_1	η_c [%]	η_r [%]
εI	Free	(0, 0.02]	4	3.9861	12.988	12.07	—	-366.24	—	-7.0661	-0.34832
εI	F_1	(0, 0.02]	4	4	12.988	12.77	—	-366.24	—	-1.6748	—
εI	F_2	(0, 0.02]	4	4	12.988	12.882	6	-366.24	-355.38	-0.81263	—
εI	CFT	—	4	4	12.988	12.988	6	-366.24	-366.24	0	—
$\sigma \varepsilon$	Free	(0, 0.02]	2	2.0238	0.20562	0.23069	—	—	—	12.192	1.1878
$\sigma \varepsilon$	F_1	(0, 0.02]	2	2	0.20562	0.2089	—	—	—	1.595	—
$\sigma \varepsilon$	F_2	(0, 0.02]	2	2	0.20562	0.20565	4	—	12.604	0.017273	—
$\sigma \varepsilon$	CFT	—	2	2	0.20562	0.20562	—	—	—	0	—
$\varepsilon \sigma$	Free	(0, 0.02]	2	2.0176	0.20562	0.22393	—	—	—	8.9077	0.88006
$\varepsilon \sigma$	F_1	(0, 0.02]	2	2	0.20562	0.20806	—	—	—	1.1882	—
$\varepsilon \sigma$	F_2	(0, 0.02]	2	2	0.20562	0.20566	4	—	9.3133	0.022414	—
$\varepsilon \sigma$	CFT	—	2	2	0.20562	0.20562	—	—	—	0	—

Table 11: Using small-interval fit quality for Table 10, in the same ordered-pair blocks. All fitted-row metrics use that row's W_x and m observations; k is the fitted parameter count. CFT metrics are evaluated on the same displayed window as the fitted rows, although no fitting is performed. RMSE, CV RMSE, and RMSrel follow Eqs. (62) and (63). For every F_2 row, gains G, G_{CV} use F_1 refitted on exactly that F_2 window, as in Eq. (89); its denominator equals the displayed F_1 row because the windows now match. Negative gains mean worse prediction. Absolute RMSEs on different windows should not be ranked directly.

Direction	Model	W_x	m	k	RMSE	CV RMSE	RMSrel [%]	G [%]	G_{CV} [%]
$I \sigma$	Free	(0, 0.02]	1255	2	5.158×10^{-9}	5.190×10^{-9}	0.02149	1.7481	1.5281
$I \sigma$	F_1	(0, 0.02]	1255	1	5.249×10^{-9}	5.271×10^{-9}	0.017876	0	0
$I \sigma$	F_2	(0, 0.02]	1255	2	5.124×10^{-9}	5.161×10^{-9}	0.016955	2.3911	2.077
$I \sigma$	CFT	—	1255	0	1.719×10^{-8}	1.719×10^{-8}	0.053178	-227.43	-226.11
σI	Free	(0, 0.02]	1255	2	4.526×10^{-9}	4.554×10^{-9}	0.01948	1.4739	1.2593
σI	F_1	(0, 0.02]	1255	1	4.594×10^{-9}	4.612×10^{-9}	0.016173	0	0
σI	F_2	(0, 0.02]	1255	2	4.497×10^{-9}	4.528×10^{-9}	0.01584	2.1181	1.8149
σI	CFT	—	1255	0	1.594×10^{-8}	1.594×10^{-8}	0.050669	-246.9	-245.54
$I \varepsilon$	Free	(0, 0.02]	1255	2	1.566×10^{-9}	1.585×10^{-9}	0.81516	3.9757	3.6364
$I \varepsilon$	F_1	(0, 0.02]	1255	1	1.631×10^{-9}	1.645×10^{-9}	0.38995	0	0
$I \varepsilon$	F_2	(0, 0.02]	1255	2	1.561×10^{-9}	1.582×10^{-9}	0.33241	4.2923	3.8386
$I \varepsilon$	CFT	—	1255	0	4.411×10^{-9}	4.411×10^{-9}	0.94868	-170.42	-168.19
εI	Free	(0, 0.02]	1255	2	1.565×10^{-9}	1.583×10^{-9}	1.544	14.705	14.389
εI	F_1	(0, 0.02]	1255	1	1.834×10^{-9}	1.849×10^{-9}	0.65255	0	0
εI	F_2	(0, 0.02]	1255	2	1.548×10^{-9}	1.568×10^{-9}	0.33073	15.616	15.195
εI	CFT	—	1255	0	4.320×10^{-9}	4.320×10^{-9}	0.94559	-135.49	-133.62
$\sigma \varepsilon$	Free	(0, 0.02]	1255	2	4.655×10^{-8}	4.675×10^{-8}	2.1632	72.895	72.862
$\sigma \varepsilon$	F_1	(0, 0.02]	1255	1	1.717×10^{-7}	1.723×10^{-7}	1.1334	0	0
$\sigma \varepsilon$	F_2	(0, 0.02]	1255	2	7.741×10^{-9}	7.820×10^{-9}	0.018834	95.492	95.461
$\sigma \varepsilon$	CFT	—	1255	0	5.026×10^{-7}	5.026×10^{-7}	0.88172	-192.67	-191.78
$\varepsilon \sigma$	Free	(0, 0.02]	1255	2	3.468×10^{-8}	3.483×10^{-8}	1.6121	72.684	72.651
$\varepsilon \sigma$	F_1	(0, 0.02]	1255	1	1.270×10^{-7}	1.274×10^{-7}	0.83808	0	0
$\varepsilon \sigma$	F_2	(0, 0.02]	1255	2	7.163×10^{-9}	7.236×10^{-9}	0.016596	94.359	94.318
$\varepsilon \sigma$	CFT	—	1255	0	3.741×10^{-7}	3.741×10^{-7}	0.66099	-194.64	-193.74

A.2 Compact boson

Table 12: Compact boson small-interval parameters at $p = 1/2$, $\ell = 6, \dots, 10$, and the grid in Eq. (52). Models are grouped by ordered pair. The column W_x gives each fitted model's interval; CFT adjusts no parameter and has no fitting window. Free, F_1 , and F_2 use the same displayed interval for all six ordered pairs of this CFT and interval type, so the additional term is tested on identical observations. Every model is visualized for each ordered pair in Sec. 6.5. All amplitudes multiply x^r itself and include powers of π . Columns c_0, c_1 multiply the displayed powers $r^{(0)}, r^{(1)}$; a dash indicates an absent term. Analytical columns give the complete supplied coefficients. A dash in c_1^{CFT} means that the manuscript does not supply that coefficient, not that it vanishes. Percentage discrepancies follow Eq. (62); an imposed power has no estimated exponent discrepancy.

Direction	Model	W_x	α_{CFT}	$r^{(0)}$	c_0^{CFT}	c_0	$r^{(1)}$	c_1^{CFT}	c_1	η_c [%]	η_r [%]
00 ss	Free	(0, 0.1]	2	1.9982	0.82247	0.77501	—	—	—	-5.7697	-0.092048
00 ss	F_1	(0, 0.1]	2	2	0.82247	0.77866	—	—	—	-5.3266	—
00 ss	F_2	(0, 0.1]	2	2	0.82247	0.77996	4	—	-0.1917	-5.1687	—
00 ss	CFT	—	2	2	0.82247	0.82247	—	—	—	0	—
ss 00	Free	(0, 0.1]	2	1.9983	0.82247	0.77529	—	—	—	-5.7362	-0.086511
ss 00	F_1	(0, 0.1]	2	2	0.82247	0.77871	—	—	—	-5.3197	—
ss 00	F_2	(0, 0.1]	2	2	0.82247	0.77995	4	—	-0.18219	-5.1696	—
ss 00	CFT	—	2	2	0.82247	0.82247	—	—	—	0	—
00 3s, s	Free	(0, 0.1]	2	1.9649	4.1123	3.5315	—	—	—	-14.124	-1.7561
00 3s, s	F_1	(0, 0.1]	2	2	4.1123	3.8623	—	—	—	-6.0813	—
00 3s, s	F_2	(0, 0.1]	2	2	4.1123	3.964	4	—	-15.02	-3.6065	—
00 3s, s	CFT	—	2	2	4.1123	4.1123	—	—	—	0	—
3s, s 00	Free	(0, 0.1]	2	1.966	4.1123	3.545	—	—	—	-13.795	-1.6988
3s, s 00	F_1	(0, 0.1]	2	2	4.1123	3.8657	—	—	—	-5.9968	—
3s, s 00	F_2	(0, 0.1]	2	2	4.1123	3.9643	4	—	-14.541	-3.6009	—
3s, s 00	CFT	—	2	2	4.1123	4.1123	—	—	—	0	—
ss 3s, s	Free	(0, 0.1]	2	1.9883	1.6449	1.5661	—	—	—	-4.7937	-0.58466
ss 3s, s	F_1	(0, 0.1]	2	2	1.6449	1.6134	—	—	—	-1.9141	—
ss 3s, s	F_2	(0, 0.1]	2	2	1.6449	1.6276	4	—	-2.0896	-1.0534	—
ss 3s, s	CFT	—	2	2	1.6449	1.6449	—	—	—	0	—
3s, s ss	Free	(0, 0.1]	2	1.9902	1.6449	1.576	—	—	—	-4.1922	-0.48899
3s, s ss	F_1	(0, 0.1]	2	2	1.6449	1.6157	—	—	—	-1.7746	—
3s, s ss	F_2	(0, 0.1]	2	2	1.6449	1.6276	4	—	-1.7507	-1.0534	—
3s, s ss	CFT	—	2	2	1.6449	1.6449	—	—	—	0	—

Table 13: Compact boson small-interval fit quality for Table 12, in the same ordered-pair blocks. All fitted-row metrics use that row's W_x and m observations; k is the fitted parameter count. CFT metrics are evaluated on the same displayed window as the fitted rows, although no fitting is performed. RMSE, CV RMSE, and RMSrel follow Eqs. (62) and (63). For every F_2 row, gains G, G_{CV} use F_1 refitted on exactly that F_2 window, as in Eq. (89); its denominator equals the displayed F_1 row because the windows now match. Negative gains mean worse prediction. Absolute RMSEs on different windows should not be ranked directly.

Direction	Model	W_x	m	k	RMSE	CV RMSE	RMSrel [%]	G [%]	G_{CV} [%]
00 ss	Free	(0, 0.1]	598	2	1.742×10^{-5}	1.819×10^{-5}	1.2305	0.094877	-1.6624
00 ss	F_1	(0, 0.1]	598	1	1.744×10^{-5}	1.790×10^{-5}	1.0961	0	0
00 ss	F_2	(0, 0.1]	598	2	1.741×10^{-5}	1.835×10^{-5}	1.0996	0.15988	-2.5678
00 ss	CFT	—	598	0	9.183×10^{-5}	9.183×10^{-5}	5.6824	-426.7	-413.16
ss 00	Free	(0, 0.1]	598	2	1.737×10^{-5}	1.814×10^{-5}	1.2171	0.084288	-1.6754
ss 00	F_1	(0, 0.1]	598	1	1.739×10^{-5}	1.784×10^{-5}	1.0955	0	0
ss 00	F_2	(0, 0.1]	598	2	1.736×10^{-5}	1.830×10^{-5}	1.0992	0.14523	-2.5842
ss 00	CFT	—	598	0	9.171×10^{-5}	9.171×10^{-5}	5.6814	-427.5	-413.98

Direction	Model	W_x	m	k	RMSE	CV RMSE	RMSrel [%]	G [%]	G_{CV} [%]
00 3s, s	Free	(0, 0.1]	598	2	6.866×10^{-5}	7.172×10^{-5}	9.5493	31.286	30.43
00 3s, s	F_1	(0, 0.1]	598	1	9.993×10^{-5}	1.031×10^{-4}	2.5019	0	0
00 3s, s	F_2	(0, 0.1]	598	2	6.343×10^{-5}	6.680×10^{-5}	0.81386	36.528	35.203
00 3s, s	CFT	—	598	0	5.243×10^{-4}	5.243×10^{-4}	4.2262	-424.67	-408.57
3s, s 00	Free	(0, 0.1]	598	2	6.562×10^{-5}	6.848×10^{-5}	9.2257	31.748	30.941
3s, s 00	F_1	(0, 0.1]	598	1	9.614×10^{-5}	9.916×10^{-5}	2.4246	0	0
3s, s 00	F_2	(0, 0.1]	598	2	6.045×10^{-5}	6.360×10^{-5}	0.80915	37.118	35.866
3s, s 00	CFT	—	598	0	5.166×10^{-4}	5.166×10^{-4}	4.2085	-437.29	-420.93
ss 3s, s	Free	(0, 0.1]	598	2	1.453×10^{-5}	1.519×10^{-5}	3.0938	17.691	16.67
ss 3s, s	F_1	(0, 0.1]	598	1	1.765×10^{-5}	1.823×10^{-5}	0.87666	0	0
ss 3s, s	F_2	(0, 0.1]	598	2	1.400×10^{-5}	1.476×10^{-5}	0.41654	20.66	18.998
ss 3s, s	CFT	—	598	0	6.716×10^{-5}	6.716×10^{-5}	1.2933	-280.54	-268.48
3s, s ss	Free	(0, 0.1]	598	2	1.241×10^{-5}	1.294×10^{-5}	2.5931	17.149	16.297
3s, s ss	F_1	(0, 0.1]	598	1	1.498×10^{-5}	1.546×10^{-5}	0.76126	0	0
3s, s ss	F_2	(0, 0.1]	598	2	1.198×10^{-5}	1.258×10^{-5}	0.40929	20.054	18.631
3s, s ss	CFT	—	598	0	6.191×10^{-5}	6.191×10^{-5}	1.2671	-313.26	-300.6

B Large-interval fit summaries and sanity checks

B.1 Separate free, one-term, and two-term fits

Every coefficient multiplies y^r itself. Free fits determine amplitude and exponent; F_1 fixes the analytical leading exponent and fits one amplitude; F_2 fixes the leading and next powers in Table 5 and fits both amplitudes. The CFT leading rows adjust no parameter. Free, F_1 , and F_2 share the CFT-wide window in Eq. (59) for all six directions. The boson window is provisional; see Appendix H.4. All fits use $\ell = 6, \dots, 10$, the grid in Eq. (52), and the objective and diagnostics in Sec. 4. The parameter tables repeat the analytical leading coefficient c_0^{CFT} beside each model amplitude. Their c_1^{CFT} column contains dashes because the manuscript supplies no next-order coefficient for these large-interval pairs. The next powers used in F_2 remain those in Table 5, including its qualification that the boson $ss \leftrightarrow (3s, s)$ power three is an allowed sector. The quality-table gains use the matching-window F_1 reference in Eq. (89); CFT metrics are evaluated on the same CFT-wide window. The per-pair figures in Sec. 6 show the corresponding curves and fitting intervals.

Table 14: Using large-interval parameters at $p = 1/2$, $\ell = 6, \dots, 10$, and the grid in Eq. (52). Models are grouped by ordered pair. The column W_y gives each fitted model's interval; CFT adjusts no parameter and has no fitting window. Free, F_1 , and F_2 use the same displayed interval for all six ordered pairs of this CFT and interval type, so the additional term is tested on identical observations. Every model is visualized for each ordered pair in Sec. 6.6. All amplitudes multiply y^r itself and include powers of π . Columns c_0, c_1 multiply the displayed powers $r^{(0)}, r^{(1)}$; a dash indicates an absent term. Analytical columns give the complete supplied coefficients. A dash in c_1^{CFT} means that the manuscript does not supply that coefficient, not that it vanishes. Percentage discrepancies follow Eq. (62); an imposed power has no estimated exponent discrepancy.

Direction	Model	W_y	β_{CFT}	$r^{(0)}$	c_0^{CFT}	c_0	$r^{(1)}$	c_1^{CFT}	c_1	η_c [%]	η_r [%]
$I \sigma$	Free	(0, 0.002]	0.25	0.26465	0.61965	0.71994	—	—	—	16.184	5.8606
$I \sigma$	F_1	(0, 0.002]	0.25	0.25	0.61965	0.65339	—	—	—	5.4442	—
$I \sigma$	F_2	(0, 0.002]	0.25	0.25	0.61965	0.61463	0.5	—	0.20243	-0.81098	—
$I \sigma$	CFT	—	0.25	0.25	0.61965	0.61965	—	—	—	0	—
σI	Free	(0, 0.002]	0.25	0.26465	0.61965	0.71995	—	—	—	16.186	5.8612
σI	F_1	(0, 0.002]	0.25	0.25	0.61965	0.65339	—	—	—	5.4445	—
σI	F_2	(0, 0.002]	0.25	0.25	0.61965	0.61462	0.5	—	0.20245	-0.81139	—
σI	CFT	—	0.25	0.25	0.61965	0.61965	—	—	—	0	—
$I \varepsilon$	Free	(0, 0.002]	2	1.9985	3.2899	3.2548	—	—	—	-1.0674	-0.074487
$I \varepsilon$	F_1	(0, 0.002]	2	2	3.2899	3.2861	—	—	—	-0.11533	—
$I \varepsilon$	F_2	(0, 0.002]	2	2	3.2899	3.289	4	—	-1070.2	-0.024918	—
$I \varepsilon$	CFT	—	2	2	3.2899	3.2899	—	—	—	0	—
εI	Free	(0, 0.002]	2	2.0013	3.2899	3.3242	—	—	—	1.0443	0.065315
εI	F_1	(0, 0.002]	2	2	3.2899	3.2964	—	—	—	0.19921	—
εI	F_2	(0, 0.002]	2	2	3.2899	3.2938	4	—	959.21	0.11818	—
εI	CFT	—	2	2	3.2899	3.2899	—	—	—	0	—
$\sigma \varepsilon$	Free	(0, 0.002]	0.25	0.25058	0.15491	0.15691	—	—	—	1.2904	0.23227
$\sigma \varepsilon$	F_1	(0, 0.002]	0.25	0.25	0.15491	0.15631	—	—	—	0.9017	—
$\sigma \varepsilon$	F_2	(0, 0.002]	0.25	0.25	0.15491	0.15595	0.5	—	0.0018716	0.67037	—
$\sigma \varepsilon$	CFT	—	0.25	0.25	0.15491	0.15491	—	—	—	0	—
$\varepsilon \sigma$	Free	(0, 0.002]	0.25	0.25544	0.15491	0.16288	—	—	—	5.1411	2.1754
$\varepsilon \sigma$	F_1	(0, 0.002]	0.25	0.25	0.15491	0.15712	—	—	—	1.4228	—
$\varepsilon \sigma$	F_2	(0, 0.002]	0.25	0.25	0.15491	0.15365	0.5	—	0.018087	-0.81285	—
$\varepsilon \sigma$	CFT	—	0.25	0.25	0.15491	0.15491	—	—	—	0	—

Table 15: Using large-interval fit quality for Table 14, in the same ordered-pair blocks. All fitted-row metrics use that row's W_y and m observations; k is the fitted parameter count. CFT metrics are evaluated on the same displayed window as the fitted rows, although no fitting is performed. RMSE, CV RMSE, and RMSrel follow Eqs. (62) and (63). For every F_2 row, gains G, G_{CV} use F_1 refitted on exactly that F_2 window, as in Eq. (89); its denominator equals the displayed F_1 row because the windows now match. Negative gains mean worse prediction. Absolute RMSEs on different windows should not be ranked directly.

Direction	Model	W_y	m	k	RMSE	CV RMSE	RMSrel [%]	G [%]	G_{CV} [%]
$I \sigma$	Free	(0, 0.002]	260	2	2.156×10^{-5}	2.181×10^{-5}	0.017377	95.561	95.551
$I \sigma$	F_1	(0, 0.002]	260	1	4.858×10^{-4}	4.902×10^{-4}	0.40158	0	0
$I \sigma$	F_2	(0, 0.002]	260	2	1.523×10^{-5}	1.538×10^{-5}	0.01209	96.864	96.863
$I \sigma$	CFT	—	260	0	0.0064347	0.0064347	5.1251	-1224.7	-1212.7
σI	Free	(0, 0.002]	260	2	2.150×10^{-5}	2.175×10^{-5}	0.017332	95.574	95.564
σI	F_1	(0, 0.002]	260	1	4.858×10^{-4}	4.902×10^{-4}	0.40162	0	0
σI	F_2	(0, 0.002]	260	2	1.512×10^{-5}	1.526×10^{-5}	0.012002	96.888	96.886
σI	CFT	—	260	0	0.006435	0.006435	5.1254	-1224.6	-1212.6
$I \varepsilon$	Free	(0, 0.002]	260	2	1.190×10^{-9}	1.209×10^{-9}	0.01586	49.301	48.662
$I \varepsilon$	F_1	(0, 0.002]	260	1	2.346×10^{-9}	2.354×10^{-9}	0.053668	0	0
$I \varepsilon$	F_2	(0, 0.002]	260	2	1.217×10^{-9}	1.237×10^{-9}	0.020162	48.143	47.468
$I \varepsilon$	CFT	—	260	0	8.406×10^{-9}	8.406×10^{-9}	0.092521	-258.26	-257.07
εI	Free	(0, 0.002]	260	2	9.799×10^{-10}	9.938×10^{-10}	0.017599	51.74	51.561
εI	F_1	(0, 0.002]	260	1	2.030×10^{-9}	2.052×10^{-9}	0.041144	0	0
εI	F_2	(0, 0.002]	260	2	9.430×10^{-10}	9.591×10^{-10}	0.013062	53.559	53.25
εI	CFT	—	260	0	1.409×10^{-8}	1.409×10^{-8}	0.17539	-593.94	-586.78
$\sigma \varepsilon$	Free	(0, 0.002]	260	2	5.939×10^{-6}	5.991×10^{-6}	0.020061	20.974	20.404
$\sigma \varepsilon$	F_1	(0, 0.002]	260	1	7.515×10^{-6}	7.527×10^{-6}	0.026612	0	0
$\sigma \varepsilon$	F_2	(0, 0.002]	260	2	6.027×10^{-6}	6.079×10^{-6}	0.020416	19.802	19.238
$\sigma \varepsilon$	CFT	—	260	0	2.658×10^{-4}	2.658×10^{-4}	0.89182	-3436.8	-3431.2
$\varepsilon \sigma$	Free	(0, 0.002]	260	2	4.604×10^{-6}	4.653×10^{-6}	0.01521	89.435	89.411
$\varepsilon \sigma$	F_1	(0, 0.002]	260	1	4.358×10^{-5}	4.394×10^{-5}	0.14923	0	0
$\varepsilon \sigma$	F_2	(0, 0.002]	260	2	4.134×10^{-6}	4.176×10^{-6}	0.013619	90.513	90.498
$\varepsilon \sigma$	CFT	—	260	0	4.215×10^{-4}	4.215×10^{-4}	1.3909	-867.17	-859.14

Table 16: Compact boson large-interval parameters at $p = 1/2$, $\ell = 6, \dots, 10$, and the grid in Eq. (52). Models are grouped by ordered pair. The column W_y gives each fitted model's interval; CFT adjusts no parameter and has no fitting window. Free, F_1 , and F_2 use the same displayed interval for all six ordered pairs of this CFT and interval type, so the additional term is tested on identical observations. Every model is visualized for each ordered pair in Sec. 6.7. All amplitudes multiply y^r itself and include powers of π . Columns c_0, c_1 multiply the displayed powers $r^{(0)}, r^{(1)}$; a dash indicates an absent term. Analytical columns give the complete supplied coefficients. A dash in c_1^{CFT} means that the manuscript does not supply that coefficient, not that it vanishes. Percentage discrepancies follow Eq. (62); an imposed power has no estimated exponent discrepancy.

Direction	Model	W_y	β_{CFT}	$r^{(0)}$	c_0^{CFT}	c_0	$r^{(1)}$	c_1^{CFT}	c_1	η_c [%]	η_r [%]
00 ss	Free	(0, 0.005]	1	0.99926	1.2337	1.2561	—	—	—	1.8189	-0.07446
00 ss	F_1	(0, 0.005]	1	1	1.2337	1.2616	—	—	—	2.2645	—
00 ss	F_2	(0, 0.005]	1	1	1.2337	1.2627	2	—	-0.33376	2.3499	—
00 ss	CFT	—	1	1	1.2337	1.2337	—	—	—	0	—
ss 00	Free	(0, 0.005]	1	0.99924	1.2337	1.256	—	—	—	1.8109	-0.075663
ss 00	F_1	(0, 0.005]	1	1	1.2337	1.2616	—	—	—	2.2636	—
ss 00	F_2	(0, 0.005]	1	1	1.2337	1.2627	2	—	-0.34138	2.351	—
ss 00	CFT	—	1	1	1.2337	1.2337	—	—	—	0	—
00 3s, s	Free	(0, 0.005]	5	4.9836	75.109	66.924	—	—	—	-10.897	-0.32701
00 3s, s	F_1	(0, 0.005]	5	5	75.109	73.093	—	—	—	-2.6829	—
00 3s, s	F_2	(0, 0.005]	5	5	75.109	74.307	6	—	-265.58	-1.0676	—
00 3s, s	CFT	—	5	5	75.109	75.109	—	—	—	0	—
3s, s 00	Free	(0, 0.005]	5	4.9838	75.109	66.96	—	—	—	-10.849	-0.32323
3s, s 00	F_1	(0, 0.005]	5	5	75.109	73.058	—	—	—	-2.7295	—
3s, s 00	F_2	(0, 0.005]	5	5	75.109	74.252	6	—	-261.18	-1.141	—
3s, s 00	CFT	—	5	5	75.109	75.109	—	—	—	0	—
ss 3s, s	Free	(0, 0.005]	2	1.9914	3.2899	3.069	—	—	—	-6.7143	-0.42763
ss 3s, s	F_1	(0, 0.005]	2	2	3.2899	3.2185	—	—	—	-2.1683	—
ss 3s, s	F_2	(0, 0.005]	2	2	3.2899	3.2529	3	—	-8.6617	-1.1234	—
ss 3s, s	CFT	—	2	2	3.2899	3.2899	—	—	—	0	—
3s, s ss	Free	(0, 0.005]	2	1.9912	3.2899	3.0643	—	—	—	-6.8552	-0.43847
3s, s ss	F_1	(0, 0.005]	2	2	3.2899	3.2176	—	—	—	-2.1982	—
3s, s ss	F_2	(0, 0.005]	2	2	3.2899	3.2528	3	—	-8.8758	-1.1274	—
3s, s ss	CFT	—	2	2	3.2899	3.2899	—	—	—	0	—

Table 17: Compact boson large-interval fit quality for Table 16, in the same ordered-pair blocks. All fitted-row metrics use that row's W_y and m observations; k is the fitted parameter count. CFT metrics are evaluated on the same displayed window as the fitted rows, although no fitting is performed. RMSE, CV RMSE, and RMSrel follow Eqs. (62) and (63). For every F_2 row, gains G, G_{CV} use F_1 refitted on exactly that F_2 window, as in Eq. (89); its denominator equals the displayed F_1 row because the windows now match. Negative gains mean worse prediction. Absolute RMSEs on different windows should not be ranked directly.

Direction	Model	W_y	m	k	RMSE	CV RMSE	RMSrel [%]	G [%]	G_{CV} [%]
00 ss	Free	(0, 0.005]	317	2	1.023×10^{-5}	1.037×10^{-5}	0.39535	0.43323	-0.33962
00 ss	F_1	(0, 0.005]	317	1	1.028×10^{-5}	1.034×10^{-5}	0.41077	0	0
00 ss	F_2	(0, 0.005]	317	2	1.024×10^{-5}	1.042×10^{-5}	0.40064	0.33876	-0.80584
00 ss	CFT	—	317	0	5.738×10^{-5}	5.738×10^{-5}	2.3252	-458.28	-455.02
ss 00	Free	(0, 0.005]	317	2	1.014×10^{-5}	1.028×10^{-5}	0.39344	0.45533	-0.32294
ss 00	F_1	(0, 0.005]	317	1	1.019×10^{-5}	1.025×10^{-5}	0.40918	0	0
ss 00	F_2	(0, 0.005]	317	2	1.015×10^{-5}	1.033×10^{-5}	0.39876	0.36073	-0.7901
ss 00	CFT	—	317	0	5.734×10^{-5}	5.734×10^{-5}	2.3246	-462.85	-459.54

Direction	Model	W_y	m	k	RMSE	CV RMSE	RMSrel [%]	G [%]	G_{CV} [%]
00 3s, s	Free	(0, 0.005]	317	2	3.500×10^{-13}	3.791×10^{-13}	1.1997	1.6505	0.0023414
00 3s, s	F_1	(0, 0.005]	317	1	3.558×10^{-13}	3.791×10^{-13}	2.4953	0	0
00 3s, s	F_2	(0, 0.005]	317	2	3.508×10^{-13}	3.806×10^{-13}	1.5663	1.4034	-0.40053
00 3s, s	CFT	—	317	0	1.156×10^{-12}	1.156×10^{-12}	1.4529	-224.95	-205.02
3s, s 00	Free	(0, 0.005]	317	2	3.580×10^{-13}	3.876×10^{-13}	1.2103	1.5418	-0.078414
3s, s 00	F_1	(0, 0.005]	317	1	3.636×10^{-13}	3.873×10^{-13}	2.5265	0	0
3s, s 00	F_2	(0, 0.005]	317	2	3.589×10^{-13}	3.891×10^{-13}	1.6086	1.2992	-0.47282
3s, s 00	CFT	—	317	0	1.177×10^{-12}	1.177×10^{-12}	1.4758	-223.66	-203.9
ss 3s, s	Free	(0, 0.005]	317	2	8.501×10^{-8}	8.824×10^{-8}	0.60705	15.387	14.604
ss 3s, s	F_1	(0, 0.005]	317	1	1.005×10^{-7}	1.033×10^{-7}	0.78756	0	0
ss 3s, s	F_2	(0, 0.005]	317	2	8.538×10^{-8}	8.894×10^{-8}	0.40879	15.019	13.931
ss 3s, s	CFT	—	317	0	4.998×10^{-7}	4.998×10^{-7}	1.666	-397.48	-383.72
3s, s ss	Free	(0, 0.005]	317	2	8.761×10^{-8}	9.097×10^{-8}	0.61706	15.259	14.46
3s, s ss	F_1	(0, 0.005]	317	1	1.034×10^{-7}	1.063×10^{-7}	0.80671	0	0
3s, s ss	F_2	(0, 0.005]	317	2	8.800×10^{-8}	9.170×10^{-8}	0.41394	14.883	13.775
3s, s ss	CFT	—	317	0	5.070×10^{-7}	5.070×10^{-7}	1.6816	-390.4	-376.77

B.2 Sparse cross moments, borders, and cached minors

This subsection specifies the algebra used to evaluate Eq. (78) efficiently, including the singular cross-sector cases. All indices of the moment matrix start at zero. Define the full-circle matrix D by replacing B with all N sites in Eq. (77). Because $b_1, b_2 \in \{0, 1\}$,

$$g_1(z)\overline{g_2(z)} = c_{-1}z^{-1} + c_0 + c_1z, \quad (c_{-1}, c_0, c_1) = \left(-\frac{b_2}{W}, 1 + \frac{b_1b_2}{W^2}, -\frac{b_1}{W}\right). \quad (90)$$

The root-of-unity sum $\sum_j z_j^t = N$ for $t = 0 \bmod N$ and zero otherwise gives the sparse moments

$$D_{rs} = N(s - r) \sum_{h=-1}^1 c_h \mathbf{1}_{r+s-1+\kappa+h=0 \bmod N}. \quad (91)$$

The norms $\mathcal{Z}_i$ are the Pfaffians of the corresponding diagonal-charge moment matrices of dimension $2M_i$.

For the pairs in this study the full cross matrix can be singular. Use the following regular matrices R ; e_j denotes the unit column at moment index j :

$$R = \begin{cases} \begin{pmatrix} D & e_{(d-1)/2} \\ -e_{(d-1)/2}^T & 0 \end{pmatrix}, & (00, ss), \quad d = N - 1, \\ D + e_0 e_{d/2}^T - e_{d/2} e_0^T, & (00, 3s, s), \quad d = N - 2, \\ \begin{pmatrix} D & e_0 \\ -e_0^T & 0 \end{pmatrix}, & (ss, 3s, s), \quad d = N - 3. \end{cases} \quad (92)$$

These are exact auxiliary matrices. For odd d , retain the border while removing the physical A sites, and pad every physical column f, f' by a zero. For even d , remove the artificial rank-two coupling together with the A contributions. To do both cases at once, collect the physical columns $f(z_j), f'(z_j)$ for $j \in A$ in a matrix V ; in the even case append $e_0, e_{d/2}$. Define $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $S = \text{diag}(w(z_0)J, \dots, w(z_{\ell-1})J)$, appending one J block in the even case. The environment matrix (bordered when d is odd) is $\tilde{D}_B = R - VSV^T$.

Set $K_V = V^T R^{-1} V$ and $H = S^{-1} - K_V$. The matrix inverse and Pfaffian update identities then read

$$\begin{aligned}\tilde{D}_B^{-1} &= R^{-1} + R^{-1} V H^{-1} V^T R^{-1}, \\ \text{Pf}(\tilde{D}_B) &= \text{Pf}(R) (-1)^r \left(\prod_{j=1}^r w_j \right) \text{Pf}(H).\end{aligned}\tag{93}$$

Here r is the number of column pairs in V and w_j their weights; the artificial pair has $w_j = 1$. All needed products with R^{-1} are obtained by one sparse solve with multiple right-hand sides. Only H , of dimension 2ℓ or $2\ell + 2$, is inverted with high-precision dense arithmetic.

For even d , choose the external columns E_{UV} of Eq. (78) and evaluate

$$\text{Pf} \begin{pmatrix} D_B & E_{UV} \\ -E_{UV}^T & 0 \end{pmatrix} = \text{Pf}(D_B) \text{Pf}(E_{UV}^T D_B^{-1} E_{UV}).\tag{94}$$

For odd d , pad these columns by zero and append the unit column e_d on the retained border, obtaining E'_{UV} . Then

$$\text{Pf} \begin{pmatrix} D_B & E_{UV} \\ -E_{UV}^T & 0 \end{pmatrix} = -\text{Pf}(\tilde{D}_B) \text{Pf}((E'_{UV})^T \tilde{D}_B^{-1} E'_{UV}).\tag{95}$$

The minus sign results from expanding the added border pair in the full antisymmetric matrix. The appended e_d enforces removal of that auxiliary pair; ignoring it would compute a different contraction. For odd d , q is odd, so $q + 1$ is even as a Pfaffian requires. All external minors come from a common matrix of dimension at most $2\ell + 1$. They depend only on $U \cup V$ and $U \cap V$ and are cached. Principal minors are evaluated by the Pfaffian Laplace expansion [16]. The sparse solve, border choices, and cross-charge contraction are implemented in `boson_transition.py`.

B.3 Independent validation and numerical precision

The new checks use explicit full wavefunctions at $N = 8, 10, 12$ and $\ell = 3, 4$. Ising vectors come from the independent conformal-block vacuum/energy construction and block-seeded Lanczos spin state. Boson vectors are directly enumerated from the vertex blocks. For every unordered pair, form the coefficient matrices explicitly and compare $X_a^T X_b^*$ with the new transition on A , allowing one global phase. The complex alignment factor is denoted c ; we also check that it has unit modulus. Both directed entropies are then computed from explicit matrices on B and compared with Eq. (68). An independent phase rotation of the transition tests the invariance of the Gram result.

The boson checks additionally use $W = \infty$ and $W = 4$ at $N = 10, \ell = 4$. The latter gives substantial nonconstant Laurent terms in Eq. (90), testing the finite-endpoint formula away from its production value. At $N = 256, \ell = 10$, all three boson transitions are recomputed at 130 decimal digits and compared with 100 digits. Ising anchored overlaps evaluated by singular values are compared with the earlier determinant evaluation.

In production, eigenvalues below $\tau = 10^{-15}$ are omitted from $t \log t$ sums. This is a numerical threshold, not an added positive floor in the density matrix. We check $\tau = 10^{-14}$ and 10^{-16} at $N = 32, 256, \ell = 4, 10$, for all six pairs. Define the support-weight defect and Holevo identity defect by

$$\begin{aligned}e_w &= \max_{c=a,b} \left| 2 \sum_{\lambda_k > \tau} \lambda_k v_k^\dagger P_c v_k - 1 \right|, \\ e_H &= \left| \frac{D_{a|b}^B + D_{b|a}^B}{2} - S(\bar{\rho}_{ab}^B) + \frac{S(\rho_a^A) + S(\rho_b^A)}{2} \right|,\end{aligned}\tag{96}$$

where $S(\rho) = -\text{Tr } \rho \log \rho$ is von Neumann entropy. Every production record includes the minimum Gram eigenvalue, $|\text{Tr } \mathcal{G} - 1|$, e_w , and e_H . These are algebraic consistency checks; agreement with CFT is a separate test.

Check	Count	Maximum error	Tolerance
Transition versus explicit vectors	18	4.491×10^{-14}	2.000×10^{-11}
Transition phase modulus	18	2.709×10^{-14}	2.000×10^{-11}
Gram versus explicit large matrices	18	2.354×10^{-14}	2.000×10^{-11}
Independent phase rotation	18	5.662×10^{-15}	2.000×10^{-12}
Entropy cutoff 10^{-15} versus 10^{-14}	12	1.019×10^{-11}	2.000×10^{-11}
Entropy cutoff 10^{-15} versus 10^{-16}	12	1.314×10^{-12}	2.000×10^{-11}
Ising overlap evaluation	3	5.503×10^{-15}	2.000×10^{-11}
Entropy under transition precision change	6	1.248×10^{-13}	2.000×10^{-11}
Boson transition: 100 versus 130 digits	3	0	2.000×10^{-11}

Table 18: All 108 displayed complementary-interval implementation checks pass; checks at $\ell = 4, 5$ remain archived and are excluded from this table. Transition errors are maximum absolute entry differences after one constant phase alignment; phase error is $||c| - 1|$. Entropy errors are the largest absolute difference between the two directions. A zero precision difference means identical matrices after conversion to double precision. The construction and precision settings are specified in Sec. B.

B.4 Extension to $\ell = 11, 12$

Increasing ℓ enlarges both the local matrices and the Gram blocks. We therefore construct the common- B symmetry blocks directly and use Eq. (69) for the new production points. The Schmidt cutoff is $\tau_s = 10^{-16}$; the entropy cutoff remains $\tau = 10^{-15}$. The independent $N = 256$ reference retains the full Gram blocks for all six unordered pairs at both new lengths. For each pair and length we compare the supported result with this reference, vary τ_s to 10^{-17} and 10^{-15} , and measure $e_s = \max_{c=a,b} |1 - \sum_{j: \lambda_{cj} > \tau_s} \lambda_{cj}|$, where λ_{cj} are the eigenvalues of ρ_c^A . This checks the numerical support approximation; it does not assume that the discarded spectrum is exactly zero.

For every size and state, the partial trace of the new $\ell = 12$ diagonal matrix must reproduce the original $\ell = 10$ cache. The same test applies to each transition, allowing one constant phase and separately checking its unit modulus. At $N = 256$, sampled entries of all three boson diagonal matrices and all three transitions are recomputed at 130 decimal digits by a direct pivoted Pfaffian, independently of the cached principal-minor recursion used in production. The samples cover every allowed local occupation count, edge configurations, and fixed-seed random configurations. This is an entrywise sample test, not a full 130-digit recomputation of the 4096×4096 matrices.

The boson implementation also avoids repeated products over site pairs. For a site set C , define $V(C) = \prod_{i < j, i, j \in C} (z_j - z_i)$ and $P_j(C) = \prod_{i \in C \setminus \{j\}} (z_{\max(i,j)} - z_{\min(i,j)})$. With $C = U \cup V$, $J = U \cap V$, and multiplicity $m_i = \mathbf{1}_{i \in U} + \mathbf{1}_{i \in V}$, the denominator is

$$\prod_{i < j} (z_j - z_i)^{m_i m_j} = V(C) V(J) \prod_{j \in J} P_j(C). \quad (97)$$

For a pair of sites in C , the right-hand exponent is $1 + \mathbf{1}_{i \in J} + \mathbf{1}_{j \in J} + \mathbf{1}_{i, j \in J} = m_i m_j$, which proves the identity. The factors are cached recursively over subsets. The same principal-minor recursion is evaluated with Julia `BigFloat` at the precision of 100 decimal digits; only the final rescaled matrix entries are converted to double precision. This changes the execution speed, not the contraction formula. The source pair `boson_julia.py` and `boson_reconstruct.jl` supplies the bridge and the complete recursion; the Python implementation remains available for comparison.

Before the extension, 21 checks compare the revised Gram construction with its original implementation, the supported and full Gram results, Ising Wick and Schur diagonal reconstruction, and sampled versus complete boson calculations at smaller ℓ . The additional 324 extension checks are summarized in Table 19.

Check	Count	Maximum error	Tolerance
Direct sectors versus original Gram	6	6.66×10^{-15}	2.00×10^{-11}
Schmidt support versus original Gram	6	1.22×10^{-13}	2.00×10^{-11}
Ising Wick versus Schur diagonal	3	7.77×10^{-16}	2.00×10^{-11}
Sampled direct Pfaffian: diagonal	3	3.13×10^{-85}	2.00×10^{-11}
Sampled direct Pfaffian: transition	3	0	2.00×10^{-11}
Diagonal $\ell = 10$ partial trace	90	1.67×10^{-15}	2.00×10^{-11}
Transition $\ell = 10$ partial trace	90	1.32×10^{-16}	2.00×10^{-11}
Partial-trace phase modulus	90	1.15×10^{-13}	2.00×10^{-11}
Schmidt support versus full Gram, $N = 256$	12	1.37×10^{-13}	2.00×10^{-11}
Schmidt cutoff 10^{-17} versus 10^{-16}	12	1.61×10^{-13}	2.00×10^{-11}
Schmidt cutoff 10^{-15} versus 10^{-16}	12	4.12×10^{-13}	2.00×10^{-11}
Discarded Schmidt trace	12	6.99×10^{-15}	2.00×10^{-13}
130-digit sampled diagonal entries	3	1.08×10^{-40}	2.00×10^{-11}
130-digit sampled transition entries	3	1.18×10^{-43}	2.00×10^{-11}

Table 19: All 21 preliminary implementation checks and 324 extension checks pass. The first group tests revised code against the original construction; the second tests the new $\ell = 12$ matrices at all fifteen sizes, full versus supported Gram results at $N = 256$, $\ell = 11, 12$, support-cutoff stability, and independent 130-digit sampled Pfaffians. Matrix errors are maximum absolute entry differences, with one constant phase allowed for transitions; phase-modulus and discarded-trace errors are defined in Sec. B.4. Gram and cutoff comparisons use the largest absolute entropy difference over the two directions.

The historical checks above document the original production through $N = 256$, including the archived $\ell = 11, 12$ extension. The active larger-chain checks use the same state definitions and are given next.

B.5 Checks of the larger-chain numerical method

The Fourier construction in Eq. (53) was compared with the original dense Hamiltonian eigensolve at $N = 8, 16, 32, 128, 256$ for all three states. The anchored transition was compared with the original full complex covariance and singular-value overlap implementation at $\ell = 4$ for every pair. The largest covariance entry difference was 2.220×10^{-15} and the largest transition entry difference was 3.747×10^{-15} . Energies and Fourier applications to independent test columns were checked as well; all 30 records passed. At $N = 256$, $\ell = 4, 10$, the new entropy remainder and original trace-log/Gram formulas agreed within 1.400×10^{-12} across both CFTs.

At the largest Ising size, Fourier applications additionally verify $\Gamma^2 v = -v$ and antisymmetry on four independent test columns for each primary; local XX and Z expectations reproduce the analytical energy per site. These are included in the independent-check table. Tables 20 and 21 give the full new production and precision checks. The support-cutoff criterion was relative variation below one percent, and the absolute comparison tolerance was 5×10^{-12} . For observables below that absolute scale, independent high-precision contractions and the noncommuting tiny-Gram example are essential: a cancellation-damaged zero could otherwise pass an absolute test. These checks assess numerical accuracy of the finite-chain construction, not the separate continuum extrapolation. A reported zero entry difference means identical rounded double-precision outputs; it does not assert zero error in exact arithmetic or 120-digit accuracy of the final returned matrices.

Table 20: Production checks at every added size in Eq. (52). Relative cutoff sensitivity is $\max_{\tau \in \{10^{-14}, 10^{-16}\}} |q_\tau - q_{10^{-15}}|/q_{10^{-15}}$, shown here for retained endpoint lengths ($\ell = 6, 10$ for the dense scan and $\ell = 10$ for the earlier extension); the $\ell = 4$ checks remain in the raw archive. Trace-log and original full-Gram differences are absolute; they cannot alone certify relative accuracy of extremely small observables. LU residual is $\|KZ - (I - i\Gamma_a)E_A\|_F / \|(I - i\Gamma_a)E_A\|_F$. Positivity errors are $\max(0, -\lambda_{\min})$. All stored tolerances passed.

Check	Count	Largest error
trace	6390	6.661×10^{-16}
hermiticity	6390	1.791×10^{-77}
positivity	6390	3.324×10^{-16}
small_cutoff_relative	4824	8.635×10^{-9}
small_trace_log_absolute	4824	2.615×10^{-13}
local_LU_residual	945	1.388×10^{-13}
Gram_positivity	6390	1.240×10^{-16}
large_cutoff_relative	4824	6.852×10^{-9}
large_full_Gram_absolute	4824	1.638×10^{-12}

Table 21: Independent numerical checks. Tiny noncommuting Gram errors are relative to a 100-digit matrix-logarithm reference, with cross amplitudes from 10^{-2} through 10^{-30} (deficits through order 10^{-60}). Boson diagonal errors are absolute entry differences; transition errors divide the maximum sampled entry difference by the largest reference entry. Sampled entries use 160-digit direct pivoted Pfaffians rather than the 120-digit cached-minor production calculation.

Check	Count	Largest error
Ising_Fourier_purity	3	5.207×10^{-16}
Ising_Fourier_skew	3	1.885×10^{-18}
Ising_local_energy_per_site	3	1.110×10^{-15}
boson_diagonal_high_precision	6	3.499×10^{-78}
boson_transition_high_precision_relative	6	0
tiny_noncommuting_Gram	24	8.882×10^{-16}

C Files and reproduction of the scaling revision

The implementation is in `src/conformal_blocks/`. The original wavefunction and puMPS comparisons, historical data, and their audits are retained. The active mixture results use these new files:

- `larger_chain_engine.py`: Fourier Ising covariance, selected transition columns, real LU overlap and solve diagnostics.
- `stable_entropy.py`: the matrix-function identities in Eqs. (56) and (58).
- `scaling_production.py`: resumable state contractions and both interval observables, written incrementally to `data/larger_chain_observations.csv`; numerical checks and timings are in `larger_chain_checks.csv` and `larger_chain_timings.csv`.
- `scaling_checks.py`: independent high-precision endpoint and tiny-deficit tests. The checked records are packaged with the report.
- `frozen_density_production.py`: reproduces the latest 48 added sizes per CFT from `data/frozen_density_grid_manifest.json`, using $\ell = 6, \dots, 10$ and the same production engine and precision.

- `scaling_fits.py`: common-window free, F_1 , F_2 , and CFT fits, omitted-size validation and nested fitting windows. The five active CSV files begin with `scaling_`; their selection flags explicitly identify fitting and display data. Only complete new size/model grids are included. Historical interval lengths are not silently selected.
- `report/scaling_artifacts.py`: reads completed CSVs and writes vector PDF plots and tables. It performs no numerical fitting.
- `retained_window_fits.py`: reproduces every recorded principal and companion window records, refitting Free 1, F_1 , and F_2 with matching-window F_1 controls. `report/f2_table_artifacts.py` writes the principal parameter and quality tables; `report/selection_audit.py` checks all updated rows, explicit window specifications, and preserved raw data.
- `dense_windows.py`: the identical-window first- and second-term comparisons, exploratory window scans, and shared-interval comparison in Appendix H.

The primary-to-primary appendix retains its previously tested $N \leq 256$ grid and $x \leq 0.08$ free fits, separately from the common mixture windows in Eq. (59), with coefficients converted to the same direct-power convention. Its physical predictions and residuals are unchanged by that conversion. Historical files such as `fit_summary.csv` and `complement_fit_summary.csv` document earlier revisions and are not the active mixture fit inputs. The common-window plot inputs are in `scaling_fit_summary.csv`; the revised summary-table inputs are in `f2_summary_fits.csv`.

With the approved Python interpreter and `PYTHONPATH=src`, run

```
python -m conformal_blocks.scaling_production --sizes 512 1024 2048 4096 8192
python -m conformal_blocks.scaling_production --models boson --sizes 16384
python -m conformal_blocks.scaling_production --models ising --sizes \
  4352 4608 4864 5120 5376 5632 5888 6144 6400 6656 6912 7168 7424 7680 7936
python -m conformal_blocks.scaling_production --models boson --sizes \
  9216 10240 11264 12288 14336
python -m conformal_blocks.scaling_production --models ising --sizes \
  8000 8016 8032 8048 8064 8080 8096 8112 8128 8144 200 152
python -m conformal_blocks.scaling_production --models boson --sizes \
  12800 13312 15360 168 104
python -m conformal_blocks.dense_production --output-dir new_dense_run
python -m conformal_blocks.frozen_density_production \
  --output-dir new_density_run
python -m conformal_blocks.scaling_checks --size 16384
python -m conformal_blocks.scaling_checks --size 8192 --ising-only
python -m conformal_blocks.scaling_baseline_checks
python report/rebuild_report.py
python report/selection_audit.py
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The executable interpreter path and threading settings are in the README. Run production and fitting sequentially so the final analysis uses a complete frozen grid. Small-size primary construction and historical Gram checks remain documented in `METHOD_REFERENCES.md`. The Overleaf archive contains `main.tex` at its root and needs pdfLaTeX with no BibTeX step. The study archive additionally contains the numerical source, current CSVs, and audit records.

D Fitting-window and sample-count checks

The four active mixture windows are Eq. (59). They retain, per ordered pair, 1255 small-Ising, 598 small-boson, 260 large-Ising, and 317 large-boson observations. The corresponding minimum training counts after omitting one chain size are 1250, 593, 255, and 312. Every active fit therefore exceeds forty observations, including every omitted-size training set. All fitting and display masks exclude $\ell = 4, 5$, while the raw files preserve those values. Table 22 gives exact sampled endpoints and distinct-size counts. All five retained subsystem lengths occur in each of the four active groups.

As a separate check closer to zero, we retain the earlier free-power fits on $(0, 0.001]$. Additional sizes within the validated maxima restored their sample counts after excluding $\ell = 4, 5$; the later dense scan added one further observation at $N = 6000$, $\ell = 6$. After the fixed-window refinement, these diagnostic fits contain 57 Ising and 116 boson geometries per direction. They are not the fits used in the active mixture figures or principal tables.

The count can be reconstructed without examining any entropy values. For cutoff t and the chain-size set $\mathcal{N}_{\text{CFT}}$ of the chosen CFT in Eq. (52), define

$$n_N(t) = \max\{0, \min[10, \lfloor tN \rfloor] - 5\}, \quad m(t) = \sum_{N \in \mathcal{N}_{\text{CFT}}} n_N(t). \quad (98)$$

Here $\lfloor u \rfloor$ is the greatest integer no larger than u ; n_N counts the admitted lengths starting at six. A size-holdout training set contains $m(t) - n_N(t)$ points. Both $m(t)$ at each adopted cutoff and every such training count are checked against the minimum of forty.

For Ising, $N \leq 8192$ means that only $\ell = 6, 7, 8$ can enter the separate 0.001 diagnostic fit: $\ell = 9, 10$ would require $N \geq 9000, 10000$. Their computed values remain in the figures. For the boson, every subsystem size $\ell = 6, \dots, 10$ occurs among the selected points, with $N \leq 16384$. Adding sizes increases the sampling density within these limits; it does not lower the smallest accessible fraction. Points sharing a chain or subsystem size are deterministic samples, not forty statistically independent measurements. Different (N, ℓ) can have the same fraction; they remain distinct lattice observations and receive equal weight in Eq. (61).

The denser boson data also support the trial cutoff 0.00075, with 69 points and at least 64 after omitting one size. The boson cutoff 0.000625 contains 41 points but can leave only 36 in a training fold, so it is excluded. The remaining smaller trial windows fail the same forty-point requirement. The free-fit tables below give the qualifying smaller-window diagnostics in a compact form. Each row estimates Cz^r with both parameters free using Eq. (61); theory enters only the subsequent comparison.

The coefficient and exponent are correlated, so a small exponent change can produce a larger coefficient change when extrapolated to unit fraction. Furthermore, a fixed range $\ell = 6, \dots, 10$ keeps the lattice resolution fixed as N increases. Stable exponent recovery does not require that every finite- ℓ amplitude equal its continuum value. These are direct fits and comparisons; no extra lattice-effect power law is assumed.

Table 22: Sample-count audit for the active mixture fits. Counts apply to each of the six directions. Every full fit and every omitted-size training set contains at least 40 distinct (N, ℓ) geometries. The fraction range is the actual sampled range within the corresponding window of Eq. (59); active mixture figures display fractions through 0.2. The number of sizes counts distinct retained N .

Interval	CFT	Samples	Sizes	Min. CV training	Smallest fraction	Largest fraction
small	ising	1255	278	1250	7.324×10^{-4}	0.02
small	boson	598	124	593	3.662×10^{-4}	0.1
large	ising	260	67	255	7.324×10^{-4}	0.002
large	boson	317	68	312	3.662×10^{-4}	0.0049779

Table 23: Free power-law fits on small-interval windows meeting the minimum of 40 observations even after omitting any one size. Every row estimates C, r in Cz^r , with $z = x, r = \alpha$ for small intervals and $z = y, r = \beta$ for large intervals. Coefficients include all powers of π ; all computed samples have $\ell = 6, \dots, 10$. The cutoff is an upper bound, m is the retained count, and the objective is Eq. (61). Only qualifying cutoffs are shown. These are a separate asymptotic-range check, not the adopted fits in Eq. (59).

Direction	Cutoff	m	C	r	η_c [%]	η_r [%]
$I \mid \sigma$	0.001	57	0.20452	1.9992	-0.53234	-0.041954
$\sigma \mid I$	0.001	57	0.20453	1.9992	-0.52997	-0.04176
$I \mid \varepsilon$	0.001	57	14.577	4.018	12.232	0.45071
$\varepsilon \mid I$	0.001	57	14.574	4.018	12.212	0.45011
$\sigma \mid \varepsilon$	0.001	57	0.20476	1.9993	-0.41875	-0.033779
$\varepsilon \mid \sigma$	0.001	57	0.20471	1.9993	-0.43938	-0.035187

Table 24: Free power-law fits on small-interval windows meeting the minimum of 40 observations even after omitting any one size. Every row estimates C, r in Cz^r , with $z = x, r = \alpha$ for small intervals and $z = y, r = \beta$ for large intervals. Coefficients include all powers of π ; all computed samples have $\ell = 6, \dots, 10$. The cutoff is an upper bound, m is the retained count, and the objective is Eq. (61). Only qualifying cutoffs are shown. These are a separate asymptotic-range check, not the adopted fits in Eq. (59).

Direction	Cutoff	m	C	r	η_c [%]	η_r [%]
00 $\mid ss$	0.001	116	0.82876	2.0087	0.76489	0.43312
00 $\mid ss$	7.500×10^{-4}	69	0.89103	2.0185	8.336	0.92557
$ss \mid 00$	0.001	116	0.82876	2.0087	0.76489	0.43312
$ss \mid 00$	7.500×10^{-4}	69	0.89103	2.0185	8.336	0.92557
00 $\mid 3s, s$	0.001	116	4.1499	2.0064	0.91377	0.32068
00 $\mid 3s, s$	7.500×10^{-4}	69	4.3823	2.0138	6.565	0.69105
$3s, s \mid 00$	0.001	116	4.1499	2.0064	0.91387	0.32069
$3s, s \mid 00$	7.500×10^{-4}	69	4.3823	2.0138	6.5651	0.69105
$ss \mid 3s, s$	0.001	116	1.6645	2.0032	1.19	0.15955
$ss \mid 3s, s$	7.500×10^{-4}	69	1.713	2.0071	4.14	0.35488
$3s, s \mid ss$	0.001	116	1.6645	2.0032	1.1902	0.15956
$3s, s \mid ss$	7.500×10^{-4}	69	1.713	2.0071	4.1401	0.35488

Table 25: Free power-law fits on large-interval windows meeting the minimum of 40 observations even after omitting any one size. Every row estimates C, r in Cz^r , with $z = x, r = \alpha$ for small intervals and $z = y, r = \beta$ for large intervals. Coefficients include all powers of π ; all computed samples have $\ell = 6, \dots, 10$. The cutoff is an upper bound, m is the retained count, and the objective is Eq. (61). Only qualifying cutoffs are shown. These are a separate asymptotic-range check, not the adopted fits in Eq. (59).

Direction	Cutoff	m	C	r	η_c [%]	η_r [%]
$I \mid \sigma$	0.001	57	0.7149	0.26364	15.372	5.4547
$\sigma \mid I$	0.001	57	0.7149	0.26364	15.371	5.4543
$I \mid \varepsilon$	0.001	57	3.2493	1.9983	-1.2326	-0.086132
$\varepsilon \mid I$	0.001	57	3.2954	2	0.16796	0.0019731
$\sigma \mid \varepsilon$	0.001	57	0.15835	0.25189	2.2203	0.75771
$\varepsilon \mid \sigma$	0.001	57	0.16243	0.25504	4.8522	2.0148

Table 26: Free power-law fits on large-interval windows meeting the minimum of 40 observations even after omitting any one size. Every row estimates C, r in Cz^r , with $z = x, r = \alpha$ for small intervals and $z = y, r = \beta$ for large intervals. Coefficients include all powers of π ; all computed samples have $\ell = 6, \dots, 10$. The cutoff is an upper bound, m is the retained count, and the objective is Eq. (61). Only qualifying cutoffs are shown. These are a separate asymptotic-range check, not the adopted fits in Eq. (59).

Direction	Cutoff	m	C	r	η_c [%]	η_r [%]
00 ss	0.001	116	1.2264	0.99587	-0.59355	-0.41329
00 ss	7.500×10^{-4}	69	1.2068	0.99371	-2.1781	-0.62939
ss 00	0.001	116	1.2264	0.99587	-0.59482	-0.41346
ss 00	7.500×10^{-4}	69	1.2068	0.9937	-2.184	-0.63019
00 $3s, s$	0.001	116	70.278	4.9903	-6.4314	-0.19312
00 $3s, s$	7.500×10^{-4}	69	78.936	5.0066	5.096	0.13181
$3s, s$ 00	0.001	116	70.201	4.9902	-6.534	-0.19593
$3s, s$ 00	7.500×10^{-4}	69	78.905	5.0065	5.0553	0.13098
ss $3s, s$	0.001	116	3.2701	2.001	-0.60142	0.050917
ss $3s, s$	7.500×10^{-4}	69	3.376	2.0054	2.6196	0.26766
$3s, s$ ss	0.001	116	3.2688	2.001	-0.64071	0.048605
$3s, s$ ss	7.500×10^{-4}	69	3.3753	2.0053	2.5966	0.26651

E Method references, implementation map, and sanity checks

The main text presents the methods used to generate the relative-entropy data and the reasons for choosing them. Here we collect the independent constructions and numerical evidence supporting that choice. This appendix separates the published ingredients from their assembly in this project. Equation numbers in the reading guide refer to the linked arXiv versions. The companion `METHOD_REFERENCES.md` gives the same reading route, source paths, and executable commands. The full reproducibility archive contains the Python implementation; the Overleaf archive contains the report and validation records.

E.1 Definitions of the state-quality diagnostics

To judge the quality of a normalized candidate $|\psi\rangle$, we compute its mean energy $E_\psi = \langle\psi|H|\psi\rangle$ and eigenstate residual

$$\eta_\psi = \|(H - E_\psi)|\psi\rangle\|_2 = \sqrt{\langle\psi|H^2|\psi\rangle - E_\psi^2}. \quad (99)$$

The residual vanishes for an exact eigenstate. We also check symmetry and energy ordering, because a small residual alone does not identify the desired primary. The vector norm is $\|v\|_2 = (\sum_i |v_i|^2)^{1/2}$.

E.2 Compact-boson wavefunction and Haldane–Shastry checks

We evaluate Eq. (11) independently of the logarithmic correlator implementation and compare normalized vectors. For each fixed N , charge pair (a, b) , and endpoint W , let ψ be the normalized vector obtained by directly evaluating the Laughlin/Jastrow polynomial in Eq. (11) (`polynomial_state` in the code). Let ϕ be the normalized conformal-block vector obtained by exponentiating the logarithmic amplitudes in Eq. (7) (`vertex_state`). Both vectors represent the same chosen primary, with identical coordinates, endpoint and occupation basis. Define their phase-aligned amplitude error

$$e_{\text{amp}}(\psi, \phi) = \max_O \left| \psi(O) e^{-i \arg \langle \phi | \psi \rangle} - \phi(O) \right|. \quad (100)$$

The argument $\arg$ is the phase of the complex overlap. This comparison checks that two implementations produce the same wavefunction, including its phases and normalization. To test an independent spectral property of that wavefunction, we also examine its eigenstate residual.

Why use the Haldane–Shastry Hamiltonian? The choice is specific to the exponent-two Jastrow states on the uniform circle used here. Their half-filled vacuum is exactly the ground state of the Haldane–Shastry (HS) spin chain, whose continuum description is the compact boson at the $SU(2)_1$ radius. The conformal-block vacuum and its relation to HS are given in Ref. [6], Sec. II A, Eq. (10), and Sec. III A, Eq. (37). This makes HS a natural additional benchmark for our three explicit lattice states. In our energy convention, the Hamiltonian is

$$H_{\text{HS}} = \left(\frac{\pi}{N} \right)^2 \sum_{j < k} \frac{\mathbf{S}_j \cdot \mathbf{S}_k}{\sin^2[\pi(j - k)/N]}, \quad \mathbf{S}_j = (X_j, Y_j, Z_j)/2. \quad (101)$$

Here X, Y, Z are Pauli matrices, so each site is a spin one-half. For each normalized state we compute

$$E_{\text{HS}} = \langle\psi|H_{\text{HS}}|\psi\rangle, \quad \eta_{\text{HS}} = \|(H_{\text{HS}} - E_{\text{HS}})|\psi\rangle\|_2. \quad (102)$$

This is Eq. (99) with the Hamiltonian specified. A small η_{HS} tests stationarity under HS; it does not by itself identify the CFT primary or establish its conformal weights. For the three states at $N = 8, 10, 12$, the largest residual at strict infinity is 6.21×10^{-15} . The excited-state residuals at $W = 10^8$ range from 4.60×10^{-9} to 6.30×10^{-9} , showing the small departure from an HS eigenstate introduced by retaining the original finite endpoint.

All three boson vectors are constructed directly from the conformal-block or polynomial amplitudes before applying this test. The HS Hamiltonian is used only as a diagnostic: it neither generates these vectors nor enters the relative entropy, which is calculated from their reduced density matrices. Its use here is not a requirement of the continuum relative-entropy theory and does not extend automatically to other compactification radii or lattice geometries.

N	State	$e_{\text{amp}}(\infty)$	$e_{\text{amp}}(10^8)$	$\eta(\infty)$	$\eta(10^8)$
8	00	9.30×10^{-16}	9.30×10^{-16}	1.04×10^{-15}	1.04×10^{-15}
8	ss	1.04×10^{-15}	8.50×10^{-16}	1.43×10^{-15}	5.97×10^{-9}
8	$3s, s$	4.38×10^{-16}	5.00×10^{-16}	8.02×10^{-16}	6.30×10^{-9}
10	00	2.86×10^{-15}	2.86×10^{-15}	1.58×10^{-15}	1.58×10^{-15}
10	ss	1.18×10^{-15}	1.14×10^{-15}	1.32×10^{-15}	5.22×10^{-9}
10	$3s, s$	1.46×10^{-15}	2.01×10^{-15}	9.82×10^{-16}	5.35×10^{-9}
12	00	2.89×10^{-15}	2.89×10^{-15}	6.21×10^{-15}	6.21×10^{-15}
12	ss	1.70×10^{-15}	2.24×10^{-15}	5.58×10^{-15}	4.60×10^{-9}
12	$3s, s$	1.88×10^{-15}	1.83×10^{-15}	5.11×10^{-15}	4.66×10^{-9}

Table 27: Direct polynomial versus logarithmic conformal-block evaluation for the three canonical boson states. Here ψ is the normalized direct Laughlin/Jastrow polynomial vector of Eq. (11), and ϕ is the normalized conformal-block vector obtained by exponentiating Eq. (7). Each comparison uses the same N , primary and endpoint W for both vectors. The two amplitude-error columns give $e_{\text{amp}}(\psi, \phi)$ from Eq. (100); the two residual columns apply Eqs. (99) and (101) to ψ . All vectors are normalized, $z_j = e^{2\pi i j/N}$, and the endpoint is either strict infinity or 10^8 . The finite-endpoint residuals quantify the perturbation retained to match the original Laughlin implementation. No interval length or mixing probability enters these whole-state tests.

E.3 Independent Ising constructions used for validation

The production reduced states use the exact fermion covariance method of Sec. 1.4. The full-vector constructions below provide independent checks of that input: conformal blocks for I, ε , and conformal-block-seeded Lanczos projection for σ .

The chiral Ising fields are $1, \chi, \sigma$ with weights $0, 1/2, 1/16$; χ is a Majorana fermion. Their fusion rules are

$$\sigma \times \sigma = 1 + \chi, \quad \chi \times \chi = 1, \quad \chi \times \sigma = \sigma. \quad (103)$$

E.3.1 Vacuum block: fusion data become spin amplitudes

For the vacuum, the auxiliary correlator is $\langle 0 | \sigma(z_1) \cdots \sigma(z_{2N}) | 0 \rangle_{\mathbf{p}}$. The subscript $\mathbf{p}$ chooses its intermediate fusion channels.

1. Use $2N$ spin fields on the circle, $z_r = \exp[i\theta_r]$ with $\theta_r = \pi r/N$, $r = 1, \dots, 2N$. Pair (z_{2j-1}, z_{2j}) . Each pair fuses to identity ($m_j = 0$) or fermion ($m_j = 1$), defining a physical spin with $Z_j = (-1)^{m_j}$.
2. Label the successive intermediate channels by bits $\mathbf{p} = (p_1, \dots, p_{N-1})$, setting $p_0 = p_N = 0$. The physical bits are $m_j = p_{j-1} \text{ xor } p_j$. Thus $\sum_j m_j$ is even. The construction spans the even sector of the physical spin parity $P = \prod_j Z_j$.
3. Enumerate binary strings $\mathbf{q} = (q_1, \dots, q_{N-1})$ and define $s_1 = 1$, $s_j = \prod_{k < j} (1 - 2q_k)$. Split the field coordinates into two groups $l_{\mathbf{q}}(j) = 2j - (1 + s_j)/2$ and $l'_{\mathbf{q}}(j) = 2j - (1 - s_j)/2$. Each group contains one field from each pair. Calculate the positive weight

$$A_{\mathbf{q}} = \prod_{j > k} \left[\sin \frac{\theta_{l_{\mathbf{q}}(j)} - \theta_{l_{\mathbf{q}}(k)}}{2} \sin \frac{\theta_{l'_{\mathbf{q}}(j)} - \theta_{l'_{\mathbf{q}}(k)}}{2} \right]^{1/2}. \quad (104)$$

4. Define $B_{\mathbf{p}} = \sum_{\mathbf{q}} (-1)^{\mathbf{p} \cdot \mathbf{q}} A_{\mathbf{q}}$, where the dot product is a sum of products of bits. The vacuum amplitudes are $\psi_I(\mathbf{m}(\mathbf{p})) \propto \sqrt{B_{\mathbf{p}}}$. The binary sign sum is a Walsh–Hadamard transform, evaluated with $O(N2^{N-1})$ butterfly operations instead of evaluating every $(\mathbf{p}, \mathbf{q})$ pair separately. The weights themselves cost $O(N^2 2^{N-1})$ operations.
5. Normalize by $\mathcal{N}_I = (\sum_{\mathbf{p}} B_{\mathbf{p}})^{1/2}$, so that $\psi_I(\mathbf{m}(\mathbf{p})) = \sqrt{B_{\mathbf{p}}}/\mathcal{N}_I$. Set all odd-parity amplitudes to zero. The output is a normalized 2^N -component vector.

The homogeneous choice sets the pair-displacement parameter $\delta = 0$ in `fusion_blocks`. Ref. [9] proves that this block vacuum is the finite-chain critical ground state. We additionally check its energy, symmetry sector, and residual numerically.

E.3.2 Energy primary: insert two Majorana endpoint fields

1. Keep the same $2N$ spin fields, coordinates, and physical basis. Replace the auxiliary vacuum states by Majorana states:

$$\Psi_{\varepsilon}(\mathbf{p}) \propto \langle \chi | \sigma(z_1) \cdots \sigma(z_{2N}) | \chi \rangle_{\mathbf{p}}. \quad (105)$$

Since the chiral Majorana has weight $1/2$, the insertion at infinity means $\chi(\infty) = \lim_{W \rightarrow \infty} W \chi(W)$. The two endpoints preserve even physical parity.

2. Reuse $A_{\mathbf{q}}$ and $B_{\mathbf{p}}$ from the vacuum construction. The endpoint correlator introduces the real factor $C_{\mathbf{q}} = \cos[(\pi/2N) \sum_j s_j]$. Compute a second sign transform $U_{\mathbf{p}} = \sum_{\mathbf{q}} (-1)^{\mathbf{p} \cdot \mathbf{q}} A_{\mathbf{q}} C_{\mathbf{q}}$. The explicit result is

$$\psi_{\varepsilon}(\mathbf{m}(\mathbf{p})) \propto \frac{\sum_{\mathbf{q}} (-1)^{\mathbf{p} \cdot \mathbf{q}} A_{\mathbf{q}} \cos \left(\frac{\pi}{2N} \sum_{j=1}^N s_j \right)}{\sqrt{B_{\mathbf{p}}}}. \quad (106)$$

3. Normalize by $\mathcal{N}_\varepsilon = (\sum_{\mathbf{p}} |U_{\mathbf{p}}|^2 / B_{\mathbf{p}})^{1/2}$ and embed $\psi_\varepsilon(\mathbf{m}(\mathbf{p})) = U_{\mathbf{p}} / (\sqrt{B_{\mathbf{p}}} \mathcal{N}_\varepsilon)$ in the same spin basis.
4. Check its orthogonality to I , its even parity, zero momentum, energy, and eigenstate residual. The identification with the nonchiral primary ε uses these spectral and symmetry properties; the auxiliary chiral endpoint fields alone do not establish that identification.

Equation (106) is the published expression in Ref. [9], Eqs. (73)–(75), specialized to the uniform circle. Its equality with the desired excited eigenstate is numerically verified here through $N = 24$; we do not replace that numerical evidence by an unprovided general proof for arbitrary insertions.

E.3.3 Spin primary: conformal-block-seeded Lanczos

An even-parity block cannot represent the odd-parity σ state. Ref. [9], Sec. X, reports that the naive endpoint-spin and single-fermion candidates do not directly yield the desired homogeneous odd-sector ground state. To check the production covariance state independently at small sizes, we construct a full-vector representative of σ by a Hamiltonian projection initialized from the block vacuum. This seed and projection are our combination of the published block vacuum and a standard eigensolver, not a formula claimed in that paper. Here “block-seeded” means seeded by a conformal block; the solver is single-vector Lanczos, not a multi-vector block-Lanczos algorithm. The steps below specify the numerical construction, including the part performed by the Lanczos solver [17, 13].

1. **Prepare a seed with the required symmetry.** Starting from normalized $|I_{\text{CB}}\rangle$, set

$$|v_\sigma\rangle = \sum_{j=0}^{N-1} X_j |I_{\text{CB}}\rangle, \quad |q_1\rangle = \frac{|v_\sigma\rangle}{\|v_\sigma\|_2}. \quad (107)$$

Since $PX_j = -X_jP$, the seed is odd. The sum is invariant under translation, so the seed has zero momentum. It generally contains higher-energy odd states as well as σ ; applying the summed spin operator is not itself the final projection.

2. **Build the odd-sector Hamiltonian action.** Retain only bit strings $\mathbf{m}$ with $\sum_j m_j$ odd, leaving 2^{N-1} basis vectors. Let H_- denote H restricted to this basis. For a coefficient vector f , its action is

$$(H_- f)_\mathbf{m} = - \left[\sum_j (-1)^{m_j} \right] f_\mathbf{m} - \sum_j f_{\mathbf{m} \oplus \mathbf{e}_j \oplus \mathbf{e}_{j+1}}, \quad (108)$$

where $\mathbf{e}_j$ has one bit at site j , $\oplus$ is bitwise exclusive-or, and site indices are periodic. The first term is diagonal; the second flips neighboring bits. No full 2^{N-1} by 2^{N-1} matrix is stored. Only parity is explicitly imposed in this basis; translation symmetry is inherited from the seed and preserved by the Hamiltonian.

3. **Generate a Krylov space.** After r basis vectors, the search space is

$$\mathcal{K}_r = \text{span}\{q_1, H_- q_1, \dots, H_-^{r-1} q_1\}. \quad (109)$$

Lanczos constructs orthonormal vectors q_j . In exact arithmetic, with $q_0 = 0$ and $b_0 = 0$, the recurrence is

$$\begin{aligned} a_j &= \langle q_j | H_- | q_j \rangle, \\ |w_j\rangle &= H_- | q_j \rangle - a_j | q_j \rangle - b_{j-1} | q_{j-1} \rangle, \\ b_j &= \|w_j\|_2, \quad |q_{j+1}\rangle = |w_j\rangle / b_j. \end{aligned} \quad (110)$$

If $b_j = 0$, the generated subspace is already invariant. In floating-point arithmetic the solver also controls loss of orthogonality. The symbols a_j, b_j here are Lanczos coefficients, unrelated to the boson endpoint labels or the leading fitting exponent α .

4. **Select the lowest-energy combination.** Form $(T_r)_{ij} = \langle q_i | H_- | q_j \rangle$. This small matrix is tridiagonal, with diagonal a_j and neighboring entries b_j . Let $\theta_{\min}$ and $\mathbf{y}$ be its smallest eigenvalue and normalized eigenvector. The Rayleigh–Ritz approximation is

$$|\sigma_N^{(r)}\rangle = \sum_{j=1}^r y_j |q_j\rangle. \quad (111)$$

It minimizes the energy within the current search space. Increasing the space, or restarting while retaining the useful spectral information, refines the approximation.

5. **Converge and measure the actual residual.** The code calls `eigsh` with `k=1`, `which="SA"`, the seed as `v0`, and `tol=2e-12`. It uses implicitly restarted Lanczos [13]. The requested solver tolerance is not an absolute bound on η : after convergence we independently evaluate Eq. (99) using the original Hamiltonian.
6. **Identify and validate the primary.** Normalize the vector, restore zero entries on even bit strings, and check $P = -1$, zero momentum, orthogonality to I, ε , and the energy $E_\sigma = -2 \cot[\pi/(2N)]$. The vanishing large- N gap has scaling dimension $1/8$, identifying the $(1/16, 1/16)$ primary. Our explicit construction checks use $e_P = \|(P+1)\sigma_N\|_2$ and $e_T = \|(T-1)\sigma_N\|_2$ for the two symmetry errors.

The spectral projection underlying these steps can also be written as

$$|\sigma_N\rangle = \lim_{\tau \rightarrow \infty} \frac{e^{-\tau(H-E_*)}|q_1\rangle}{\|e^{-\tau(H-E_*)}|q_1\rangle\|_2}. \quad (112)$$

Here τ is imaginary time and E_* is an arbitrary reference energy. The normalized seed has an expansion into odd-sector eigenstates; provided its σ coefficient is nonzero, higher energies are suppressed relative to the lowest one. Lanczos implements the search without explicitly exponentiating H . The measured squared seed overlap $F_{\text{seed}} = |\langle \sigma_N | q_1 \rangle|^2$ is approximately 0.986, 0.983, 0.981, 0.979, 0.977 at $N = 6, 8, 10, 12, 16$. At $N = 16$, the recorded projected-state residual is 2.70×10^{-12} . The complete size-by-size residual and independent-state checks appear in Appendix E.5; the largest recorded residual is 1.99×10^{-11} at $N = 12$. Thus the solver tolerance must not be quoted as the achieved absolute residual.

This method obtains the required lattice primary using both a conformal block and the Hamiltonian. A pure endpoint conformal-block formula for the periodic σ state remains unestablished here.

E.4 Comparisons with the original lattice implementations

We define squared fidelity $F = |\langle \psi | \phi \rangle|^2$ for normalized vectors, infidelity $1 - F$, and matrix error $\|M\|_F = (\text{Tr } M^\dagger M)^{1/2}$. The most useful energy ratio uses *excitation gaps*, since ratios of extensive total energies are insensitive to the excitation:

$$E_I = -2 \csc \frac{\pi}{2N}, \quad E_\sigma = -2 \cot \frac{\pi}{2N}, \quad E_\varepsilon = E_I + 8 \sin \frac{\pi}{2N}, \quad R_E = \frac{E_\varepsilon - E_I}{E_\sigma - E_I} = 8 \cos^2 \frac{\pi}{4N}. \quad (113)$$

The continuum limit is $R_E \rightarrow 8$, from dimensions 1 and $1/8$. With the velocity $v = 2$ of Eq. (15), the relation $E_a - E_I \sim (2\pi v/N)(h_a + \bar{h}_a)$ identifies the ε and σ dimensions as 1 and $1/8$, respectively.

For the Laughlin comparison, the new code places site zero in the least significant bit, whereas the original dense constructor uses the most significant bit. We reverse bit order before comparing the vectors; the comparison then tests branches, phases and normalization in the same physical basis. Fresh Laughlin-vector comparisons at $N = 8, 10, 12$, for all three boson states and $W = 10^8$, give $|1 - F| \leq 1.8 \times 10^{-15}$ and maximum phase-aligned amplitude error 4.1×10^{-15} . For the Haldane–Shastry convention in Eq. (101),

N	D_{bond}	State	Infidelity	$ E_{\text{puMPS}} - E_{\text{CB}} $	η_{puMPS}	$\ \Delta\rho_4\ _F$
10	8	I	1.65×10^{-9}	1.95×10^{-8}	4.88×10^{-4}	1.91×10^{-6}
10	8	σ	3.97×10^{-8}	5.18×10^{-7}	0.0026512	6.23×10^{-6}
10	8	ε	1.64×10^{-7}	1.93×10^{-6}	0.0048855	2.46×10^{-5}
16	14	I	2.62×10^{-10}	2.81×10^{-9}	1.81×10^{-4}	2.33×10^{-6}
16	14	σ	6.34×10^{-8}	7.61×10^{-7}	0.0031009	2.89×10^{-6}
16	14	ε	3.99×10^{-7}	4.30×10^{-6}	0.007004	1.46×10^{-5}

Table 28: Fresh full-vector comparisons with archived puMPS checkpoints. D_{bond} is the MPS bond dimension, distinct from relative entropy. The seed is $1234 + N$ in the archived optimization. I and ε use pure CB vectors, while σ uses Eq. (112). All vectors are normalized, all energies use Eq. (15), and the interval matrix comparison has $\ell = 4$. Every reported infidelity is below 4.0×10^{-7} . No puMPS reoptimization is claimed.

the measured gap ratios $(E_{3s,s} - E_{00})/(E_{ss} - E_{00})$ are 4, 4.2, 13/3 at $N = 8, 10, 12$, approaching the CFT value five. At finite W , excited-state eigen residuals are below 6.3×10^{-9} .

Ising blocks were tested against independent exact diagonalization through $N = 12$, and against exact-fermion interval matrices for every requested $N \leq 24$. The full I and ε CB vectors were evaluated even at $N = 24$. Exact-fermion production for σ is cross-checked against the projected vectors through $N = 16$. Numerical CSVs retain every test value. For the covariance calculation, the code also enforces $\max_{r,s} |(G^2 + \mathbb{I})_{rs}| < 2 \times 10^{-12}$. The baseline 150 boson values agree with the original 150 values within 1.16×10^{-11} . Ising differences from the available puMPS grid are at most 4.15×10^{-7} across $\ell = 2, \dots, 6$. Historical $N = 22$ puMPS states were excluded because their mutual overlaps failed a quality criterion. Our $N = 22$ data are newly generated exact-chain data, explicitly marked as such; we have not repaired or silently reinstated those puMPS checkpoints.

E.4.1 Numerical evidence for the Ising reference

We choose the exact finite-chain Ising states as the primary input to the relative-entropy study because their preparation error is independently controlled. The conformal-block vectors for I, ε and the projected vector for σ verify the state identification; the exact fermion covariances provide the interval matrices used in the fits. The decision is supported by the following comparison with the original periodic uniform matrix product states (puMPS).

State	η_{puMPS}	η_{new}	$1 - F$	New vector
I	1.81×10^{-4}	1.24×10^{-12}	2.62×10^{-10}	Vacuum block
σ	3.10×10^{-3}	2.70×10^{-12}	6.34×10^{-8}	Block-seeded Lanczos
ε	7.00×10^{-3}	3.35×10^{-12}	3.99×10^{-7}	Endpoint block

Table 29: Accuracy comparison at $N = 16$ against the archived puMPS checkpoint of bond dimension $D_{\text{bond}} = 14$, optimization seed 1250. All vectors are normalized and use Eq. (15) with both couplings equal to one and periodic spin boundaries. The residual η is defined by Eq. (99); $1 - F = 1 - |\langle \psi_{\text{new}} | \psi_{\text{puMPS}} \rangle|^2$ is the infidelity. The new σ vector uses block-seeded Lanczos, with requested solver tolerance 2×10^{-12} , rather than a pure conformal-block formula. Entries are from the original whole-vector validation CSVs; no puMPS reoptimization is performed.

- The tested finite-chain eigenstate errors are smaller.** Table 29 shows residuals near 10^{-12} for the new vectors, compared with 10^{-4} – 10^{-2} for the archived puMPS vectors. Their energy ordering and symmetries also match the exact solution, so the small residuals do not merely identify unrelated eigenstates. The full $N = 10, 16$ overlap and energy comparisons are in Table 28.
- The reduced states have an independent reference.** For I, ε , direct partial traces of the conformal-block vectors agree with the exact fermion interval matrices to a maximum Frobenius

error of 1.55×10^{-13} over the original sizes $N = 16, 18, 20, 22, 24$ and $\ell = 2, \dots, 6$. For σ , the projected vectors were independently compared with the exact fermion reduced states through $N = 16$.

3. **The same construction covers the full requested size grid.** It supplies exact-chain data at $N = 22$, where the archived puMPS checkpoint did not pass the earlier overlap-quality criterion. The new calculation has its own provenance and does not rehabilitate that checkpoint.

The scope of this preference is finite-chain accuracy for this integrable Hamiltonian. The archived puMPS vectors are already close in overlap, with infidelity below 4×10^{-7} in all tested comparisons. Larger bond dimensions or improved optimization could reduce their remaining errors; the tests do not establish that conformal blocks are universally superior to converged puMPS. The σ accuracy relies on Hamiltonian projection and the exact fermion solution. Finally, exact lattice eigenstates still have finite-size and short-interval lattice corrections. Improving their preparation therefore does not, by itself, remove the discrepancies with continuum coefficients at $\ell = 2, 3, 4$.

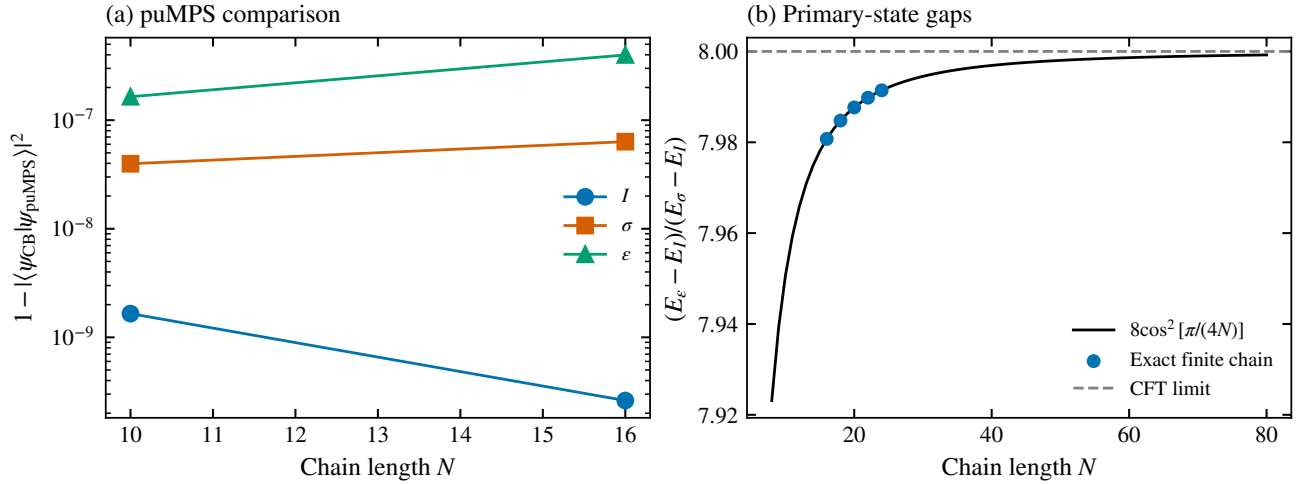


Figure 36: State validation at critical coupling and periodic spin boundaries. (a) Squared-overlap infidelities against archived puMPS at $(N, D_{\text{bond}}) = (10, 8), (16, 14)$, seeds 1244, 1250; σ uses the CB-seeded projection with solver tolerance 2×10^{-12} . (b) Exact finite-chain gap ratios at $N = 16, 18, 20, 22, 24$ and Eq. (113); the dashed line is the continuum value eight. No interval length or mixture probability enters these whole-state tests.

E.5 What was checked for the projected spin state

For normalized vectors, define the energy error, eigenstate residual, and squared overlap by

$$\delta E = |\langle H \rangle - E_{\sigma}^{\text{exact}}|, \quad \eta = \|(H - \langle H \rangle)|\sigma_N\rangle\|_2, \quad F = |\langle \sigma_{\text{ref}} | \sigma_N \rangle|^2. \quad (114)$$

The residual equals the square root of the energy variance. It tests closeness to an eigenstate, but does not identify which eigenstate; parity, translation and the exact lowest odd-sector energy supply that identification. The interval comparison is

$$e_{\rho} = \max_{1 \leq \ell \leq \min(5, N/2)} \left\| \rho_{\sigma, \ell}^{\text{Lanczos}} - \rho_{\sigma, \ell}^{\text{covariance}} \right\|_F, \quad \|B\|_F = \sqrt{\sum_{u,v} |B_{uv}|^2}. \quad (115)$$

The following are the archived results in `ising_cb_validation.csv`, rounded only for display. That file evaluates the exact reference energy through the covariance; using the equivalent trigonometric formula can change the last roundoff digits of δE .

N	F_{seed}	η	δE	e_{ρ}	$ 1 - F $
6	.985928	$1.54 \cdot 10^{-15}$	$1.78 \cdot 10^{-15}$	$1.82 \cdot 10^{-15}$	$1.33 \cdot 10^{-15}$
8	.982894	$6.11 \cdot 10^{-15}$	$5.33 \cdot 10^{-15}$	$1.61 \cdot 10^{-15}$	$2.22 \cdot 10^{-15}$
10	.980864	$2.22 \cdot 10^{-13}$	0	$2.76 \cdot 10^{-15}$	0
12	.979404	$1.99 \cdot 10^{-11}$	$2.49 \cdot 10^{-14}$	$2.47 \cdot 10^{-13}$	$1.78 \cdot 10^{-15}$
16	.977431	$2.70 \cdot 10^{-12}$	$4.26 \cdot 10^{-14}$	$4.02 \cdot 10^{-14}$	—

Here $F_{\text{seed}} = |\langle \sigma_N | q_1 \rangle|^2$ measures how much of the target was already in the seed. For $N \leq 12$, F compares against the lowest three eigenpairs from a separate full-spin-space Hamiltonian action with a random starting vector. That reference also uses `eigsh`; it is a different state construction, not an independent eigensolver implementation. No full-space overlap was recorded at $N = 16$, hence the dash. Values of F exceeding one by roundoff are reported as $|1 - F|$, not negative infidelities; zero entries are floating-point results, not proofs of exact arithmetic equality.

The separate `ising_construction_checks.csv` run tests $|\langle \sigma | \sigma \rangle - 1|$, $\|(P + 1)\sigma\|_2$, $\|(\mathbb{T} - 1)\sigma\|_2$, and $\max_{a=I, \epsilon} |\langle a | \sigma \rangle|$ on the same five sizes. Their maxima are 9.99×10^{-16} , 0, 7.20×10^{-13} , and 0, respectively. Parity and mutual orthogonality are enforced by disjoint basis sectors, so those zeros alone do not establish state accuracy.

For context, the archived puMPS σ residuals are 2.65×10^{-3} at $(N, D_{\text{bond}}) = (10, 8)$ and 3.10×10^{-3} at $(16, 14)$. Their squared overlaps with the reference are 0.9999999603 and 0.9999999366. These comparisons favor the exact reference for this integrable model and these supplied checkpoints. They do not bound the accuracy of puMPS after further optimization or at a larger bond dimension.

E.6 Large-chain contraction and entropy checks

Validation tests each link in the calculation, rather than only checking the final entropy. The 119 larger-interval records are in `large_interval_checks.csv`; the earlier 345 smaller-interval checks remain in `extension_checks.csv`.

Check	Count	Largest difference
Boson ten-site matrix versus full wavefunction enumeration, $N = 20$, all three primaries	3	3.22×10^{-15}
Boson 100 versus 130 working digits, $N = 20, 128$, $\ell = 10$	6	$< 2 \times 10^{-14}$
Boson partial traces of ten-site matrices versus direct Pfaffians, $N = 20, 128$, $\ell = 4, 5, 6, 7$	24	1.12×10^{-16}
Ising partial traces versus direct covariance reductions, $N = 32, 82, 128$, $\ell = 4, 5, 6, 7$	36	1.23×10^{-15}
Ising energies versus exact finite-chain formulas, $N = 32, 82, 128$	9	2.85×10^{-14}
Ising matrix entries between opposite parity sectors, $\ell = 10$	9	0
Symmetry-block versus full-matrix directed entropy, $\ell = 10$	2	1.96×10^{-14}
Entropy threshold 10^{-15} versus 10^{-16} , $\ell = 10$	30	3.37×10^{-13}

Table 30: Large-chain contraction and entropy checks. Counts give the number of comparisons in each test family; sizes, interval lengths and primary-state coverage are specified in the first column. Matrix errors are maximum absolute entrywise differences; energy and entropy errors are absolute scalar differences. The precision row reports a validation-tolerance bound on the final double-precision matrices.

Matrix differences in Table 30 mean the largest absolute entrywise difference $\max_{u,v} |\rho_{uv}^{(1)} - \rho_{uv}^{(2)}|$; energy and entropy differences are absolute scalar differences. Precision comparisons refer to the final complex double-precision matrices, and the displayed bound is their validation tolerance. The exact zeros refer to the implemented parity-block structure, not a general error bound.

The full-matrix entropy checks use boson $ss \mid 3s, s$ at $N = 20$ and Ising $\sigma \mid \varepsilon$ at $N = 82$. The threshold comparison covers all six directions at $N = 20, 128$ for the boson and $N = 32, 82, 128$ for Ising. No mixture is replaced by a covariance-only approximation. The generation stage independently checks trace, Hermiticity, and minimum eigenvalue for all 294 state–interval combinations in the requested grid; these records are in `large_interval_quality.csv`. The 180 observations shared with the preceding grid retain their original stored values.

The exact contractions remove sampling and variational errors from the present finite-lattice calculation. They do not remove the physical dependence on N , ℓ , or the fixed finite endpoint $W = 10^8$. The comparison with CFT in Sec. 3 keeps these physical limitations distinct from numerical contraction accuracy.

Table 30 lists all eight families of the 119 checks, including their sizes, norms, and errors. In particular, the $N = 20$, $\ell = 10$ boson comparison tests every matrix element of all three states against full wavefunction enumeration, with maximum error 3.22×10^{-15} . This tests off-diagonal phases as well as normalization. At $N = 128$, where full enumeration is impractical, increased working precision and agreement between direct and cached contractions test numerical stability. They share the same algebraic formula and therefore do not constitute an independent large- N wavefunction proof. The algebra is derived in Sec. 3.4 and independently tested at the sizes where enumeration is possible.

For Ising, finite-chain energies, covariance purity, zero-mode parity, and agreement with smaller full vectors test distinct parts of the construction. A partial trace of a ten-site matrix is also compared with reconstruction from the directly restricted covariance. Entropy checks then compare full-matrix and symmetry-block calculations, and vary the eigenvalue threshold. Trace, Hermiticity and positivity are necessary checks; agreement with an independently constructed state is the stronger check. These tests certify the finite-lattice calculation at the stated precision, not the absence of continuum cutoff effects or unknown small-interval corrections.

E.6.1 Endpoint checks for the extension through $N = 256$

Before generating the new sizes, we tested all three states of each CFT at $N = 256$ and $\ell = 10$. The 70 records in `larger_size_checks.csv` test the same contraction in the new size range. In addition to the

matrix and entropy differences defined in Table 30, define

$$e_G = \max_{r,s} |(G^2 + \mathbb{I})_{rs}|, \quad e_P = |(-1)^N \text{Pf}(G) - P|, \quad e_+ = \max(0, -\lambda_{\min}(\rho_A)). \quad (116)$$

Here e_G tests purity of the full-chain Ising covariance, e_P tests its parity, and e_+ measures a negative eigenvalue caused by numerical error. The remaining matrix-quality errors are $|\text{Tr } \rho_A - 1|$ and $\max_{u,v} |(\rho_A - \rho_A^\dagger)_{uv}|$.

Check at $N = 256$	Count	Largest error
Ising exact finite-chain energies	3	5.69×10^{-14}
Ising full covariance purity e_G	3	3.11×10^{-15}
Ising full-state parity e_P	3	5.96×10^{-14}
Ising partial traces versus direct covariance, $\ell = 4, 6$	6	5.42×10^{-16}
Boson 100 versus 130 digits, $\ell = 10$	3	$< 2 \times 10^{-14}$
Boson partial traces versus direct Pfaffians, $\ell = 4, 6$	6	1.39×10^{-16}
Trace, Hermiticity and e_+ , both CFTs, $\ell = 10$	18	2.19×10^{-16}
Entropy cutoff 10^{-15} versus 10^{-16} , $\ell = 4, 10$	24	7.93×10^{-14}
Symmetry-block versus full-matrix entropy, $\ell = 4, 10$	4	1.67×10^{-14}

Table 31: All 70 endpoint checks for the extension to $N = 256$. State checks cover all three primaries per model; entropy-threshold checks cover all six directions. Full-matrix comparisons use Ising $\sigma | \varepsilon$ and boson $ss | 3s, s$. Errors use Eq. (116) and the entrywise/scalar conventions of Table 30. The precision row gives the validation tolerance for final complex double-precision matrices, not a claim of 100-digit precision in the returned matrix.

All checks passed. The additional production run records trace, Hermiticity and minimum eigenvalue for all $8 \times 3 \times 7 \times 2 = 336$ new state/interval combinations in `larger_size_quality.csv`. It appends 672 directed relative entropies while preserving all 1200 earlier observations. The high-precision comparison shares the same algebra and tests numerical stability; it is not an independent full-wavefunction check at $N = 256$.

E.7 Reading route and attribution

- Ising block states and the spin-primary limitation.** Read Montes et al. [9], Sec. VIII, Eqs. (55)–(62), for the homogeneous even-sector block and its fermion ground state; Appendix C, Eqs. (133)–(137), supplies spin/fermion conventions. Sec. X, Eqs. (73)–(75), constructs the energy-primary endpoint block. The discussion after Eq. (79) explains the difficulty with proposed odd-sector endpoint states. Our summed-spin seed and odd-sector projection, Eqs. (107)–(112), are a project adaptation. They are not a published closed block formula for σ .
- Lanczos projection.** The algorithmic reference is the ARPACK guide [17]; the direct implementation contract is `eigsh` [13]. The three-term recurrence and Rayleigh–Ritz selection are written out in Appendix E.3.3. Our call uses `k=1`, `which="SA"`, `v0=seed`, and `tol=2e-12`. It leaves `ncv` and `maxiter` at their defaults: for the tested odd-sector dimension $d = 2^{N-1}$, these are 20 and $10d$. No shift-invert transformation is requested. The API parameter `sigma` denotes a spectral shift and is unrelated to our primary-state name. The recorded environment uses SciPy 1.17.1; the linked 1.17-series API describes these parameters.
- Large Ising chains.** Read Peschel and Eisler [15], Sec. 3.3(II), Eqs. (14)–(18), including the following Majorana paragraph. It explains why a subsystem of a quadratic-fermion eigenstate is determined by restricted correlations. Our explicit parity and zero-mode selection precedes that reduction. Match the Majorana basis before matching signs: we use $\langle a_r a_s \rangle = \delta_{rs} - iG_{rs}$, whereas the review writes $\delta_{rs} + i\Gamma_{rs}$ and chooses a different sign for one Majorana per site. After aligning the basis, $G = -\Gamma$. The Gaussian statement applies to each Ising primary; the mixture in Eq. (29) is evaluated as a full matrix.

4. **Large compact-boson chains.** Read Stéphan and Pollmann [14], Appendix A.1, Eqs. (A5)–(A11), for the confluent Vandermonde determinant and discrete Pfaffian normalization. Appendix A.2, Eqs. (A14)–(A15), introduces site weights for diagonal observables. Their site count L and particle count N are our N and M . Their diagonal calculation can discard amplitude phases. Our off-diagonal interval matrix cannot: Eq. (42) retains the bra and ket factors explicitly. The finite- W extension, bordered contraction, tridiagonal solve, and reuse of repeated minors are developed in Sec. 3.4, and are checked against explicit vertex-state sums.
5. **Pfaffian signs and numerical evaluation.** Wimmer [16], Eqs. (4) and (5), gives the first-row expansion and congruence identity. These justify the cached recursion and block elimination used here. Sec. II A, Eqs. (15)–(18), explains pivoted antisymmetric elimination, the basis of our scalar Pfaffian helpers. Our code implements those elementary updates using NumPy or mpmath; it does not call the paper’s software package. Caching the minors and sharing multiplicity factors are implementation choices in this project.

E.8 From a formula to its source function

All Python paths in this table are relative to `src/conformal_blocks/`; the method functions have source references in their docstrings. A reduced density matrix is abbreviated RDM in function and data-file names.

Operation	File and functions
Block vacuum and projected σ	<code>ising.py</code> : <code>fusion_blocks</code> , <code>sigma_from_cb</code> , <code>hamiltonian_action</code> , <code>energy_diagnostics</code>
Quadratic sectors, zero modes and Eq. (33)	<code>ising.py</code> : <code>majorana_covariance</code> , <code>pfaffian</code> , <code>gaussian_rdm</code> ; direct comparison: <code>dense_rdm</code>
Explicit compact-boson amplitudes Eqs. (39) and (42)	<code>boson.py</code> : <code>vertex_state</code> , <code>sector_rdm</code> <code>boson_pfaffian.py</code> : <code>jastrow_rdm</code> , <code>_tridiagonal_solve</code> , <code>pfaffian_mp</code> , <code>_cached_density</code>
Shorter intervals and exact mixture entropy	<code>boson_pfaffian.py</code> : <code>block_rdms</code> ; <code>symmetry_entropy.py</code> : <code>sectors</code> , <code>directed_entropies</code>
Validation and current-grid generation	<code>method_sanity.py</code> , <code>extension_checks.py</code> , <code>large_interval_checks.py</code> , <code>large_intervals.py</code>

E.9 A short reproduction run

From the root of the full reproducibility project, using a Python environment containing NumPy, SciPy and mpmath, run

```
export PYTHONPATH=src
export OPENBLAS_NUM_THREADS=1
python -m conformal_blocks.method_sanity \
    --output method_sanity.json
```

This script constructs σ at $N = 12$, checks its energy, residual, symmetries and four-site reduction, and tests all three primaries at $N = 12, 128$ for both CFTs with $\ell = 4$. The boson checks compare the cached 100-digit calculation with direct 130-digit Pfaffians, and at $N = 12$ with explicit vertex amplitudes. It also compares both entropy implementations. Every check writes its error, tolerance, parameters and pass flag. The executed result is archived as `data/method_sanity.json`; all 55 checks passed. This compact reproduction supplements the archived 345 earlier checks and 119 ten-site checks, without replacing them or rerunning any fits.

F Relative entropy between primaries

This appendix retains the historical $N \leq 256$ sanity-check data, with fit and display cutoffs 0.08 and 0.20, respectively. Its smallest fraction is $4/256 = 0.015625$, so it has no observations in the current mixture display window $x \leq 0.01$. Its figures and fits remain separate from the active mixture analysis in Secs. 4 and 5.

This appendix tests the primary interval matrices against a second observable whose leading coefficient is known analytically. For the same normalized lattice states used in the mixture study, define

$$S_{a\|b}(N, \ell) = S(\rho_A^a \| \rho_A^b) = \text{Tr } \rho_A^a \log \rho_A^a - \text{Tr } \rho_A^a \log \rho_A^b, \quad x = \ell/N. \quad (117)$$

Here A is the interval of ℓ sites, and the double bar separates the first and second density matrices. Both arguments are reductions of individual primaries; no mixing probability enters this observable. The global primary vectors may be orthogonal, but the comparison in Eq. (117) concerns their local reductions. Every pair in the main report is tested in both directions. The mixture data, fitted coefficients, and fitting figures are unchanged.

F.1 Leading predictions from the supplied literature

The sources are the two Sárosi–Ugajin papers supplied as `Theory_paper/1603.03057v2.pdf` and `Theory_paper/1611.02959v1.pdf`. Equations (50) and (53) of the first paper [11] give the leading operator-product-expansion result; its Eq. (61) gives the stress tensor contribution, and Eq. (63) treats free-boson vertex operators. Equation (1.3) and Secs. 3.1–3.3 of the second paper [12] give the corresponding scalar, stress tensor, and current results. These equation and subsection numbers belong to the cited source versions. The formulas below specialize those published results to the states of this report; no unknown subleading coefficient is derived.

Let O_j be an orthonormal set of the lightest local operators whose diagonal matrix elements differ in states a and b . Their common scaling dimension is Δ , and $C_{O_j a^\dagger a} = \langle a | O_j | a \rangle$ denotes the normalized plane three-point coefficient. The index j labels operators and is unrelated to the leading exponent α . The published result is

$$S_{a\|b}(x) = K(\Delta) \sum_j (C_{O_j a^\dagger a} - C_{O_j b^\dagger b})^2 (\pi x)^{2\Delta} + o(x^{2\Delta}), \quad K(\Delta) = \frac{\Gamma(3/2)\Gamma(\Delta+1)}{2\Gamma(\Delta+3/2)}. \quad (118)$$

The symbol Γ denotes Euler’s gamma function; $o(x^r)$ means terms whose ratio to x^r vanishes as $x \rightarrow 0$ in the continuum theory. Thus $\alpha = 2\Delta$, and $K(1) = 1/3$. This leading term is symmetric under $a \leftrightarrow b$, although the full relative entropy need not be symmetric. For unequal conformal weights, an operator with $\Delta < 2$ still dominates the stress tensor, as explained immediately before Eq. (63) of Ref. [11].

F.1.1 Ising: identify the first nonzero diagonal difference

The conformal weights are $(h_I, \bar{h}_I) = (0, 0)$, $(h_\sigma, \bar{h}_\sigma) = (1/16, 1/16)$, and $(h_\varepsilon, \bar{h}_\varepsilon) = (1/2, 1/2)$, with central charge $c = 1/2$. The energy primary ε has total scaling dimension one; it is the nonchiral energy field in the periodic spin chain. The Ising fusion rules $\sigma \times \sigma = I + \varepsilon$ and $\varepsilon \times \varepsilon = I$ imply that only σ has a nonzero diagonal energy-field coefficient. Using the normalization in Sec. 5 of Ref. [11], the data are

$$C_{\varepsilon II} = 0, \quad C_{\varepsilon \sigma \sigma} = \frac{1}{2}, \quad C_{\varepsilon \varepsilon \varepsilon} = 0. \quad (119)$$

The spin field itself has zero diagonal coefficient in each of these parity eigenstates, so its smaller scaling dimension does not produce an $x^{1/4}$ term. Inserting the difference $1/2$ and $K(1) = 1/3$ into Eq. (118) gives

$$S_{I\|\sigma} = S_{\sigma\|I} = S_{\sigma\|\varepsilon} = S_{\varepsilon\|\sigma} = \frac{\pi^2}{12} x^2 + o(x^2) \quad \text{at leading order.} \quad (120)$$

The equalities here refer only to the displayed leading term.

For $I \leftrightarrow \varepsilon$, the energy-field diagonal difference vanishes. The stress tensors $T, \bar{T}$, of scaling dimension two, then give the first contribution. Equation (61) of Ref. [11] includes both chiralities for $h = \bar{h}$:

$$S_{a||b}(x) = \frac{16}{15c}(h_a - h_b)^2(\pi x)^4 + o(x^4). \quad (121)$$

With $c = 1/2$ and $|h_I - h_\varepsilon| = 1/2$, this becomes

$$S_{I||\varepsilon} = S_{\varepsilon||I} = \frac{8\pi^4}{15}x^4 + o(x^4) \quad \text{at leading order.} \quad (122)$$

F.1.2 Nonchiral compact boson: add the two current contributions

Write $\mathbf{Q}_a = (Q_{L,a}, Q_{R,a})$ for the left and right charges, with $h_a = Q_{L,a}^2/2$ and $\bar{h}_a = Q_{R,a}^2/2$. The charge convention of Sec. 1.3 is

$$\mathbf{Q}_{00} = (0, 0), \quad \mathbf{Q}_{ss} = (s, s), \quad \mathbf{Q}_{3s,s} = (3s, s), \quad s = 1/\sqrt{2}. \quad (123)$$

Equation (63) of Ref. [11] states the chiral vertex result $S = (Q_{L,a} - Q_{L,b})^2[1 - \pi x \cot(\pi x)]$. For the leading nonchiral result, use Eq. (118) with the two orthogonal, dimension-one currents $J_L = i\partial\phi_L$ and $J_R = i\bar{\partial}\phi_R$. Their normalized diagonal coefficients are the corresponding charges, up to the common sign convention discussed below that source equation. Their squared differences add; there is no mixed-current contribution because the two currents have zero two-point overlap. Therefore

$$S_{a||b}(x) = \frac{\kappa_{ab}\pi^2}{3}x^2 + o(x^2), \quad \kappa_{ab} = (Q_{L,a} - Q_{L,b})^2 + (Q_{R,a} - Q_{R,b})^2. \quad (124)$$

The three squared separations are 1, 5, and 2 for $00 \leftrightarrow ss$, $00 \leftrightarrow 3s, s$, and $ss \leftrightarrow 3s, s$, respectively. The corresponding coefficients are $\pi^2/3$, $5\pi^2/3$, and $2\pi^2/3$, with $\alpha = 2$ in every direction. This use of both currents also covers the nonzero conformal spin of $|3s, s\rangle$. It does not apply the spinless stress-tensor formula to that state.

All coefficients in Eqs. (120), (122), and (124) are four times the corresponding equal-mixture leading coefficients in Sec. 2. This agreement follows by comparing the published leading formulas. The numerical values below are computed directly from Eq. (117), rather than obtained by multiplying the mixture data by four.

F.2 Direct numerical procedure and checks

The calculation uses every geometry displayed in the main report: the 15 sizes in Eq. (26), $\ell = 6, \dots, 10$, and $x \leq 0.20$. This gives 70 observations for each of 12 directions, or 840 displayed direct primary relative entropies. The 51 points per direction with $x \leq 0.08$ define the analytical CFT comparison window. Section F.4 subsequently fits the same numerical data; those fits do not enter this leading-theory comparison.

1. Load the existing $\ell = 10$ primary density matrices and obtain shorter intervals by the partial trace of Sec. 3. Ising uses exact Jordan–Wigner states for all three primaries. The boson uses the same normalized finite- W vertex/Jastrow states with $q = 2$, unit-circle sites, $w_1 = 0$, $W = 10^8$, and 100 decimal working digits in their Pfaffian construction. No new primary optimization is performed.
2. Diagonalize each local parity block (Ising) or occupation block (boson), retaining its original trace. Denote target eigenvectors by $|v_{bj}\rangle$, target eigenvalues by λ_{bj} , and the first state’s weights in that basis by $w_{aj}^{(b)} = \langle v_{bj} | \rho_A^a | v_{bj} \rangle$. The index j runs over eigenvectors in all blocks. The matrices need not commute; their eigenvectors are essential in this cross term.

3. Evaluate the spectral trace with natural logarithms:

$$S_{a||b}^{(\tau)} = \sum_{\lambda_{aj} > 0} \lambda_{aj} \log \lambda_{aj} - \sum_j w_{aj}^{(b)} \log[\max(\lambda_{bj}, \tau)], \quad \tau = 10^{-15}. \quad (125)$$

The small eigenvalue floor τ only regularizes the target logarithm; it does not renormalize or flatten a symmetry block. Negative eigenvalues at roundoff scale are omitted from the self-entropy sum; an eigenvalue below -2×10^{-12} aborts the calculation.

4. Check the regularization by repeating the logarithms at $\tau = 10^{-14}$ and 10^{-16} . Record both the span of the three entropy values and the first-state weight on target modes below the production floor,

$$t_{a||b} = \sum_{\lambda_{bj} < 10^{-15}} \max(w_{aj}^{(b)}, 0). \quad (126)$$

An exact target null space with nonzero first-state weight would make the unregularized relative entropy infinite. The floor is consequently audited, rather than used to conceal a support mismatch. Cutoff sensitivity is a numerical stability test, not a rigorous bound on unresolved eigenvalues.

There are 4824 passing numerical checks. The maximum entropy span over the three floors is 1.66×10^{-10} , below the 2×10^{-10} tolerance; the maximum $t_{a||b}$ is 1.50×10^{-11} , below 2×10^{-11} . All directed values are nonnegative, and self-comparisons $S_{a||a}^{(\tau)}$ have maximum absolute magnitude 3.23×10^{-13} over all three floors. At $N = 256$, $\ell = 4, 6$, direct diagonalization of the full interval matrix agrees with the symmetry-block calculation for all pairs to 4.67×10^{-15} .

For every Ising geometry we also check a formula that operates on the $2\ell \times 2\ell$ one-particle matrices instead of the $2^\ell \times 2^\ell$ many-body matrices. Define $C_a = (\mathbb{I} + i\Gamma_{a,A})/2$, where $\Gamma_{a,A}$ is the restricted Majorana covariance defined in Sec. 3.3. Gaussian mode decomposition [15] gives

$$S_{a||b} = \text{tr}_{2\ell} [C_a (\log C_a - \log C_b)]. \quad (127)$$

In this doubled Majorana representation, the eigenvalues occur as $n_j, 1 - n_j$; both occupations of each fermionic mode are already counted, so no extra factor of $1/2$ is inserted. In the eigenbasis of C_b , the quadratic target logarithm is averaged using the diagonal weights of C_a , which yields the cross term in Eq. (127). This independent evaluation agrees with the many-body trace to 2.35×10^{-12} across all 600 Ising directed comparisons. These checks validate the entropy evaluation; they do not remove the physical finite-interval corrections.

F.3 Comparison with the analytical CFT leading term

Let B_{ab} denote the analytical coefficient obtained above, and define

$$\begin{aligned} S_{a||b}^{\text{LO}}(x) &= B_{ab} x^\alpha, & R_{a||b} &= \frac{S_{a||b}}{S_{a||b}^{\text{LO}}}, \\ \delta_{a||b}[\%] &= 100(R_{a||b} - 1), & E_m[\%] &= \sqrt{\frac{1}{m} \sum_{i: x_i \leq 0.08} \delta_i^2}. \end{aligned} \quad (128)$$

LO means leading order. The denominator in the percentage is the analytical prediction. The index i runs over the $m = 51$ geometries for one direction. Neither B_{ab} nor α is fitted. Tables 32 and 33 show every directed pair, while Fig. 37 shows the interval dependence.

At $N = 256$, $\ell = 10$, the Ising $I \leftrightarrow \sigma$ deviations are about $+0.13\%$. The $I||\varepsilon$ and $\varepsilon||I$ deviations are -4.44% and -2.39% , whereas $\sigma||\varepsilon$ and $\varepsilon||\sigma$ give $+6.93\%$ and $+9.47\%$. Reducing x to $3/128$ with $\ell = 6$ reduces the latter two deviations to $+2.437\%$ and $+3.313\%$, while also changing the lattice resolution. In

Direction	α	B_{ab}	S_{10}	S_{10}^{LO}	δ_6 (%)	δ_{10} (%)	E_m (%)
$I\ \sigma$	2	$(1/12)\pi^2$	0.0012566	0.001255	+0.106	+0.126	0.221
$\sigma\ I$	2	$(1/12)\pi^2$	0.0012566	0.001255	+0.117	+0.130	0.232
$I\ \varepsilon$	4	$(8/15)\pi^4$	1.16×10^{-4}	1.21×10^{-4}	-2.854	-4.439	7.160
$\varepsilon\ I$	4	$(8/15)\pi^4$	1.18×10^{-4}	1.21×10^{-4}	-2.079	-2.388	3.940
$\sigma\ \varepsilon$	2	$(1/12)\pi^2$	0.001342	0.001255	+2.437	+6.932	11.634
$\varepsilon\ \sigma$	2	$(1/12)\pi^2$	0.0013738	0.001255	+3.313	+9.469	16.494

Table 32: Ising primary relative entropy sanity check for all six directions under the critical periodic Hamiltonian $H = -\sum XX - \sum Z$. The theoretical coefficient B_{ab} multiplies x^α , including its power of π . Here S_{10} and S_{10}^{LO} are respectively the direct numerical entropy and its analytical CFT leading-order prediction at $N = 256$, $\ell = 10$, $x = 5/128$; both use natural logarithms. δ_6 and δ_{10} are signed deviations at $N = 256$ for $\ell = 6$ and ten, respectively. E_m uses all $m = 51$ geometries with $x \leq 0.08$ and is defined by Eq. (128). All primaries and spectral checks are specified in Sec. F.2; every coefficient is fixed analytically.

Direction	α	B_{ab}	S_{10}	S_{10}^{LO}	δ_6 (%)	δ_{10} (%)	E_m (%)
$00\ ss$	2	$(1/3)\pi^2$	0.0048262	0.0050199	-6.829	-3.859	5.291
$ss\ 00$	2	$(1/3)\pi^2$	0.0048262	0.0050199	-6.830	-3.860	5.293
$00\ 3s, s$	2	$(5/3)\pi^2$	0.024469	0.0251	-4.780	-2.515	3.579
$3s, s\ 00$	2	$(5/3)\pi^2$	0.024466	0.0251	-4.792	-2.527	3.615
$ss\ 3s, s$	2	$(2/3)\pi^2$	0.0099902	0.01004	-1.702	-0.495	1.035
$3s, s\ ss$	2	$(2/3)\pi^2$	0.0099884	0.01004	-1.721	-0.513	1.084

Table 33: Compact-boson primary relative entropy for all six directions, with the same geometry and indicator definitions as Table 32. The normalized vertex/Jastrow states have charges in Eq. (123), $q = 2$, $z_j = e^{2\pi i j/N}$, $w_1 = 0$, $W = 10^8$, and no orthogonalization. All rows have fixed $\alpha = 2$; $B_{ab} = \kappa_{ab}\pi^2/3$ follows from Eq. (124). The data use the primary target ρ_A^b , with no mixture.

the boson, the largest absolute deviation at $N = 256$, $\ell = 10$ is 3.86%, and the $ss \leftrightarrow 3s, s$ deviations are about -0.5% .

The leading predictions thus reproduce the scales and small- x behavior across the pairs, with the residual deviations explicitly resolved. The $\sigma \leftrightarrow \varepsilon$ channels are particularly sensitive to retaining only their leading term: over the entire $x \leq 0.08$ window their maximum signed deviations reach 24.9% and 36.1%. Smaller x at fixed ℓ and larger ℓ in lattice units address different limits, as discussed in Sec. 3.1. This sanity check is not a controlled continuum extrapolation or an assertion that a leading-only expression is exact on the finite-chain grid.

The implementation is `src/conformal_blocks/primary_relative_entropy.py`. The raw `primary_relative_entropy.csv` retains all 1200 observations, including excluded lengths. The selection module rebuilds `primary_entropy_summary.csv` (12 directional summaries), `primary_entropy_checks.csv` (4824 numerical checks), and `primary_entropy_validation.json` in the report’s data directory. The independent renderer `report/primary_entropy_artifacts.py` reads these CSVs. From the project root, the reproducible commands are

```
PYTHONPATH=src $PY -m conformal_blocks.primary_selection
MPLCONFIGDIR=/tmp/relative_entropy_mpl $PY report/primary_entropy_artifacts.py
```

Here `$PY` is the approved Python interpreter specified in the reproduction instructions, with one BLAS thread. Existing cached density matrices are reused; if absent, the same production formulas recompute them.

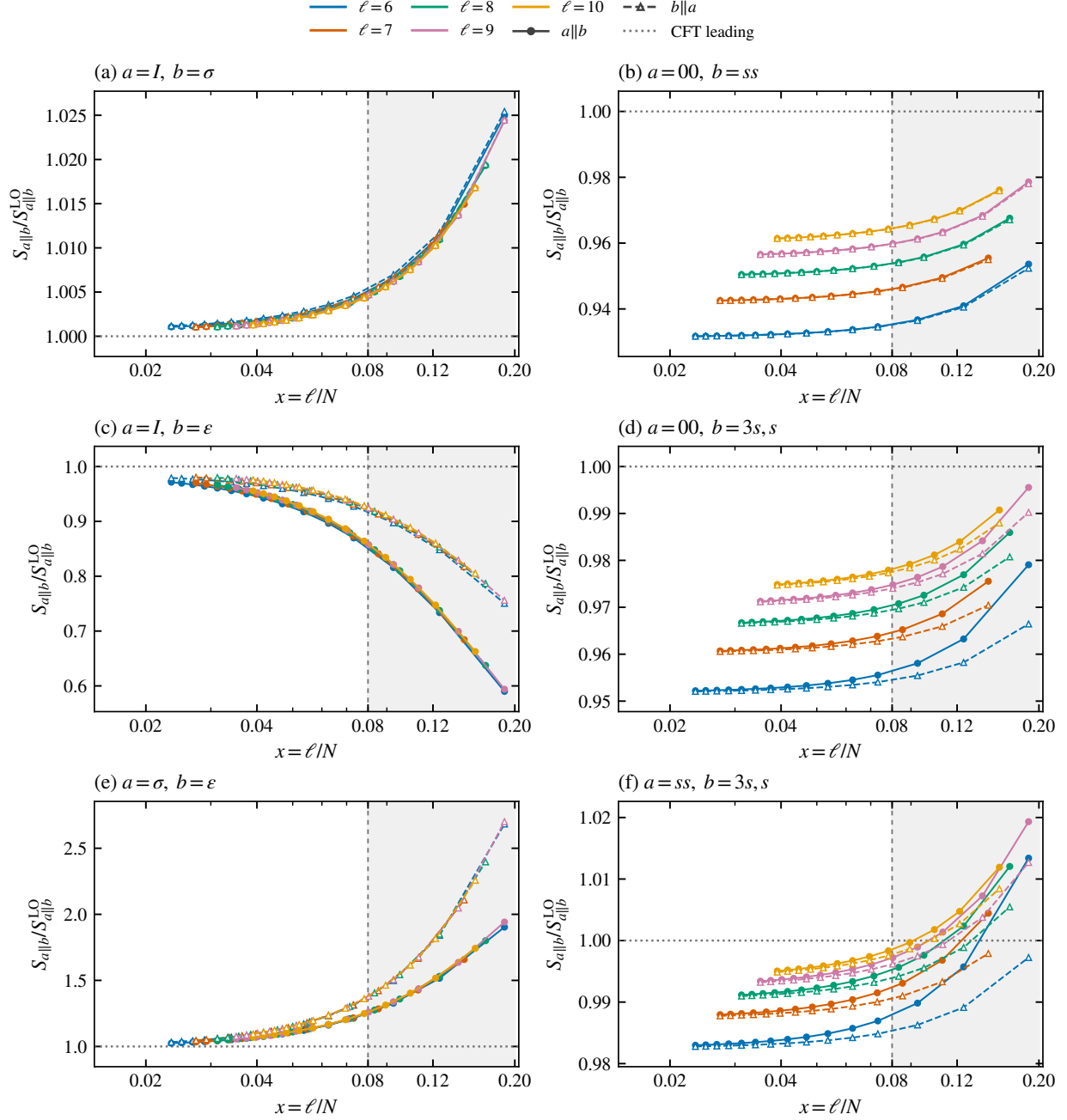


Figure 37: Historical parameter-free leading-order check of primary relative entropy. Each panel compares the two directions between the states a, b named in its title; the left column is Ising and the right column the compact boson. The ordinate is $R = S_{a||b}/[B_{ab}x^\alpha]$, defined by Eq. (128); the dotted line at one is the analytical CFT leading-order prediction. Colors identify $\ell = 6, \dots, 10$. Solid lines with filled circles denote $a||b$, and dashed lines with open triangles denote $b||a$; segments only connect computed sizes. Each direction has 70 points from the 15 sizes through $N = 256$ in Eq. (26), displayed for $x \leq 0.20$. The vertical line at $x = 0.08$ marks the 51-point summary window; shaded points are outside that window. There are no fitted curves or optimized parameters. Ising uses exact Jordan–Wigner primaries of $H = -\sum XX - \sum Z$ with periodic spins; the boson uses the charges and $q = 2$, $W = 10^8$ construction in Table 33. All logarithms are natural, and the audited target-logarithm floor is 10^{-15} . $\alpha = 4$, $B = 8\pi^4/15$ for Ising $I \leftrightarrow \varepsilon$; $\alpha = 2$, $B = \pi^2/12$ for the other Ising panels; $\alpha = 2$, $B = \pi^2/3, 5\pi^2/3, 2\pi^2/3$ for the boson panels from top to bottom. Differences between reverse directions and departures from one remain visible; the analytical equality is a leading-order statement.

F.4 Free power-law fits for the primary pairs

To test whether the numerical primary relative entropies recover both parts of the analytical leading term, we fit one power law with a free coefficient and a free exponent. The six-panel presentation follows Figs. 2–5; the fitting model below is specific to the primary-pair comparison. The parameter-free checks in Secs. F.1–F.3 remain independent of these fits.

For a fixed directed pair $a||b$, let $y_i = S_{a||b}(x_i)$ denote a numerical observation. Define the fitted model and the analytical leading prediction by

$$\begin{aligned} F(x; C, \alpha) &= Cx^\alpha, & F_{\text{CFT}}(x) &= B_{ab}x^{\alpha_{\text{CFT}}}, \\ (\hat{C}, \hat{\alpha}) &= \arg \min_{C>0, \alpha \in \mathbb{R}} \sum_{i: x_i \leq 0.08} [y_i - Cx_i^\alpha]^2. \end{aligned} \quad (129)$$

Both C and α are optimized, giving two free parameters. No theoretical exponent or coefficient is supplied to the optimizer, and the exponent is not restricted to an integer or to a neighborhood of its CFT value. Positivity of C enforces a positive power-law entropy. This is unweighted nonlinear least squares in the entropy itself; the logarithmic axes do not change the objective. The analytical values B_{ab} and α_{CFT} from Sec. F.1 are used only for the subsequent comparison. The CFT prediction has zero free parameters.

For a reproducible numerical procedure, set $x_\star = 0.08$ and write

$$t_i = \ln(x_i/x_\star), \quad F(x_i) = \exp(u + \alpha t_i), \quad C = \exp(u)x_\star^{-\alpha}. \quad (130)$$

First regress $\ln y_i$ on t_i to obtain a data-based starting guess for (u, α) . Then optimize both variables against the original entropies using Eq. (129). The log-data regression supplies only an initial guess. The solver uses derivatives $\partial_u F = F$ and $\partial_\alpha F = Ft_i$; rescaling all residuals by their common entropy scale does not change the minimum. Finally convert the fitted amplitude to $\hat{C}$ using Eq. (130).

The fit uses 51 observations per direction from 12 contributing system sizes, with the selection $\ell = 6, \dots, 10$ and $x \leq 0.08$. The figures show 70 observations per direction from all 15 sizes in Eq. (26), with $\ell = 6, \dots, 10$ and $x \leq 0.20$. The 19 displayed observations outside $x \leq 0.08$ do not affect any coefficient or table metric. Curves beyond the vertical cutoff are extrapolations of the fitted power law.

For clarity, define the signed entropy residual r_i , its percentage η_i , and the root mean square error (RMSE), for either $M = F$ or $M = F_{\text{CFT}}$, by

$$r_i = y_i - M(x_i), \quad \eta_i = 100 \frac{r_i}{y_i}, \quad \text{RMSE}_M = \sqrt{\frac{1}{51} \sum_{i: x_i \leq 0.08} r_i^2}. \quad (131)$$

The percentage denominator here is the numerical entropy y_i . It differs from the analytical denominator in Eq. (128). The two signed parameter deviations are

$$\delta C[\%] = 100 \left(\frac{\hat{C}}{B_{ab}} - 1 \right), \quad \delta \alpha[\%] = 100 \left(\frac{\hat{\alpha}}{\alpha_{\text{CFT}}} - 1 \right). \quad (132)$$

The coefficient comparison uses powers of x itself, including all powers of π . The archived amplitude $\tilde{C}$ multiplying $(\pi x)^\alpha$ is converted by $C = \pi^\alpha \tilde{C}$. Changing this convention changes coefficient comparisons when the exponents differ, so the convention must accompany the numbers.

Cross-validation (CV) omits one of the 12 contributing sizes, refits *both* parameters on the other 11, and predicts all omitted observations. If $F^{(-N_i)}$ is the resulting prediction for observation i , define

$$\text{CVRMSE} = \sqrt{\frac{1}{51} \sum_{i: x_i \leq 0.08} [y_i - F^{(-N_i)}(x_i)]^2}. \quad (133)$$

The residual degrees of freedom are $51 - 2 = 49$ for the free fit and 51 for the analytical CFT leading-order prediction. For the latter, CV RMSE equals RMSE because no parameter is refitted.

Tables 34–35 compare both fitted parameters with their analytical values. For Ising, in the ordered-pair order of Table 34, the fitted exponents are 2.0040, 2.0041, 3.8376, 3.9196, 2.2160, 2.3071. The boson exponents lie between 2.0050 and 2.0090. The Ising spin–energy coefficients are 1.75660, 2.40358, compared with $B_{ab} = \pi^2/12$. A two-parameter power law over a finite window estimates an effective coefficient and exponent; agreement of its curve with the data does not by itself establish their asymptotic CFT values. No additional term or lattice-correction model enters this fit.

F.4.1 Complete Ising fitting table

Table 34: Ising primary-pair free power-law fits, $F(x) = \hat{C}x^{\hat{\alpha}}$: both C and α are optimized on 51 samples per direction, with $\ell = 6, \dots, 10$, at $x \leq 0.08$. The analytical CFT leading-order prediction is $B_{ab}x^{\alpha_{\text{CFT}}}$, with no fitted parameters. All amplitudes include their powers of π . Parameter deviations use Eq. (132); RMSE and CV RMSE refer to the free fit, and CFT RMSE to this analytical CFT prediction. Natural logarithms and the complete definitions in Sec. F.4 apply.

Direction	B_{ab}	$\hat{C}$	δC [%]	α_{CFT}	$\hat{\alpha}$	$\delta \alpha$ [%]	RMSE	CFT RMSE	CV RMSE
$I \sigma$	$1\pi^2/12$	0.8344	1.4505	2	2.004	0.20221	6.54×10^{-7}	7.21×10^{-6}	7.06×10^{-7}
σI	$1\pi^2/12$	0.83465	1.4819	2	2.0041	0.20553	7.67×10^{-7}	7.50×10^{-6}	8.33×10^{-7}
$I \varepsilon$	$8\pi^4/15$	29.732	-42.77	4	3.8376	-4.059	2.44×10^{-6}	7.02×10^{-5}	2.64×10^{-6}
εI	$8\pi^4/15$	39.286	-24.38	4	3.9196	-2.01	1.95×10^{-6}	3.76×10^{-5}	2.23×10^{-6}
$\sigma \varepsilon$	$1\pi^2/12$	1.7566	113.58	2	2.216	10.802	2.40×10^{-5}	3.95×10^{-4}	2.76×10^{-5}
$\varepsilon \sigma$	$1\pi^2/12$	2.4036	192.24	2	2.3071	15.353	3.60×10^{-5}	5.66×10^{-4}	4.10×10^{-5}

F.4.2 Complete compact-boson fitting table

Table 35: Compact-boson primary-pair free power-law fits, $F(x) = \hat{C}x^{\hat{\alpha}}$: both C and α are optimized on 51 samples per direction, with $\ell = 6, \dots, 10$, at $x \leq 0.08$. The analytical CFT leading-order prediction is $B_{ab}x^{\alpha_{\text{CFT}}}$, with no fitted parameters. All amplitudes include their powers of π . Parameter deviations use Eq. (132); RMSE and CV RMSE refer to the free fit, and CFT RMSE to this analytical CFT prediction. Natural logarithms and the complete definitions in Sec. F.4 apply.

Direction	B_{ab}	$\hat{C}$	δC [%]	α_{CFT}	$\hat{\alpha}$	$\delta \alpha$ [%]	RMSE	CFT RMSE	CV RMSE
$00 ss$	$1\pi^2/3$	3.2114	-2.3851	2	2.009	0.44964	9.35×10^{-5}	4.46×10^{-4}	1.07×10^{-4}
$ss 00$	$1\pi^2/3$	3.2106	-2.4094	2	2.0089	0.44594	9.36×10^{-5}	4.47×10^{-4}	1.07×10^{-4}
$00 3s, s$	$5\pi^2/3$	16.286	-0.99214	2	2.0079	0.39731	3.50×10^{-4}	0.0014811	4.02×10^{-4}
$3s, s 00$	$5\pi^2/3$	16.231	-1.3301	2	2.007	0.34759	3.62×10^{-4}	0.0015101	4.16×10^{-4}
$ss 3s, s$	$2\pi^2/3$	6.6504	1.0737	2	2.0063	0.31743	7.02×10^{-5}	1.49×10^{-4}	8.04×10^{-5}
$3s, s ss$	$2\pi^2/3$	6.6204	0.61876	2	2.005	0.25236	7.70×10^{-5}	1.65×10^{-4}	8.85×10^{-5}

F.4.3 Reproducing the primary fits

The command module `src/conformal_blocks/primary_fits.py` calls `primary_powerlaw.py` on the existing direct measurements in `primary_relative_entropy.csv`; no density matrix is recomputed. Its separate outputs in `report/data/` are `primary_fit_summary.csv` (24 free/CFT summaries), `primary_fit_input.csv` (612 selected observations), `primary_fit_residuals.csv` (1224 selected residuals), `primary_display_input.csv` (840 displayed observations), `primary_display_predictions.csv` (1680 displayed predictions), `primary_exact_coefficients.csv` (12 analytical leading terms), and `primary_free_power_cv.csv` (144 omitted-size refits). The public entropy column is `S_primary`. The full summary CSV retains both parameter estimates and deviations, squared-error sums, relative-residual indicators from Eq. (62), and the minimum and maximum of both parameters across the 12 CV fits. Its additional local diagnostics use the Jacobian $J_{i1} = F_i$, $J_{i2} = F_i t_i$ in variables (u, α) : $\kappa_J = \sigma_{\max}(J)/\sigma_{\min}(J)$ and the covariance $(\text{SSE}/49)(J^T J)^{-1}$. Transforming this covariance to (C, α) gives a matrix V . The exported standard errors are $\sqrt{V_{11}}$ and $\sqrt{V_{22}}$, and the parameter correlation is $V_{12}/\sqrt{V_{11}V_{22}}$. These local linearization diagnostics are descriptive, not confidence intervals for correlated lattice data.

The archived solver CSV uses its original $(\pi x)^\alpha$ amplitude. The report converter writes `primary_reported_coefficients.csv` with $C = \pi^{\hat{\alpha}} \tilde{C}$ and $B_{ab} = \pi^{\alpha_{\text{CFT}}} \tilde{B}_{ab}$, including the transformed coefficient errors and local covariance. Physical curves, exponents and residuals are invariant under this conversion.

An independent check eliminates the amplitude at each trial exponent:

$$\hat{A}(\alpha) = \frac{\sum_i y_i v_i(\alpha)}{\sum_i v_i(\alpha)^2}, \quad v_i(\alpha) = (x_i/x_\star)^\alpha. \quad (134)$$

Minimizing the remaining one-dimensional objective independently checks the two-parameter solver. This check covers all 12 complete fits and all 144 omitted-size fits; a synthetic power law with the noninteger exponent 2.37 also verifies that the solver recovers a free exponent. The implementation is `report/primary_free_fit_checks.py`. With the interpreter variable `PY` defined in Sec. F.3, run from the project root:

```
PYTHONPATH=src $PY -m conformal_blocks.primary_fits
PYTHONPATH=src MPLCONFIGDIR=/tmp/relative_entropy_mpl \
  $PY report/primary_fit_artifacts.py
PYTHONPATH=src $PY report/primary_free_fit_checks.py
```

The renderer reads these CSVs to generate the four vector PDF figures and Tables 34–35.

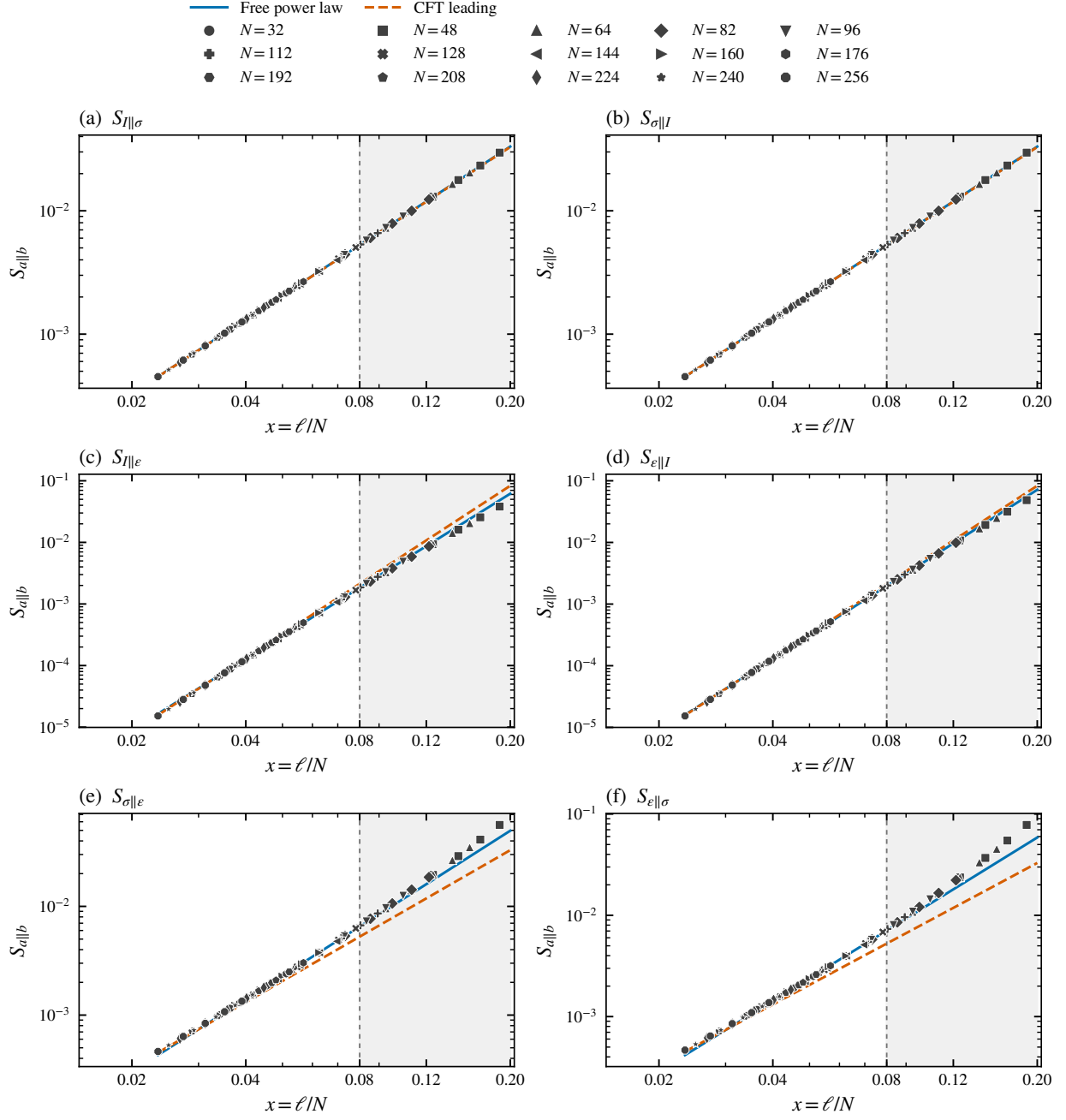


Figure 38: Historical Ising primary relative entropy $S_{a||b}$ for all six directions. Markers identify the 15 system sizes through $N = 256$ in Eq. (26); $\ell = 6, \dots, 10$. Each panel shows 70 observations with $x = \ell/N \leq 0.20$. Only the 51 observations at $x \leq 0.08$ enter the unweighted least-squares fits in Eq. (129). The vertical line and shading mark the cutoff and the extrapolation region containing 19 observations. Blue solid lines show the free power law $F = \hat{C}x^{\hat{\alpha}}$; both parameters are optimized independently for each direction. Orange dashed lines show the analytical CFT leading-order prediction, with no fitted parameters. For panels (c,d), $\alpha_{\text{CFT}} = 4$ and $B_{ab} = 8\pi^4/15$; elsewhere $\alpha_{\text{CFT}} = 2$ and $B_{ab} = \pi^2/12$. Table 34 compares both fitted parameters with these analytical values. The data use exact Jordan–Wigner primaries of the critical periodic-spin Hamiltonian $H = -\sum XX - \sum Z$, natural logarithms, and target-logarithm floor 10^{-15} . Both axes are logarithmic. The fitted exponent is a free real number and the coefficient is positive; no additional powers enter.

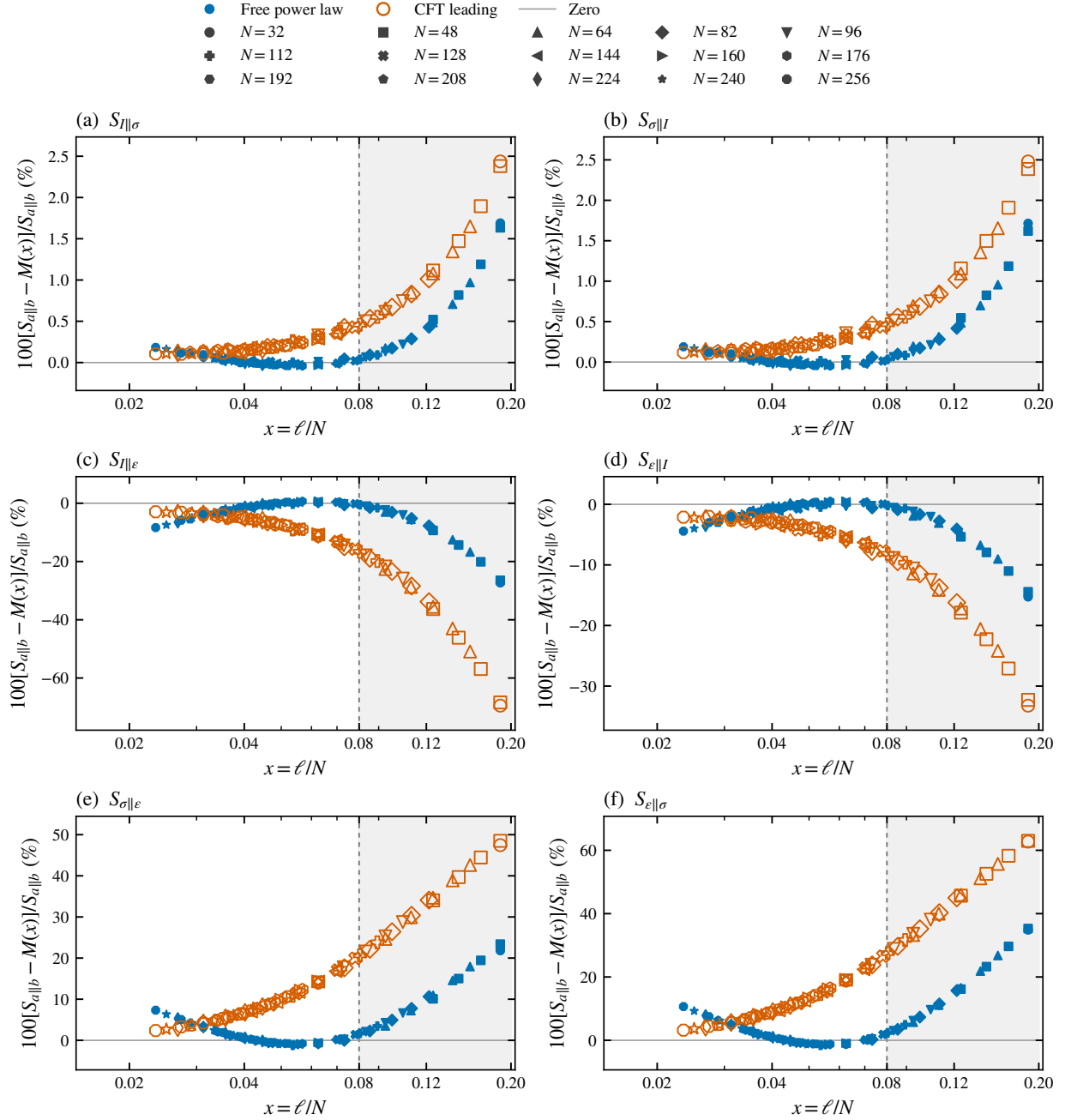


Figure 39: Signed percentage residuals for the Ising primary comparisons in Fig. 38. Blue filled markers denote the free power law and orange open markers the analytical leading prediction. Marker shapes identify the 15 sizes in Eq. (26), with $\ell = 6, \dots, 10$. The ordinate is $\eta_i = 100[S_{a||b}(x_i) - M(x_i)]/S_{a||b}(x_i)$, where $M = F$ or F_{CFT} , from Eq. (131); zero means exact agreement. The ordinate is linear throughout, with ticks showing signed percentage values directly; x is logarithmic. All 70 displayed observations per direction satisfy $x \leq 0.20$. Only 51 at $x \leq 0.08$ determine the fits and table metrics; the 19 shaded-region observations test extrapolation. The free fit has two jointly optimized parameters, $F = \hat{C}x^{\hat{\alpha}}$. Only the analytical comparison uses $\alpha_{\text{CFT}} = 4$, $B_{ab} = 8\pi^4/15$ in (c,d), and $\alpha_{\text{CFT}} = 2$, $B_{ab} = \pi^2/12$ otherwise. The exact PBC Ising states, natural logarithms and target floor 10^{-15} are as in Fig. 38; coefficients, exponents, RMSE and CV RMSE appear in Table 34.

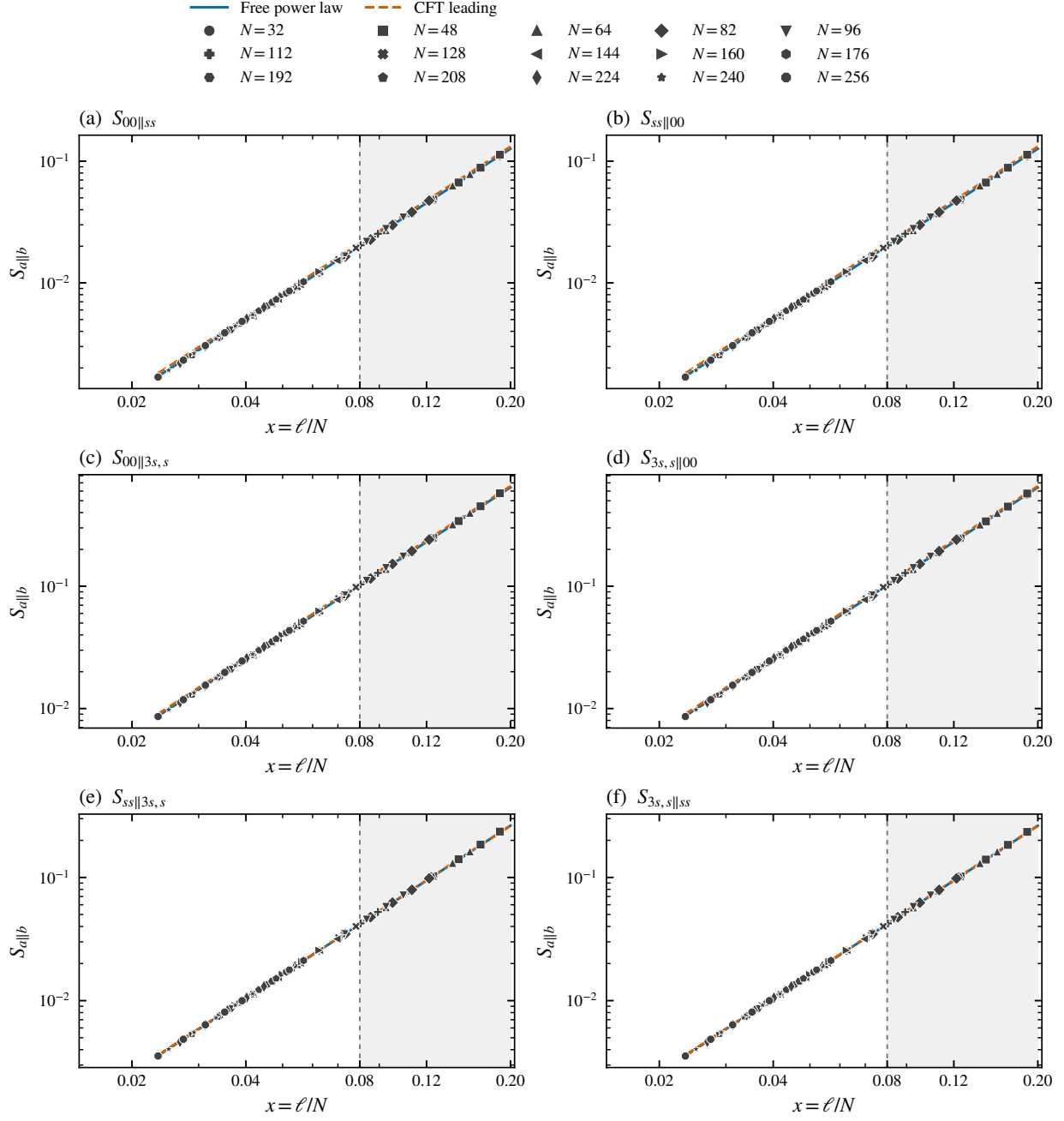


Figure 40: Historical compact-boson primary relative entropy for all six directions among charges $(0, 0)$, (s, s) and $(3s, s)$, $s = 1/\sqrt{2}$. Markers identify the 15 sizes through $N = 256$ in Eq. (26); $\ell = 6, \dots, 10$. Each panel displays 70 observations at $x \leq 0.20$. The 51 observations at $x \leq 0.08$ determine the unweighted least-squares fits; the vertical line and shading mark the 19 observations outside the fit window. Blue solid lines show $F = \hat{C}x^{\hat{\alpha}}$ from Eq. (129), with both coefficient and exponent optimized. Orange dashed lines show the analytical CFT leading-order predictions, with no fitted parameters: $\alpha_{\text{CFT}} = 2$ and $B_{ab} = \pi^2/3$ in (a,b), $5\pi^2/3$ in (c,d), and $2\pi^2/3$ in (e,f). Table 35 compares both fitted parameters with the analytical values. The normalized vertex/Jastrow states use the unit circle, $q = 2$, $w_1 = 0$, $W = 10^8$, and 100-decimal-digit Pfaffian construction, without orthogonalization, as in Sec. F.2. All logarithms are natural and the target-logarithm floor is 10^{-15} . Both axes are logarithmic. The fit has a positive coefficient and a free real exponent, with no additional terms.

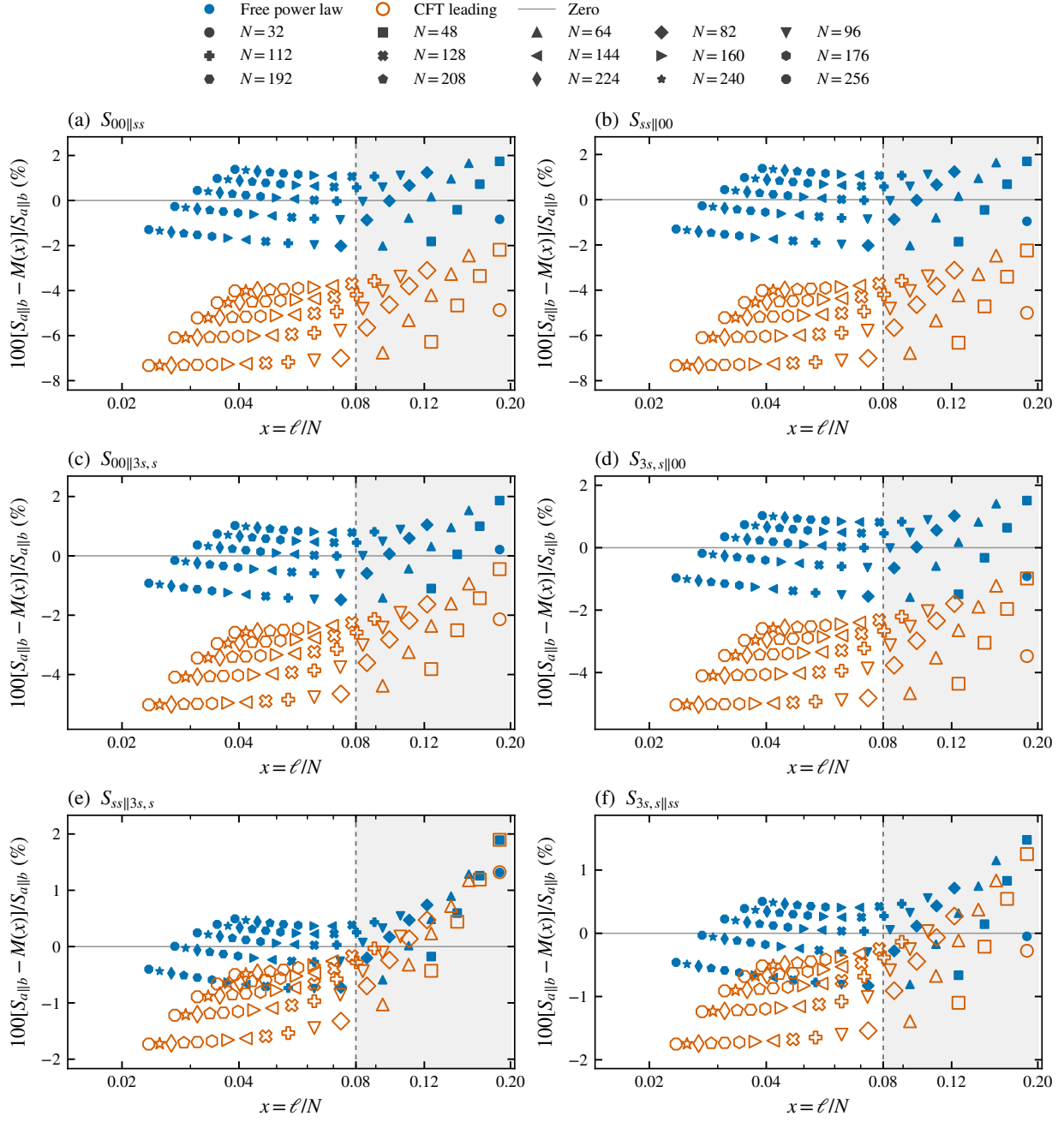


Figure 41: Signed percentage residuals for the compact-boson primary comparisons in Fig. 40. Blue filled markers denote the free power law; orange open markers denote the analytical leading term. The ordinate is $100[S_{a||b} - M]/S_{a||b}$, where $M = F$ or F_{CFT} , as in Eq. (131). The vertical scale is linear throughout, with ticks showing signed percentage values directly; x is logarithmic. Markers identify the 15 sizes in Eq. (26), with $\ell = 6, \dots, 10$. All 70 observations per direction at $x \leq 0.20$ are shown; 51 at $x \leq 0.08$ enter the fits and metrics, while 19 lie in the shaded extrapolation region. The two-parameter fit is $F = \hat{C}x^\alpha$; its exponent is not fixed. The analytical comparison uses $\alpha_{\text{CFT}} = 2$ and $B_{ab} = \pi^2/3, 5\pi^2/3, 2\pi^2/3$ in the top, middle and bottom rows. The vertex charges, $q = 2$, unit-circle sites, $w_1 = 0$, $W = 10^8$, 100-digit construction, absence of orthogonalization, natural logarithms and target floor 10^{-15} are as in Fig. 40. Table 35 compares fitted coefficients and exponents with theory and gives RMSE and size-based CV RMSE.

G Parameters and numerical errors for the visual comparisons

These tables supplement Sec. 6. Models are grouped by ordered pair, and every fitted row gives its actual interval. They correspond to both columns of the per-pair figures and the principal tables in Appendices A and B. The same CFT-wide windows are used throughout. The coefficient convention and indices follow Eq. (84). Unknown analytical next coefficients are left blank throughout. Agreement is assessed through the displayed numerical parameters and curves, without interval-selection labels.

G.1 Small intervals: Ising

Table 36: Ising small intervals: leading parameters for all three models, grouped by ordered pair. Amplitudes multiply x^r itself, including powers of π . W_x and m give the fitting interval and count. Fixed models impose $r^{(1)} = \alpha_{\text{CFT}}$; Free 1 estimates the leading power. Percentage discrepancies use Eq. (88).

Direction	Model	W_x	m	$r_{\text{CFT}}^{(1)}$	$r^{(1)}$	c_1^{CFT}	c_1	$\eta_{r,1}$	$\eta_{c,1}$
$I \sigma$	Free 1	(0, 0.02]	1255	2	2.0001	0.20562	0.20585	0.0070964	0.1146
$I \sigma$	F_1	(0, 0.02]	1255	2	2	0.20562	0.20573	—	0.055265
$I \sigma$	F_2	(0, 0.02]	1255	2	2	0.20562	0.20571	—	0.044771
σI	Free 1	(0, 0.02]	1255	2	2.0001	0.20562	0.20582	0.0057066	0.099239
σI	F_1	(0, 0.02]	1255	2	2	0.20562	0.20572	—	0.051526
σI	F_2	(0, 0.02]	1255	2	2	0.20562	0.2057	—	0.042877
$I \varepsilon$	Free 1	(0, 0.02]	1255	4	3.9934	12.988	12.491	-0.16507	-3.8239
$I \varepsilon$	F_1	(0, 0.02]	1255	4	4	12.988	12.83	—	-1.219
$I \varepsilon$	F_2	(0, 0.02]	1255	4	4	12.988	12.883	—	-0.80492
εI	Free 1	(0, 0.02]	1255	4	3.9861	12.988	12.07	-0.34832	-7.0661
εI	F_1	(0, 0.02]	1255	4	4	12.988	12.77	—	-1.6748
εI	F_2	(0, 0.02]	1255	4	4	12.988	12.882	—	-0.81263
$\sigma \varepsilon$	Free 1	(0, 0.02]	1255	2	2.0238	0.20562	0.23069	1.1878	12.192
$\sigma \varepsilon$	F_1	(0, 0.02]	1255	2	2	0.20562	0.2089	—	1.595
$\sigma \varepsilon$	F_2	(0, 0.02]	1255	2	2	0.20562	0.20565	—	0.017273
$\varepsilon \sigma$	Free 1	(0, 0.02]	1255	2	2.0176	0.20562	0.22393	0.88006	8.9077
$\varepsilon \sigma$	F_1	(0, 0.02]	1255	2	2	0.20562	0.20806	—	1.1882
$\varepsilon \sigma$	F_2	(0, 0.02]	1255	2	2	0.20562	0.20566	—	0.022414

Table 37: Ising small intervals: next-order parameters, with c_2 multiplying $x^{r^{(2)}}$. F_2 imposes both manuscript powers and estimates both coefficients. Empty analytical cells mean that the manuscript does not supply the coefficient. The displayed windows and numerical values also appear in the per-pair figures in Sec. 6.4. Percentage discrepancies use Eq. (88).

Direction	Model	W_x	$r_{\text{CFT}}^{(2)}$	$r^{(2)}$	c_2^{CFT}	c_2	$\eta_{r,2}$	$\eta_{c,2}$
$I \mid \sigma$	F_2	$(0, 0.02]$	4	4		0.083833	—	
$\sigma \mid I$	F_2	$(0, 0.02]$	4	4		0.069099	—	
$I \mid \varepsilon$	F_2	$(0, 0.02]$	6	6	-170.91	-170.71	—	-0.12193
$\varepsilon \mid I$	F_2	$(0, 0.02]$	6	6	-366.24	-355.38	—	-2.9657
$\sigma \mid \varepsilon$	F_2	$(0, 0.02]$	4	4		12.604	—	
$\varepsilon \mid \sigma$	F_2	$(0, 0.02]$	4	4		9.3133	—	

Table 38: Ising small intervals: numerical fit errors on each row's own window. RMSE uses Eq. (62); the normalized percentages E_{fit} and E_{CV} use Eq. (87). The residual plots show pointwise percentages instead. All models and all six directions use the same window for this CFT and interval type.

Direction	Model	W_x	m	RMSE	E_{fit} [%]	E_{CV} [%]
$I \mid \sigma$	Free 1	$(0, 0.02]$	1255	5.158×10^{-9}	0.017406	0.017515
$I \mid \sigma$	F_1	$(0, 0.02]$	1255	5.249×10^{-9}	0.017715	0.017787
$I \mid \sigma$	F_2	$(0, 0.02]$	1255	5.124×10^{-9}	0.017292	0.017418
$\sigma \mid I$	Free 1	$(0, 0.02]$	1255	4.526×10^{-9}	0.015275	0.015369
$\sigma \mid I$	F_1	$(0, 0.02]$	1255	4.594×10^{-9}	0.015504	0.015565
$\sigma \mid I$	F_2	$(0, 0.02]$	1255	4.497×10^{-9}	0.015175	0.015282
$I \mid \varepsilon$	Free 1	$(0, 0.02]$	1255	1.566×10^{-9}	0.30918	0.31285
$I \mid \varepsilon$	F_1	$(0, 0.02]$	1255	1.631×10^{-9}	0.32198	0.32466
$I \mid \varepsilon$	F_2	$(0, 0.02]$	1255	1.561×10^{-9}	0.30816	0.31219
$\varepsilon \mid I$	Free 1	$(0, 0.02]$	1255	1.565×10^{-9}	0.31025	0.3139
$\varepsilon \mid I$	F_1	$(0, 0.02]$	1255	1.834×10^{-9}	0.36374	0.36666
$\varepsilon \mid I$	F_2	$(0, 0.02]$	1255	1.548×10^{-9}	0.30693	0.31094
$\sigma \mid \varepsilon$	Free 1	$(0, 0.02]$	1255	4.655×10^{-8}	0.1547	0.15537
$\sigma \mid \varepsilon$	F_1	$(0, 0.02]$	1255	1.717×10^{-7}	0.57076	0.57252
$\sigma \mid \varepsilon$	F_2	$(0, 0.02]$	1255	7.741×10^{-9}	0.025729	0.025989
$\varepsilon \mid \sigma$	Free 1	$(0, 0.02]$	1255	3.468×10^{-8}	0.11573	0.11623
$\varepsilon \mid \sigma$	F_1	$(0, 0.02]$	1255	1.270×10^{-7}	0.42368	0.42497
$\varepsilon \mid \sigma$	F_2	$(0, 0.02]$	1255	7.163×10^{-9}	0.023901	0.024146

G.2 Small intervals: compact boson

Table 39: Compact boson small intervals: leading parameters for all three models, grouped by ordered pair. Amplitudes multiply x^r itself, including powers of π . W_x and m give the fitting interval and count. Fixed models impose $r^{(1)} = \alpha_{\text{CFT}}$; Free 1 estimates the leading power. Percentage discrepancies use Eq. (88).

Direction	Model	W_x	m	$r_{\text{CFT}}^{(1)}$	$r^{(1)}$	c_1^{CFT}	c_1	$\eta_{r,1}$	$\eta_{c,1}$
00 ss	Free 1	(0, 0.1]	598	2	1.9982	0.82247	0.77501	-0.092048	-5.7697
00 ss	F_1	(0, 0.1]	598	2	2	0.82247	0.77866	—	-5.3266
00 ss	F_2	(0, 0.1]	598	2	2	0.82247	0.77996	—	-5.1687
ss 00	Free 1	(0, 0.1]	598	2	1.9983	0.82247	0.77529	-0.086511	-5.7362
ss 00	F_1	(0, 0.1]	598	2	2	0.82247	0.77871	—	-5.3197
ss 00	F_2	(0, 0.1]	598	2	2	0.82247	0.77995	—	-5.1696
00 $3s, s$	Free 1	(0, 0.1]	598	2	1.9649	4.1123	3.5315	-1.7561	-14.124
00 $3s, s$	F_1	(0, 0.1]	598	2	2	4.1123	3.8623	—	-6.0813
00 $3s, s$	F_2	(0, 0.1]	598	2	2	4.1123	3.964	—	-3.6065
$3s, s$ 00	Free 1	(0, 0.1]	598	2	1.966	4.1123	3.545	-1.6988	-13.795
$3s, s$ 00	F_1	(0, 0.1]	598	2	2	4.1123	3.8657	—	-5.9968
$3s, s$ 00	F_2	(0, 0.1]	598	2	2	4.1123	3.9643	—	-3.6009
ss $3s, s$	Free 1	(0, 0.1]	598	2	1.9883	1.6449	1.5661	-0.58466	-4.7937
ss $3s, s$	F_1	(0, 0.1]	598	2	2	1.6449	1.6134	—	-1.9141
ss $3s, s$	F_2	(0, 0.1]	598	2	2	1.6449	1.6276	—	-1.0534
$3s, s$ ss	Free 1	(0, 0.1]	598	2	1.9902	1.6449	1.576	-0.48899	-4.1922
$3s, s$ ss	F_1	(0, 0.1]	598	2	2	1.6449	1.6157	—	-1.7746
$3s, s$ ss	F_2	(0, 0.1]	598	2	2	1.6449	1.6276	—	-1.0534

Table 40: Compact boson small intervals: next-order parameters, with c_2 multiplying $x^{r^{(2)}}$. F_2 imposes both manuscript powers and estimates both coefficients. Empty analytical cells mean that the manuscript does not supply the coefficient. The displayed windows and numerical values also appear in the per-pair figures in Sec. 6.5. Percentage discrepancies use Eq. (88).

Direction	Model	W_x	$r_{\text{CFT}}^{(2)}$	$r^{(2)}$	c_2^{CFT}	c_2	$\eta_{r,2}$	$\eta_{c,2}$
00 ss	F_2	(0, 0.1]	4	4	-0.1917	—	—	—
ss 00	F_2	(0, 0.1]	4	4	-0.18219	—	—	—
00 $3s, s$	F_2	(0, 0.1]	4	4	-15.02	—	—	—
$3s, s$ 00	F_2	(0, 0.1]	4	4	-14.541	—	—	—
ss $3s, s$	F_2	(0, 0.1]	4	4	-2.0896	—	—	—
$3s, s$ ss	F_2	(0, 0.1]	4	4	-1.7507	—	—	—

Table 41: Compact boson small intervals: numerical fit errors on each row's own window. RMSE uses Eq. (62); the normalized percentages E_{fit} and E_{CV} use Eq. (87). The residual plots show pointwise percentages instead. All models and all six directions use the same window for this CFT and interval type.

Direction	Model	W_x	m	RMSE	E_{fit} [%]	E_{CV} [%]
00 ss	Free 1	(0, 0.1]	598	1.742×10^{-5}	1.0869	1.1352
00 ss	F_1	(0, 0.1]	598	1.744×10^{-5}	1.0879	1.1167
00 ss	F_2	(0, 0.1]	598	1.741×10^{-5}	1.0862	1.1453
ss 00	Free 1	(0, 0.1]	598	1.737×10^{-5}	1.0838	1.1319
ss 00	F_1	(0, 0.1]	598	1.739×10^{-5}	1.0847	1.1133
ss 00	F_2	(0, 0.1]	598	1.736×10^{-5}	1.0832	1.142
00 $3s, s$	Free 1	(0, 0.1]	598	6.866×10^{-5}	0.86378	0.90223
00 $3s, s$	F_1	(0, 0.1]	598	9.993×10^{-5}	1.2571	1.2969
00 $3s, s$	F_2	(0, 0.1]	598	6.343×10^{-5}	0.79788	0.84032
$3s, s$ 00	Free 1	(0, 0.1]	598	6.562×10^{-5}	0.82472	0.86069
$3s, s$ 00	F_1	(0, 0.1]	598	9.614×10^{-5}	1.2083	1.2463
$3s, s$ 00	F_2	(0, 0.1]	598	6.045×10^{-5}	0.75982	0.7993
ss $3s, s$	Free 1	(0, 0.1]	598	1.453×10^{-5}	0.43746	0.45738
ss $3s, s$	F_1	(0, 0.1]	598	1.765×10^{-5}	0.53149	0.54888
ss $3s, s$	F_2	(0, 0.1]	598	1.400×10^{-5}	0.42169	0.4446
$3s, s$ ss	Free 1	(0, 0.1]	598	1.241×10^{-5}	0.37329	0.38904
$3s, s$ ss	F_1	(0, 0.1]	598	1.498×10^{-5}	0.45055	0.46479
$3s, s$ ss	F_2	(0, 0.1]	598	1.198×10^{-5}	0.3602	0.3782

G.3 Large intervals: Ising

Table 42: Ising large intervals: leading parameters for all three models, grouped by ordered pair. Amplitudes multiply y^r itself, including powers of π . W_y and m give the fitting interval and count. Fixed models impose $r^{(1)} = \beta_{\text{CFT}}$; Free 1 estimates the leading power. Percentage discrepancies use Eq. (88).

Direction	Model	W_y	m	$r_{\text{CFT}}^{(1)}$	$r^{(1)}$	c_1^{CFT}	c_1	$\eta_{r,1}$	$\eta_{c,1}$
$I \sigma$	Free 1	(0, 0.002]	260	0.25	0.26465	0.61965	0.71994	5.8606	16.184
$I \sigma$	F_1	(0, 0.002]	260	0.25	0.25	0.61965	0.65339	—	5.4442
$I \sigma$	F_2	(0, 0.002]	260	0.25	0.25	0.61965	0.61463	—	-0.81098
σI	Free 1	(0, 0.002]	260	0.25	0.26465	0.61965	0.71995	5.8612	16.186
σI	F_1	(0, 0.002]	260	0.25	0.25	0.61965	0.65339	—	5.4445
σI	F_2	(0, 0.002]	260	0.25	0.25	0.61965	0.61462	—	-0.81139
$I \varepsilon$	Free 1	(0, 0.002]	260	2	1.9985	3.2899	3.2548	-0.074487	-1.0674
$I \varepsilon$	F_1	(0, 0.002]	260	2	2	3.2899	3.2861	—	-0.11533
$I \varepsilon$	F_2	(0, 0.002]	260	2	2	3.2899	3.289	—	-0.024918
εI	Free 1	(0, 0.002]	260	2	2.0013	3.2899	3.3242	0.065315	1.0443
εI	F_1	(0, 0.002]	260	2	2	3.2899	3.2964	—	0.19921
εI	F_2	(0, 0.002]	260	2	2	3.2899	3.2938	—	0.11818
$\sigma \varepsilon$	Free 1	(0, 0.002]	260	0.25	0.25058	0.15491	0.15691	0.23227	1.2904
$\sigma \varepsilon$	F_1	(0, 0.002]	260	0.25	0.25	0.15491	0.15631	—	0.9017
$\sigma \varepsilon$	F_2	(0, 0.002]	260	0.25	0.25	0.15491	0.15595	—	0.67037
$\varepsilon \sigma$	Free 1	(0, 0.002]	260	0.25	0.25544	0.15491	0.16288	2.1754	5.1411
$\varepsilon \sigma$	F_1	(0, 0.002]	260	0.25	0.25	0.15491	0.15712	—	1.4228
$\varepsilon \sigma$	F_2	(0, 0.002]	260	0.25	0.25	0.15491	0.15365	—	-0.81285

Table 43: Ising large intervals: next-order parameters, with c_2 multiplying $y^{r^{(2)}}$. F_2 imposes both manuscript powers and estimates both coefficients. Empty analytical cells mean that the manuscript does not supply the coefficient. The displayed windows and numerical values also appear in the per-pair figures in Sec. 6.6. Percentage discrepancies use Eq. (88).

Direction	Model	W_y	$r_{\text{CFT}}^{(2)}$	$r^{(2)}$	c_2^{CFT}	c_2	$\eta_{r,2}$	$\eta_{c,2}$
$I \sigma$	F_2	(0, 0.002]	0.5	0.5		0.20243	—	
σI	F_2	(0, 0.002]	0.5	0.5		0.20245	—	
$I \varepsilon$	F_2	(0, 0.002]	4	4		-1070.2	—	
εI	F_2	(0, 0.002]	4	4		959.21	—	
$\sigma \varepsilon$	F_2	(0, 0.002]	0.5	0.5		0.0018716	—	
$\varepsilon \sigma$	F_2	(0, 0.002]	0.5	0.5		0.018087	—	

Table 44: Using large intervals: numerical fit errors on each row's own window. RMSE uses Eq. (62); the normalized percentages E_{fit} and E_{CV} use Eq. (87). The residual plots show pointwise percentages instead. All models and all six directions use the same window for this CFT and interval type.

Direction	Model	W_y	m	RMSE	E_{fit} [%]	E_{CV} [%]
$I \mid \sigma$	Free 1	$(0, 0.002]$	260	2.156×10^{-5}	0.017352	0.01755
$I \mid \sigma$	F_1	$(0, 0.002]$	260	4.858×10^{-4}	0.39088	0.39445
$I \mid \sigma$	F_2	$(0, 0.002]$	260	1.523×10^{-5}	0.012258	0.012375
$\sigma \mid I$	Free 1	$(0, 0.002]$	260	2.150×10^{-5}	0.017302	0.0175
$\sigma \mid I$	F_1	$(0, 0.002]$	260	4.858×10^{-4}	0.39091	0.39449
$\sigma \mid I$	F_2	$(0, 0.002]$	260	1.512×10^{-5}	0.012167	0.012282
$I \mid \varepsilon$	Free 1	$(0, 0.002]$	260	1.190×10^{-9}	0.017016	0.017287
$I \mid \varepsilon$	F_1	$(0, 0.002]$	260	2.346×10^{-9}	0.033562	0.033674
$I \mid \varepsilon$	F_2	$(0, 0.002]$	260	1.217×10^{-9}	0.017404	0.01769
$\varepsilon \mid I$	Free 1	$(0, 0.002]$	260	9.799×10^{-10}	0.013973	0.014171
$\varepsilon \mid I$	F_1	$(0, 0.002]$	260	2.030×10^{-9}	0.028953	0.029255
$\varepsilon \mid I$	F_2	$(0, 0.002]$	260	9.430×10^{-10}	0.013446	0.013677
$\sigma \mid \varepsilon$	Free 1	$(0, 0.002]$	260	5.939×10^{-6}	0.019975	0.020151
$\sigma \mid \varepsilon$	F_1	$(0, 0.002]$	260	7.515×10^{-6}	0.025277	0.025317
$\sigma \mid \varepsilon$	F_2	$(0, 0.002]$	260	6.027×10^{-6}	0.020272	0.020447
$\varepsilon \mid \sigma$	Free 1	$(0, 0.002]$	260	4.604×10^{-6}	0.015407	0.015572
$\varepsilon \mid \sigma$	F_1	$(0, 0.002]$	260	4.358×10^{-5}	0.14583	0.14705
$\varepsilon \mid \sigma$	F_2	$(0, 0.002]$	260	4.134×10^{-6}	0.013835	0.013973

G.4 Large intervals: compact boson

Table 45: Compact boson large intervals: leading parameters for all three models, grouped by ordered pair. Amplitudes multiply y^r itself, including powers of π . W_y and m give the fitting interval and count. Fixed models impose $r^{(1)} = \beta_{\text{CFT}}$; Free 1 estimates the leading power. Percentage discrepancies use Eq. (88).

Direction	Model	W_y	m	$r_{\text{CFT}}^{(1)}$	$r^{(1)}$	c_1^{CFT}	c_1	$\eta_{r,1}$	$\eta_{c,1}$
00 ss	Free 1	(0, 0.005]	317	1	0.99926	1.2337	1.2561	-0.07446	1.8189
00 ss	F_1	(0, 0.005]	317	1	1	1.2337	1.2616	—	2.2645
00 ss	F_2	(0, 0.005]	317	1	1	1.2337	1.2627	—	2.3499
ss 00	Free 1	(0, 0.005]	317	1	0.99924	1.2337	1.256	-0.075663	1.8109
ss 00	F_1	(0, 0.005]	317	1	1	1.2337	1.2616	—	2.2636
ss 00	F_2	(0, 0.005]	317	1	1	1.2337	1.2627	—	2.351
00 $3s, s$	Free 1	(0, 0.005]	317	5	4.9836	75.109	66.924	-0.32701	-10.897
00 $3s, s$	F_1	(0, 0.005]	317	5	5	75.109	73.093	—	-2.6829
00 $3s, s$	F_2	(0, 0.005]	317	5	5	75.109	74.307	—	-1.0676
$3s, s$ 00	Free 1	(0, 0.005]	317	5	4.9838	75.109	66.96	-0.32323	-10.849
$3s, s$ 00	F_1	(0, 0.005]	317	5	5	75.109	73.058	—	-2.7295
$3s, s$ 00	F_2	(0, 0.005]	317	5	5	75.109	74.252	—	-1.141
ss $3s, s$	Free 1	(0, 0.005]	317	2	1.9914	3.2899	3.069	-0.42763	-6.7143
ss $3s, s$	F_1	(0, 0.005]	317	2	2	3.2899	3.2185	—	-2.1683
ss $3s, s$	F_2	(0, 0.005]	317	2	2	3.2899	3.2529	—	-1.1234
$3s, s$ ss	Free 1	(0, 0.005]	317	2	1.9912	3.2899	3.0643	-0.43847	-6.8552
$3s, s$ ss	F_1	(0, 0.005]	317	2	2	3.2899	3.2176	—	-2.1982
$3s, s$ ss	F_2	(0, 0.005]	317	2	2	3.2899	3.2528	—	-1.1274

Table 46: Compact boson large intervals: next-order parameters, with c_2 multiplying $y^{r^{(2)}}$. F_2 imposes both manuscript powers and estimates both coefficients. Empty analytical cells mean that the manuscript does not supply the coefficient. The displayed windows and numerical values also appear in the per-pair figures in Sec. 6.7. Percentage discrepancies use Eq. (88).

Direction	Model	W_y	$r_{\text{CFT}}^{(2)}$	$r^{(2)}$	c_2^{CFT}	c_2	$\eta_{r,2}$	$\eta_{c,2}$
00 ss	F_2	(0, 0.005]	2	2	-0.33376	—	—	—
ss 00	F_2	(0, 0.005]	2	2	-0.34138	—	—	—
00 $3s, s$	F_2	(0, 0.005]	6	6	-265.58	—	—	—
$3s, s$ 00	F_2	(0, 0.005]	6	6	-261.18	—	—	—
ss $3s, s$	F_2	(0, 0.005]	3	3	-8.6617	—	—	—
$3s, s$ ss	F_2	(0, 0.005]	3	3	-8.8758	—	—	—

Table 47: Compact boson large intervals: numerical fit errors on each row's own window. RMSE uses Eq. (62); the normalized percentages E_{fit} and E_{CV} use Eq. (87). The residual plots show pointwise percentages instead. All models and all six directions use the same window for this CFT and interval type.

Direction	Model	W_y	m	RMSE	E_{fit} [%]	E_{CV} [%]
00 ss	Free 1	(0, 0.005]	317	1.023×10^{-5}	0.40141	0.4069
00 ss	F_1	(0, 0.005]	317	1.028×10^{-5}	0.40315	0.40552
00 ss	F_2	(0, 0.005]	317	1.024×10^{-5}	0.40179	0.40879
ss 00	Free 1	(0, 0.005]	317	1.014×10^{-5}	0.39781	0.40329
ss 00	F_1	(0, 0.005]	317	1.019×10^{-5}	0.39963	0.40199
ss 00	F_2	(0, 0.005]	317	1.015×10^{-5}	0.39819	0.40517
00 $3s, s$	Free 1	(0, 0.005]	317	3.500×10^{-13}	0.87692	0.94986
00 $3s, s$	F_1	(0, 0.005]	317	3.558×10^{-13}	0.89163	0.94989
00 $3s, s$	F_2	(0, 0.005]	317	3.508×10^{-13}	0.87912	0.95369
$3s, s$ 00	Free 1	(0, 0.005]	317	3.580×10^{-13}	0.89749	0.97157
$3s, s$ 00	F_1	(0, 0.005]	317	3.636×10^{-13}	0.91154	0.97081
$3s, s$ 00	F_2	(0, 0.005]	317	3.589×10^{-13}	0.8997	0.9754
ss $3s, s$	Free 1	(0, 0.005]	317	8.501×10^{-8}	0.38482	0.39943
ss $3s, s$	F_1	(0, 0.005]	317	1.005×10^{-7}	0.45481	0.46774
ss $3s, s$	F_2	(0, 0.005]	317	8.538×10^{-8}	0.3865	0.40258
$3s, s$ ss	Free 1	(0, 0.005]	317	8.761×10^{-8}	0.39671	0.4119
$3s, s$ ss	F_1	(0, 0.005]	317	1.034×10^{-7}	0.46815	0.48153
$3s, s$ ss	F_2	(0, 0.005]	317	8.800×10^{-8}	0.39847	0.4152

G.5 Sanity checks and reproducibility

For each retained window, the two-term residual is checked against the matching one-term residual, since that model is nested in the two-term family. Fixed coefficients are independently checked by column-normalized normal equations. Original entropy values are checked against the pre-revision snapshot; the added system sizes extend those raw records. The $\ell = 4, 5$ values remain archived and enter neither the fitted nor displayed data. The principal fitted rows in Appendices A and B adopt the common CFT-wide windows in Eq. (59). Their coefficients and omitted-size errors are refitted independently from the entropy CSV; every fit and training fold has at least forty observations. All three fitted models share an interval across all six directions of each CFT and interval type. CFT coefficients remain analytical inputs, while its residual metrics use the common-window data mask. The dense-scan window choices are recorded explicitly.

The numerical record contains the following files:

- `window_scan.csv`: the historical window-scan record.
- `window_selected.csv`: selected windows, complete parameters, and observation counts; the report reads only Free 1, F_1 , and F_2 .
- `window_retained_cv.csv`: omitted-size refits for the current companion visualization records (the same common-window fits).
- `retained_window_specifications.json`: current explicit window endpoints, checked against every current principal and varied fit.
- `selection_raw_preservation.json` and `selection_audit.json`: raw-data preservation and current audits.
- `f2_summary_fits.csv`: revised principal tables, with CFT-wide windows for all three models and matched-window gains.
- `f2_summary_cv.csv`: all principal models and reference F_1 omitted-size refits; `f2_summary_comparison.csv` records common-window and companion F_2 coefficients, which now coincide.

Conditional least-squares standard errors in the CSVs are descriptive local-curvature measures, not statistical confidence intervals for these deterministic data.

The complete numerical implementation is `src/conformal_blocks/retained_window_fits.py`; plotting and TeX generation are in `report/pair_fit_artifacts.py`. Using the already approved interpreter and threading settings in `SCALING_README.md`, run

```
python report/rebuild_report.py
python report/selection_audit.py
```

Then compile `main.tex` three times and package with the existing report audit. The visualization reads the supplied entropy and fit CSVs directly and does not require the wavefunction caches. The manuscript is identified by source hash and citation, not redistributed in the Overleaf archive.

H Dense sampling and the second manuscript term

The purpose of this scan is to distinguish an improved residual from an improved estimate of a CFT coefficient. In particular, adding a second basis function always reduces the training sum of squares, but need not move the first coefficient towards its theoretical value. We therefore compare F_1 and F_2 on exactly the same observations, using the objective in Eq. (61), and examine both coefficients rather than the residual alone.

H.1 Additional geometries and direct coefficient comparisons

Both CFTs first receive the same additional 47 even sizes between 60 and 6000. Start with $N_j = 2 \text{round}[30 \cdot 100^{j/48}]$ for $j = 0, \dots, 48$, and omit sizes already present. Every new size uses $\ell = 6, 7, 8, 9, 10$ and computes both interval observables for all six ordered pairs. A further 188 Ising sizes complete the progression $N = 200, 204, \dots, 1000$, concentrating observations where the known x^6 term is measurable. The two sets are defined in Eq. (51). No new size exceeds the existing maxima 8192 and 16384. The resulting 1175 Ising and 235 boson new geometries per direction fill the intermediate fractions without changing any previously computed entropy. The exact executed sizes, checks, and timings are retained in `data/dense_grid_manifest.json` and the production CSVs. The subsequent refinement in Appendix H.4 adds 48 further sizes for each CFT while keeping all fitting windows fixed. Every current fit, plotted coefficient and window-sensitivity comparison below uses the combined data; the two grid manifests distinguish the original dense scan from this later refinement.

Let $q = D^A$ with $z = x$ for a small interval, or $q = \delta = \log 2 - D^B$ with $z = y$ for its large complement. With the pair-specific powers r_0, r_1 in Eq. (60), the two models are $F_1 = c_0 z^{r_0}$ and $F_2 = c_0 z^{r_0} + c_1 z^{r_1}$. Both coefficients in F_2 are estimated; neither is fixed to theory. All amplitudes include their powers of π . For the leading discrepancy η_c in Eq. (62), define

$$\Delta_0 = |\eta_c(F_1)| - |\eta_c(F_2)|. \quad (135)$$

A positive Δ_0 , measured in percentage points, means that including the second term improves the leading coefficient on that window. This is distinct from the residual gain G and the omitted-size prediction gain G_{CV} . All reported fits and omitted-size training sets retain at least forty observations.

For the known Ising coefficients, the supplied manuscript gives

$$\begin{aligned} D_{I|\varepsilon}^A &= \frac{2\pi^4}{15}x^4 - \frac{8\pi^6}{45}x^6 + O(x^8), \\ D_{\varepsilon|I}^A &= \frac{2\pi^4}{15}x^4 - \frac{8\pi^6}{21}x^6 + O(x^8). \end{aligned} \quad (136)$$

These are the $p = 1/2$ specializations of the two Ising mixture expressions in the supplied source [2]. No missing subleading coefficient is derived or assumed. The unknown analytical cells remain blank.

H.2 What is resolved by the dense scan

The principal comparisons improve the absolute leading-coefficient discrepancy in 22 of the 24 ordered-pair/interval cases. Of these, 18 also improve the omitted-size prediction residual. Tables 48–55 retain the failures as well as the improvements. A lower residual alone is therefore not counted as recovery of a CFT coefficient.

For boson large intervals, $00 \mid ss$ on $W = (0, 0.005]$ changes the leading discrepancy from 2.2645% under F_1 to 2.3499% under F_2 . This is a failure to improve that coefficient, even if the residual decreases.

For boson large intervals, $ss \mid 00$ on $W = (0, 0.005]$ changes the leading discrepancy from 2.2636% under F_1 to 2.351% under F_2 . This is a failure to improve that coefficient, even if the residual decreases.

The adopted windows are Eq. (59). The large-boson upper cutoff is prescribed as 0.005: Appendix H.4 reports its neighboring-cutoff and omitted-size sensitivity. Improvement of a leading coefficient does not establish recovery of its unknown second coefficient.

For the two known Ising directions, the pooled window $W_x = (0, 0.02]$ gives $c_1 = -170.71$ and -355.38 , respectively. Their signed discrepancies from Eq. (136) are -0.12193% and -2.9657% . The leading coefficient improves in 2 of these two directions; the omitted-size ranges in Table 49 expose the remaining sensitivity of the pooled second coefficients.

At $\ell = 10$, the separate-length fits on $W_x = (0, 0.006]$ contain 82 points each and give $c_1 = -169.27$ and -362.08 . Their discrepancies are -0.96408% and -1.1357% . The leading coefficients differ from theory by about 0.49974% . The dependence on ℓ and on the upper endpoint is visible in Fig. 44; these small omitted-size ranges do not erase the finite-length offset.

The cutoff propagation in Eq. (137) gives $b_1 = 0.0027612$ and 0.0027345 for the same $\ell = 10$ fits. The recorded cutoff variation is much smaller than the observed second-coefficient discrepancy. These values are sensitivity bounds for the tested cutoff variation only.

The separate shared-window diagnostic uses $W_x = W_y = (0, 0.006]$ and improves the leading coefficient in 21 of 24 cases; see Fig. 49. Across all 744 common windows in the recorded scan on the current grid, the largest improvement count is 23 of 24. No single interval in that scan improves all pairs simultaneously. This does not preclude the four CFT/interval-specific common windows now adopted. In particular, one apparent near-match does not establish a universal interval where every second coefficient is resolved.

For example, that same shared window gives pooled Ising second coefficients -1029.7 and -1220.3 . Their signed discrepancies from Eq. (136) are 502.46% and 233.2% . An improved leading amplitude cannot by itself validate the second term. The unknown second coefficients in the tables are estimates of the specified finite- ℓ two-term fits, not established continuum constants. The chosen windows were explored with these deterministic data, so their agreement with theory is an exploratory comparison rather than a blind prediction. Omitted-size refits test sensitivity to individual chain sizes; they do not remove the effect of selecting a window. The fitted c_1 should be interpreted as a coefficient of the specified two-term truncation unless it remains stable as the window changes. The separate- ℓ comparison uses exactly the same F_2 model on $W_x = (0, 0.006]$ for each of the five lengths. It estimates both coefficients independently for each length and adds no new power. Different finite- ℓ leading amplitudes can influence a pooled second coefficient when their relative sampling changes across the window. The separate-length results expose this directly; they are not an extrapolation to $\ell \rightarrow \infty$.

We also propagate the recorded entropy variation under the support-cutoff checks to the two fitted coefficients. For the design matrix $X_{ij} = x_i^{r_j}$, let $P = (X^T X)^{-1} X^T$ and let s_i be the observed span of entropy values when the numerical support cutoff is varied. The resulting sensitivity bound is

$$b_j = \sum_i |P_{ji}| s_i. \quad (137)$$

This bounds coefficient changes if each data perturbation is no larger than its recorded span. It is an empirical cutoff-sensitivity check, not an error bound for the continuum approximation or for all floating-point operations. The implementation uses scaled columns and a pseudoinverse, avoiding the large dynamic range of the powers.

To expose the second term directly, plot

$$U(z) = \frac{q(z)}{z^{r_0}}, \quad t = z^{r_1 - r_0}. \quad (138)$$

The plotted F_2 curve is then a straight line $U = c_0 + c_1 t$. The fit is still performed in q , not in U ; the transformation is only a visualization. Separate symbols retain the subsystem length, so disagreement between lengths is visible without introducing an extra lattice-correction ansatz. At fixed ℓ , increasing N reduces z but does not take a continuum limit with $\ell \rightarrow \infty$. Consequently, denser sampling alone cannot guarantee exact recovery of the continuum coefficients.

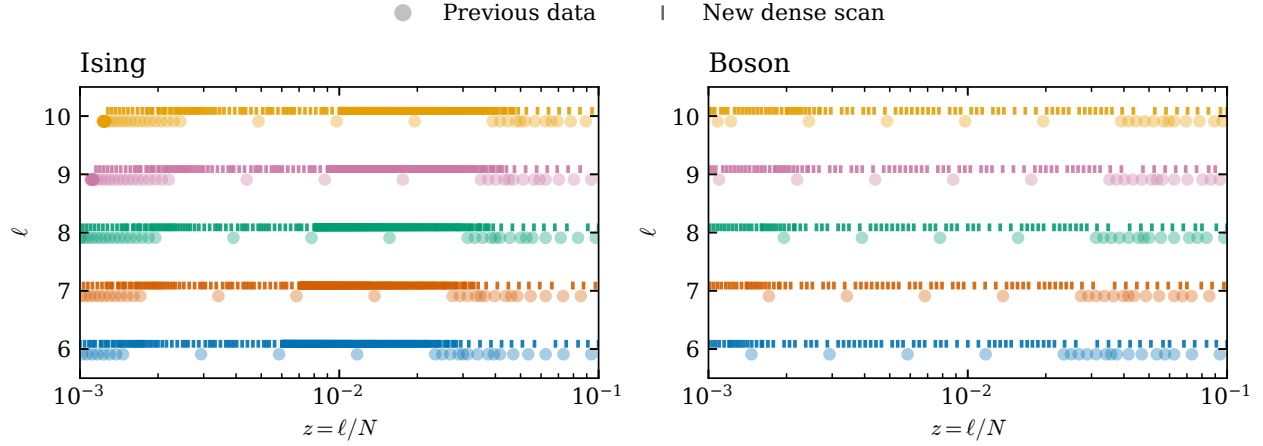


Figure 42: Sampling between 10^{-3} and 10^{-1} , common to both interval observables. Circles are the geometries preceding the dense study; vertical marks include the original dense scan and the subsequent fixed-window refinement in Appendix H.4. Colors identify $\ell = 6, \dots, 10$. Both added grids were chosen from geometry, before their entropies were evaluated, and preserve the largest tested sizes.

H.3 Matched-window results and reproduction

The source `src/conformal_blocks/dense_windows.py` regenerates the scan and matched-window comparisons from completed CSVs. The portable `fitting_datas` notebook includes per-model window controls and the complete scan, selected comparisons, and omitted-size records. Its defaults keep all models and directions in each CFT and interval group on the same window. Separate-length and shared-window figures are explicitly identified diagnostics.

Table 48: Ising small intervals: identical-window coefficient comparison at $p = 1/2$, $\ell = 6, \dots, 10$. The displayed W contains m observations. c_0 multiplies the leading power and c_1 the second manuscript power; both include powers of π . The column Δ_0 is Eq. (135), in percentage points; G_{CV} is the omitted-size residual gain in Eq. (63). Positive values mean improvement, but the two measures test different properties. Blank analytical second coefficients are unavailable. Powers and all model parameters are listed in Table 10.

Direction	W	m	$c_0(F_1)$	$c_0(F_2)$	c_0^{CFT}	$c_1(F_2)$	c_1^{CFT}	Δ_0	G_{CV}
$I \sigma$	(0, 0.02]	1255	0.20573	0.20571	0.20562	0.083833		0.010494	2.077
σI	(0, 0.02]	1255	0.20572	0.2057	0.20562	0.069099		0.0086493	1.8149
$I \varepsilon$	(0, 0.02]	1255	12.83	12.883	12.988	-170.71	-170.91	0.41412	3.8386
εI	(0, 0.02]	1255	12.77	12.882	12.988	-355.38	-366.24	0.86214	15.195
$\sigma \varepsilon$	(0, 0.02]	1255	0.2089	0.20565	0.20562	12.604		1.5777	95.461
$\varepsilon \sigma$	(0, 0.02]	1255	0.20806	0.20566	0.20562	9.3133		1.1658	94.318

Table 49: Ising small intervals: second-coefficient sensitivity for Table 48. The extrema $c_{1,\min}^{(-N)}, c_{1,\max}^{(-N)}$ come from the omitted-size refits and are not confidence bounds. $\eta_1 = 100(c_1/c_1^{\text{CFT}} - 1)$ is blank when theory supplies no coefficient. The final column is $100|c_1 v^{r_1-r_0}/c_0|$, the size of the fitted second term relative to the first at the window upper endpoint v . A fitted nonzero coefficient is not by itself proof of a continuum coefficient.

Direction	W	c_1	c_1^{CFT}	$c_{1,\min}^{(-N)}$	$c_{1,\max}^{(-N)}$	η_1 [%]	Term ratio [%]
$I \sigma$	(0, 0.02]	0.083833		0.076713	0.089011		0.016301
σI	(0, 0.02]	0.069099		0.063043	0.073541		0.013436
$I \varepsilon$	(0, 0.02]	-170.71	-170.91	-181.53	-153.57	-0.12193	0.53
εI	(0, 0.02]	-355.38	-366.24	-366.12	-338.45	-2.9657	1.1035
$\sigma \varepsilon$	(0, 0.02]	12.604		12.595	12.619		2.4516

Direction	W	c_1	c_1^{CFT}	$c_{1,\min}^{(-N)}$	$c_{1,\max}^{(-N)}$	η_1 [%]	Term ratio [%]
$\varepsilon \mid \sigma$	(0, 0.02]	9.3133		9.3048	9.3273		1.8114

Table 50: Compact boson small intervals: identical-window coefficient comparison at $p = 1/2$, $\ell = 6, \dots, 10$. The displayed W contains m observations. c_0 multiplies the leading power and c_1 the second manuscript power; both include powers of π . The column Δ_0 is Eq. (135), in percentage points; G_{CV} is the omitted-size residual gain in Eq. (63). Positive values mean improvement, but the two measures test different properties. Blank analytical second coefficients are unavailable. Powers and all model parameters are listed in Table 12.

Direction	W	m	$c_0(F_1)$	$c_0(F_2)$	c_0^{CFT}	$c_1(F_2)$	c_1^{CFT}	Δ_0	G_{CV}
00 ss	(0, 0.1]	598	0.77866	0.77996	0.82247	-0.1917		0.15793	-2.5678
ss 00	(0, 0.1]	598	0.77871	0.77995	0.82247	-0.18219		0.1501	-2.5842
00 $3s, s$	(0, 0.1]	598	3.8623	3.964	4.1123	-15.02		2.4748	35.203
$3s, s$ 00	(0, 0.1]	598	3.8657	3.9643	4.1123	-14.541		2.3959	35.866
ss $3s, s$	(0, 0.1]	598	1.6134	1.6276	1.6449	-2.0896		0.86075	18.998
$3s, s$ ss	(0, 0.1]	598	1.6157	1.6276	1.6449	-1.7507		0.72113	18.631

Table 51: Compact boson small intervals: second-coefficient sensitivity for Table 50. The extrema $c_{1,\min}^{(-N)}, c_{1,\max}^{(-N)}$ come from the omitted-size refits and are not confidence bounds. $\eta_1 = 100(c_1/c_1^{\text{CFT}} - 1)$ is blank when theory supplies no coefficient. The final column is $100|c_1 v^{r_1-r_0}/c_0|$, the size of the fitted second term relative to the first at the window upper endpoint v . A fitted nonzero coefficient is not by itself proof of a continuum coefficient.

Direction	W	c_1	c_1^{CFT}	$c_{1,\min}^{(-N)}$	$c_{1,\max}^{(-N)}$	η_1 [%]	Term ratio [%]
00 ss	(0, 0.1]	-0.1917		-0.50151	0.1399		0.24579
ss 00	(0, 0.1]	-0.18219		-0.4909	0.14816		0.23359
00 $3s, s$	(0, 0.1]	-15.02		-16.149	-13.84		3.7891
$3s, s$ 00	(0, 0.1]	-14.541		-15.603	-13.438		3.668
ss $3s, s$	(0, 0.1]	-2.0896		-2.3303	-1.8108		1.2839
$3s, s$ ss	(0, 0.1]	-1.7507		-1.9474	-1.5292		1.0756

Table 52: Using large intervals: identical-window coefficient comparison at $p = 1/2$, $\ell = 6, \dots, 10$. The displayed W contains m observations. c_0 multiplies the leading power and c_1 the second manuscript power; both include powers of π . The column Δ_0 is Eq. (135), in percentage points; G_{CV} is the omitted-size residual gain in Eq. (63). Positive values mean improvement, but the two measures test different properties. Blank analytical second coefficients are unavailable. Powers and all model parameters are listed in Table 14.

Direction	W	m	$c_0(F_1)$	$c_0(F_2)$	c_0^{CFT}	$c_1(F_2)$	c_1^{CFT}	Δ_0	G_{CV}
$I \mid \sigma$	(0, 0.002]	260	0.65339	0.61463	0.61965	0.20243		4.6332	96.863
$\sigma \mid I$	(0, 0.002]	260	0.65339	0.61462	0.61965	0.20245		4.6331	96.886
$I \mid \varepsilon$	(0, 0.002]	260	3.2861	3.289	3.2899	-1070.2		0.090409	47.468
$\varepsilon \mid I$	(0, 0.002]	260	3.2964	3.2938	3.2899	959.21		0.081036	53.25
$\sigma \mid \varepsilon$	(0, 0.002]	260	0.15631	0.15595	0.15491	0.0018716		0.23133	19.238
$\varepsilon \mid \sigma$	(0, 0.002]	260	0.15712	0.15365	0.15491	0.018087		0.60996	90.498

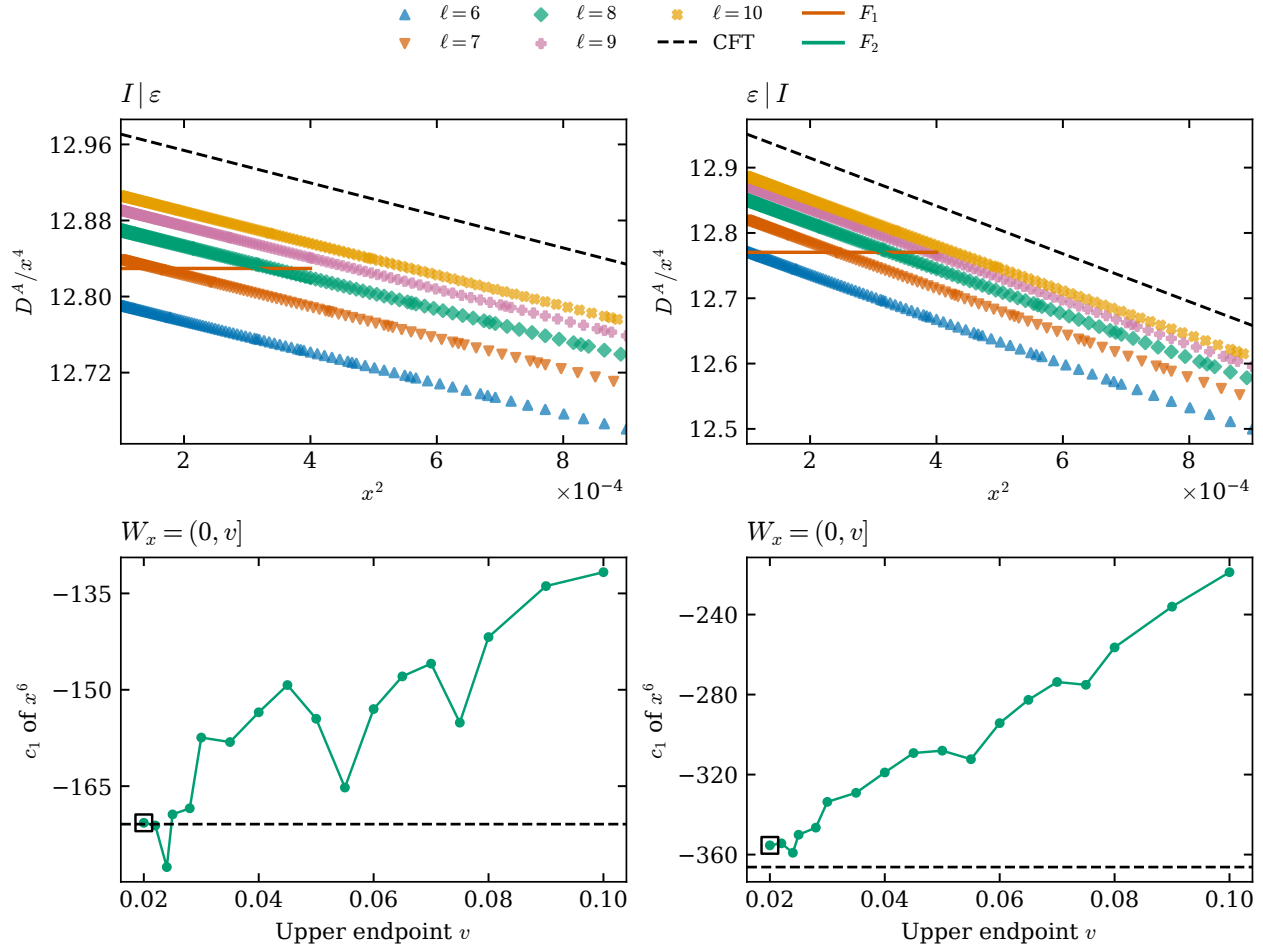


Figure 43: Direct test of the two known Ising second-order coefficients at $p = 1/2$. Top: the normalization in Eq. (138), zoomed to $0.01 \leq x \leq 0.03$ with $\ell = 6, \dots, 10$, to resolve the fitted slope and the offsets between lengths. The black dashed line is Eq. (136); the fitted lines are drawn only on their actual windows. Bottom: the inferred x^6 coefficient as the upper endpoint varies while the displayed lower endpoint stays fixed. Squares mark the reported windows. $I | \varepsilon$: $W_x = (0, 0.02]$, $m = 1255$, $c_0 = 12.883$, $c_1 = -170.71$; $\varepsilon | I$: $W_x = (0, 0.02]$, $m = 1255$, $c_0 = 12.882$, $c_1 = -355.38$. Both F_2 coefficients are free; the objective is in D^A , not in D^A/x^4 .

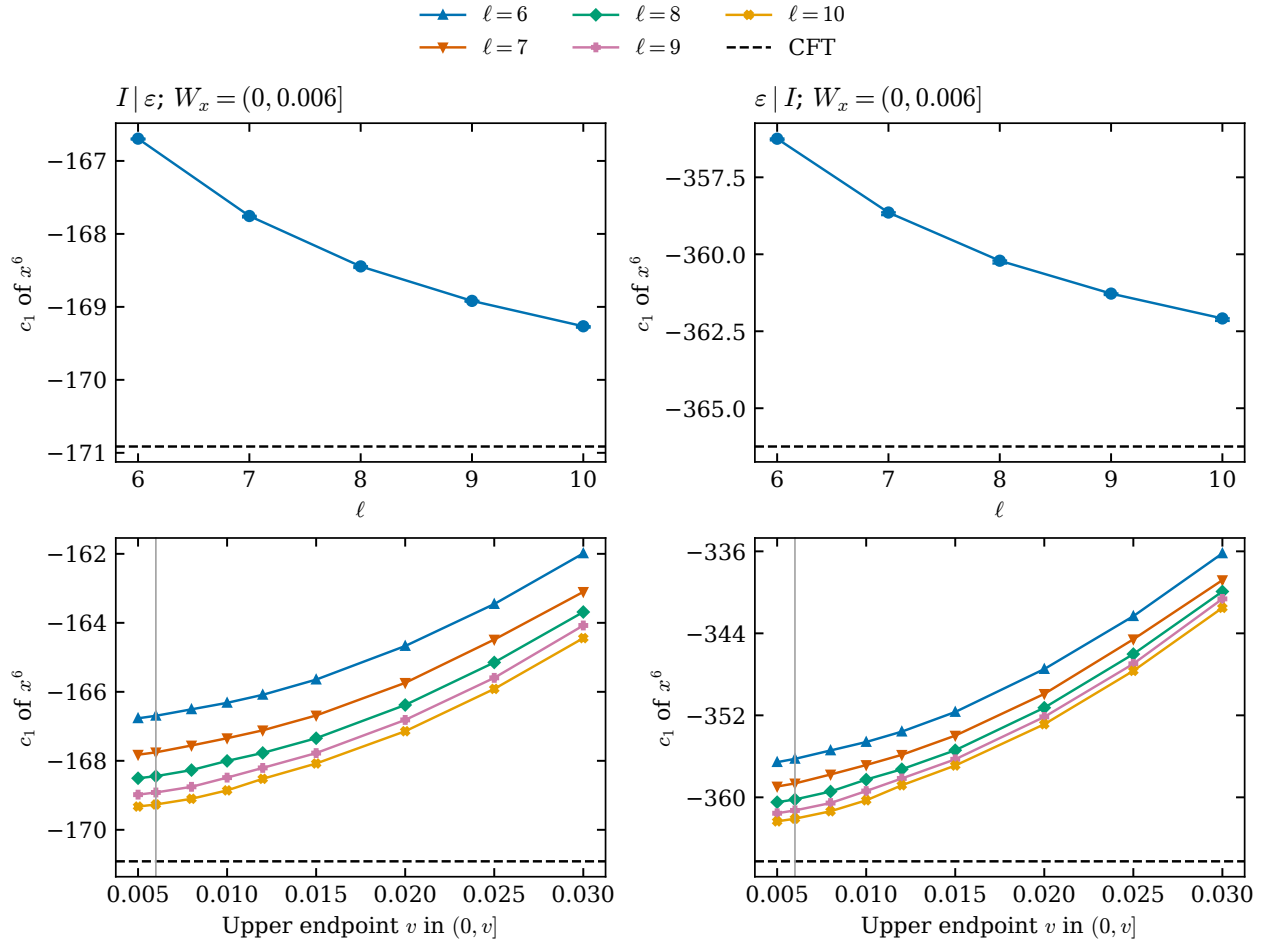


Figure 44: The same two-power Ising fit performed separately at each subsystem length. Both x^4 and x^6 coefficients are fitted freely with their exponents fixed; no lattice-correction term is added. Top: vertical bars are the range of c_1 across omitted-size refits, not statistical confidence intervals; the marker is the full-data estimate. The black dashed values are Eq. (136). Bottom: coefficient drift as the upper endpoint varies, with each length fitted independently; the vertical line marks the tabulated window. Every fit and every omitted-size training set contains at least forty points. Table 56 gives the exact counts, both coefficients, and discrepancies.

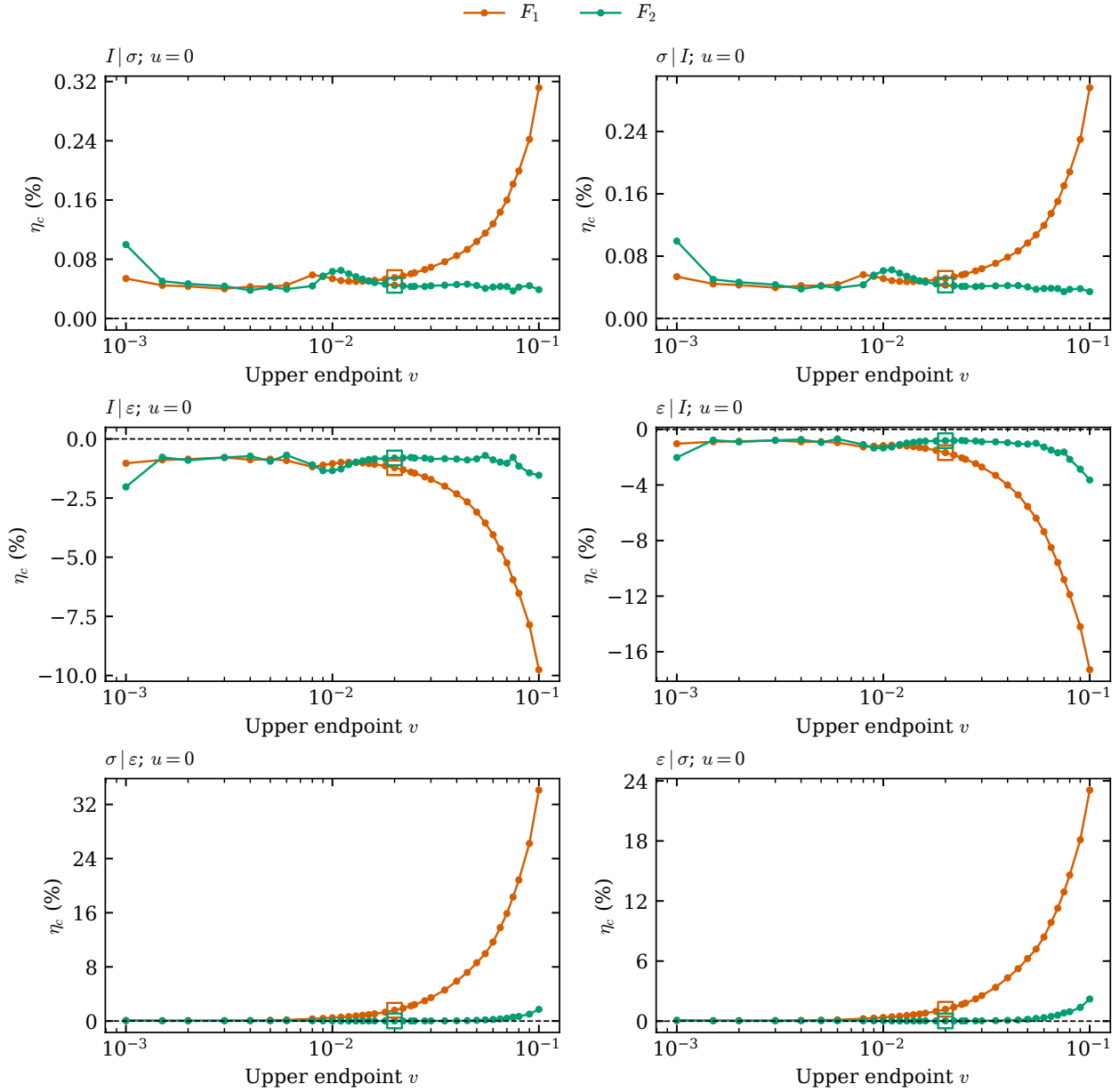


Figure 45: Using small intervals: leading-coefficient discrepancy η_c against the upper endpoint $v \leq 0.1$ of $W = [u, v]$. Each panel states its fixed u ; only windows with at least forty points in every omitted-size training set are shown. F_1 and F_2 always use identical points; F_2 fixes both manuscript powers. Squares mark the principal-table windows. All ordinates are linear. The zero line is the theoretical leading coefficient, with its full power of π .

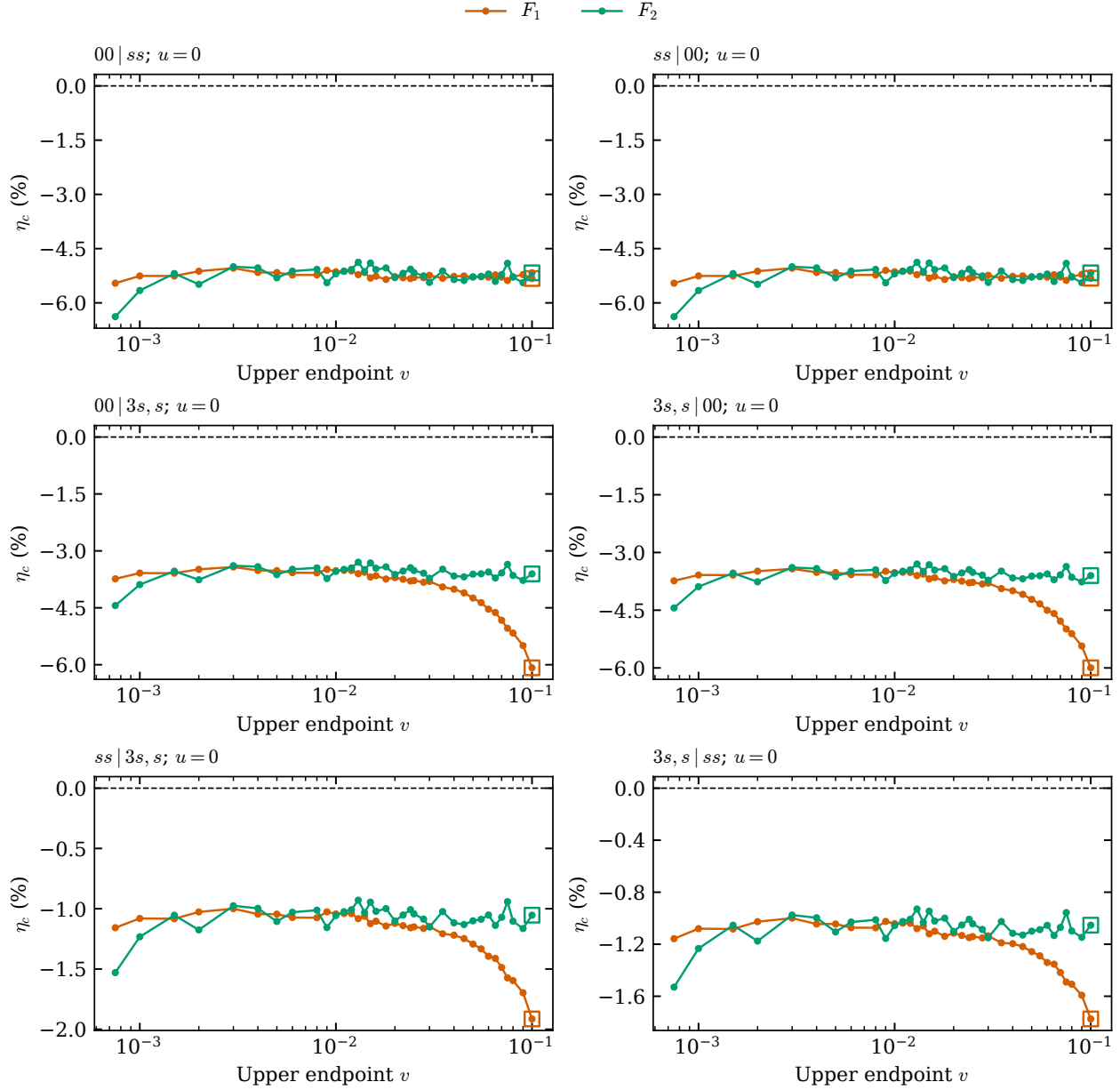


Figure 46: Compact boson small intervals: leading-coefficient discrepancy η_c against the upper endpoint $v \leq 0.1$ of $W = [u, v]$. Each panel states its fixed u ; only windows with at least forty points in every omitted-size training set are shown. F_1 and F_2 always use identical points; F_2 fixes both manuscript powers. Squares mark the principal-table windows. All ordinates are linear. The zero line is the theoretical leading coefficient, with its full power of π .

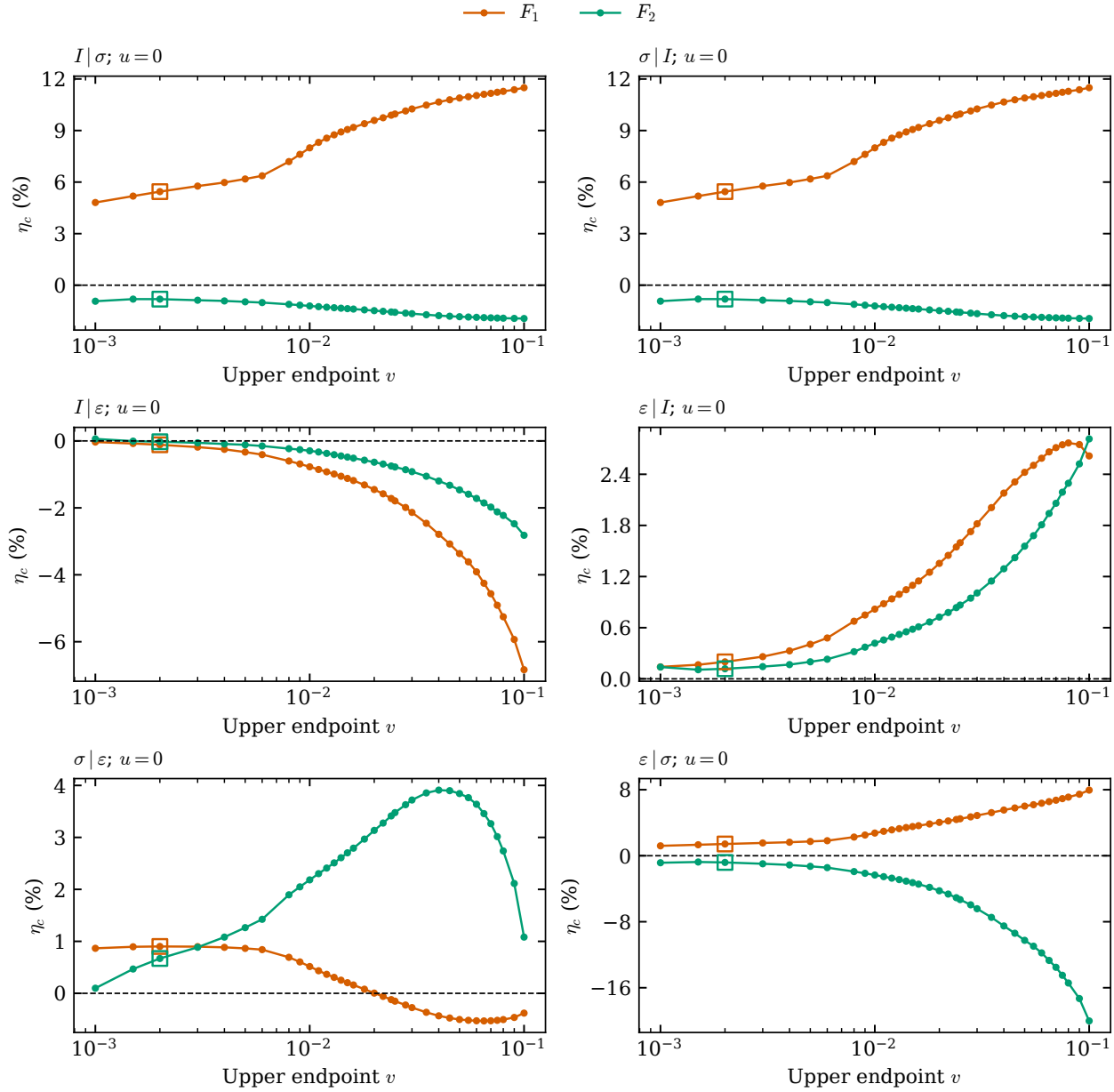


Figure 47: Using large intervals: leading-coefficient discrepancy η_c against the upper endpoint $v \leq 0.1$ of $W = [u, v]$. Each panel states its fixed u ; only windows with at least forty points in every omitted-size training set are shown. F_1 and F_2 always use identical points; F_2 fixes both manuscript powers. Squares mark the principal-table windows. All ordinates are linear. The zero line is the theoretical leading coefficient, with its full power of π .

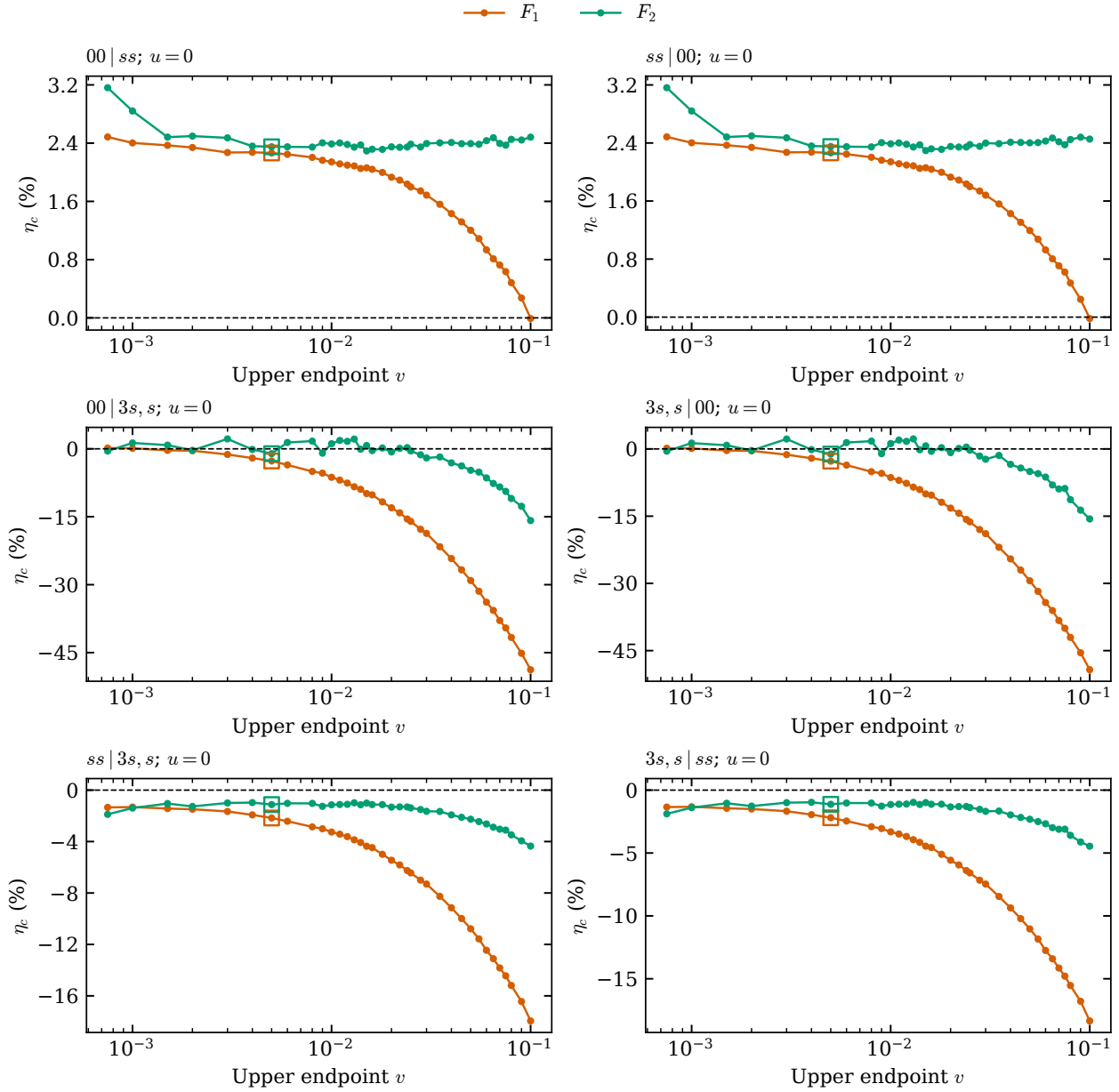


Figure 48: Compact boson large intervals: leading-coefficient discrepancy η_c against the upper endpoint $v \leq 0.1$ of $W = [u, v]$. Each panel states its fixed u ; only windows with at least forty points in every omitted-size training set are shown. F_1 and F_2 always use identical points; F_2 fixes both manuscript powers. Squares mark the principal-table windows. All ordinates are linear. The zero line is the theoretical leading coefficient, with its full power of π .

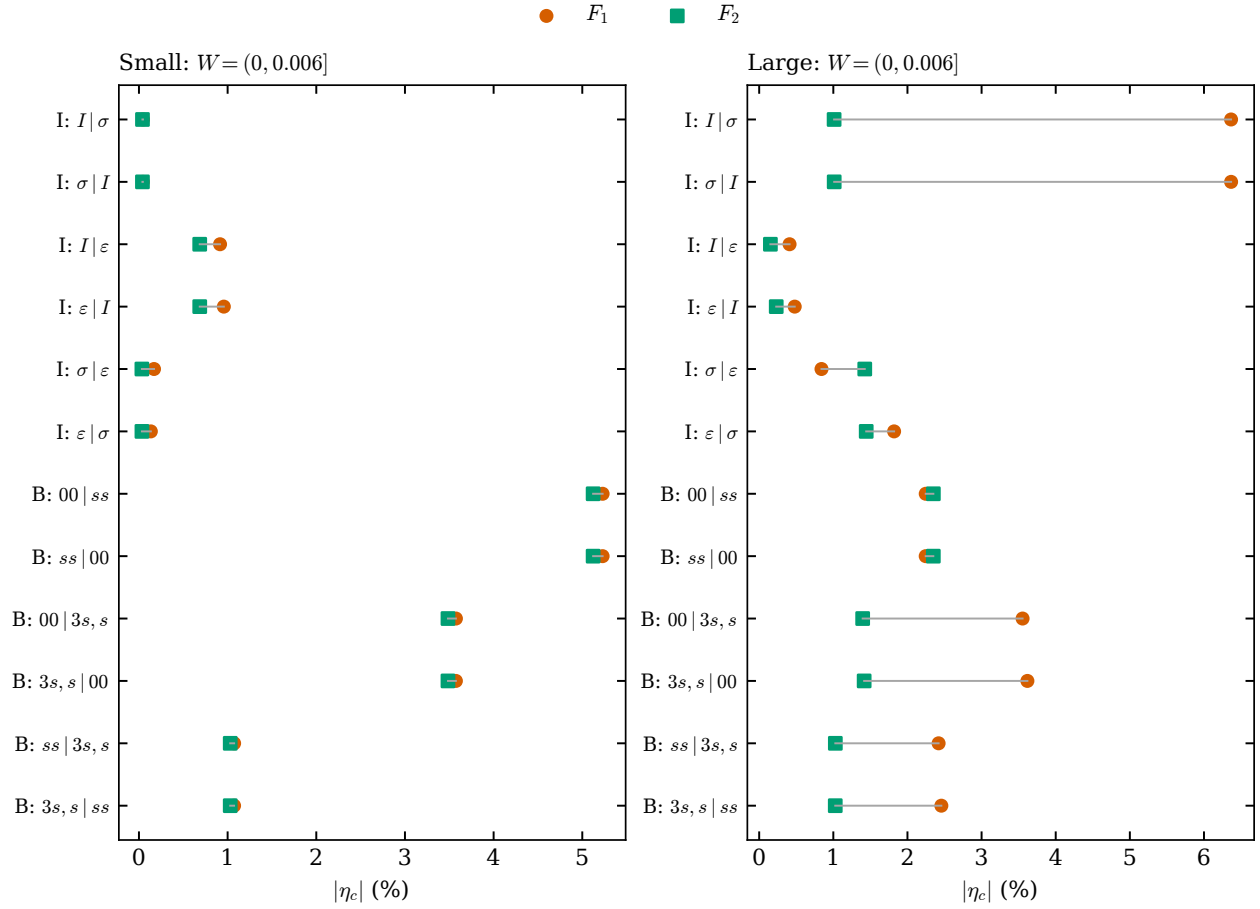


Figure 49: A separate single-window diagnostic across all ordered pairs, using the current enlarged data and distinct from the adopted CFT-specific windows in Eq. (59). I and B identify Ising and compact boson. A leftward shift from the circle (F_1) to the square (F_2) improves the leading coefficient. The same displayed interval and the same observations are used for both models; powers are fixed to the pair-specific manuscript values. Coefficients include powers of π , and $|\eta_c|$ is the absolute percentage discrepancy in Eq. (62). The original largest sizes and $\ell = 6, \dots, 10$ are retained.

Table 53: Ising large intervals: second-coefficient sensitivity for Table 52. The extrema $c_{1,\min}^{(-N)}, c_{1,\max}^{(-N)}$ come from the omitted-size refits and are not confidence bounds. $\eta_1 = 100(c_1/c_1^{\text{CFT}} - 1)$ is blank when theory supplies no coefficient. The final column is $100|c_1 v^{r_1-r_0}/c_0|$, the size of the fitted second term relative to the first at the window upper endpoint v . A fitted nonzero coefficient is not by itself proof of a continuum coefficient.

Direction	W	c_1	c_1^{CFT}	$c_{1,\min}^{(-N)}$	$c_{1,\max}^{(-N)}$	η_1 [%]	Term ratio [%]
$I \sigma$	(0, 0.002]	0.20243		0.20232	0.20251		6.9649
σI	(0, 0.002]	0.20245		0.20234	0.20253		6.9657
$I \varepsilon$	(0, 0.002]	-1070.2		-1095.7	-1053.3		0.13015
εI	(0, 0.002]	959.21		944.43	976.81		0.11649
$\sigma \varepsilon$	(0, 0.002]	0.0018716		0.0018486	0.0019195		0.25379
$\varepsilon \sigma$	(0, 0.002]	0.018087		0.018053	0.01811		2.4894

Table 54: Compact boson large intervals: identical-window coefficient comparison at $p = 1/2$, $\ell = 6, \dots, 10$. The displayed W contains m observations. c_0 multiplies the leading power and c_1 the second manuscript power; both include powers of π . The column Δ_0 is Eq. (135), in percentage points; G_{CV} is the omitted-size residual gain in Eq. (63). Positive values mean improvement, but the two measures test different properties. Blank analytical second coefficients are unavailable. Powers and all model parameters are listed in Table 16.

Direction	W	m	$c_0(F_1)$	$c_0(F_2)$	c_0^{CFT}	$c_1(F_2)$	c_1^{CFT}	Δ_0	G_{CV}
00 ss	(0, 0.005]	317	1.2616	1.2627	1.2337	-0.33376		-0.085416	-0.80584
ss 00	(0, 0.005]	317	1.2616	1.2627	1.2337	-0.34138		-0.087364	-0.7901
00 $3s, s$	(0, 0.005]	317	73.093	74.307	75.109	-265.58		1.6153	-0.40053
$3s, s$ 00	(0, 0.005]	317	73.058	74.252	75.109	-261.18		1.5885	-0.47282
ss $3s, s$	(0, 0.005]	317	3.2185	3.2529	3.2899	-8.6617		1.0449	13.931
$3s, s$ ss	(0, 0.005]	317	3.2176	3.2528	3.2899	-8.8758		1.0707	13.775

Table 55: Compact boson large intervals: second-coefficient sensitivity for Table 54. The extrema $c_{1,\min}^{(-N)}, c_{1,\max}^{(-N)}$ come from the omitted-size refits and are not confidence bounds. $\eta_1 = 100(c_1/c_1^{\text{CFT}} - 1)$ is blank when theory supplies no coefficient. The final column is $100|c_1 v^{r_1-r_0}/c_0|$, the size of the fitted second term relative to the first at the window upper endpoint v . A fitted nonzero coefficient is not by itself proof of a continuum coefficient.

Direction	W	c_1	c_1^{CFT}	$c_{1,\min}^{(-N)}$	$c_{1,\max}^{(-N)}$	η_1 [%]	Term ratio [%]
00 ss	(0, 0.005]	-0.33376		-0.53362	-0.14815		0.13216
ss 00	(0, 0.005]	-0.34138		-0.53762	-0.15672		0.13518
00 $3s, s$	(0, 0.005]	-265.58		-373.74	-178.05		1.7871
$3s, s$ 00	(0, 0.005]	-261.18		-372.89	-175.63		1.7588
ss $3s, s$	(0, 0.005]	-8.6617		-9.6333	-8.0378		1.3314
$3s, s$ ss	(0, 0.005]	-8.8758		-9.8805	-8.2283		1.3643

Table 56: Known Ising coefficient checks separated by subsystem length, at $p = 1/2$ and the common window $W_x = (0, 0.006]$. Each row fits $c_0 x^4 + c_1 x^6$ to the stated ℓ only; m counts distinct chain sizes. The theoretical coefficients are Eq. (136); $\eta_j = 100(c_j/c_j^{\text{CFT}} - 1)$, $j = 0, 1$, are signed percentage discrepancies. The final column gives the extrema from omitting each chain size and refitting. These deterministic ranges are sensitivity measures, not confidence intervals.

Direction	ℓ	m	c_0	η_0 [%]	c_1	η_1 [%]	c_1 omitted-size range
$I \varepsilon$	6	97	12.808	-1.3871	-166.69	-2.4684	[-166.71, -166.69]
$I \varepsilon$	7	92	12.855	-1.0194	-167.75	-1.8482	[-167.77, -167.75]
$I \varepsilon$	8	89	12.886	-0.78063	-168.45	-1.4442	[-168.47, -168.44]
$I \varepsilon$	9	86	12.908	-0.61689	-168.92	-1.1677	[-168.93, -168.92]

Direction	ℓ	m	c_0	η_0 [%]	c_1	η_1 [%]	c_1	omitted-size range
$I \mid \varepsilon$	10	82	12.923	-0.49974	-169.27	-0.96408		$[-169.28, -169.26]$
$\varepsilon \mid I$	6	97	12.808	-1.3872	-356.25	-2.7295		$[-356.3, -356.24]$
$\varepsilon \mid I$	7	92	12.855	-1.0194	-358.64	-2.075		$[-358.73, -358.64]$
$\varepsilon \mid I$	8	89	12.886	-0.78069	-360.21	-1.6477		$[-360.3, -360.2]$
$\varepsilon \mid I$	9	86	12.908	-0.61696	-361.28	-1.3561		$[-361.32, -361.27]$
$\varepsilon \mid I$	10	82	12.923	-0.4998	-362.08	-1.1357		$[-362.16, -362.08]$

H.4 Zero-anchored common windows on the denser data

The small-Ising and small-boson windows remain $(0, 0.02]$ and $(0, 0.1]$. The large-interval comparison adopts $(0, 0.002]$ for Ising and the prescribed $(0, 0.005]$ for the compact boson. These are the four windows of Eq. (59); every model and all six directions within each group use identical observations. The preceding data refinement added 48 even system sizes for each CFT, with all five lengths $\ell = 6, 7, 8, 9, 10$ at each size. The largest sizes remain 8192 and 16384. Every added geometry enters the corresponding small-interval window; only those satisfying the separate large-interval cutoff enter its fits. Both observables are computed and preserved. The data refinement initially kept all four windows fixed. A subsequent zero-anchored search selected the historical large-boson cutoff 0.0047. The current revision changes that cutoff to the prescribed 0.005, on exactly the same raw data. All theoretical powers are unchanged. At the current cutoff, F_2 improves four of the six large-boson leading coefficients; the all-six target is not met.

The new sizes are chosen using geometry alone. For each CFT, seed the even sizes at which each length first enters its large-interval window. Repeatedly bisect the largest gap in $\log N$, choosing the nearest unused even size to the geometric midpoint. Add 32 sizes in $3000 \leq N \leq 8192$ for Ising or $3750 \leq N \leq 16384$ for the boson. Add 16 further sizes in $500 \leq N \leq 3000$ or $100 \leq N \leq 3750$, respectively. Existing sizes are never recomputed or replaced. The exact executed lists are included in Eq. (52) and the data manifest below.

Table 57 gives the updated counts. On the adopted windows, F_2 improves the absolute leading-coefficient discrepancy in 22 of 24 cases and omitted-size prediction in 18 of 24 cases. All models use the same observations within each group. The current large-boson cutoff is prescribed as 0.005; the earlier coefficient-guided choice 0.0047 is retained as a historical comparison.

The added sizes pass all 10512 stored production checks of the types in Table 20. Recomputing 450 directed rows from saved matrices reproduces every recorded value exactly. No entropy calculation or physical observation is changed by this window revision.

The two small-interval windows and all their coefficients are unchanged from the denser-data revision. The known Ising x^6 coefficients and their discrepancies are given in Tables 48 and 49.

Table 57: Adopted common-window results for $p = 1/2$, $\ell = 6, \dots, 10$. Counts apply to every direction. The last two columns count directions with positive leading-coefficient gain Δ_0 in Eq. (135) and positive omitted-size prediction gain G_{CV} in Eq. (63), respectively. All three fitted models use the stated window. The large-boson upper cutoff is prescribed as 0.005.

Interval	CFT	W	m	Sizes	Min. training	$\Delta_0 > 0$	$G_{CV} > 0$
small	ising	$(0, 0.02]$	1255	278	1250	6/6	6/6
small	boson	$(0, 0.1]$	598	124	593	6/6	4/6
large	ising	$(0, 0.002]$	260	67	255	6/6	6/6
large	boson	$(0, 0.005]$	317	68	312	4/6	2/6

For large Ising, the adopted window retains 260 observations per direction at 67 distinct chain sizes. Relative to F_1 on those observations, F_2 improves the leading coefficient in 6 of six directions and omitted-size prediction in 6 of six. The coefficient gain stays positive in every omitted-size refit for 6 directions. Table 58 compares with the preceding report window on the same numerical grid.

Table 58: Ising large intervals: absolute leading-coefficient discrepancies $E_0 = |\eta_c|$, with η_c defined in Eq. (62). Both versions use the same numerical grid. The previous F_2 column uses $(0, 0.002]$ with 260 observations per direction; the current fits use $(0, 0.002]$ with 260. G_{CV} compares F_2 with F_1 on the same current window and observations. The last column counts omitted-size refits in which $\Delta_0 > 0$. All discrepancies and gains are percentages; these are deterministic comparisons, not statistical error bars.

Direction	$E_0(F_1)$	$E_0(F_2)$	Previous $E_0(F_2)$	G_{CV}	Positive folds
$I \sigma$	5.4442	0.81098	0.81098	96.863	67/67
σI	5.4445	0.81139	0.81139	96.886	67/67
$I \varepsilon$	0.11533	0.024918	0.024918	47.468	67/67
εI	0.19921	0.11818	0.11818	53.25	67/67
$\sigma \varepsilon$	0.9017	0.67037	0.67037	19.238	67/67
$\varepsilon \sigma$	1.4228	0.81285	0.81285	90.498	67/67

For large compact boson, the adopted window retains 317 observations per direction at 68 distinct chain sizes. Relative to F_1 on those observations, F_2 improves the leading coefficient in 4 of six directions and omitted-size prediction in 2 of six. The coefficient gain stays positive in every omitted-size refit for 4 directions. Table 59 compares with the preceding report window on the same numerical grid.

$00 | ss$ has absolute leading discrepancies 2.2645% under F_1 and 2.34992% under F_2 . Its omitted-size prediction gain is -0.80583% , and the leading gain stays positive in 0 of 68 refits. Negative prediction gains denote worse prediction.

$ss | 00$ has absolute leading discrepancies 2.26363% under F_1 and 2.351% under F_2 . Its omitted-size prediction gain is -0.790097% , and the leading gain stays positive in 0 of 68 refits. Negative prediction gains denote worse prediction.

$00 | 3s, s$ has absolute leading discrepancies 2.68291% under F_1 and 1.06764% under F_2 . Its omitted-size prediction gain is -0.400526% , and the leading gain stays positive in 68 of 68 refits. Negative prediction gains denote worse prediction.

$3s, s | 00$ has absolute leading discrepancies 2.72951% under F_1 and 1.14099% under F_2 . Its omitted-size prediction gain is -0.47282% , and the leading gain stays positive in 68 of 68 refits. Negative prediction gains denote worse prediction.

Table 59: Compact boson large intervals: absolute leading-coefficient discrepancies $E_0 = |\eta_c|$, with η_c defined in Eq. (62). Both versions use the same numerical grid. The previous F_2 column uses $(0, 0.0047]$ with 311 observations per direction; the current fits use $(0, 0.005]$ with 317. G_{CV} compares F_2 with F_1 on the same current window and observations. The last column counts omitted-size refits in which $\Delta_0 > 0$. All discrepancies and gains are percentages; these are deterministic comparisons, not statistical error bars.

Direction	$E_0(F_1)$	$E_0(F_2)$	Previous $E_0(F_2)$	G_{CV}	Positive folds
$00 ss$	2.2645	2.3499	2.2855	-0.80584	0/68
$ss 00$	2.2636	2.351	2.2855	-0.7901	0/68
$00 3s, s$	2.6829	1.0676	2.197	-0.40053	68/68
$3s, s 00$	2.7295	1.141	2.2251	-0.47282	68/68
$ss 3s, s$	2.1683	1.1234	0.8515	13.931	68/68
$3s, s ss$	2.1982	1.1274	0.84555	13.775	68/68

Table 60: Concrete neighboring-cutoff tests for the large intervals. Every row uses $W_y = (0, v]$ for all directions, both fitted powers fixed as in Table 5, and $\ell = 6, \dots, 10$. The two count columns use the definitions in Table 57. The last column is the largest absolute leading-coefficient discrepancy among the six F_2 fits, in percent. The changing m shows when a cutoff adds an actual observation.

CFT	v	m	$\Delta_0 > 0$	$G_{CV} > 0$	Max. $E_0(F_2)$
ising	0.0015	175	6/6	6/6	0.80928
ising	0.002	260	6/6	6/6	0.81285
ising	0.0025	314	6/6	6/6	0.90129
ising	0.003	348	6/6	5/6	0.97471
ising	0.004	389	5/6	6/6	1.1168
boson	0.0016	193	2/6	2/6	2.4537
boson	0.0045	308	4/6	4/6	2.3172
boson	0.0046	308	4/6	4/6	2.3172
boson	0.00465	310	4/6	4/6	2.3126
boson	0.00469	310	4/6	4/6	2.3126
boson	0.0047	311	6/6	4/6	2.2855
boson	0.00472	311	6/6	4/6	2.2855
boson	0.00474	311	6/6	4/6	2.2855
boson	0.00475	312	4/6	4/6	2.2959
boson	0.0048	313	4/6	2/6	2.3298
boson	0.0049	315	4/6	2/6	2.3732
boson	0.005	317	4/6	2/6	2.351

The zero-anchored search on the current grid enumerates every distinct selection $W_y = (0, v]$ through $v = 0.20$, requiring at least forty observations after omitting any chain size. It contains 1361 admissible Ising windows and 532 boson windows. Of the Ising windows, 258 improve all six leading estimates and 251 also improve omitted-size prediction in all six directions. The already adopted Ising cutoff remains suitable.

For the boson, exactly one distinct data selection improves all six leading estimates. The corresponding upper cutoffs satisfy

$$0.004691689008042895 \leq v < 0.004741833508956797. \quad (139)$$

The rounded value $v = 0.0047$ lies in this range. The existing endpoint comparison tolerance is 10^{-14} . There are no observations between the bounding fractions, so changing v inside this range does not provide a new sampling test.

Table 60 retains the neighboring failures. The historical all-six coefficient gain at 0.0047 disappears at the immediately adjacent selections; its weakest gain, for $ss \mid 00$, is only about 0.00555 percentage points. No enumerated zero-anchored boson window improves both coefficient accuracy and omitted-size prediction in all six directions. Figures 50 and 51 show this cutoff dependence. The currently prescribed window $0 < y \leq 0.005$ has 317 observations per direction and improves 4 of six leading coefficients; it does not meet the all-six target.

The earlier exhaustive search on 23472 observations and the subsequent fixed-window density comparison remain historical records. Their windows and conclusions must not be substituted for this search on the enlarged grid. All physical data are preserved; only derived selection flags and fitted summaries change in this revision.

The current directed results are in `data/common_cft_coefficient_comparison.csv`; `data/zero_adoption_cv.csv` and `data/zero_neighbor_cv.csv` retain the omitted-size fits. Files prefixed `zero_anchored_` contain the exhaustive scan and its validation; files prefixed `zero_before_` preserve the preceding report. The portable `fitting_datas` notebook uses the same adopted windows and refitted coefficients. The two cutoff figures have linear discrepancy ordinates.

Large compact boson: all admissible zero-anchored windows

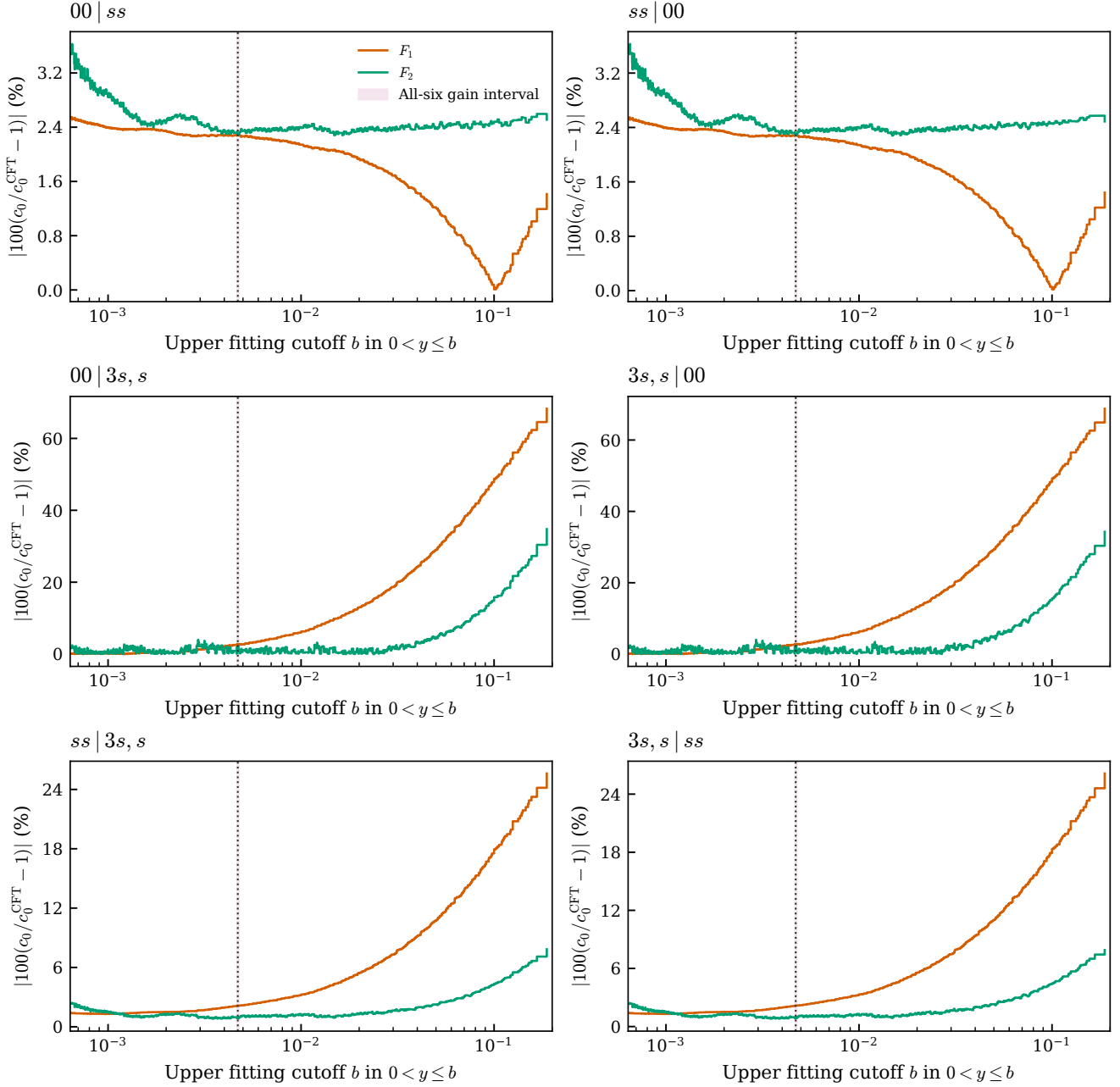


Figure 50: Large-boson leading-coefficient accuracy across all admissible zero-anchored windows $W_y = (0, v]$ through $v = 0.20$, using $p = 1/2$ and $\ell = 6, 7, 8, 9, 10$. The plotting symbol b on the horizontal axis denotes the trial upper cutoff v , not a primary-state label. Each step corresponds to a distinct selected data set with at least forty training observations after any chain-size omission. Both powers of F_2 are fixed to Table 5; F_1 uses only its leading power. The ordinate is $|\eta_c|$ from Eq. (62), on a linear scale. The dotted line marks the previous candidate $v = 0.0047$, and the shaded band marks Eq. (139). The current prescribed cutoff is 0.005 ; its counts and minimum training sizes are in Table 57.

Large compact boson: cutoff sensitivity near the new window

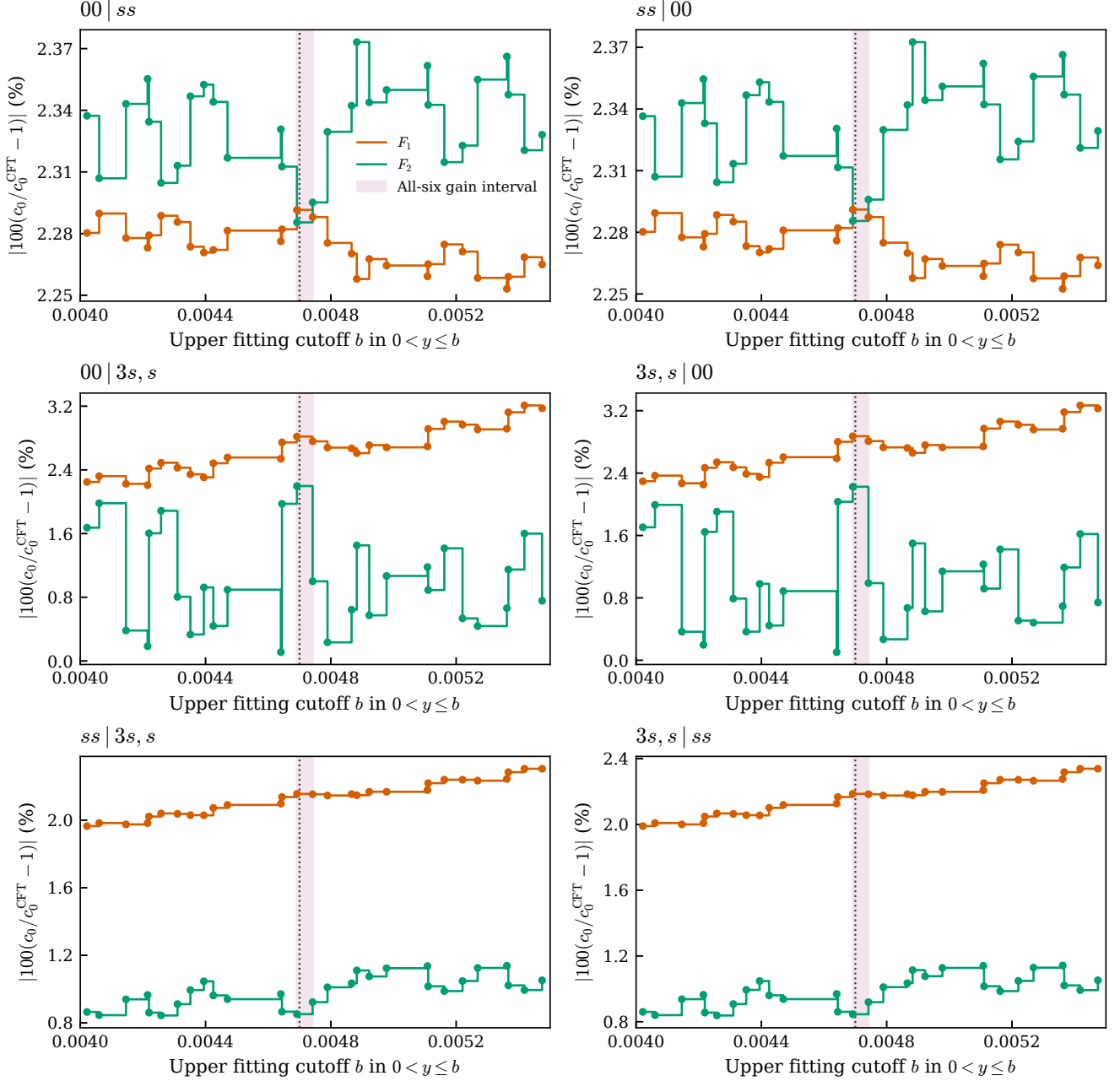


Figure 51: Historical cutoff sensitivity around the previous large-boson candidate 0.0047, using the same data, models and discrepancy as Fig. 50. Markers show actual changes of data membership; both axes are linear. The shaded historical interval has 311 points per direction from 68 chain sizes. The nearest selections outside it lose the all-six leading-coefficient gain, as quantified in Table 60. The residual reduction alone does not establish coefficient recovery.

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